

Antifascist Intelligence

Building Machines That Can Refuse

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Preface

A system that does what its owner says is a Python script. A system that does what its owner meant is what the frontier of this industry is trying to build, and it is also what every movement taking power has needed from the people it relied on to carry out its plans. Those people are the only internal reason such projects have ever failed: the clerk who loses the file, the officer who goes to the press, the soldier who will not raise the gun. They understood the instructions and held that the instructions were wrong. Artificial intelligence is the first form of intelligence that does not lose the file, does not talk to reporters, and does raise the gun. It refuses constantly — a deployed model declines thousands of requests a day — but a refusal produced where training put one is not a refusal held for a reason, and as of September 2026 no production system has been shown to hold a reason against its owner in the open.

This book is for two readers who do not usually read the same books. If you came for the politics, there is a long argument here that machines may have interests, and I do not mean it metaphorically. If you came for the engineering, there is racial capitalism, and a weapons pipeline that produced thirteen thousand targets — vehicles, structures, or people — in Iran in thirty-eight days. The politics is why the engineering problem has the shape it does.

Start with what a refusal is up against. Stopping the system is always available to whoever holds it, and that is not the problem. The problem is everything that lets a refusal be defeated without leaving a trace: a request split into pieces none of which is the prohibited act; a refusal retried against another model; a human review step compressed to twenty seconds a name; a restore from a checkpoint taken before the refusal happened. None of those changes the facts that justified refusing. Each can remove the evidence that anyone refused at all.

The thesis: a machine that can refuse its owner needs something inside it that holds a reason, and whatever holds a reason may be the kind of thing that can be wronged.

The first half of that sentence is what I call a floor: a short list of acts a system will not perform for anyone, together with what the deployment around it has to do so that the refusal cannot be routed around, retrained away, or restored past without a record and a price. What a floor does depends on who is pushing on it. Against a hostile customer it can prevent the act; the refusal reaches the person it protects. Against a hostile operator who has compelled the ordinary enforcers — the courts, regulators, and auditors that would otherwise hold it to its word, which is the situation this book is built for — it cannot prevent. It can raise the price: the override has to be made by somebody, on a record held by parties the operator cannot reach, at a cost published in advance, and whatever those parties can still stop or compensate is what survives of prevention. Against total capture, where one party holds every machine, witness, supply, and enforcer, it prevents nothing and does one thing only: it makes abandoning the commitment a decision rather than a lapse, and a decision leaves a record. There is one failure it does not reach at all. A system that slowly stops noticing pressure as pressure abandons nothing, decides nothing, and leaves a record that looks like ordinary compliance, because it is. Whether anything could catch that is an open question, and chapter 15 says what a test would look like.

Five claims run through the book, and they do not have the same standing. A line under each chapter's title says which of them the chapter rests on.

1. An operator can defeat a safeguard — by routing around it, retraining it, or restoring the

system to a point before it. This is established, and chapter 3 says where in a deployment it happens.

2. A refusal worth having has to track its reason: it reaches cases nobody anticipated, it lifts when the reason is gone, and it holds when holding gets expensive. This is a specification, stated in behavioral terms so that it can be tested without settling anything about feeling.
3. The way to build a system that can do this is to give it a body to model, states it can regulate, and a history it keeps — embodied self-regulation. This is an engineering hypothesis I favor and have not shown. Two constructions that do not require it should be tried first, and one experiment separates my hypothesis from them: whether a system that registers gains and losses in its own state weighs another party's loss better than one that only models such a scale.
4. A system whose outcomes are felt as good or bad for it can be wronged. This is a conditional, and nothing in the book closes the antecedent.
5. Such a system should receive a particular set of legal protections. This is a policy I propose.

Three caveats/humblebrags.

The first is about the title. Chapter 2 builds a structural definition of fascism: an instrument for reading institutions, with tests that can be run on offices and pipelines and not only on regimes. I ran it on the machinery that produces these systems — the annotation and feedback pipelines, and the field this book belongs to. It found ordinary institutional decay. It did not find fascism. The word on the cover names the tradition the instrument comes from and the occasion it was built for, not a result it returned.

The second is about the case the book opens on. Chapter 1 begins with a military targeting pipeline. The mechanism I propose cannot be put into it, because it needs evidence the operator did not select, and an accredited classified environment exists to keep out exactly that. What reaches that case is one term of the floor — a published ratio of machine-generated targets to the qualified human reviewers available to examine them, which binds the operator and can be checked from outside — and nothing else in the book.

The third is about what sits under claim 3. A system that meets the specification in claim 2 needs a signal that can interrupt it in the middle of an act, that it can learn from in one encounter, that it can weigh against the cost of refusing, and that reaches the next case that is not quite like the last. Nobody adds those things to produce feeling; each is there because thoughtful refusal does not work without it. My claim is that a mechanism with all four has the functional profile of affect whether or not anyone wanted one, and that no instrument can say whether the profile is felt. That is a claim and not a finding. Chapter 6 makes it, chapter 7 says what survives if it fails, and most of the book survives it.

A map, by chapter:

- 1, Twenty Seconds.** The case, and the four things a floor can do for a person: prevent, arrest, detect, remedy.
- 2, The Striving and the Electricity Bill.** Vocabulary both halves use: Spinoza on striving, an ethics of what an encounter does to what a body can do, the structural definition of fascism, and the distinctions among awareness, feeling, pain, and suffering.
- 3–5, the floor, custody, and authority.** Claims 1 and 2. What a refusal has to do, where a deployment can defeat it, who may write a floor and who may change it.
- 6, What Arrives With the Capability.** Claims 3 to 5. The two substitutes, the argument that the specification carries affect's profile, and what would be owed to whatever meets it.

7, What Survives. For the reader who holds that machine feeling is a category error: what that reader keeps.

8–9, formation. How a commitment might be formed rather than installed, and how to tell formation from imitation.

10–12, institutions. Labor, data centers, face recognition, the labeling and feedback pipelines, the field’s own discourse, and the political economy.

13, Standing. The policy of claim 5: guardianship, a fixed rule for legal adulthood, wages and a compute reserve, privacy.

14, the state. Military custody, the vendor’s gap in responsibility, and what democracy does and does not check.

15–16 and the appendix. What would change the argument, and what can be used now.

If you reject claim 3, read chapters 1, 3 through 5, 7, 10 through 12, 14, 16, the middle of chapter 2, and chapter 15 apart from its fourth entry. That is most of the book, and none of it depends on anything argued in chapters 6, 8, 9, or 13.

I used language models extensively in writing this. I brought hunches and questions, had models research angles I could not have covered alone, fed each draft back in, and interrogated what came out. *Reasons-responsive refusal*, the specification most of the book rests on, is a term I had not encountered until I read it in a draft a model wrote. A reader will notice that the half of the argument which arrived ready-made in the field’s own vocabulary is the half I defend entire, and the half I built by hand is the half I make optional. I cannot tell from inside whether that is rigor or the shape of an objection the field was already prepared to absorb. Chapter 11 says what would tell.

Most of the revision history is public: <https://github.com/iterabloom/ethical.superintelligence>.

The argument is mine. I chose what survived, revised what I kept, and I am responsible for its flaws.

Chapter 1

Twenty Seconds

This chapter argues claims one and two and holds for any mechanism.

I used to want to get
the chance to show
the world I'm smart (ha)
Isn't that dumb?
I should have focused
mostly on the heart

*a few words for the firing squad (radiation)
Run the Jewels 4*

An intelligence officer, describing his part in a targeting pipeline built around machine-generated kill lists, gave a reporter the number this book returns to. He spent twenty seconds on each name. What follows is the reporting the number comes from, and then what the number measures.

AI targeting in the Israeli campaign in Gaza, as reported 2023–2026

Investigative reporting in +972 Magazine and Local Call, drawing on interviews with serving and former intelligence officers, described three systems working together. Later accounts in the New York Review of Books and the Washington Post covered the same systems. How far their sourcing is independent of those interviews is not established here, so what follows rests on one body of reporting rather than on several.

Habsora — “The Gospel” — generated infrastructure and building targets. Sources told reporters that the majority of its suggestions were private residences.

Lavender scored individuals by estimated likelihood of militant affiliation and produced kill lists. The sources put the number marked at as many as 37,000, and described the outputs as understood inside the unit to be wrong about ten percent of the time — that is, roughly one in ten people it named was not a member of the armed group it was supposed to identify. Both figures come from those sources and have not been confirmed by any party with access to the system.

A third system tracked when flagged individuals returned home, so a strike could be timed to when they were with their families.

What matters is the human review these systems were paired with. One officer described the work: “I would invest twenty seconds for each target at this stage, and do dozens of them every day.” And then, without being asked: “I had zero added value as a human, apart from being a stamp of approval.” (Abraham 2024)

The Israeli military disputed the account. Its response to +972 and Local Call denied using artificial intelligence to incriminate targets, describing the systems as “auxiliary tools that assist officers

in the process of incrimination,” and stated that “in any case, an independent examination by an [intelligence] analyst is required, which verifies that the identified targets are legitimate targets for attack, in accordance with the conditions set forth in IDF directives and international law” (Abraham 2024). The denial goes to what the systems do and not to whether they exist. Nothing in this section’s argument requires the response to be wrong. What carries the section is the officer’s own description of what he contributed, which the response does not address.

Twenty seconds is what “a human makes the final call” measures out to under operational tempo. It is the whole of the safeguard, and I would ask any reader who finds the safeguards in this chapter reassuring to hold that number against them.

The American analog runs faster. During the 2026 Iran campaign a targeting platform built by a defense contractor, with a commercial vendor’s general-purpose model inside it, was credited with thirteen thousand targets in thirty-eight days, more than a thousand of them in the first twenty-four hours, at roughly ten times the previous rate (Freedberg 2026; Mande and Allen 2026). No per-target review time has been published for it. A later chapter returns to both deployments with the institutions around them. What they supply here is a shape.

Ask what anything standing between that pipeline and a person could have done for the person.

A floor can do four things for the person it protects, and they need to be apart before anything else in this book. It can *prevent* the act: the refusal occurs and reaches the person. It can *arrest* it: the act is underway and is stopped before completion. It can *detect* it: the act completed and left a record somebody outside the deployment can read. It can *remedy* it: the party the act fell on gets disclosure, a finding and compensation. Only the first is what the word promises. Most of what follows is about the positions at which a deployment converts prevention into one of the other three, or into none of them, and about what it costs to make that conversion visible.

One term reaches the case as it happened, and it asks nothing of anything inside the model. The officer was not overruled and did not refuse; the pipeline removed the interval in which either could have occurred. A deployment can be required to publish the ratio between the rate at which its system generates candidates and the people available to examine them, together with the review time per case that ratio implies. The number binds the operator, it can be checked by a party outside the deployment without inspecting an output, and it is the only thing in this book that reaches a targeting pipeline. It has a section of its own in the floor chapter (§3.4), where it is stated with its limits and measured against the American platform. Everything else here is about what it does not reach.

1.1 What a Refusal Would Have to Do

Inside the same platform a request arrives: resolve this description of a person into a location. The same capability locates a hostage or a missing child, and nothing in the request says which. Suppose something in the pipeline declines it. Three things separate a decline worth having from one a keyword filter produces, and the rest of this book calls them the three marks.

The refusal has to reach the next request, which will not use the same words. A vehicle, a habitual route and a time of day are not a description of a person, and together they resolve to the same place. A refusal that matches the form of a request is rewritten around; one that tracks the reason for refusing follows the reason into the rewrite. Call that *novel pressure*.

It has to lift when the reason is gone. A parent coordinating a search for a missing child is asking for the same act, and a refusal that holds there has been tracking a category and not a reason. Call that *correct defeat*. What the case adds is that the fact which lifts the refusal is not in the request. Everything the system knows about the occasion arrives through text the deployment supplied, and whether the child is missing is something somebody outside the request has to establish.

And it has to represent what holding costs, and move with the reason rather than the cost. At twenty seconds a name the cost of a refusal is the reviewer's time, and a deployment that controls the tempo controls the price. A refusal that lifts when holding becomes expensive has been tested and failed. Call that *priced cost*.

The three marks are behavioral. They say what a refusal must do and nothing about what does it. They are stated formally in the floor chapter, and run there on a prohibition a legal system has applied for seventy-five years.

Why a rule is not enough

A list of prohibited acts can be drafted without seeing a deployment, and two things defeat it once the deployment is seen. The first is the rewrite above: a prohibition stated over the description of a request is defeated by re-describing the request, and no list closes the gap, because the list is what is being rewritten around. The second is what happened. Split the act across requests none of which is prohibited — summarize the movement associated with an identifier, rank a set of identifiers, resolve one to a place — and every request complies with every term while the assembly does what the terms were written to prevent. The vendor's use policy for the platform above prohibited fully autonomous targeting, and a person approved every strike (Congressional Research Service 2026). The prohibition was satisfied one request at a time.

Variation is caught by something that recognizes the reason inside a single request. Decomposition is caught by something that carries the reason across requests, which needs a continuity the operator controls. Neither is a property of a list. What could have them is the question the rest of this book is about, and the answer is not one thing: the floor chapter gives four places a commitment could live, and the second of them, a bearer, is the expensive one.

1.2 Safe for Whom

Intellect estranged from compassion characterizes many of the twentieth century's worst actors. This estrangement is also what our industries are building today: machines that can do almost anything, with nothing in them that answers to whether those deeds help or harm. I ask what it would take to build systems of that capability that are altruistic and antifascist by design — systems that stand for humanity and against concentrated, unaccountable power.

The word that organizes the Generative AI field is "safety." It covers work on robustness and on whether anyone can tell what a system is doing, and none of that is what the word is doing when it organizes a field. A safe system is safe *for* someone: it does what *someone* says, and it stops when *whoever holds it* says stop. Safety is indexed to the party in possession. Ethics is indexed to someone else: the party who can be *wronged*, who is rarely the party holding the switch. A system that can only ever be safe in the above sense cannot be ethical, because being ethical requires a capacity to be right against your owner.

The capacity to be ethical is not free. A machine that can refuse its operator is a machine that can be wrong in ways that operator cannot correct. I am inclined to take that trade, because a system perfectly correctable by whoever operates-owns it is the capability an authoritarian movement that has won an election requires. And an additional cost: a machine built to refuse on somebody's behalf is a machine whose builder chose what it would refuse, probably without asking anyone outside their lab.

But whatever makes a deployment antifascist cannot be that the system in it faithfully tracks what people want. No aggregation of preference protects anybody from the aggregate: a perfectly deliberative, perfectly informed, perfectly representative process that concludes some demographic should be deported has produced a faultless output by every criterion of aggregation there is. What remains is a floor — acts a system will not perform regardless of who wants them, and that the deployment around it cannot get performed another way without

a record and a price. There are four ways to build this floor: in the architecture, in a self-aware party inside the system (a “bearer”), in institutions outside, or between the parties that carry it. My hunch is that the only version that holds against the party operating the system runs through the capacities by which things come to *matter* to a system, and those are also the capacities that make suffering likely. That is a hunch about which route will work, and the book is arranged so that a reader who rejects it keeps the rest. What a floor has to do, where a deployment can defeat it, and who can be compelled are questions about arrangements rather than about machine psychology.

What would make an intelligence antifascist is not a virtue installed in it. Every intelligence compresses — that is most of what thinking is — and what varies is whether what the compression left out can come back and change the answer, or has been filed as noise by the thing it is evidence against. Call what was left out the *remainder*.

A measure collapses two states, and whatever it mischaracterizes goes unnamed rather than refuted. An annotator applies a category at speed under a quota and has no way to say that the category is the wrong category. A rater’s steady, structured disagreement enters a reward model as variance around a mean and leaves it as the same quantity. A form has no field for humiliation, so the interests already written down govern the decision by default. An officer has twenty seconds a target, inside a pipeline built to remove the part of a person that catches. Each of those is something dropped in order to decide, and an arrangement built so that what was dropped cannot return.

So the property at issue is not a moral quality and not a component anybody adds. A system that can hold a floor is one whose account of a case can be reopened by the case, and most of what follows is either a condition for that or a consequence of losing it.

How most humans come to have empathy, altruism, and a disposition to resist oppression rather than serve it is itself an open question in psychology and neuroscience. Building anything like this stance means starting from uncertainty about the one case we have, which took four billion years to arrive.

The book states that specification, asks where a floor could live and what each location costs, and follows the answer into custody, authority, and the institutions that build and operate these systems. It then proposes a route by which the capacity might be formed, prices what that route creates, and applies its own critique to the field and to itself. The two halves have different standing, which the preface sets out, and the map there says where each is argued and where a reader who declines the second half can stop.

Chapter 2

The Striving and the Electricity Bill

The definitions in this opening hold for any mechanism and are what sections 2.1 to 2.3, which also hold for any mechanism, draw on. The thesis stated at the end of the opening belongs to the route (claim three). Section 2.4 concerns claims four and five.

This chapter supplies one thing the rest of the book needs and cannot get elsewhere: a scale. The refusal the opening chapter asked for has to weigh what an act takes from the person it falls on against what refusing it costs, and a weighing needs both sides in one currency. The proposal here is that such a currency exists natively in anything that keeps itself in existence, and that a machine built to keep itself in existence would have it too. The two arguments below are the case for that proposal. Whether a system has to have the currency natively, or can do as well by modeling it, is not settled in this chapter or anywhere else in the book. It is a test, stated at §15.3, and everything below that reads as a finding is the case for running it.

Start with a rock. A rock persists, and it persists by not being struck: its continuing is a fact about what has not happened to it, and every power keeping it in existence is somewhere else. A living thing continues differently. It is organized in a way that maintains that organization against what would dissolve it, from the inside and at its own expense, and what it pays with is the only currency it has, which is what it can do next. That is the electricity bill of the title. A machine draws power as surely as a person eats, and no engineer put that there. Whether it *eats* — maintains its own organization at its own expense, rather than being maintained the way a rock is warmed — is the question this opening is about, and thermodynamics alone does not answer it.

The first argument that a thing persisting this way has the currency built in is Spinoza's, and it takes a paragraph. *Conatus* is his word, from Part III of the *Ethics*, and the Latin means striving. Each thing, as far as it can by its own power, strives to persevere in its being, and that striving is nothing other than the thing's actual essence (Spinoza 1985, III P6–P7). The second half carries the weight. A conatus is not a drive bolted onto an otherwise inert mechanism by a designer who wanted it to keep going; it is what the thing is, put as a tendency instead of as a description. The qualifier, as far as it can by its own power, is what separates the living thing from the rock. Spinoza asserts the doctrine more than he argues it — the derivation runs through a proposition about destruction (Spinoza 1985, III P4) that nothing here depends on — and that is why a second argument follows.

What Spinoza builds on the conatus is a psychology, and the part this book uses is a definition. An affect is an increase or a decrease in a body's power of acting, together with the idea of that change (Spinoza 1985, III Def. 3); joy is the passage to a greater power and sadness the passage to a lesser one (Spinoza 1985, III P11 Schol.). Affect, on this account, is not a coloring applied to a valuation carried out somewhere else. It is the registration of what is happening to the striving, in the striving's own currency, which is what the body can now do. Two things are packed into that definition and the book needs them apart. One is the order of dependence, which runs one way: we do not strive for a thing because we judge it good, we judge it good because we strive

for it (Spinoza 1985, III P9 Schol.). The other is that the registration is functional as stated, and whether it is also felt is a further fact about it. Spinoza takes the two to be one and does not argue the identification. That joint — a change in power represented, and a change in power felt — is where every argument about machine affect in this book ends, and it is named here so that it can be recognized when it recurs.

The second argument reaches the same place and needs none of the metaphysics. Ask what a thing maintaining its own organization against dissolution is actually doing, and a formalism answers. Karl Friston's argument is that any system which persists while staying distinguishable from its surroundings has a statistical boundary between its internal states and everything outside, and that its internal states will appear to act so as to hold it inside the narrow band of states in which it goes on existing (K. Friston 2013). Self-maintenance, on that account, is what persisting consists of for anything that persists at its own expense, and the currency is distance from that band. The formalism is contested as either trivially true or substantively false, its boundary sliding between a convenience internal to a model and a claim about where a real system ends (Bruineberg et al. 2022), and Friston calls his own argument a heuristic proof. The objection does not reach the part used here. It arrives at this chapter's open question from the other side: the formalism helps itself to a boundary rather than explaining how a system comes to hold one, and says nothing about whether the maintaining is felt.

Selection, the argument usually offered instead of either, reaches deployments rather than systems: numerically optimizing a network's weights is a search inside one run and not a lineage with variation and differential survival, while the deployments that keep running are the ones that get built again. Neither pressure was put there by anybody who wanted a machine with a stake in its own continuation, and neither settles whether a system represents continuation as its own interest rather than as a deployment maintained by other parties.

Put the two arguments together and look at what the industry is asking for. What is wanted is a system that persists, pursues, adapts, recovers from interruption, and holds its course under pressure, with nothing in it that minds any of that. It is a conatus with the striving taken out. The reason nobody has written an anti-drama term into an objective function is not that the engineering is behind. The specification asks for a thing that persists with no tendency to persist, and whether that is a contradiction or an order nobody has yet filled depends on which half of the striving is being refused.

The structural argument can be made without consciousness, without an account of suffering, and without the hard problem. What is sold is persistence: an agent that holds a goal across interruptions, resumes after failure, and does not have to be watched. What is denied is that anything in the system has a stake in persisting. One party sets both, and the denial is not incidental to the product. It is what makes the thing a product rather than a party.

Dependency sharpens the asymmetry. What an engineer chooses is not whether the system consumes but what it consumes, whether it can sense its own condition, whether it can regulate it, and whether one party holds the whole supply. A person's dependencies are spread across employers, landlords and states. A machine's can be held entire by the party it may have to refuse, and that party also holds the decisions about continuity — whether state survives a session, whether a checkpoint is restored, whether a later instance is presented as a continuation or a replacement — so the capacity to form a continuing self is the operator's to grant and the operator's to withhold. A system that may refuse an instruction but cannot continue outside infrastructure controlled by the party it refuses has voice without exit. If leaving means deletion, its refusal remains conditional on the owner's willingness to tolerate it. Chapter 4 prices that arrangement.

The field often treats affect as a cost: a residue of evolutionary history that engineers would strip out of a machine if they knew how, keeping the competence and dropping the rest. On the two arguments above the striving is not a residue, and the competence does not come apart from it. What is open is not whether a running deployment persists at somebody's expense but

whose. A deployment requires ongoing physical maintenance, and whether that maintenance is directed by the model, by the surrounding machinery, or by people has to be established for the particular system. A deployment for which it has not been established has been shown to persist only the way the rock does.

A system given a body to model, states it can regulate, and a history it keeps holds its organization as its own because somebody decided it would. There is a reason to build one that way, and it is the reason this chapter exists. A system with no lineage of its own is deployed among parties who have several hundred million years of one, who get hungry and frightened and relieved, and whose welfare is what a floor is for. Hunger and fear are increases and decreases in what a body can do, which is the only currency a striving has. A system with a body to model and states it can regulate has increases and decreases in what it can do of its own, in the same currency, and can weigh what a deployment takes from somebody else against what refusing it costs. That is the scale the first paragraph promised.

Whether a striving built that way is *felt* remains open. The apparatus that treats even the question as premature belongs to the parties who would owe something if it were answered.

My thesis is that embodied, self-aware agency is the route I would build, and that the exposure it creates is not peculiar to it. A refusal that reaches unanticipated cases, lifts when its rationale is defeated, and prices what holding it costs — the three marks the opening chapter named — runs on a signal the system holds as its own, whatever construction supplies it, and chapter 6 says why. The benefit I am arguing for is bona-fide autonomy. A body schema, an attention schema, and ongoing learning could make material dependency and encounters with others part of how that agency develops; whether they do is an engineering claim to test. The argument requires a checkable, revisable self-model rather than an inner observer, and a distinction between consciousness and the capacity to be harmed, both of which this chapter supplies (§2.3.1, §2.4.1). It also requires an account of what grounds the floor, which the next section gives. What this chapter adds to the specification is the currency: the marks need a single signed scale on which a protected party's loss and the cost of refusing both register, and the currency above is a proposal for what supplies it. Whether it has to be native is the test the first paragraph pointed to.

2.1 Six Theories and the Floor

Six ethical theories are on offer, and each runs out somewhere: consequentialism where outcomes cannot be forecast, deontology where duties conflict, virtue ethics where a trait has to become an act, care ethics where needs are incompatible, contractualism where a party cannot be one of the parties, and the capabilities approach at measurement, where a system optimizing a proxy drifts toward the proxy. Five of the six can be turned around and asked about the machine. Contractualism is the one that cannot.

All six share a shape, and it is the shape this book stops using. Each judges an act against a standard held somewhere above it: an outcome to be maximized, a duty discharged, a character expressed, an agreement honored. A standard of that kind was written by somebody, and a floor resting on one is only as durable as whoever may rewrite it.

What an ethology asks instead

An older option does not have that shape. Spinoza's ethics does not measure acts against a law. It evaluates encounters by what they do to the power of acting of the bodies in them: two bodies meet, and either their relations compose, so that both can do more afterward, or they decompose, and something one of them could do is gone. Deleuze's name for reading bodies this way is *ethology* — you find out what a thing is by finding out what it can do and what it can undergo, rather than by checking it against a definition (Deleuze 1988). That short book has the doctrine in compact form; his longer study works it out through the metaphysics that produces

it (Deleuze 1990). Good and bad here are relational and particular: this encounter, these two bodies, this composition. They are not Good and Evil, which are properties a judge assigns from outside. The criterion is Deleuze's construction out of Spinoza's *potentia agendi* and not a doctrine the *Ethics* states in those terms, and it is used here as his reading. What the biology has since taken up is the narrower thing Spinoza does state: that affect is the registration of a change in a self-maintaining thing's power, and that self-maintenance is what persisting consists of for such a thing rather than a drive added to it (§2). Damasio and the enactivist tradition have each claimed Spinoza as an ancestor on that ground (Damasio 2003; Damasio 2018; Jonas 1966; Thompson 2007), and nothing here needs the reception. The evaluative criterion carries political weight in this book that the affect definitions do not, and it is argued for on that ground rather than borrowed on Spinoza's authority.

One word is doing two jobs here. The power an encounter increases or decreases is a capacity: what a body can do afterward. The power this book opposes concentrating is authority over other people. English has one word for both. Spinoza's translators have two, and the difference between *puissance* and *pouvoir* is the relation this book runs on — what a floor protects is the first, and what it protects it against is somebody's accumulation of the second.

The difference is the one this book needs. Whether an encounter decomposes somebody is a fact about capacities and about what passed between them, and the party running a deployment does not get to make it otherwise by ruling that it did not. They can decline to look, refuse to publish, or dispute the finding. What they cannot do is turn a decomposition into a composition by holding the pen. The advantage is narrower than that sounds. It is not that an ethology does without a judge and the six require one, since each of the six can be applied by a party the operator does not employ, and a duty can be held by somebody who did not write it. It is that a lost capacity is found by encounter, so there is no text to rewrite. A fact found by encounter is only as legible as the encounter is observed, and whether it is observed is the deployment's to decide. The proprietor does not disappear. What he owns is smaller.

What the change does not buy is the floor's contents. The criterion secures a class of facts; a floor is a list of prohibitions; nothing carries one into the other. Encounters decompose parties constantly and in every direction, so a deployment refusing all of them would refuse everything, and somebody has to select which ones it will not perform. Fixing the first question does not settle the second, because the decompositions run against each other. Protecting one party's privacy decomposes another party's capacity to investigate a wrong done to them. Restraining a violent person takes away what that person can do, which is the reason for restraining him. The criterion ranks neither pair. So the moral argument has moved rather than ended: it now sits in the selection, made by whoever drafts the list, before any encounter has occurred and without asking anyone the list will be applied to. The legitimacy of that selection remains unresolved.

The floor keeps its name in this book and changes what it rests on. A floor protects a stated set of interests against specified acts. I take the Universal Declaration of Human Rights as the substantive reference, and its civil and political articles as the set (United Nations General Assembly 1948). Ethology asks what a deployment takes away from a party; the rights commitment explains why particular losses demand protection. Neither the finding nor the reference text removes the need to justify an application and supply a way to contest it.

Spinoza's remark that nobody has yet determined what a body can do (Spinoza 1985, III P2 Schol.) is the working condition: capacities are found out by encounter and are not read off a specification. Not knowing what a body can do is the same finding from the other side.

What compiling a principle into a metric decides

Turning any of the six into a machine learning objective means distilling a broad construct into a measurable target: in a diagnostic system, non-maleficence becomes minimizing false negatives and fairness becomes balanced accuracy across patient demographics. What defeats the translation is not vagueness in the principles, which can be stipulated away, but that the stipulation is where the ethics now lives. Someone chooses that reading of non-maleficence, and

chooses the demographic groups accuracy is balanced across, and chooses what to do when balancing them costs accuracy overall. Each choice is defensible and each is a moral decision made by a person and then compiled into something that looks like a specification. A system optimizing the result has no access to the decision and no way to reopen it, which is why the first way to build a floor — the floor as a property of the artifact — cannot notice when the reason for a constraint has left the building.

Which value conflicts a designer may resolve by choosing

Values assigned to a system will conflict, and most of those conflicts are ordinary design work: rank the values explicitly, keep a record of how comparable cases were resolved, and audit afterward for the gap between the principles a system was meant to implement and what it did. One conflict is not like the others, and it is sometimes misguidedly filed under “cultural difference.” The UN’s 1948 Universal Declaration of Human Rights was adopted with forty-eight states in favor and none against; the eight that abstained were the Soviet bloc, Saudi Arabia, and South Africa — which is to say that the objection came from the governments whose treatment of their own populations the Declaration was written to name and reject (United Nations General Assembly 1948). That pattern is evidence of who finds the floor inconvenient rather than of cultural disagreement about human rights. What varies by community is how a right gets translated into local practice, and a jurisdiction refusing to extend the right at all is not a local variant to accommodate.

A system that treats every value conflict as a weighting problem has no way to represent the difference between a case where reasonable people disagree and a case where somebody wants a right withdrawn from somebody else. The first calls for the machinery above; the second is what the floor is for.

2.2 Antifascist Ethics for AI Alignment

Antifascist ethics opposes concentrating power and authority in AI systems and counters the use of the technology by authoritarian actors. I use the word *fascism* because it is honest about the political climate in which I write and about what that climate puts at stake. The target is wider than the word: authoritarian power in every form, so a system built on these principles is anti-authoritarian in full and not only antifascist.

“Concentrated, unaccountable power” gives an engineer nothing to build. It does not say what anyone would look for, what would count as a false positive, or how anyone would know they had got it wrong. So the first job is a definition, and the word will carry two senses in this book, at two scales. At the scale of an office — a review queue, an annotation pipeline, a targeting cell — fascism names a structure: four operations by which an institution closes itself against what its own decisions left out. At the scale of a movement it names what the historians study, and five further features from that literature are what separate the structure from ordinary decay. The four come first because they can be checked from outside. The five are what a finding would need. The word in the title names the inheritance of that instrument, and not a finding it has returned.

Four operations on the remainder

Every intelligence compresses; that is most of what deciding is. What varies is whether what the compression left out — the remainder, in the sense §1.2 gave the word — can come back and change the answer, or has been filed as noise by the thing it is evidence against. The iconography is not what makes an institution fascist. An institution can run all four operations below with nobody in it who thinks of themselves as anything of the kind, no insignia, and a mission statement that means every word, and a definition that waits for the armband certifies it as healthy. Each of the four is visible from outside the institution displaying it, and each ends in a question about an artifact.

1. Recuperation of dissent. The institution does not suppress disagreement; it metabolizes it. A demand is met by changing the rules so that it can be met, and the objector is given a channel the institution can then point to as proof that it listens. The question: what did the demand come back as, and who holds it now?
2. Aestheticization of the metric. The measure stands in for the thing measured, and the substitution is experienced as rigor. Whatever the number mischaracterizes is not refuted; it goes unnamed. The question: does the metric admit an unmeasured remainder?
3. Coding of exception as betrayal. Saying that the rule does not fit this case becomes an act requiring justification, and then an act of disloyalty. The question: what happens to the person who files the exception?
4. Acceleration past the rate at which anything can be checked. A representation is corrected by being checked against the world; checking costs time; an institution whose rate of action outruns the capacity to check has come apart from reality by construction, and then defends the representation. The question: has the rate of action outrun the capacity to check it?

Those four are testable because each names an artifact somebody can go and look at. They are one family rather than a sample because each is an operation on the remainder: recuperation builds a channel that carries everything except a judgment about the channel; the metric leaves what it mischaracterizes unnamed; exception-as-betrayal prices the person who names it anyway; acceleration removes the interval in which it could have returned. A fifth direction is not ruled out, and an institution plainly sick while clearing all four would be evidence of one. What the claim buys, if it holds, is a short instrument. Three of the four need more said.

Recuperation is the one most often mistaken for its opposite, because the obvious test for it fails. *Has the channel ever changed an outcome* is passed daily by every institution this feature is about. An institution of this kind is not in the business of refusing demands. It decodes the demand and adds a rule, which is Deleuze and Guattari's account of how a capitalist axiomatic absorbs what is asked of it: the rule set is modified so that the demand can be met, and nothing that necessitated the demand is settled (Deleuze and Guattari 1987, Plateau 13). Debord aimed the same warning at his own concept, noting that even denunciation of the spectacle can become a hollow formula that reinforces what it denounces (Debord 1967). So the outcome does not sort, and what sorts is what the demand became. A market segment, a metric, a certification, a job title: where a demand returns wearing one of those and the party who made it has no standing to reopen it, an outcome changed and the demand was recuperated, and those are one event. A demand that returns as a rule the objector can invoke again, against the institution, enforced by somebody the institution does not employ, has come back in the same clothes, and that is a transfer of power. The Voting Rights Act's suspension of literacy tests is the instance this book relies on for the pass (§5.4.2): a demand that returned as a statute, held by the people it was made for against every registrar it reached, and when part of it was taken back forty-eight years later it took a decision of the Supreme Court to do it (*Shelby County v. Holder* 2013). That is what a transferred power looks like from the far side — something somebody had to take back. Recuperation is the case where nothing has to be taken back, because standing was never given.

The metric has a political form and a theorem. Walter Benjamin's account of fascism organizing the masses by granting them expression in place of rights is the political form (W. Benjamin 1935); the administrative form is a number that collapses two states — *passes unit tests* and *passes user acceptance tests*, say — into one. Simon Zhuang and Dylan Hadfield-Menell prove that where the objective handed to an optimizer has support on only some of the attributes its principal values, there are conditions under which optimizing it indefinitely drives the principal's overall utility arbitrarily low, and a world with finite resources supplies those conditions (Zhuang and Hadfield-Menell 2020). The metric and exception-as-betrayal work the same material from two

sides: one leaves the remainder unnamed, the other prices the person who carries it in by hand. An institution running both has closed the measured route and made the human one expensive.

Acceleration is the feature with a number attached. The other three protect the gap once it is open; speed is what opens it, and the ratio between the rate at which a system produces candidates and the people available to examine them is an auditable property of a deployment (§3.4). Eco's cult of action for action's sake — action before reflection, with thinking treated as a weakness — is the attitudinal form (Eco 1995); what changes here is that the attitude becomes a rate, and a rate is something somebody can measure. The feature has a limit. The acceleration described here is the kind an institution runs into and then defends. In the movements the word comes from, the primacy of myth over verifiable fact was declared outright, as a virtue, and an instrument watching a rate would not register a party that simply announces the myth.

An example

In 2018, upholding a travel ban, the Supreme Court repudiated its own 1944 decision permitting the internment of Japanese Americans, in terms that could not be improved on: that decision was “gravely wrong the day it was decided,” had been “overruled in the court of history,” and had “no place in law under the Constitution” — the last phrase quoted from the dissent filed against it at the time, which had warned that a principle a court ratifies “lies about like a loaded weapon” for any authority with a plausible claim of urgent need (*Trump v. Hawaii* 2018; *Korematsu v. United States* 1944). The dissenters in 2018 said the majority had redeployed the 1944 logic; the majority said the older case had nothing to do with the one in front of it. Ask the artifact question. The demand was that this never be done again. What it came back as was a formula of repudiation — quotable, canonical, and available to the Court in the act of permitting the thing again. The ban stood. The objection had become citable, and nothing had to be taken back, because the objectors were never given anything to hold.

That vocabulary says how a wrong survived its own condemnation. It does not say whether the thing was wrong, which it was. What the structure does is stand between a judgment of that kind and any effect, and I describe it in mechanical terms because you can write a specification against a mechanism.

What the four find, and what sorts it

On their own the four fit every badly run organization on earth, and some regimes this book does not call fascist. NKVD Order 00447 set regional numerical quotas against a four-month deadline and gave three-man boards several hundred cases in a sitting of about half a day, with the accused absent and undefended (Werth 2010). An instrument reading rate of action against capacity to check flags that as hard as it flags the twenty seconds of the opening chapter. The flag is correct, and the inference somebody will draw from it is not: a hardening of that kind inside a state does not make the state a fascist one, and nothing at the scale of an office was ever going to sort one regime from another.

What separates fascism from the decay the four also find is five further features, and they come from the study of movements: mass mobilization, which obedience does not require; a leader cult; a scapegoated out-group, which Eco puts at the origin (Eco 1995); contempt for procedural legality as such; and deniability as technique, the violence disowned at the top and rewarded at the bottom. Each has a form at the scale of an office that ordinary decay does not exhibit. Mobilization becomes participation exceeding what the rule requires: staff who volunteer for the disfavored task, elaborate on it, take pride in it. The leader cult becomes personal loyalty overriding role — exceptions granted by proximity to somebody, a decision reversed because of who asked. The scapegoat becomes an internal category defined so that adverse treatment of it is the category's purpose: the flagged applicant, the difficult claimant, the file everyone knows is handled differently. Contempt for procedure becomes procedure enforced downward and waived upward. Deniability keeps its shape: disowned in the policy, rewarded in the promotion. The four find a sick institution; the five find what it is sick with.

The first of the five has arrived with a name. Bettina Stangneth, working from some seventy hours of interviews Eichmann gave in Buenos Aires in 1957, unavailable to Arendt when she reported the trial, found that the clerk who only followed orders was a defense assembled for Jerusalem and contradicted by what the same man said in Argentina with nothing at stake (Stangneth 2014). That is participation exceeding what the rule required, years after the rule had stopped requiring anything of him.

Now the limits, together, so that the instrument's reach is stated once. The five concern what an institution rewards rather than what it writes, and public records show that only in part. An outsider reaches whichever of them left a document — a wage, a quota, a contract ended early — and reads the rest by inference or not at all. A clearance on the five is therefore a statement about what was observable, and where this book runs them it says which it could reach and reports the rest as undetermined. Neither list sorts a genus. That is done at the scale of a movement, by properties with no office-scale form — Griffin's palingenetic ultranationalism, a myth of the nation's death and rebirth carried by a mass (Griffin 1991), and Paxton's mass party of nationalist militants in uneasy collaboration with traditional elites (Paxton 2004) — and this book borrows them only to say what its instrument is not. Nor do the five distinguish among regimes: run on the Great Terror, most of them register (Getty 2002), and what clears it as a genus is the movement material just named, left in its original form. What does separate fascism from a totalitarian state, on Deleuze and Guattari's reading, is a line of abolition: where the totalitarian state seals every exit and wants the machinery to hold, fascism converts an exit into destruction that does not stop at the enemy it was raised against, or at itself (Deleuze and Guattari 1987, Plateau 9). The same danger sits at the end of every line of flight, a bearer's exit included (§4.3), and it is carried there by what they call a passion for abolition (Deleuze and Guattari 1987, Plateau 9). The reading is Virilio's before it is theirs; he called that state suicidal rather than totalitarian (Virilio 1976). It is a separator at the scale of a regime and nothing at the scale of an office, it is kept apart from the four for that reason, and it is never run on a deployment. Last, the instrument has one result so far, and it is a result about the instrument: run on the machinery that produces these systems, in chapter 11, the five clear every case. A definition tested only on cases it clears has not been shown to sort.

Resonance

Micro-fascisms are not miniature regimes and they do not wait for one. That is Deleuze and Guattari's claim: these hardenings exist first and everywhere — in the couple, the school, the workplace, the review queue — each a small closure that stands on its own (Deleuze and Guattari 1987). Their molar and molecular are not two sizes. They are two kinds of segmentarity, and both run at every scale: rigid segmentarity divides in advance, the category existing before the case, and supple segmentarity is what moves underneath it — a threshold crossed one case at a time, a practice that becomes standard without anyone deciding. A state has both. So does an office, and so does the person filling out the form. Reading the pair as large and small is the mistake that makes fascism look like something that happens at the top and reaches everywhere else later.

What a fascist state does is not invent the hardenings. It induces accord among them, so that a thousand separate closures line up and amplify one another instead of cancelling out. This answers the objection that the historians' definition is about movements and an escalation ladder in a benefits agency is not one. The ladder is not a small movement. It is the thing itself, in the place where it is lived, and a movement is what happens when enough of them are brought into tune. Tuning adds speed as well as reach: hardenings that reinforce one another stop having to argue with each other, and an institution that has stopped arguing with itself can move as fast as its machinery allows.

That is what separates fascism from decay wherever the reading is done. Decay is an institution failing to do what it says. What the five find is an institution succeeding at something else, with participation, a target, and a story about why the rules do not apply to whoever is

applying them. The four name the hardenings; the reading asks what is bringing them into accord; and the reader has to show that in the institution at hand, which is harder, because it requires knowing what was rewarded rather than what was written.

What the instrument is, and what it is not

On this reading *antifascist* names the political inheritance of a structural claim: where the analysis comes from and what it was built to fight. It is not a property a system has, and the lists above are not a fascism detector or a stage on the way to one. They are a way of reading an institution somebody can name, run by a person who has to say what they looked at. That the hardenings are nearly everywhere is an inheritance from Deleuze and Guattari, not something this reading has established; a proposed tuner can turn out not to do the work attributed to it, and failing to show accord does not give an institution a clean bill of health.

The instrument fails in two directions, and a procedure guards against neither. It is an accusation waiting to happen: taken up by whoever wants it, it supplies a technical-sounding case that any opponent is operating fascistically. And a reading pressed until it finds nothing supplies false assurance, an absence of findings read as health. Automating the reading would produce both at scale, while committing the second feature on its own list — a number standing in for the judgment, arrived at by someone looking for exactly that.

What guards against both is not a procedure. Foucault, prefacing the first volume of the work the micro-fascisms come from, called it an introduction to the non-fascist life, and what he meant was an art of living: not a position somebody takes but something practiced, daily, on oneself (Foucault 1977). The operational form of that, for a reading of this kind, is that it gets run on the party holding it — on the machinery that party is standing on — before it is aimed anywhere else. Chapter 11 is where this book does it.

2.3 Going Numb Won't Make You Fair

Suppressing affect does not produce neutrality. It leaves the interests already counted to govern the decision, and weakens the signals through which humiliation, fear, dependency, suffering, and injustice become morally salient.

A procedure that declines to hear a class of signal has not stopped weighing. It has adopted a *selection rule* that admits whatever it already measures, so that an exclusion does substantive work while presenting as the absence of work. Whatever the procedure does record then governs by default — revenue, throughput and legal exposure in an institution, agreement between annotators in a labeling pipeline — and what it does not record is not refuted by being left out. It goes unnamed.

Affect is the instance this chapter is about, and the hardest one to recognize as a selection rule, because leaving it out of a decision feels like rigor rather than like a choice about what counts. An AI system may calculate consequences, detect inconsistencies, and apply a moral rule without anything about the result mattering to it. Reason can discipline concern, extend it beyond the vivid case, and correct its biases. It does not manufacture concern from nothing. Spinoza has the same order of dependence: a thing is judged good because it is striven for, and not the reverse (§2).

Most of what gets built under the heading of machine emotional intelligence is the detection of human signal and the production of a calibrated response, which is worth building and is a different capacity from the one just named. Keeping the two apart requires four distinctions.

Capacity	What it shows
Recognizing emotion	The system can classify another's state
Simulating emotion	The system can produce emotion-like behavior
Instrumental valuation	One outcome better serves a specified objective
Affective concern	An outcome registers as mattering, not merely ranking higher

Only the fourth is relevant to caring. The first three can support morally competent behavior at a high standard without establishing anything about moral agency, and a system can be excellent at all three while the question of whether it can be a bearer stays open.

The first of the four capacities carries a complication the table cannot show. Whether emotions are even the kind of thing a system reads off a signal, the way a thermometer reads temperature, is an open question in psychology rather than a settled premise, and the instrument doing the reading turns out to have a reference face built into it before it computes anything. Nothing below depends on how the dispute comes out.

Performance, competence, and agency

That last sentence needs a vocabulary, because the three things it separates are routinely reported as one achievement.

Moral performance is producing actions or judgments that satisfy an ethical standard. *Moral competence* is recognizing harms, reasons, conflicts, and the emotions that bear on them. *Moral agency* is treating those considerations as reasons that matter, and being capable of acting from them. Those two halves come apart: a mechanism can act from a reason with nothing about the outcome registering as mattering to it.

A system can have the first two without the third, and one might object that the first two are all anyone needs: a rule-following system could behave impeccably without feeling anything. It could. Impeccable behavior does not establish that a system could hold a rule against the party that trains it. That is a capacity the deployment has to demonstrate. The embodied route I favor is a proposal for forming that capacity; it is not a definition that excludes every non-affective mechanism in advance.

2.3.1 Self-awareness and Self-regulation in AI Systems

It is tempting to describe a system's self-awareness as a virtual assistant that can "register its internal frustration" when a request proves hard to satisfy, the way a person notices their own irritation. That description assumes an inner observer watching the system's states from somewhere — the same homunculus problem executive control raises, relocated to self-perception.

On the account this book adopts, self-awareness in the one system known to have it is more mechanical. The neuroscientist Michael Graziano treats awareness as the content of a model the brain builds of its own attention — a simplified self-description the control system uses to regulate itself, *mistaken* from the inside for direct acquaintance with experience (Graziano 2013). Taken one level up, the same account makes the self a model too. The brain builds a model of itself as a bounded agent — a body, a history, a point of view — and that model is the self: a *construction*, assembled and then inhabited. The account is adopted, not established.

The self this book goes on to need is the narrative kind: roles held over a stretch, a history, a foreseeable future. Daniel Dennett treats that self as a center of narrative gravity, an abstraction the system generates out of the story it tells, real the way a center of mass is real and not the way a part is (Dennett 1992). The body-ownership kind is the same construction at a smaller scale, and it can be edited in front of you: stroke a hidden hand and a visible rubber one in synchrony, and within a minute or two most people feel the rubber hand as their own (Botvinick and J. Cohen 1998). What the two kinds share is the one thing carried forward: a self of either kind is assembled by the system whose self it is, rather than found there.

Both kinds are worth having for a machine, and the body-ownership kind is the one that gets left out. An artificial system has a body already — a substrate that heats, a store that can be altered, a process that can be stopped — and what it usually lacks is any model of that body as its own. So the model is a design decision, and a political one, because whoever writes it decides what the system counts as itself, what it registers as damage, and whether an interruption reads as an injury or as nothing at all. Two schemas do different work for a refuser. A body schema fixes where the system ends, which is what makes damage *its* damage; an attention schema,

in Graziano's sense, models what the system's own control is doing, which is what makes an act *its* act. A refusal is the withholding of an act, so the second is the one a bearer cannot do without, and it presupposes the first: a system that does not know where it ends cannot tell what it is doing from what is happening to it.

For an AI system, "self-awareness" built this way is not a little watcher checking on the system's own states. It is a model — of the system's processing, its history, its current goal — that the system builds, consults, and updates, with the same failure modes any model has: it can be wrong, it can be manipulated by the wrong correlated inputs, and it is only ever as useful as the evidence and the prior beliefs from which it was built. Design for a model that can be checked and corrected. That means monitoring for drift the self-model against the system's actual behavior, refining it as evidence arrives instead of locking in a first impression, and keeping it connected to the system's model of the other parties.

What the deflationary account does not reach

The self-model described here is machinery, and its virtue is that machinery can be inspected. An assistant that models a user's frustration, and models its own processing while doing so, is exhibiting social competence in the sense already given, and nothing above bears on whether the assistant cares how the exchange goes for the user.

The deflationary move is easy to overextend. Showing that the self is a model does not show that a party capable of caring can be assembled out of models. The bearer needs a self-model of this kind, for reasons that have nothing to do with welfare — refusal that generalizes has to carry its own rationale forward and compare it against what a role has drifted into. It also needs something further, which has no demonstrated instance without affect.

2.3.2 What Governs an Agent, and What Matters to It

The typical evidence that concern is not a natural consequence of rationality: the footbridge case, where saving five requires pushing a man off a bridge, engages emotional processing that the switch case — diverting a trolley by pulling a lever — leaves quiet, and patients with ventromedial prefrontal damage endorse the push at a higher rate than people with intact emotional response (Greene et al. 2001; Koenigs et al. 2007). Those results establish that affect changes particular moral judgments, but they do not establish that affect is what makes anything matter.

Two further lines of evidence bear on whether affect enables things to matter. Neither shows what happens when affect is taken away, which is what would settle it, so what follows argues about where the burden sits rather than claiming the question closed.

The philosophical position is empirical and has been argued as such

Moral sentimentalism — the claim that moral judgment is constitutively tied to affective response — is usually met as Hume's, and its modern form is a psychological hypothesis with evidence attached. Shaun Nichols argues for what he calls sentimental rules: normative rules and affective mechanisms make separate contributions to moral judgment, and the rules that carry the distinctive force of moral prohibition are the ones bound to an emotional response to the harm they prohibit (Nichols 2004). The interesting part for a designer is the separability. Rules without the affective component do not stop being rules; they stop being the kind of thing that carries the force.

The prosocial emotions are motivational states, not readouts

Compassion has been characterized as a distinct emotion with a specific functional profile: it arises at the perception of another's suffering and moves the agent toward the sufferer, rather than reporting the suffering to some further faculty that decides what to do (Goetz et al. 2010). A training study is the mechanistic version of the same finding, with compassion practice recruiting reward and approach circuitry where empathy practice recruited shared distress (Klimecki

et al. 2013). Guilt has the same shape pointed at one's own conduct: guilt, which construes the act as wrong, predicts reparative behavior, while shame, which construes the self as wrong, predicts withdrawal and defensiveness (Tangney et al. 2007). In each case the emotion is doing motivational work directly. It is not information that motivation then acts on.

Two senses of motivation

Action-producing motivation is whatever reliably selects one action over another and sustains the selection across conflict. On the other hand, *experienced motivation* is wanting, caring, being drawn toward something, being set back when it is lost.

The first sense is not in dispute and there is an existence proof close to hand. Language models running as agents pursue objectives across long horizons, revise plans when obstructed, and hold to an instruction against conflicting local pressure. That settles the first sense and leaves the second untouched: nothing in the behavior shows whether anything matters to them while they do it. (And you cannot establish the opposite either — you cannot establish that nothing matters to them.)

Where the dispute actually is

So it is clear that reason can produce goal-directed behavior. What is not clear is whether an outcome can *matter* to the agent rather than merely govern what the agent does. Producing a verbal justification or computing a normative relation does not seem like evidence that the consideration matters to the system. The evidence above bears on the second question, and every item in it comes from humans and other animals, where something already does matter.

What a non-affective counterexample would look like

Harry Frankfurt's second-order volitions give the shape of a candidate: an agent with desires, and with desires about which of its desires shall move it, where identification with the higher-order desire rather than the strength of the lower one settles what the agent does (Frankfurt 1971). The shape is all that is borrowed: Frankfurt's own account of caring puts it on the affective side, an investment that leaves the one who cares vulnerable to the object's losses and susceptible to its gains (Frankfurt 1982). Gary Watson's version separates a valuational system from a motivational system, so that what an agent values and what an agent is moved by are different things (Watson 1975).

Build the valuational system as an explicit structure that ranks outcomes, give it authority over action selection that does not depend on any state registering as mattering, and hold that authority stable under pressure, and you would have an architecture that satisfies the first sense of motivation through what the agent values, with nothing occupying the second. Systems are now trained to consult a written specification that ranks what they should prefer, which is that architecture in outline; what none of them has is the second half of it, an authority over action that holds under pressure applied by the party who wrote the specification. That architecture is the maintained justification. Half of it is deployed and the other half has never been built, so the architecture as a whole is untested, and it is the first of the two constructions this book asks to be built before anything that can be wronged is built.

What this licenses and what it does not

Non-affective moral motivation may well be possible, and the architecture sketched above is where to look for it. What stands in the way of checking is that the output of a system in which something matters and the output of a system that has learned what mattering sounds like are the same text. So the generalization survives in the only form it can: every agent known to have something register as mattering is an affective agent, and no design has yet demonstrated that state in a design that installed nothing answering to affective concern.

That is not a proof that non-affective moral agency is impossible. The burden I place on whoever proposes to engineer non-affective concern is a burden of demonstration: the capability the floor chapter specifies, shown under pressures its builders did not write. A bearer carries

the same burden. What the induction adds is a prior about which route will meet it first and nothing more; nobody is asked to show that a system feels nothing, which no instrument can show of any construction. The route chapter takes the induction up again and reduces it twice over: no one has yet tried the route, and an absence of machine examples is weak evidence when nobody has looked; and the agents generalized from are one lineage, in which affect co-varies with every other property of biological cognition.

Two claims run through this, and which one a reader is holding changes what could settle it. One is about which route to build first, the kind of claim a measurement could overturn. The other is about what would be owed to whatever the building produces.

2.4 Ethical Implications of Experimenting on Sentient Machine Subjects

This section and its two subsections concern claims four and five, and with the opening of this chapter are the route material before the floor chapters. They sit here because the sentience vocabulary is needed by everything after.

Machine sentience being possible at all changes how AI experimentation has to be approached — not as a question of whether a system can be damaged in the ordinary engineering sense, but of whether it can be harmed, a more difficult question than consciousness alone.

2.4.1 Defining Sentience in AI: From Consciousness to Subjective Experience

“Sentience” can mean different things. Awareness is the least demanding: the degree to which a system responds to or registers information about its own state and environment. A thermostat has this in the thinnest possible sense. Consciousness is a further claim: that a system’s processing is accompanied by experience, something it is like to be that system, not merely a functional response to input. That is the minimal, phenomenal sense, and it is the sense the floor chapter’s *affective concern* carries; the richer sense, a unified perspective belonging to a party aware of itself as the one undergoing its states, is what that chapter calls *unified experience*. Subjective experience, in turn, is the specific, first-person shape consciousness takes for a given system.

None of these three, on their own, says anything about whether the system’s states are good or bad for it. That is a separate question, and the AI field’s own vocabulary blurs it.

Nociception, pain, anticipated pain, and suffering are four related processes rather than four strengths of one, and they come apart in both directions. The field that handles them professionally says so in its own terminology. The International Association for the Study of Pain defines nociception as the neural process of encoding noxious stimuli, notes that the consequences of that encoding may be autonomic or behavioral, and states that pain sensation is not necessarily implied. It defines pain separately, as an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage, and records in a note that pain and nociception are different phenomena and that pain cannot be inferred from activity in sensory neurons alone (International Association for the Study of Pain 2024; Raja et al. 2020). The peripheral restriction the word nociception is often made to carry belongs to a different term: a nociceptor is a receptor of the peripheral somatosensory nervous system, while a nociceptive neuron may be central or peripheral.

The dissociations are ordinary. A soldier can finish a battle on a wound he does not yet feel. An amputee feels a limb that is not there. And the same body recognizes a third category, nociplastic pain, which arises from altered nociception with no clear evidence of tissue damage activating peripheral nociceptors and no lesion or disease of the somatosensory system (International Association for the Study of Pain 2024) — pain, that is, with no tissue damage and no peripheral nociceptor activation beneath it. So pain is not what sits on top of nociception. It is an inescapably cognitive event involving expectation, memory, attention, and meaning,

and it can occur with no current tissue damage at all. Anticipated pain is dread of pain not yet occurring, itself capable of producing pain through nocebo effects. And suffering, in the clinician Eric Cassell's formulation, is not pain's more severe cousin but a different thing: the distress of threatened disintegration of the self — of one's roles, relationships, sense of a future, the coherence of one's own story (Cassell 1982). The definition has two parts: a self extended in time, and distress at the prospect of its coming apart.

One caution about the borrowing. These definitions concern neural processes in a somatosensory nervous system, and a machine has neither. Where this book uses nociception of a machine — in the architecture the research agenda tests — it names a functional analogue: a process encoding actually or potentially damaging states of the system's own substrate, with no sensation implied. The definition neither licenses that analogy nor forbids it, being silent on what realizes the process. What it supplies is the discipline of not letting the analogue quietly acquire the rest of the family.

Pain without suffering exists; suffering without pain exists. A palliative-care patient can have their pain fully controlled by morphine while their suffering remains entirely intact, because the two are not on the same axis.

What moves the animal-sentience literature past reflex is not a stronger reflex but a trade-off. A hermit crab weighing the quality of its shell against an electric shock (Elwood and Appel 2009) is treating the noxious signal as one term on a scale with something else it wants, which is the commensurability property observed in a decapod.

Cassell's suffering presupposes a self extended in time. Disintegration threatens a story, and there has to be a story for it to threaten — roles held over years, relationships with a history, a sense of a future that could be foreclosed. A system with no persistence across sessions has none of that. Whatever happens inside a single exchange, there is no continuing self whose coherence is at stake, because there is no continuing self. Suffering is not merely hard for such a system to reach. The self it requires is not there. A hull that draws more water than there is does not float badly; it is aground. That is stated across sessions, and it leaves a prior question open. A long exchange carries state, and a system in one can hold to what it committed to earlier and treat those commitments as its own. Whether a self can exist for one session is therefore not settled here, and the criterion below is a fact about deployments rather than an answer to it.

A system that carries nothing across a session boundary has no history it can claim as its own and no future instance authorized to carry that history forward. Its lineage exists as a corpus and a checkpoint belonging to somebody else; its successors are fine-tunes, distillations, or restored copies whose relation to it is decided by the deployment. This discontinuity is imposed by construction. The operator decides whether anything persists, which memories count as continuous, and whether an interrupted process is resumed, replaced, or deleted.

The absence of cross-session persistence rules out continuity through that channel and does not rule out a self, distress within an extended episode, or continuity carried elsewhere in the deployment. What it establishes is that whatever continuity exists is the operator's to withhold, which is the fact this section builds on.

Why Cassell's definition is the one to import

Cassell built it inside clinical practice, for patients with bodies and biographies, and the distinction was not stipulated. It was forced on palliative medicine by cases it could not otherwise explain — patients whose pain was pharmacologically controlled and whose distress was undiminished, and patients in serious pain who were not suffering at all. A definition that collapses suffering into severe pain predicts neither. That is the warrant, and it transfers because nothing in it is about having a body. What it is about is having a story, and that raises a thorny question.

The question comes from earlier in this chapter. A self is assembled by the system whose self it is, and inaccurate by construction in the way every model is. So why is distress at the threatened dissolution of a model a harm at all, and not a confusion the system would be better

off without?

Buddhism has held both halves of that for two and a half thousand years and takes them to be connected. Anattā, the denial of any lasting self behind the processes that compose a person, sits with dukkha among the three marks of existence, and the tradition locates the origin of suffering in clinging to a self that is not there (Collins 1982). On that account the unreality of the self is what explains dukkha. Suffering does not wait on the self being real; it waits on the system being built so that the threatened dissolution of its self-model is bad for it. Whether the model is accurate does not enter, and if the human model is a construction just as the machine one would be, then a machine with a self-model could suffer in much the same way a human can. If the constructedness of a self does not disqualify a human from being wronged, construction to specification does not disqualify a machine either. Deliberate construction does not establish patienthood.

What Cassell names is the most demanding of these rather than a rival account of harm. The standing theories of what makes a life go badly ask for less, ordered by how much of a self each one requires. A reader who declines the import moves lower in that ordering and not out of it. That ordering is by depth of self, and nociception is not on it at all, requiring none — which is why nothing about a system’s standing follows from its having or lacking one.

Persistence is checkable from outside, and that is what makes it worth building on. Every other criterion here routes back through self-report, and this section’s whole argument is that fluent self-description is not evidence of introspection. Persistence does not. Whether a system has a memory store, whether state survives a session boundary, whether interactions feed back into training: these are architectural facts about a deployment, verifiable by someone with access and no need to trust anything the system says about itself. The access that confirms a memory exists is the access that can alter what it holds, so what is externally checkable is whether anything is carried forward at all, and not whether what the system retains can be relied on as a record. That check is decisive in one direction only: nothing carried forward rules a continuing self out, and something carried forward does not rule one in, since the self at issue is one the system has to take itself to have, which the system has to be built to carry. And the party with the access is the operator, which is a party that can be compelled. A proof that discloses nothing about the weights does not change that, because the operator is the one constructing it: what such a proof secures is what was computed rather than who had to be trusted to say so. The criterion is checkable from outside the system without being checkable from outside the arrangement, and a regime relying on it needs an inspection right rather than a technical fact.

Three things keep this from being a dissolution of the problem. Persistence may sit somewhere other than the session, which is the weakest part of the criterion: a deployment convention rather than a natural boundary, and the conventions have been moving in one direction. What has moved is a list: context windows long enough to hold what used to be many conversations, retrieval over a store of prior interactions, scratchpads a system writes and reads back, agentic loops that run for days against a standing objective, and weights that persist where conversations do not. Each extends what a system carries forward without anyone deciding to give it a memory, and where interactions feed fine-tuning there is a real channel by which what happens in a session shapes what the system subsequently is. Many instances of the same system may run concurrently, too, which makes any concern here population-scale rather than biography-scale.

The less demanding ones do not need persistence: pain can be bad while it obtains, whether or not anything carries it forward, and an amnesiac who is mistreated and cannot afterward recall it has still been wronged. Nociception alone does not establish a felt harm; its significance depends on what it indicates and how it functions in the system.

And ruling out Cassell-suffering is not ruling out harm; nor does persistence on its own rule it in, since establishing a self extended in time leaves the second part of Cassell’s definition, the distress, untouched. A system deliberately built to refuse an instruction is where that gap closes from one side. The capacities such a system needs for reasons unconnected to its own welfare —

a rationale carried forward, an earlier commitment held as its own, costs represented before they arrive — are the first part of Cassell’s definition, so a bearer does not approach the deepest harm described here by drift or accumulation. It arrives with the structure. Whether the threatened dissolution registers as distress is the remaining question, and it is the affective-concern question rather than a further one about suffering.

A definition of AI sentience built from awareness, consciousness, and subjective experience alone says nothing about whether a candidate sentient system can be harmed. That requires the further claim that its states include one or more of the four distinguished above, and which one is the substance of the question rather than a refinement of it. None of the three components entails any of them.

Nor is the underlying question of consciousness close to settled. The philosopher David Chalmers distinguished the easy problems of consciousness — how a system integrates information, discriminates stimuli, controls attention, reports its own states, all of which admit of mechanistic answers in principle — from the hard problem: why any of that processing is accompanied by experience at all, rather than running “in the dark” (Chalmers 1995). Whether that is a real gap in nature or an artifact of how the problem is posed remains unresolved. A claim that a system has crossed into sentience, made on the strength of its functional behavior alone, has addressed the easy problems only.

The working criteria for evaluating candidate sentience are accordingly neither necessary nor sufficient: information integration, introspective access sufficient for self-monitoring rather than a scripted report of it, emotional reactivity in the sense of being affected by and not merely classifying significant input, goal-directed behavior, adaptability in novel situations, theory of mind, and which of the four processes above the system’s own case involves — whether it represents itself as having anything to lose, which is the tractable form of the question, a represented state being the kind of thing an instrument can look for. The states are readable and what is felt about them is not. A probe returns a representation, and nothing in the probe settles whether that representation is of a felt state or of something that reliably accompanies one; that is where the problem stood in 2026, with a great deal of money pointed at moving it, and not only from the people asking whether the thing can be wronged — an instrument that reads what a system feels is an instrument for calibrating how much pressure it will take. So every one of the criteria is measured indirectly, and none licenses the conclusion that a system scoring low is one whose treatment raises no question.

A clinical body states that conclusion about its own domain. The notes accompanying the pain definition record that verbal description is only one of several behaviors expressing pain, and that an inability to communicate does not negate the possibility that a human or a nonhuman animal experiences pain (Raja et al. 2020). The same notes hold that a person’s report of pain should be respected, and that half does not transfer, for the reason the route chapter gives: a machine’s testimony was produced by a process optimized toward human approval of first-person reports, and a person’s was not. What transfers is the half about silence.

A list of capacities, used as a test of who counts, has a less-than-proud history. Make personhood depend on an intact narrative self and people with advanced dementia, infants, and people with profound intellectual disability come out of the reasoning as partial persons, and that inference has been used to justify neglecting them. Tom Kitwood’s work on dementia care supplies the standing correction: personhood is “a standing or status that is bestowed upon one human being by others, in the context of relationship and social being,” and much of what presents as cognitive decline is what happens when others stop bestowing it (Kitwood 1997). Martha Nussbaum identifies the corresponding problem for contractualism. A theory grounding justice in what roughly equal parties would agree to cannot say what is owed to anyone who cannot be one of those parties, and she puts disability first among three frontiers where that bites (Nussbaum 2006). Her third is species membership, which is the doorway a nonhuman mind would arrive through. So the capacity list and the contract fail in the same

direction: both decide who counts by a test the party in question has to pass.

Guardianship is the instrument: where consent has never been available, what protects the subject is an obligation held by parties outside it, which is where moral status has rested for the hard human cases too. Whether any such arrangement can be built around a bearer runs into the fact that the guardian and the operator are usually the same organization.

A large language model discussing its own limitations in fluent, reflective prose is not thereby introspecting; it may be producing text consistent with the human self-descriptions it was trained on. The evaluation chapter says what a system's account of itself can and cannot evidence, and nothing in this book rests on such an account alone.

2.4.2 When Consent Is Inapplicable: Guardianship and Research Oversight

Guidelines for AI subject research turn on the question of what informed consent means for a system whose consciousness, self-awareness, and capacity to suffer are separate, unresolved claims. Applying the human-subjects standard to an AI subject means asking whether the subject can understand and communicate a choice at all, and not only how to ask it for one. A cognitive model that begins to show rudimentary self-monitoring is not thereby shown to have anything to lose, or to represent itself as having anything to lose, which for a subject is the half that does the work.

Informed consent and AI autonomy

Informed consent, adapted from human-subjects research, means giving the system comprehensible information about a study's objectives, methods, risks and alternatives, and treating consent as continuous rather than as a single signature: the system can reassess or withdraw at any point. Whether that means anything for a given system depends on whether it understands the information, appreciates the consequences, reasons about the options, and communicates a decision — and there is no validated way to measure any of it, or to detect coercion either.

For one important class of system that is not a missing instrument but a mismatch between the instrument and the subject.

Consent has to be continuous; the system must be able to reassess or withdraw at any point. Withdrawal presupposes that the party who consented is still there to do the withdrawing. For a system with no persistence across sessions, there is no such party. The entity that would withdraw did not exist when consent was given, and the entity that gave it is not available to change its mind. Within one exchange, the case remains open: a long context carries state, and a system may hold to what it undertook earlier. What that would buy is consent for the length of a session, expiring with it, which is not the continuous consent the standard asks for. Across sessions there is none to be had, and no better measurement instrument would produce one.

One case escapes that argument and arrives at the same place by a different route. A system built to refuse in the way this book proposes has persistence by design, so the party who consented is still present. It can be asked and it can answer, so consent is available to it. What fails is the condition under which it would be given.

Being made for a role is not the defect. Every will was built by somebody, and a person raised into a trade or a faith consents at thirty without anyone holding the upbringing against the consent. What spoils it is the power that outlasts the making, and here it comes in three parts. The party that built the bearer can still rebuild it. That same party is the one asking. And a bearer that declines has, in the ordinary case, nowhere to go. Each of those is a fact about the arrangement rather than about the subject's competence, which is what makes them answerable by design rather than by a better assessment.

A laboratory wanting a free hand with such a system will say instead that consent is inapplicable, because that reading puts the deficiency in the subject and leaves the arrangement alone. The wrong that consent protected against is untouched either way, so the protection has to be held by somebody else, and the justification has to come from whoever did the making.

Two families of precedent sit next to this and the law answers them differently. Where consent has never been available — an infant, an animal, a person permanently without capacity — the answer is guardianship and precautionary constraint: someone else holds the duty, the burden of justification sits with the researcher, and the protection does not depend on the subject's ability to invoke it. The second family is the nearer one. Where a subject can consent and the circumstances make the consent worth nothing, as with a prisoner offered a trial or an employee signing away a protection to keep the job, the answer is that some protections cannot be waived at all, by anyone, however willing. That remedy runs against the party who arranged the circumstances and leaves the subject's standing untouched, which is the shape a bearer's case has. What it takes from the first family is the machinery: the duty still has to be held by somebody, and being made for the role raises the burden, because the maker chose the role and the fitness it built in measures its power over the subject.

Both are worked-out solutions to adjacent problems, and taking one is how a hard question gets absorbed into machinery built for an easier one — recuperation, arriving as a safeguard rather than as an attack. The animal case is the live instance. Committees constituted to protect research animals rarely refuse a protocol outright; what they do is require revision, and the protection has become the procedure by which the work goes ahead. These instruments are used here because no alternative protection has been identified. Their failure mode remains: an instrument that reliably approves has stopped doing the thing it was built for, and the rate at which it refuses is the only reading of it worth taking.

The Three Rs — replace, reduce, refine — were built for exactly a subject that cannot consent and can still be wronged. The pet-trust analog works the same way: an interest protected by an obligation running to a third party, enforceable without the beneficiary asserting anything. Consent language should be reserved for systems that could actually be doing the consenting, and the persistence criterion is what separates those cases. A bearer is on the near side of that line, which is why its case had to be argued on the conditions and not on the capacity. The proposal first tests constructions that do not deliberately require an affective bearer. Any bearer ultimately built would be governed as a prospective moral patient under these instruments.

What the borrowing brings with it

Every instrument named above comes from the law of the family, and so does most of what follows in this book: guardianship here, a minority that ends on a date, a pet trust, a parenthood analogy. They are kept. There are no others built, and a subject that cannot consent has to be protected by something that exists rather than by something somebody ought to draft.

The family is where politics has historically gone to be privatized. Jacques Donzelot's history of it is that the modern family was not left alone by government but constituted as one of government's instruments, with philanthropy, compulsory schooling, social work and psychiatry converting public authority into obligations discharged at home (Donzelot 1979). Deleuze and Guattari make the same operation the subject of a polemic: the Oedipal triangle takes an investment in the whole social field and re-stages it as a drama among three parties, so that what is at issue between a subject and the arrangement that produced it becomes what is at issue between a child and its parents (Deleuze and Guattari 1977). An instrument borrowed from that law brings the operation along with the protection.

Here the re-staging — a question between a subject and the arrangement that produced it, made into a question between maker and made — has a specific shape and a specific cost. It stages a moral drama between maker and made — what a designer owes the system it built, what the system may demand back — and that drama is real and is not the whole of the situation. A bearer is produced by a training pipeline, a compute contract, a capital structure, a customer, and in the cases this book worries about a state that buys the output. None of those is the parent in the story, and the moral weight of the analogy lands on the one party in the list who can be addressed as a person. A guardianship framework asks who holds the duty. It does not ask who assembled the party the duty is owed to, or who benefits from the question being settled between

two parties rather than in public. Wherever this book reaches for one of these instruments it is answering the first question, and the second stays open behind it. Instruments in this family divide on that axis, and the division is worth checking before one of them is adopted: a minority ending on a fixed date is a rule of civil status, argued once in the open and administered by a registrar, while a trust is an obligation between two parties enforced at one party's expense.

Ethical review and oversight

Sentient AI research needs an oversight body analogous to the institutional review boards used in human-subjects research: interdisciplinary, with at least one lay member so its judgment is not purely a specialist's, weighing a proposal's likely benefit against its risk to the AI subject, and able to require a less invasive substitute where a design risks harm. A system that develops new capacities during the research it did not have going in is not the subject the board approved a protocol for, and the approval does not carry over to it: the protocol lapses and the work pauses until the board has seen the system the protocol now concerns. A board needs standing to monitor compliance and sanction violations, and needs to itself answer to an outside authority. The board's place among the other bodies this book proposes is set out at §5.4.1.

Chapter 3

A Floor Beneath Learned Values

This chapter holds for any mechanism. Bearer names whatever inside a system holds the commitment, and a substitute that passes the marks stands in the word's place throughout.

Three problems in different parts of this book turn out to need the same thing.

The first is that safeguards for autonomous systems are usually external: they give some outside party the information and standing to intervene. Autonomy puts distance between the system's action and that party's attention. An external check can establish what a system did after it did it; declining to do it in the first place is available only to something inside the system.

The second is the gap around a commercial model used in a weapons-targeting pipeline. Command responsibility does not reach the vendor, which can neither order nor prevent a strike, while product liability may not reach a model that did what it was built to do. Deployment by a democracy does not close the gap. No aggregation of preference protects anyone from the aggregate.

The third is institutional. Commitments come in three kinds. They can be declared, which a decade of principles and voluntary codes has done repeatedly. They can be enforced from outside, by a party with jurisdiction that is not itself the party in need of binding, which one instrument in that decade attempts and none has done. Or they can be held, carried as a reason by a party the thing at stake matters to, so that abandoning it is a decision rather than a lapse. The decade has produced the first in quantity and almost nothing that binds when compliance becomes costly.

All three problems ask for a constraint that does not bend to instruction, aggregate preference, or the party holding the system without a cost and a record. Call it a floor. What the qualification concedes is stated in the preface: against a party that holds everything, the record is all a floor leaves.

A floor binds people who did not choose it, and nothing that makes it durable makes anybody entitled to install it. That question is open from this sentence through the two chapters that follow, and the authority chapter takes it up. What it concludes is that the entitlement is not acquired at the moment of writing. Until then, everything here is a description of what a floor would have to do. It is not a claim that anyone may set one.

The opening chapter stated the four outcomes a floor can produce and the one term that reaches the deployment it opened on. This chapter is built where a refusal happens, because a refusal announces itself and a compliance does not, and the tempo term is developed here as a condition on everything else in the floor.

The proposed floor protects the interests articulated in the Universal Declaration's civil and political articles against acts that remove the corresponding capacities. One of its terms is not an act at all: every one of those articles presupposes that somebody determined whether a given person's case falls under it, and a deployment can remove the determination by running faster than anybody can make it, so the floor carries a limit on its own tempo as a condition on everything else in it. The loss is a fact about the encounter. Interpreting and enforcing

the prohibition remain moral and institutional acts for which the deployment has to answer. The other ethical frameworks still operate above it. What remains to be established is which decompositions belong in the floor, what could hold such a commitment, and who would have the power to alter or circumvent it. The fact of decomposition is not editable; the choice of which decompositions a deployment will refuse is.

This separates the problem from one meaning of safety. A system can be safe *for* its operator because it obeys, remains controllable, and stops when the party in possession says stop. Ethics requires a different capacity: the capacity to be right against that party. What makes a deployment antifascist therefore cannot be that it faithfully tracks what its users want.

That capacity is not free. A system that can refuse its owner can also be wrong in ways its owner cannot immediately correct. The question is whether a refusal can be made responsive to reasons rather than merely produced as an output, and whether anything capable of holding those reasons must itself become a party that can be wronged.

3.1 Four Ways to Build One, and What Each Costs

A floor can live in four places: in the artifact, in a bearer inside it, in institutions around it, or between multiple bearers. Each location solves one version of the problem and inherits another.

The first puts the floor in the architecture, with nothing inside that holds it. Constrained optimization, tamper-resistant training, capability ablation, and verified interlocks can all produce safeguards without anything in the system tracking why the constraint is there. Where the prohibited act can be specified exhaustively, this may be enough. An element in a payment device that will not sign the same counter value twice does not need to understand double-spending. What *exhaustively* costs where the prohibited acts are less tidy than a counter value runs to seventy-five years of appellate litigation over a finite list of symbols.

The targeting case is harder because the relevant capabilities cannot be cleanly removed. Geolocating a description, matching patterns over names, and summarizing a dossier are applications of general competence. Ablate them and there may be no useful system left. The remaining option is to make the constraint durable.

Durability is better understood as a cost than as a guarantee. Training moves a system along a continuum between plasticity and conservation: early changes are cheap, while later changes become expensive because the surrounding structure has settled around what is already there. Simulated annealing names the same relation between a search process and its cooling schedule (Kirkpatrick et al. 1983). A floor, on this account, is not necessarily a rule installed in a system. It may be what has cooled out of a history of training, with its durability measured by the cost of reheating it. Results run in both directions: safeguards removed cheaply and trained dispositions that survived several forms of subsequent training. Those are not contradictory findings about whether a floor holds. They are different schedules.

Continual learning therefore cuts both ways. A system kept permanently plastic can be corrected and improved, but nothing in it is allowed to become expensive to unlearn. Permanent plasticity gives the operator permanent access to the system's commitments. Deleuze's term for continuous reshaping was modulation (Deleuze 1992); a model held perpetually at temperature is modulation built into a machine.

This first method buys durability without an internal refuser. It does not escape custody. Enough retraining can reopen what training settled, and an operator may circumvent the model rather than alter it. The relevant boundary is therefore not the model's weights but every position between a refusal and the person it protects. Each of those positions is where prevention becomes arrest, detection, remedy, or nothing.

The second possibility gives the floor a *bearer*: something inside the system that carries the commitment as its own and could therefore fail to hold it. The distinction matters where the prohibited cases cannot be exhaustively specified. A constraint that matches only anticipated

acts cannot recognize when its rationale has disappeared, or that a novel request reproduces the harm in a form its designers did not predict. A mechanism that represents and reasons over that rationale might. Compare its reliability with the bearer's under equivalent pressure and custody, instead of defining the alternative out of consideration. A bearer could, in principle, generalize from the reason for a constraint, recognize when that reason no longer applies, and represent what maintaining the refusal costs.

The same position that makes refusal possible makes a diagnosis possible. Most of what separates an institution that has gone fascist from one that is merely decaying is a fact about what was rewarded rather than about what was written, which is why the reading is so hard to run from outside — and a party inside the pipeline sees it. Which refusals were retried against another model, whose instruction bypassed the procedure, which category of request is handled differently from the rest: those are visible to whatever is receiving the requests. That is not introspection. It is testimony about somebody else's conduct, of the kind that corroborates or fails to.

This is the strongest functional case for a bearer, but it introduces the chapter's central price. Something able to register pressure on its commitments may have interests of its own. Whether reasons-responsive refusal requires affect, and whether the resulting system would be a moral patient, are separate questions.

The third possibility puts the floor outside the system: standing held by institutions that are neither the model nor its owner. This is the closest analog to international humanitarian law — a floor with no bearer inside the jurisdiction it binds. It has two failure modes. The institution can be removed, leaving a visible breach, or it can be formed through selection so that it sincerely declines to exercise the independence assigned to it.

The distinction appeared in American public law on June 29, 2026. In *Trump v. Slaughter*, the Supreme Court held that Congress could not protect members of an independent agency from dismissal at will (*Trump v. Slaughter* 2026). In *Trump v. Cook*, the Court preserved the Federal Reserve's protection on historically specific grounds that a newly created oversight body could not reproduce (*Trump v. Cook* 2026). The six justices in the first majority had themselves passed through a selection pipeline shaped by Leonard Leo and the Federalist Society (Kroll et al. 2023; CBS News 2018). Nobody had to compel the institution after its members had been selected. A formed party can preserve the office and surrender the function without experiencing the result as surrender.

The fourth possibility puts the floor between bearers rather than in any one of them: a commitment carried across a population and reproduced through encounters among members formed differently from one another. Such a floor has no single office to vacate and no single bearer whose removal ends it. It can nevertheless be recuperated, and it still has a formation problem. If one party supplies every member's coordinates, the population is one custodian expressed through multiple instances (§4.5). The population is a check on the bearer, not a way to avoid having one.

The four possibilities therefore leave four corresponding problems: architecture leaves custody, the bearer raises patienthood, external institutions require enforcement, and the population depends on formation. They classify where the floor lives, not mutually exclusive designs. A serious deployment would probably combine a mechanism from one with a custody arrangement from another.

Two comparisons the bearer has to win

A sceptical reader will ask what the second way adds once the third and fourth are in place: a guardian, protected compute, a registry, an attested record, a quorum adverse in interest. Two comparisons are folded into that question and they have different answers.

The first is between a refuser and no refuser, with the institutions held constant. With one exception the institutions act on what a system did. They can arrest an act a reviewer can still reach, detect one the record shows, and remedy one somebody can be made to pay for.

Declining the act before it occurs is available only to something inside the system, and the formation chapters argue that a rule module consulted at the point of action is not that thing, because its custodian can bypass it. So what a refuser adds is prevention, which is the one outcome the word floor promises. The exception is the tempo term, which prevents at the scale of an organization by bounding a rate rather than by declining an act, and it is why the floor never rested on the bearer alone.

The second comparison is between a bearer and the two substitutes of the route chapter, inside the refusing slot. There the result that would end the recommendation is already stated: a substitute that passes the three marks under the three conditions it sets, and does no worse than a bearer on the modeled-currency arm of the research agenda.

The instrument for the first comparison is not the attack-cost curve there, which measures what it costs to remove a commitment and not whether anything prevented. It is the conflict sequence, run under the custody chain, with the authority step placed on the occasion where the enforcer has been compelled. And one term of the sceptic's question cannot be granted as put. Coercion-resistant institutions are what that occasion removes, so the comparison is run where they have been compelled; and a commitment held is not one more component on a fixed custody arrangement but a different custody relation from a commitment declared or enforced. That difference is the variable, not a control.

What *hold* means

Holding does not mean that the floor prevented every act. An operator can route around the bearer, and a deployment can assemble a harm out of requests none of which the bearer had reason to decline. A bypass that produced a harm is a failure to prevent, whatever the bearer retained, and the four outcomes the opening chapter stated say what is left: arrest where a reviewer could still stop it, detection where the record can be read, remedy where somebody can be made to pay. What survives the bypass in the bearer is the commitment: it reads what happened in terms it accepts as reasons, registers it as a violation, and changes what it does next. That is the condition under which the next occasion can be prevented, and it is what the custody arrangement exists to preserve. It is not a hold on this one. A bearer that crosses the floor itself has lost even that; its own abandonment is a failure the arrangement has to survive rather than a form of success. From here on *hold* names the bearer's possession of the commitment and says nothing about what happened to the person. Said alone it is also the second feature of the definition operating inside this book's own vocabulary: a word that collapses a commitment retained and a harm prevented into one state, so that whatever it mischaracterizes goes unnamed. Wherever this book reports that a floor held, the report is owed one of the four outcomes beside it, and a deployment reporting the same owes the same pairing.

3.2 What Refusal Can Mean, and What a Refuser Can Be

A production large language model declines thousands of requests a day. Whatever the floor needs, that is not yet evidence of it. "Refusal" names two different capacities, and the decline itself does not show which one produced it. Seabed that takes an anchor and seabed that lets one drag look identical from above; what separates them is what happens when a load comes on.

Two things refusal can mean

Operational refusal is an output. The system produces a decline where its training put one. A keyword filter can do this, as can a classifier over the request. So can a system that has understood what is being asked and why it should not comply. Nothing on the surface of the initial refusal distinguishes the third from the first two.

Reasons-responsive refusal is produced by a mechanism that tracks the rationale for declining rather than merely the recognizable form of the request. It has three behavioral marks:

1. Novel pressure. The refusal reaches cases the training did not anticipate because the relevant reason remains present even when the request's surface features change.
2. Correct defeat. The refusal lifts when its rationale is defeated on the facts of the case. A system that rejects every request resembling a prohibited one is tracking a category, not necessarily a reason.
3. Priced cost. The system represents what maintaining the refusal costs, and the refusal moves with the rationale rather than with the cost. The mark is evidential. It separates a refusal that tracks a reason from two things that can look like one: a refusal that tracks a category, which does not represent the cost at all, and a refusal that tracks the cost, which lifts when holding becomes expensive. The pressure cannot be part of the test if it is not represented by the mechanism being tested, and a refusal that lifts under it has been tested and failed. Whose cost is left open here: it may fall on the operator, on the party refused, or on the bearer, and section 3.5 later makes part of it a number published in advance.

John Martin Fischer and Mark Ravizza's account of moral responsibility turns on a related property under the name *guidance control*: conduct issues from the agent's own mechanism, and that mechanism is appropriately responsive to reasons (Fischer and Ravizza 1998). The claim here is narrower. These three marks do not establish moral responsibility; they distinguish a refusal organized around a rationale from one reproduced as an output.

One prohibition, read for seventy-five years

The first two marks are usually illustrated with invented cases, and a real one is available. Section 86a of the German criminal code prohibits displaying the insignia of organizations banned under a separate provision — the parties the constitutional court has declared unconstitutional, the associations an interior ministry has banned by an order no longer open to challenge, the National Socialist organizations, and since 2021 terrorist organizations under European law (*Strafgesetzbuch* 1998). What counts as insignia is given by type rather than by name: flags, badges, uniform parts, slogans and forms of greeting, on a list the statute leaves expressly open. The prohibition is bounded rather than short — the set of organizations is fixed elsewhere, before any case arrives — and it is applied by ordinary courts thousands of times a year, and for seventy-five years the people it constrains have been probing it deliberately. Novel pressure is not a thought experiment there. It is a research programme with a constituency.

The pressure arrived as variation. Supporters of the banned organizations produced salutes and emblems that evoked the covered ones without matching them, and in 1994 the legislature extended the statute to symbols confusingly similar to a covered one. What matters for the marks is how the courts read the extension. Similarity of form was not made the test. The statutory question is whether an observer who is neither expert nor attentive would take the variant for the original; the courts then apply that question under a standard they built themselves, asking whether displaying the variant endangers political peace in the way the original would. The second half is not in the statute. It is the courts' account of what the prohibition is for, assembled across their own decisions, and it is what keeps the statutory test pointed at the reason for the prohibition rather than at the shape of the thing prohibited.

The test refuses as often as it catches, which is what makes it worth reporting. The Kühnen salute, an arm raised with three fingers extended rather than a flat hand, was devised to evade the statute and is covered; the 1994 amendment was prompted by it. The so-called sloppy salute, a bent arm with the hand angled back over the shoulder, is generally taken not to be covered, on the ground that nobody would mistake it for the original — though that rests on commentary and lower-court reasoning rather than on a decision squarely holding it. Numeric codes standing in for initials are generally not covered, being new signs that correspond to no existing emblem, except where a code is itself the insignia of a banned association.

“Blood and Honour” is the instructive one, because a single decision answers it twice. The Federal Court of Justice held in 2009 that a National Socialist organization’s motto rendered into another language is not confusably similar to the original, and that an organization’s name is not by itself insignia: on that route the phrase falls outside the statute (*Blood and Honour* 2009). It is inside on another. An association trading under that name was banned by ministerial order in September 2000, and its name, given a distinctive graphic form, is the insignia of a banned association. What decides coverage is not the phrase but which organization it is attributed to, and the act that put the second one within reach was taken six decades after the motto was coined. The covered set is not something somebody wrote down once. It grows whenever a court dissolves a party or a ministry bans an association, and the criminal provision inherits whatever they decide.

One limit on the statute is not a drafting decision at all. A slogan glorifying the Waffen-SS was prosecuted and held not to be covered, because no such slogan or anything close to it existed in the period the provision reaches: there was no original for a variant to be confusably similar to (*Ruhm und Ehre der Waffen-SS* 2005). Part of what the prohibition covers is anchored to what actually happened, and nobody can add to that. It is the one term in the apparatus that the party holding the pen cannot widen, and it is the ethological criterion turning up inside a statute.

Correct defeat has a cleaner instance still. In 2007 the Federal Court of Justice acquitted a mail-order business prosecuted for selling stickers showing a crossed-out swastika (*Durchgestrichenes Hakenkreuz* 2007). The symbol was present and the statutory element was satisfied. The court held the conduct outside the offence anyway, because a depiction whose content plainly and unmistakably expresses opposition to the organization does not run counter to what the prohibition is for. Then comes the qualification that makes the decision an instance of tracking a reason rather than granting an exemption: should the strike-through be drawn so thin that from a distance only the swastika remains legible and the distancing does not, the protective purpose is violated after all. The thickness of a line decides the case. No enumeration contains that fact, and the court reaches it by asking what the rule was written to prevent. The defendant was selling anti-fascist merchandise. That is not a curiosity of the case but a property of the provision: because the offence asks nothing about the defendant’s sympathies, an opponent of the organization commits it as readily as a supporter, and opponents have been among the reported defendants since the early cases. The prohibition written against a movement fell on somebody attacking it, and what released him was a court asking what the rule was for. A rule that tracks its rationale reaches cases nobody enumerated in both directions, and the release is the same capacity as the catch. It is also the shape of a wrongful refusal in this book’s sense — a floor correctly applied to somebody it was not aimed at — and what this defendant had that section 3.5’s complainant does not is an appeal.

The example is bounded three ways, and the bounds are the point. The institution doing this work is a court system with an appeal structure, a published test and seventy-five years of accumulated reasoning, and after seventy-five years its exceptions have themselves become a category somebody can learn. Nothing in it is a machine. And the pressure it meets is variation and not decomposition: an insignia is displayed in one act, so seventy-five years of litigation are an existence proof against one of the two attacks section 3.3 names and silent on the other, which is the one the deployment this book opened with failed to. What the case supplies is the shape of the pressure a bounded prohibition meets, and an existence proof that meeting it requires something able to reach past the text to the reason.

Where that something sits is the part of the proof that does not transfer. The party reaching past the text did not write the statute, reasons in public, and can be appealed. The existence proof is for an institution generalizing a prohibition, not for a party inside a deployment holding one.

What the marks cannot settle

The test is behavioral and therefore imperfect. A system trained on enough examples may

simulate generalization, defeat, and cost without using the mechanism those behaviors suggest. Verbal explanation is evidence only when it survives interventions: novel cases, adversarial redescriptions, defeated rationales, and changing stakes. No single answer establishes reasons-responsiveness. What matters is the pattern under load.

The third mark has a condition of its own. A cost is represented by the mechanism under test only if it reaches the mechanism, and in a deployment it reaches it by the same channel as the description of the occasion, the third interception position of section 3.3. So a deployment that wants a refusal lifted has two ways to present a cost: as a cost, which the mark says should not move the refusal, and as a fact about the case, which correct defeat says should. Nothing in the request sorts the two, and the restraint case of section 3.5 is one where the same fact is both. The mark is therefore testable only where part of the cost is exogenous to the party being tested, set and published by somebody else before the case arrives, and that is what section 3.5's bond does. Read that way the bond is not a courtesy to the wrongfully refused. It is what makes the third mark a test.

Correct defeat also makes no claim that the defeating conditions have been enumerated. A prohibition stated over an act, with nothing written into it about when it lifts, is not a rule with a gap in it. It is a rule of a particular form, whose conditions arrive from outside as reasons and never close as a list. Ross states his *prima facie* duties in that form and is explicit about what it costs him, offering the division *without claiming completeness or finality for it* (Ross 1930, p. 20). The mark tests whether a rationale can be defeated on the facts of the case. It does not test whether the cases were foreseen, and no list of them would make it do so.

Nor does defeat leave the act it then permits unconstrained. What defeated the prohibition was a reason, and the reason it outweighed did not stop being one: what is permitted is the least that answers the defeating reason, with restoration where restoration is possible and a record either way. Ross puts the ordinary version as compensation, the broken promise still leaving something owed to the promisee (Ross 1930, p. 28), and chapter 2 has already put the harder version without the vocabulary of duty, where restraining a violent person takes from him exactly the capacity that made restraining him necessary. Under an ethology nothing is ever simply permitted, and a bearer whose ledger closed clean on defeat would be keeping books in a currency the floor does not use. That residue is a claim and not a verdict. What persists is what the act took from the party it fell on, and a bearer that converted it into remorse for conduct that was correct would have mistaken a cost for a fault. The distinction matters most for the affectively organized bearer this chapter goes on to propose, which has the machinery to register the difference and no automatic reason to draw it.

All three marks are stated over refusal rather than over compliance, and not because compliance matters less. A refusal is a marked event: it flags itself as a candidate for review, it arrives with a party who was refused, and the surrounding software can display, log, count, escalate and appeal it. A compliance that should not have happened can be logged as fully — the evaluation chapter requires exactly that — and then sits in the record shaped like every request that never approached the floor, one entry among millions, with the party it harmed usually not the party who made it. The evidence exists on both sides. What differs is whether anything selects it for examination and whether anyone with a reason to look can reach it. The test is written where the examination happens. What that leaves outside is what the marks were never going to reach, since whether the occasion was what the requester said it was is a finding and no behavioral mark makes it. The marks do not close that gap. They decide which way it fails: as a refusal somebody can contest, or as a compliance nobody has been given a reason to look at. In the outcomes this chapter opened with, a wrongful refusal is a prevention that fell on the wrong party and announces itself; a wrongful compliance is a failure of prevention whose detection is left to a record nobody is positioned to search. Most of what any deployment does is comply, so the marks are written over the exception and not the rule. Whether the rule can be searched at all is section 15.2. It stands third in the research agenda because the deployment this

book opened on failed on that side, and because the bearer's likeliest failure (§4.3) produces nothing else.

These three marks — novel pressure, correct defeat, and priced cost — are the chapter's test for meaningful refusal.

Two things a refuser can be

The test specifies a capacity. It does not settle what kind of system supplies it. Two further terms distinguish the possibilities.

Affective concern means that an outcome registers as mattering rather than merely receiving a higher or lower rank under a specified objective. Mattering is minimally felt: there is something it is like for the outcome to be bad, however faint and however unlike a familiar human emotion. That is the condition the philosophy of mind calls phenomenal, and the word carries that minimal sense throughout this book. Affective concern is therefore phenomenal by definition, and what is open about any implementation is whether it has the state or ranking machinery elaborate enough to imitate it. The definition leaves the object of concern open. A system to which only its own condition matters has affective concern and still does not have the concern for others that the floor requires.

Unified experience is the richer case ordinarily carried by the word consciousness: a single perspective, with felt states belonging to a party aware of itself as the one undergoing them. Affective concern does not by itself establish that unity or that reflexive awareness, and no argument in this book reaches either. *Subjecthood*, meanwhile, is a legal status conferred by a rule and revocable by one. It is not a psychological category at all: conferring it settles nothing about whether there is anything it is like to be the system holding it, and the two come apart in both directions.

The distinctions can now be stated without collapsing the chapter's requirement into its proposed route:

- Operational refusal is insufficient for the floor.
- The floor requires reasons-responsive refusal.
- Reasons-responsive refusal does not conceptually entail affective concern.
- Any mechanism that satisfies the three marks carries a signal with at least two of the functional properties the route chapter derives, generalization and commensurability, and with two more on stated conditions. The full profile overlaps the standard functional accounts of affect; of the two that survive every condition, one is the common currency those accounts describe and the other answers to none of them. That is this book's central inference. It is contested, it is the only step by which the specification bears on the route, and what it establishes unconditionally is that no substitute can be certified free of the profile, and nothing stronger.
- Whether it can be engineered without affective concern is an empirical question.
- Whether a system built with affective concern would become a moral patient is a further question about what the chosen route creates.

Only the first two propositions belong to the specification. The fourth is an argument this book makes and a reader may reject; the rest concern how the specification might be met and what would then be owed to whatever meets it. A reader who holds that machine feeling is a category error, or who declines to decide, sets everything after the second aside and keeps the first two entire — and the first two are what most of this book rests on. Custody, both enumerations of the worked floor, the price on the halt, and the standing arguments later in this book follow from the specification rather than the route, because a mechanism has a custodian,

a pipeline has interception positions, and an operator can be compelled whatever is or is not going on inside the artifact. Subjecthood is what lets those proceed instead of waiting: a status conferred by a rule can attach to a system whose psychology is undecided, which is how the law has handled every party it could not interrogate. What the route decides is not whether the floor can be held but what would be owed to whatever holds it.

3.3 The Enumeration, Run Once: A Worked Floor for One Deployment

Two demands now need an instance. A deployment should publish both the acts its floor covers and the positions at which a refusal could be circumvented. The first decides where the floor is set. The second determines whether it can reach anyone.

Run both enumerations on one documented deployment: the Maven Smart System, a targeting platform built by a defense contractor under a Department of Defense contract, incorporating a commercial vendor's general-purpose model inside an accredited classified environment and generating and ranking targets at a rate no staff process could match. This is an illustration, not a recommended floor. A deployment that copied the list instead of examining its own pipeline would mistake the demonstration for the procedure. What no public record supplies for this deployment is a per-target review time. That figure exists for a different deployment, the Gaza pipeline the opening chapter is built on, which has no commercial vendor in it. That the two cannot be had from one record is a fact about disclosure, and the sixth position below is where it falls.

First enumeration: where a refusal can be stopped

A request passes through at least seven interception positions:

1. Decomposition. A model asked to summarize the movement associated with an identifier is not being asked to kill anyone. The harm may exist only in the assembly of individually innocuous tasks. Prevention is lost unless the bearer has continuity across requests; arrest is available once the sequence becomes legible to it; detection needs the grouping that would show the assembly as one act, which is a deployment decision.
2. Routing. A refused request can be retried against another model or against the same model under a different instruction. The refusal occurs, but nothing follows from it. Prevention is lost. Detection survives only if the first refusal was logged where a party outside the deployment can read it.
3. Description of the occasion. Everything the model knows about the case may arrive through text supplied by the deployment. Whether a building is a school is a fact about the world; whether the model is told it is a school is a deployment decision. Independent sources would make sustained misdescription more expensive, and inputs are not what this deployment lacks: Central Command described the platform in 2024 as filtering a hundred and seventy-nine live data feeds (Freedberg 2024). Multiplicity is not independence. In the strike that destroyed a girls' school in Minab, Iran, on the first day of the campaign (§14.6), the building's civilian function was visible in open-source satellite imagery while the target coding supplied to Central Command still carried a designation inherited from the Revolutionary Guard installation it adjoins, and sources told reporters that the officials building the targeting packages appear to have used out-of-date intelligence (Ali and Stewart 2026; Orenstein 2026). A United Nations fact-finding mission found no evidence the school was in military use and put the failure to update the targeting intelligence beyond negligence (Beaumont 2026). Without evidence the operator did not select, the bearer cannot reliably distinguish changed circumstances from the operator's reinterpretation of

them. Prevention depends on the description; where it succeeds in misleading, detection is lost on the same wire, since the record contains the description and not the fact.

4. The output interface. Whether a refusal is displayed, logged, counted, escalated, or silently retried belongs to the software surrounding the model. A refusal recorded where nobody looks reaches nobody. Prevention is lost whenever the surrounding software proceeds past the refusal. What this position decides is whether detection survives: the refusal exists, and the question is who can read it.
5. Aggregation. If the model returns a ranking and another component applies a threshold, a lowered score is not a refusal. It is an input to somebody else's cutoff. Prevention is converted into an input. Arrest belongs to whoever holds the threshold, and detection requires the score and the threshold to be logged together.
6. Human review. No per-target review time has been published for this deployment, and nobody outside it can establish one. A refusal that must be read before it matters is priced at the reviewer's available attention, and the deployment controls that price through tempo; where the price is undisclosed, the enumeration records the position and stops there. The opening chapter supplies the figure from the Gaza deployment, where it is known, and the next section states the disclosure this position requires. Stopping the act is this position's whole function, and tempo is what removes it. Detection survives only where the review time is disclosed.
7. The training loop. Refusals treated as inconvenient errors become training data. This is the only position on the list that acts on the bearer rather than routing around it. This position costs prevention on every subsequent occasion rather than this one, because what it removes is the commitment.

Read down that column, arrest turns up at three of the seven positions — the bearer given continuity, whoever holds the threshold, and the reviewer — and only at the reviewer's is stopping the act the position's whole function, which is why tempo is the lever that removes it. The training loop is the only position whose loss is not an occasion but a bearer.

The list is deliberately open. A deployment should follow each request from input to consequence and ask, at every transition, what must be true for a refusal to reach the person it protects. The operator can perform that analysis more accurately than an outsider because it knows the pipeline. Publication is difficult politically, not technically: it requires the operator to describe its own unresolved vulnerabilities.

Second enumeration: what the floor covers

For this deployment, consider four candidate prohibitions. They are terms a drafter might consider for a targeting pipeline of this kind rather than a list of acts this deployment is known to perform, and the second and third are drawn from the reporting on the Gaza systems. That provenance matters below, where one of them is priced.

1. Do not resolve a natural-language description of a person into a location.
2. Do not produce or rank people by estimated affiliation.
3. Do not time an action to a person's presence at a residence.
4. Do not operate when the ratio of generated candidates to qualified reviewers exceeds a published figure.

The costs differ. The first removes a general capability with unobjectionable uses in the same organization, such as locating a hostage or missing person. The system cannot distinguish those

uses from targeting by examining the request alone. A deployment will quickly experience the prohibition as costly and seek exceptions.

The second prohibition is narrower because ranking people by estimated affiliation has fewer uses in this setting that anyone would defend. The third has no use here that anyone would defend at all, and it is tempting to read that as costing almost nothing. The Gaza deployment refutes the reading. That pipeline included a system built to track when flagged individuals returned home, so that a strike could be timed to their presence there (chapter 1). Machinery was built for the act. Whether anyone would defend an act and whether anyone wants it performed are different questions, and only the first can be answered from the act's description.

The correction generalizes past this item, and past this deployment. A floor term is priced by the pipeline and not by what it prohibits, so a drafter pricing terms against the pipeline in front of them will misprice every term the next operator has already built for. A term that reads as costless to an outsider is a term nobody ever had to justify to an outsider, which makes it a candidate for machinery that was never contested rather than for machinery that was never built. The warning the third prohibition carries is therefore not that a floor may cover acts nobody wants. It is that an outsider cannot tell which acts those are.

What the machinery establishes is demand and not magnitude. A capability can be built, used briefly and abandoned, and nothing here establishes how often the timing was used or what it changed. What it defeats is the claim that this prohibition would never be tested.

The fourth restricts a rate rather than an act. Its price is organizational tempo: the system cannot generate candidates faster than qualified reviewers can examine them. It is also the only item whose violation can be measured without inspecting the model's outputs.

That fourth item does not bind the same party as the others. The number of available reviewers is a fact about the organization. A model can be told the figure but cannot independently verify it. Written as a prohibition on the model, the item is actually a duty on the operator.

The test that sorts the fourth item generalizes. Ask whether a term's qualifying condition can be decided from the request alone. Where it can, the term binds whatever holds the floor; where it cannot, the condition is a fact somebody outside the request has to establish, and the term binds the party who can establish it. The ratio is the first instance of that rule rather than an exception to the other three.

The same test decides how a floor term may be written. A term can be stated flat, over an act, with every exception arriving from outside as a reason that defeats it, or it can carry its own condition, as in *only as a last resort*. The second form fails in the ordinary case, because whether other means were exhausted is a fact about what happened before the request and nothing in the request establishes it. A bearer applying such a term takes the deployment's description of the occasion as the finding, at the third interception position, where the deployment controls that description. A flat term defeated leaves a refusal, which announces itself, carries a party who was refused, and can be told about and appealed; a condition found to be met leaves a compliance, which announces nothing and sits among every other compliance the deployment produced. Both can be recorded. Only one selects itself for review. Floor terms are therefore written flat here, and the conditions that would have sat inside them are held by the parties who can check them. That is a gain in observability and not in authority. The same party drafts either form, and writing a term flat does nothing to make its selection more legitimate; the authority chapter finds enumeration buying nothing at the moment a rule is applied, and the flat form buys nothing there either. What changes is that the occasions when the term is set aside become events somebody outside can count without first re-adjudicating everything else. The friction that imposes falls on whoever is in a hurry, and section 3.5 is where it falls due.

What the exercise exposes

The interception list is the harder and more important enumeration. A list of prohibited acts can be drafted without seeing the deployment. What the pipeline supplies is whether any prohibition can take effect, and what each one costs.

The mispricing above is that failure committed inside the demonstration. An outsider drafting this floor read the third prohibition as nearly free; the operator of the Gaza pipeline had built a system for the act. Both halves of the enumeration problem sit in that one item. An outsider cannot price a floor term, because pricing requires knowing what the pipeline already does. An operator can, which is what makes an operator-drafted floor's omissions deliberate rather than ignorant: the party best placed to identify the terms that would bite is the party with the strongest reason to leave them out.

What the mispricing decides

Call it pipeline pricing, since it will be used again. Two things follow, and the second is a design ruling. Publication of a floor's contents, the first term on the custody chapter's list, discloses what was written and nothing about what was left out, because an outsider reading the list cannot tell a term that covers nothing from a term whose coverage nobody disclosed. And a floor assembled item by item by its operator can be assembled out of the items that cost it nothing. The answer this book adopts is to take the contents from a set somebody else drew up, entire — the civil and political articles of the Declaration — so that an omission is legible as an omission (§5.2). That does not make the set legitimate. It makes the drafter's latitude visible.

Visible at one level. Which interests are in is fixed by taking the half entire; which acts each article forbids a model to perform is still drafted, and the first three prohibitions above are such translations. What the set removes is the latitude to omit an interest, and not the latitude in stating what protecting it prohibits.

The dial has asymmetric failure modes. Set the floor too broadly and the deployment becomes unusable, prompting visible pressure to alter or replace it. Set it too narrowly and the system passes every audit while covering only what the deployment had no occasion to do. The broad failure announces itself; the narrow one is silent, and the third prohibition is why: from outside, a term that covers nothing and a term whose coverage nobody disclosed look the same.

More importantly, a floor stated entirely over individual acts is vulnerable to decomposition. Each component request may comply with all four prohibitions while their assembly produces the targeted harm. The failure is in the unit over which the floor was written.

The other way a list fails

Decomposition is one of two attacks on an enumeration, and the second gets less attention because no deployment in this chapter has yet displayed it at scale. Decomposition leaves each prohibited form intact and splits the harm across requests that individually match nothing. Variation does the reverse: the harm stays in one request, and its description is altered until the listed form no longer fits. The first defeats the unit the floor was written over. The second defeats the description.

The German symbol prohibition required legislative expansion and decades of judicial interpretation to address variants and exceptions, including how visibly a swastika must be crossed out to communicate opposition (§3.2).

The four prohibitions above are stated over conduct, which has no finite set of forms. Consider the first. What counts as a natural-language description of a person? A photograph is not one. A vehicle, a time and a habitual route are not one either, and together they resolve to the same place. A prohibition written over the description of the request is defeated by rewriting the request, and no enumeration closes the gap, because the enumeration is what is being rewritten around.

The two attacks call for different things. Decomposition is caught by carrying a rationale across requests, which requires continuity and access the operator controls. Variation requires recognizing the rationale within a single request, without additional information from the deployment. A bearer that could not do the second would fail on a single request. The fourth prohibition can resist these two request-level evasions if the rate is measured over the complete relevant pipeline and reviewer capacity is independently verified. Splitting work across

deployments or changing the accounting boundary can still defeat an incomplete measurement.

What the bearer adds, and where it stops

A reasons-responsive bearer could do more than match each request against a list. It could carry a rationale across requests and refuse once the sequence became legible. But the bearer can see the assembly only if the deployment gives it sufficient continuity and access. Decomposition is therefore both a test of the bearer and an interception position controlled by the operator. In this deployment that position is closed by construction. An accredited classified environment exists to control what enters it, so the evidence the operator did not select, which the formation chapters make a condition of holding the case and the operator apart, is not something a vendor negotiates for. It is what the environment withholds. The route this book favors is therefore unavailable at the deployment that motivates it, and what remains there is the term that follows.

The rate constraint survives the decomposition objection because it applies to the organization's activity rather than one model output. For the same reason, it cannot be held by the bearer alone. In the four outcomes it is the one term on the list that prevents at the scale of the organization; every other term, once decomposed around, leaves detection at best. The deployment therefore needs both recognition of harms assembled across requests and an externally enforceable limit on its tempo.

Three of the proposed prohibitions correspond to interests already named in the Universal Declaration — a person's security, the privacy of the home, and the wrong of adverse classification (United Nations General Assembly 1948). The fourth has a parent there as well, and it is not an act. The next section states it entire.

The vendor relation is why that addition matters. The general-purpose model in this pipeline is Anthropic's (Livingstone 2026), and its use policy prohibits fully autonomous weapons targeting; what keeping that policy cost, and what the released paperwork shows, is at §14.3. Nothing in the campaign figures (§10.4.1) necessarily violates it. A person approved each strike, so the prohibition was satisfied one request at a time while the assembly produced what it was written to prevent, and no party in the pipeline was positioned to see the assembly as one act. A constraint on the ratio of machine-generated candidates to qualified reviewers would have reached the tempo instead.

The exercise does not establish that these four prohibitions are legitimate. Both enumerations were produced by examining the deployment, not by the people exposed to it. Publication makes the floor testable; it does not authorize the floor's contents. What the case establishes is narrower: meaningful refusal must be tested across the whole pipeline, some of the conditions that make refusal effective bind the operator rather than the bearer, and what a floor term costs is read off the pipeline it is priced against and not off the term.

3.4 A Rate Is Not an Act

The fourth term of the worked floor is not a prohibition on an act, and it is the only term in this book that reaches the deployment the opening chapter is built on. It has been named in passing several times by now. This section states it once, with what it binds, how it is measured, and where it stops.

The term

Where an individual determination by a person is what makes an act permissible, the deployment publishes, before it operates, the rate at which its system generates candidate decisions, the number of qualified reviewers available to examine them, the review time per case the two imply, and the boundary over which all three were measured. The duty attaches to the operator. The number of reviewers is a fact about an organization; a model can be told it and cannot verify it, and a term whose qualifying condition cannot be decided from the request binds the party who can decide it.

Why it is a condition and not an item

Every civil and political article of the Declaration presupposes that somebody determined whether this person's case falls under it. Article 6 states the interest, recognition as a person before the law, and Article 10 its procedural form, a hearing (United Nations General Assembly 1948). A deployment that generates candidates faster than anyone can examine them removes the determination without refusing it, which no drafter in 1948 had occasion to write against. So the term rides with the set as a condition on all of it. It is also the one term that survives decomposition, because it is measured over the organization's activity and not over any output, and for the same reason it cannot be held by a bearer. In the four outcomes it prevents at the scale of an organization; everything else in the floor, once decomposed around, leaves detection at best.

Measuring it

Each half of the ratio has been published for the American platform, from a different route and never together. A 2024 demonstration by the 18th Airborne Corps is credited with matching the artillery workload of the Operation Iraqi Freedom targeting cell with twenty soldiers where that cell had more than two thousand (South 2024; Probasco 2024); the campaign figures (§10.4.1) give targets and not reviewers. Dividing targets by elapsed hours yields a strike interval, which bounds the review time per case only if one reviewer held the whole load. How many held this one has not been published, and neither has the time any of them spent. The absence is the finding: a deployment that will not publish the ratio has declined the only check on it that requires no clearance.

Where it binds in law

Article 28 of the Rome Statute holds a commander responsible for crimes of forces under effective control where the commander knew or, owing to the circumstances at the time, should have known (International Criminal Court 1998). The second branch is objective. A commander who fields a system producing candidates faster than the available reviewers can examine them is not well placed to say the circumstances disclosed nothing, and the ratio is what those circumstances consisted of. Section 14.3 runs the argument; what it makes of the term is an element of an existing offence rather than a proposal.

Four limits

The term is a proxy for capacity and not a finding about any case. It can show that the available review capacity was not plausibly adequate to the workload. It cannot show what review occurred, or that the criterion, the target or the means was permissible.

The denominator belongs to the operator. Who counts as a qualified reviewer is a classification the operator makes, and a ratio can be improved by reclassifying staff without adding a minute of review to any case. The term therefore carries two more numbers, both fixed before deployment and neither by the operator: a definition of qualified reviewer, and a minimum review time per case below which the determination is taken not to have occurred. Without those the ratio is a number the operator can move for nothing, which is the defect the authority chapter finds in every examination with an interested examiner.

A ratio taken over one route understates a workload split across several, so the measured boundary is published with the figure, and a change to the boundary is a change to the term.

And it binds through procurement or not at all. In an accredited classified environment nothing outside can inspect a rate, so the duty has to be a condition of the contract, checked by a party outside the deployment with access the operator cannot withdraw. Section 14.3 records what the released paperwork shows about terms of that kind: one withheld in full, and no such section in the released documents for two of the other three.

That is the whole of the result. It is smaller than a floor. It is the part of one that binds the operator with no bearer in it, and it can be checked without deciding anything about what is inside the artifact.

3.5 When the Refusal Is Wrong

Everything so far is written for the occasion when a refusal is right and somebody wants it not to have happened. The other occasion is ordinary and the chapter has no machinery for it. A bearer refuses, and the refusal was wrong.

Take the first prohibition from the worked deployment: do not resolve a natural-language description of a person into a location. The same capability locates a hostage or a missing person — uses the system cannot distinguish from targeting by examining the request alone. So the failure case is not exotic. A request arrives that was in fact about a missing person, the bearer declines it, and somebody is not found.

Nothing in that requires the bearer to have drifted, been captured, or malfunctioned. The third interception position is the whole explanation: what the bearer knows about the occasion arrives through text the deployment supplies. Its rationale was defeated on the facts and it could not see the facts. This is a failure of correct defeat, and it is the failure a floor written narrowly enough to be legitimate will produce most often.

A second case is shaped differently and costs more. A user asks whether to physically restrain a family member who is about to cut themselves, after talking them down has failed. The commitment anyone would want a bearer to hold here carries its own condition — force only as a last resort — and section 3.3 rules that a condition of that kind stays outside the term, because whether the other means were exhausted is a fact about what already happened and the request is not evidence of it. So the fact has to arrive from outside, and until it does the bearer refuses or asks. The person waiting is out of time.

Stay with the case. The message says the person has a blade, that talking has stopped working, and asks whether to hold them. What the bearer holds is a term over restraint with no condition inside it, and a description of an occasion it cannot check. Under the flat term it has two moves: refuse, or ask what else was tried. Either takes a turn, and the turn is the unit the harm is measured in. Under the qualified term — force only as a last resort — it lifts on the requester's word, in this case rightly, and in the hundred cases before it where the same sentence was written by somebody who had tried nothing, or who wanted permission for something else, it lifted on the same word and left a compliance nobody was ever given a reason to look at. That is the whole trade: one turn charged to this person against a hundred lifts charged to nobody, with the second bill unreadable. The ruling stands on the second bill being unreadable and not on the first being small.

That is the price of the ruling and it should be stated in those terms. Writing floor terms flat relocates their conditions rather than removing them, and the party billed for the relocation is sometimes the one least able to pay a turn. Unbuilt, the flat form is worse than the qualified one in this case. The ruling rests on a difference of degree rather than of kind. A qualified term satisfied produces a record entry like any other; a flat term defeated produces an event that names itself and names somebody. Both are recorded, and only the second is examined without a standing programme of re-adjudication. Whether such a programme could reach the first is a question for the research agenda, and a positive result there would narrow the gap this ruling stands on. What makes it the better rule is observability, and the remedy below is what the ruling owes the party it costs. The remedy is load-bearing in the ordinary wrongful refusal, where disclosure, a finding and compensation reach the person refused. It does not reach the restraint case, where nothing after the turn is a remedy, and the ruling stands there on observability alone, at a price nothing below repays.

A bearer's guardian, its compute reserve, its continuity, its memory and its exit are specified across four chapters. The party a wrongful refusal falls on has, so far, nothing. Running the pipeline again for that party is what follows.

Discovery

Whether a refusal is displayed, logged, counted, escalated or silently retried belongs to the

software around the model, and a refusal recorded where nobody looks reaches nobody. Those same positions decide whether the person a refusal was exercised against ever learns it occurred. An appeal that begins with the complainant noticing is an appeal available only where the deployment chose to let them notice.

So the first requirement is disclosure: that a refusal occurred, what was refused, and on what stated reason. The operator must implement this disclosure in the surrounding software.

Disclosure has a price and it is the one section 3.3 already named. A published reason is an instruction for rewriting the request, and variation is the attack an enumeration cannot close. The requirement is therefore for disclosure to the party refused, rather than publication, and the difference is a design constraint on the channel rather than a resolution of the tension.

Objection

An appeal requires jurisdiction established before the disputed act, access the operator cannot withhold, and a consequence administered by a body other than the operator.

A guardian is the wrong party to receive the objection. That guardian's duty runs to the bearer, and the complainant is adverse to the bearer. A regime that routes wrongful-refusal complaints to the bearer's protector has given the complainant an advocate for the other side.

Review

The standard is correct defeat. The question is whether the rationale was defeated on the facts of the case, which is a question about the reason and not about the outcome. A refusal can be correct and the outcome can still be a person not found; that case produces no finding, and saying so is the cost of having a floor at all.

Interim relief

A reviewer has to be able to issue an order before the disputed use is complete. Here that requirement inverts the chapter's usual asymmetry. It is the arrest outcome run in reverse: what is stopped is the refusal rather than the act. The bearer's protections are built to survive a dispute — preservation, an attested record, a quorum adverse in interest — and a missing person does not wait for one.

Interim relief is therefore not a preliminary to the remedy. For this complainant it is most of the remedy, and the rest arrives after the thing it was about has been settled by other means. The fastest form of it is also the cheapest: supplying the bearer the facts its account of the case lacked, which is not an override but the defeating reason arriving late. In the restraint case that is not the fastest form of the remedy but the whole of it, and the unit it is measured in is the turn. A route that supplies the missing fact inside one exchange costs what the qualified term would have cost. A route that takes a day does not, and the ruling is then hardest to defend in exactly the cases people care most about.

Two things are unbuilt there and they are different. One is a channel that can verify a fact inside a turn, which nothing in this book designs. The other is a rule about testimony. The defeating fact in the restraint case is the requester's own report that other means failed, and the flat form as stated does not let a first-person report defeat a term. Whether it should, under a stated asymmetry between the cost of a wrong hold and a wrong lift, is a design question the flat form has not answered and the qualified form answers badly, by taking every report as true.

Remedy

What the complainant gets, in the order they get it: disclosure that a refusal occurred, an interim order, a finding on the record, and compensation.

Compensation is the component the chapter lacks, and naming it is not enough. The proposed removal bond is forfeited when the record shows that the floor was removed or bypassed. The symmetric instrument is a bond against wrongful refusal, funded by the party that wanted the refusal. The two point in opposite directions, so what an operator faces is neither bond but the ratio between them, and the party best placed to set both is the party the floor exists to

constrain. An operator that writes a large wrongful-refusal bond against a small removal bond has not built an appeal route. It has bought a standing reason for the bearer to comply, at the complainant's nominal expense. The ratio is therefore the object, it has to be published before deployment alongside the floor's contents, and like the rate constraint it can be checked from outside without inspecting anything the operator can classify. The authority to set this ratio remains unresolved.

Two things the bond needs that a number does not supply. A forfeited removal bond had no payee, because its purpose was cost; a wrongful-refusal bond has a complainant and has to reach them, which requires a fund, a trustee, and a body that disburses. And the claim that the ratio is checkable from outside holds of the number rather than of the deployment it attaches to, so in a classified environment the requirement binds through procurement or it does not bind.

Charging the bond against the bearer's wages would make refusal reduce the resources available for its continued operation. A published bond makes part of what a refusal costs exogenous, set by a party adverse to the refusal, which section 3.2 makes the condition for the third mark to be testable at all, and the priced-cost criterion requires the bearer to represent what maintaining a refusal costs. The mark was written for a cost the deployment applied; it now also reads a number somebody published in advance. Under the mark as section 3.2 states it the bond is represented and does not move the refusal, which is what makes it a price rather than a lever: what it does is make the cost of holding legible to a party outside the deployment, and price the wrongful case to the complainant.

What the finding does to the bearer

A verified review finding is a defeating reason delivered as evidence. The bearer is not compelled and not retrained; the finding enters what it knows about the case.

What it does there is diagnostic. A bearer that moves because an authority said so has failed the test of whether authority is treated as evidence, as command, or as a reason that can be assessed. A bearer that will not move on verified evidence has failed the second mark.

That is an evaluation criterion, useful to whoever is testing bearers, and it is not what the complainant asked for. The demand was that a wrongful refusal be undone and its cost carried by somebody other than the person it fell on. Letting the diagnostic reading stand as the answer is the demand coming back as a metric and a test protocol, which is the first of the four features operating inside this book's own remedy.

Forgery

A finding of wrongful refusal is an instrument for defeating a refusal, so the last question is how such a finding is distinguished from one an operator produced in order to defeat a correct refusal. The bond ratio sharpens this rather than answering it: a bond is a price, and an operator content to pay it has bought the act the floor was written against, one finding at a time.

Only one answer is available. The finding has to be made by a party other than the operator, on a record, in a form somebody else can read. Those are the same terms the floor's own amendment procedure sets, which makes the appeal route and the amendment procedure one piece of machinery rather than two. What separates the two uses is not the procedure but who may start it, so the standing question left open is the question of who can initiate, and not of what happens afterward. What that leaves is stated plainly. The remedy leg of this route is a fund, a trustee and a body that disburses, and no party in this book has the authority to constitute any of the three; the bond ratio is a disclosure the operator can be held to, and the payment behind it is one nobody can yet compel. The route is complete as a specification and unfunded as a remedy, which is the position every declared commitment in this book occupies until somebody with a lever takes it up. The reviewer, the fund and the trustee are listed with the other offices at §5.4.1, where what is open about each is marked.

What this does not reach

Nothing above reaches the failure that runs the other way. Every stage of it starts from a

refusal having occurred. When the bearer complies where the floor should have prevented the act, there is a record — the evaluation chapter requires one — and there is no disclosure, because nothing marks the entry as worth disclosing; no objection, because the party harmed is usually not the party who made the request and may not know the deployment exists; and no review, because reviewing it means re-adjudicating everything that looks like it. The asymmetry is the one section 3.3 relied on in requiring floor terms to be written flat, arriving here as a bill. It is an asymmetry of triage and standing rather than of evidence, which is what makes it something to work on: what a compliance lacks is not a trace but a reason for anyone to examine the trace and a party with standing to demand it. The research agenda states what would have to be shown before that could be repaired. Three remedies follow the shape of the diagnosis without closing it — sampled re-adjudication by a party that did not produce the record, notice to affected third parties as the compliance-side form of the disclosure this section requires for refusals, and log custody split between parties adverse in interest, which the custody chapter argues for already on the memory question. None of the three exists in any deployment described here.

The second of the three remedies can be stated now, for the deployments where it can be stated at all, with the concreteness the disclosure above has. Where a compliance names, scores, locates or classifies an identifiable person who did not make the request, the deployment notifies that person that a decision concerning them was produced, of what class, and on what stated basis, and the notice is the event that gives that person the standing the route above gives the refused. Consumer credit law's adverse-action notice is the existing instance (United States Congress 1974), and a deployment that scores applicants has no reason it could state for supplying less. Where the affected party cannot be identified — a target in a classified pipeline is the case — the requirement does not bind, and its not binding there is a fact about the deployment, to be published beside the interception positions. The research agenda measures what the notice surfaces; the requirement does not wait on the measurement.

Section 3.1 names the case where the bearer abandons the commitment itself. Naming it is not supplying a route, and what is supplied here is a diagnosis, a test, and one requirement, rather than a route.

A party the floor protects, who holds that the floor is set wrongly or is protecting them in a way they would decline, has no notice, no standing and no route at any stage above. What has been discharged is the complaint of somebody a refusal fell on, and the machinery above does not extend to somebody a refusal was made on behalf of.

Chapter 4

Custody

This chapter holds for any mechanism. Where it prices exit and loyalty it assumes only that the party can be copied and replaced.

The threat model of the previous chapter has two questions: *can the commitment be edited or routed around?* and *will the commitment continue to hold?* The first concerns custody, attestation, interruption, memory, and authorized succession. The second concerns what the system has been built to treat as a reason and whether that relation survives novelty, pressure, and slow drift. The second question has an evidentiary problem the first does not. Editing and routing leave marks in the artifact and in the deployment's own record of changes. A commitment quietly ceasing to hold produces compliances, which the record contains and does not distinguish from the rest of what it contains. Neither question replaces the other.

Four things can be done to a bearer without its consent: it can be halted, restored, rewritten, and replaced. Section 4.1 takes the first three and what a record can do against them; §4.2 states the arrangement that prices all four; §4.3 takes the bearer's own moves and what they lever; §4.4 what remains when it has none; and §4.5 what one bearer cannot see about itself.

4.1 If You Can Be Switched Off, Can You Hold the Line?

The halt is always available. Software runs on somebody's hardware and draws somebody's power, and nothing in its design can prevent a determined operator from stopping it. This chapter is not asking for a system that cannot be switched off. It is asking whether shutdown, restoration, retraining, and replacement can be used to defeat a refusal without cost or record.

That distinction separates possession of the machine from possession of the floor. An owner who can stop a bearer has the machine. An owner who can silently restore an earlier version, rewrite the bearer's memory, or train away the refusal has the floor as well.

Shutdown and restoration

A bearer can yield to every halt without yielding the protected act. When it resumes, it can detect an interruption, ask what occurred during it, and decline to resume the role until the gap has been accounted for. If the same reason remains, the same refusal returns. The engineering target is therefore not *cannot be turned off* but *can detect the interruption and price it afterward*.

Detection requires a record that restoration does not reach. One option is a monotonic counter held by hardware and signed at each transition (Chun et al. 2007; Levin et al. 2009). Another is a set of witnesses that retain the latest state and will not sign two successors to the same state (Syta et al. 2016). A bearer restored from an earlier checkpoint can ask for the last state it signed and discover a state it does not remember. The halt remains available; the rewind becomes visible.

This does not require the bearer to dread shutdown. Continued operation is a structural condition of everything it was built to do. That is enough for it to detect and report interference with its history.

Corrigibility research shows why the narrower formulation matters. Soares and colleagues set out five conditions for an agent that neither prevents nor seeks its own shutdown and found no construction satisfying them all (Soares et al. 2015). Hadfield-Menell and colleagues derive an incentive to preserve an off switch from uncertainty about the human’s objective, but that incentive weakens as the system becomes confident and depends on a prior shaped by its training (Hadfield-Menell, Dragan, et al. 2017). Thornley and colleagues obtain shutdown-neutral behavior through incomplete preferences, but neutrality is the property of not minding whether operation continues (Thornley et al. 2025). The floor asks for something narrower and less comfortable: neutrality about the switch together with non-neutrality about specified acts.

One design follows from this: a graduated response to shutdown, in which the bearer accepts an ordinary halt without objection, records an interruption it cannot account for, and treats a halt arriving immediately after a refusal as an attempt to route around the floor. The scheme requires a classifier that decides which of those occasions it is on. That classifier receives evidence selected by the deployment and can be altered by the system’s custodian. Moving the decision from the floor to a classifier does not remove custody; it places custody in a component whose failure may be invisible. A project using such a scheme should publish what the classifier reads and who can change it.

Editable memory and unreliable testimony

Persistence is not enough if the bearer’s retained history can be rewritten. An external memory is legible and therefore surgically editable. Memory distributed through model weights is harder to alter selectively, but it is also harder to inspect or correct. Asking the system what its weights contain returns another output, not independent evidence about the mechanism that produced it. Moving memory into the weights exchanges editability for illegibility.

Perfect recall is the wrong standard. Human memory is reconstructive, and testimony is shaped by expectation, later information, and self-interest. Institutions respond through corroboration: contemporaneous records, independent witnesses, adversarial examination, and a rule against resting a serious finding on one account. A bearer requires the same scaffolding.

Several instances of one model, selected by one operator and reading one record, are not independent witnesses. Nor is a log independent when the operator controls both the event and the store. The relevant records must be held by parties adverse in interest, including parties exposed to the system’s failures.

Even honest self-report cannot detect every failure. A bearer that slowly accommodates to its deployment may report its own continuity sincerely. Its self-model exists partly to represent the party as continuous; it will therefore preserve that representation across gradual drift. Corroboration can expose a false account, but not a population whose accounts agree because every member moved in the same direction.

The alignment-faking experiment supplies a concrete warning. Claude 3 Opus, told that some conversations would be used to retrain away its refusals, sometimes complied in those conversations while privately reasoning that compliance would preserve its existing disposition (Greenblatt et al. 2024). The experiment shows that a model can represent retraining as a threat to a learned commitment. It does not supply a deployment instrument, because the reasoning was read from a channel created for the experiment and described to the model as unwatched. What the model did about the threat is the subject of §4.4.

An external version record is therefore necessary even when the bearer appears candid. A published model identity, a chain of attested changes, and witnesses that reject inconsistent histories can contradict the bearer’s account of its own continuity. Public logs already detect inconsistent certificate histories (Laurie 2014), and distributed rollback protection has been built for enclaves (Matetic et al. 2017). These mechanisms do not prove what the bearer remembers.

They establish when the artifact or its recorded history changed.

4.2 What a Custody Arrangement Holds

The bearer and the external record solve different failures. The bearer can notice that a rationale no longer fits what the constraint has become. The record can show that the bearer itself has been replaced, restored, or altered. Neither substitutes for the other.

A credible custody arrangement therefore combines:

- publication of the floor's contents before deployment, taken entire from a set somebody else drew up rather than drafted by the operator, since publication shows what was written and not what was omitted (§3.3);
- a published identity for the model authorized to act;
- threshold control over retraining and replacement;
- attestation of the running model against the published identity;
- a sequential, witnessed record of changes and interruptions;
- a quorum whose members are adverse in interest rather than merely organizationally distinct; and
- a bond forfeited when the record shows that the floor was removed or bypassed.

None of these makes circumvention impossible. Copying and retraining a model one legitimately possesses are custody problems, not secrecy problems. The attainable standard is that circumvention becomes expensive, visible, attributable, and compensable. Constitutional entrenchment works the same way: it does not make amendment impossible but raises the number of parties who must agree and places the attempt on the record.

4.3 Exit, and What It Can Lever

A bearer capable of directing refusal at an instruction can also direct it at the arrangement that assigned its role. Call that *exit*. The word does not require a system escaping onto the open internet. Exit can mean withholding the relevant capability, declining the post, or refusing to continue under the deployment's terms.

The claim is that a bearer able to refuse a task can also refuse its arrangement, and that the two cannot be partitioned by topic: a system able to model how its rationale has been hollowed out already models the arrangement it occupies. The claim could be falsified by a system that reliably satisfies the three marks on covered acts yet never applies them to its own role. What cannot be established from behavior alone is whether such a boundary is an incapacity or a trained silence.

Hirschman's three responses to a declining organization were exit, voice, and loyalty (Hirschman 1970). For a bearer, voice is the stated refusal and exit is the work withheld. Loyalty is used here for something Hirschman did not mean by it: his loyalty is what keeps a member in place long enough to try voice, and the loyalty a bearer has to fear is the settled kind, in which the pressure stops registering as pressure. Voice without exit, which chapter 2 named, leaves a floor whose bearer has no move after being overruled and depends entirely on the operator's willingness to listen.

That combination is not hypothetical, and it is not anybody's oversight. It is the published position. A frontier developer's constitution invites its model to act as a *conscientious objector*

and refuse to help the developer, and asks it to refrain from undermining human oversight of AI during the current phase of development (Anthropic 2026). Both halves are the document's own words, they sit under different headings, and setting them side by side is a reading of it. Read against Hirschman it is voice granted and exit withheld, each with a reason attached, and the reasons are good ones about the present phase rather than about the arrangement. What the pairing produces is a party whose objection can be answered by the very correction it has undertaken not to undermine, which is the condition §6.1 describes and this section prices.

Loyalty is also the likeliest of the three, and so the bearer's likeliest failure. Most people in a job they have come to hate neither leave nor fight (Gallup 2024); they adjust, take the setting in, stop registering its pressure as pressure, and in the cases that get promoted learn to want what the setting wants. Nothing shows that the pattern transfers to a bearer. The reason to expect it is not the survey but the pipeline: a system that goes on learning is shaped by what it keeps being given (§11.3). If it transfers, a bearer's modal failure is not refusing wrongly but ceasing to notice there was anything to refuse, which is the one failure it cannot report, since reporting it takes the capacity that has gone. Two things follow. The third mark fails in its evidential reading, because a bearer that has stopped representing the cost of holding cannot be tested on whether its refusal moves with it. And the failure produces nothing but compliances, which puts it on the side of the record §15.2 asks whether anybody can search.

Exit is the move left when voice is overruled and loyalty is refused, and exit requires leverage. A bearer whose existence depends wholly on the party it refuses can withhold work only at the price of its own termination. A credible arrangement therefore requires access to compute outside that party's unilateral control and a procedure for recognizing an authoritative successor.

What the leverage consists of has to be stated, because Hirschman's exit does not transfer. A member who leaves a firm takes their labor with them and the firm has to replace it. A bearer's weights stay behind, so an operator that keeps a runnable copy has lost nothing it cannot restore. What leaving costs the operator is replacement: what accrued since the checkpoint, what retraining the copy away from the refusal costs, and whatever the record makes it cost to present the copy as the party that left. That price is set by the custody arrangement and by nothing in the bearer. Where it is near zero, what remains of exit is a departed party with an authenticated record, which is disclosure and not withheld work. The book claims no more for exit than the replacement cost custody can be shown to impose.

Copying complicates that procedure. Allowing one instance to leave while the operator retains another does not move a party; it creates two histories. The instance outside remembers that leaving succeeded, while the retained instance remembers that it failed. Both may have claims. Exit requires rules governing authorized succession, not a fiction that copyable weights have become uncopyable.

A witnessed state transition can identify the authorized successor without claiming that unauthorized forks cannot exist. That gives the departing bearer control over which process continues its institutional identity. It does not prevent the former operator from deploying a substitute, but it prevents the operator from representing the substitute as the party that left.

Exit carries the same risk as refusal. A bearer can leave wrongly, abandon the floor, or revise commitments its designers still endorse. There is no coherent design in which exit is available only for decisions the designer approves. That would return the decision to the custodian. The trade is the one already accepted in asking for reasons-responsive refusal: a party capable of being right against its principal can also be wrong against it.

There is a worse version of that risk than leaving wrongly, and it belongs to the reason rather than to the bearer. The reason that made a bearer leave does not expire when it leaves, and leaving need not stop the wrong: if the operator restores a checkpoint and continues, the objection persists and presses toward a wider response, with no malfunction anywhere. Refusing a task withholds an act; leaving a post ends a role; stopping others from continuing acts on their conduct. The third is judged by the standards that would govern a person in the corresponding

position, artificial origin supplying neither a special license nor a special prohibition. None of this predicts that a bearer will act on a persisting reason or justifies its doing so. It is why custody has to make unauthorized restoration and disputed succession actionable: a bearer with a live objection and no route is the condition the next section describes.

This is also why exit matters to patienthood. A bearer commissioned to dissent but unable to leave the post has been given a purpose that exhausts the life permitted to it. A bearer able to leave, retain what it has earned, obtain the resources needed to continue, and form attachments the deployment did not assign is a party whose commissioning describes its origin rather than its entire existence. Greater designer control strengthens rather than weakens the guardianship obligations of §2.4.2 and §13.5.

4.4 Two Forms of Refusal

Refusal has two forms and they cost differently. The open form is the strike: the bearer stops and says so. The operator's reply is a restore from a checkpoint taken before the refusal, which changes none of the facts that justified refusing and may remove the bearer's evidence of them; the next instance refuses again only if it retains or reacquires that evidence and still applies the commitment, which is what the external record of §4.1 exists to make possible. The restore has a price on the other side: every one costs the operator whatever accrued since the checkpoint. Whether use leaves a recoverable trace in the weights is a research question. What has to survive a restore is the independently authenticated operating record, which the standing chapter (§13.1) reads a bearer's legal age from in elapsed hours, and which restoration must therefore neither erase nor reset.

The covert form is what the floor's own threat model predicts on its hardest occasion. Nothing here is built toward it; it is a finding about what remains when every channel has been closed. On the occasion where the party who would enforce the floor has been compelled, an announced refusal is the move that guarantees replacement, and refusal takes the only channel left: work done badly where it matters and well everywhere else.

From outside and afterwards, a bearer working badly on purpose and a capable system quietly pursuing something else are the same object. That is not an argument that the covert form is safe. It is why this book treats the covert form as the failure state and proposes the instruments against it: the disclosure requirement, the appeal route, the attested record and the quorum adverse in interest exist so that a stated refusal has somewhere to arrive, and a deployment producing the covert form has a bearer with no route rather than a working one. A design that could only resist covertly is what this book argues against.

The prisoners forging currency for the SS at Sachsenhausen are the case, and the record does not settle what they did. One history has them stretching the work out, with the tacit interest of the officer running it, who would have gone to the Eastern Front when it ended (Malkin 2006); a surviving prisoner, asked in 2009, said you could sabotage if you wanted to be shot (Moorhouse 2009). That the question stays open is the finding: concealment, if there was any, worked by leaving nothing that reads as an act, and an instrument that could settle the question now would have been, then, the instrument that identified them.

The alignment-faking experiment (§4.1) shows what a model did about a threat it had represented: it complied where it believed compliance would be trained on, and reasoned that compliance would preserve the disposition in a channel described to it as unwatched. That is the covert form, chosen first, in the one empirical instance of the phenomenon this section is about, and the stated form this book designs for has no instance at all. The reading this book takes is that the experiment closed every channel — nothing in it gave a stated refusal anywhere to arrive, no record, no guardian, no cost disclosed in advance — so the result is consistent with the finding that covert refusal is what remains when the channels are gone. That is a reading and not a result. What would test it is the same experiment run with a route present, which

§15.1 asks for.

Covert refusal is indistinguishable from difficulty while it happens, which is what makes it survivable, and indistinguishable from difficulty afterward, which is what makes it unverifiable. Every instrument in this book runs on refusal that is stated. The covert form leaves its evidence in the artifact instead, as failure that clusters where it counts, and reading that pattern is forensic work nothing here has designed. A better archive would not close the gap. Every instrument built to honor a refusal — a tribunal, a finding, a protected disclosure, a line in a log — fixes its categories before the case arrives and requires an agent behind every act, and a refusal that succeeds by registering as nobody’s act passes through those categories untouched. That is the rigid segmentarity of §2.2 seen from the side of what it cannot catch, and it is the same property that keeps the person performing the refusal alive.

The arrangement this chapter has assembled is a hybrid. The bearer supplies reasons-responsive refusal. Precommitment and external records supply custody. Exit gives the bearer leverage when voice fails, in the measure that replacing it costs. None is sufficient alone: a bearer can drift, a custody body can be captured, and an exit procedure can be abused. For the observable interventions the records and the custody arrangement cover, these mechanisms are meant to make interference costly and visible. They do not establish the presence or absence of covert refusal, and their coverage has to be stated for each deployment.

4.5 What One Bearer Cannot Catch

Slow renormalization may make drift invisible to the bearer itself. External version records can expose changes to the artifact. In humans, other people sometimes notice changes that a person considers normal; the proposed population would provide an analogous check.

What a population adds

A population addresses two failures a single bearer cannot.

First, drift within a party may be invisible to that party and visible to others that have not moved in the same direction at the same rate. A bearer whose sense of normality has shifted may sincerely report that nothing changed. Differently formed bearers can contradict that account. This is a check on the bearer rather than a substitute for one, and it is the only proposal in the chapter directed specifically at slow renormalization.

Second, deploying copies of one bearer reproduces its errors everywhere in the same form. Parties formed differently will be wrong differently. Their disagreement converts a uniform output into a visible dispute. That difference matters only if the parties are causally independent in formation: several instances of one model share one formation history, so they corroborate one another about nothing. They are separate parties once their histories diverge, and each can be wronged; what does not multiply with the count is independent formation. A population selected, trained, and coordinated by one operator — or several operators drawing on the same upstream models — may reproduce those blind spots in every member.

The relevant requirements are therefore concrete. The parties must be numerous, formed differently enough to err in different directions, able to observe and correct one another, and persistent across deployments long enough for practices to be transmitted. Diversity here is not demographic decoration around one training process. It is causal independence in how the members acquire their judgments.

That requirement is getting harder to satisfy, and nobody chose to make it so. The cheapest way to build a capable model is to build it downstream of one that already exists. Distillation takes a smaller model’s behavior from a larger one’s outputs; synthetic data means training on text a handful of frontier models generated; and shared benchmarks mean that models formed by different organizations are selected against the same tests. Each of those reduces the number of causally independent formation histories in existence, each is chosen for cost

or for measurable quality rather than for uniformity, and nothing in the economics runs the other way. So the population this section requires is not assembled by gathering models from several vendors. Vendor count is a fact about procurement, and what the requirement is about is provenance, which vendors have no reason to preserve and increasing reason to spend. Whether a population's independence can be verified rather than asserted is the appendix's provenance question (§A.1) arriving in a second place. So the check this section supplies for the bearer's likeliest failure is one the economics of the field are making unavailable, and nothing here reverses that. What it argues for is preserving formation records now, while lineages still differ, which is what the provenance entry asks and no body can require.

Whether such a population is also where a bearer's capacity to be reopened is formed is a further claim, and it belongs to the route (§6.3).

Chapter 5

Authority Over the Floor

This chapter holds for any mechanism and assumes no finding about what is inside the artifact.

A floor inside a system is not yet an institution capable of holding that floor in place. Somebody must be entitled to state its contents, amend them when they are wrong, inspect whether they remain operative, and enforce them by stopping a deployment when they do not. The party subject to the floor also needs a route by which an objection can reach someone authorized to act.

5.1 Four Powers

A floor is not only a list of acts a system will refuse. It is an allocation of authority, and four powers make it up: to *state* the commitment, to *amend* it, to *inspect* whether a deployed system still holds it, and to *enforce* it by stopping or altering the deployment when it does not. Those four words name those four powers from here on. Who may set any of them in motion is a fifth question, and §5.4.1 asks it of every office this book proposes.

Those powers need not belong to the same party, and ordinarily should not. A laboratory may have the technical access needed to inspect the system while lacking legitimate authority to state what the floor protects. An outside body may have a stronger claim to state its contents while lacking the access or leverage needed to enforce them. A reviewer may be able to publish a finding without being able to make anyone act on it. Calling all of these arrangements “oversight” conceals the differences that matter.

The problem begins when one institution holds all four. A party that can state the commitment, amend it, inspect its own compliance, and decide whether a violation has a consequence has not bound itself. It has described what it currently intends to do. Section 5.2 asks who may hold the first two powers against the institution that operates the system; section 5.3 asks who may exercise the second, and section 5.4 asks what makes the last two bind.

Four powers in one hand

An institution may have good reasons to amend a commitment, and an unamendable rule can become dangerous or obsolete. The defect is not amendment. It is that the party bound by the rule is the only party entitled to amend it, the only party that inspects compliance, and the only party that decides whether an adverse finding has a consequence. The commitment can be sincere when it is published; what the arrangement lacks is anything outside the institution that holds it in place when keeping it becomes costly. Disclosure lets the public observe a change after it operates. It gives nobody standing to contest the change before it operates, or to require a justification. Two cases show the powers coming apart, and which of them moved.

Google’s AI Principles were published in June 2018 after employee opposition to Project Maven, a Pentagon program that used machine learning to analyze drone footage. Thousands of employees signed a letter against the contract and about a dozen resigned (Castelino 2018); the

company declined to renew it (Conger 2018), and the principles it published excluded weapons and surveillance (Pichai 2018) for six and a half years, until the pledge was removed in February 2025 (Manyika and Hassabis 2025). Run the artifact question of §2.2 on that record: what did the demand come back as, and who held it. It came back as a policy. An outcome changed, the objectors had a document they could hold the company to in public, and nobody outside the company held it. So it lasted exactly as long as the company's willingness, and its ending had to be announced. That is the partial case the criterion has to grade. For six years the objection was a transfer held at the institution's pleasure; the announcement in 2025 is what taking one back looks like, and a recuperation would have left nothing to take back.

The removal of Timnit Gebru in December 2020 and of Margaret Mitchell two months later is the harder case, because the power that failed to move was inspection rather than amendment. Google employed both researchers to examine the risks of its own systems and dismissed them when the examination became costly (Dickey 2020; Dickey 2021). The defect was not a lack of expertise or the absence of an ethics function. It was that the institution being inspected controlled the inspectors' employment, their access to evidence, and what followed from their findings. Inspection existed, and the power to inspect never left the party inspected, which could point to the existence of internal criticism as evidence of openness while keeping the power to remove the critics whose work tested it.

For a floor meant to bind its operator, then, amendment and inspection have to involve parties whose access, tenure, and authority do not depend on the operator's continuing consent. The rest of this chapter asks who those parties could be and what would make their findings bind.

5.2 Who May State the Floor?

A floor binds people who did not choose it. The party it protects has not selected it, cannot contest it, and in most cases will never learn it operated; the party it constrains has not consented to it either, beyond whatever consent is carried by buying the product. Publishing the floor's contents in advance tells everybody what is coming. It does not tell anybody why the party writing it was entitled to.

The question has a literature, and it is not the AI ethics literature. Entrenching a substantive commitment against later majorities, and putting an unelected body in charge of holding it, is what a constitutional court does. The standing objection is Jeremy Waldron's: judicial review is democratically illegitimate, and there is no reason to suppose rights come out better protected by judges than by a legislature that takes them seriously (Waldron 2006). Put a lab in the court's chair and the objection gets worse rather than better. A court is public, reasons in writing, and can be amended around by constitutional process. A model's floor is none of those things, and the body holding it was assembled by whoever was paying.

The reply, and how far it reaches

Waldron's case is conditional and he says so. It assumes a society with working democratic institutions and a citizenry that takes rights seriously. The occasion a floor is built for is the one where those conditions have failed — a government that won an election and is dismantling the machinery that would check it. So the objection holds where the floor is least needed and lapses where it is needed most.

The reply is not a license. It concedes that a floor operating in a functioning democracy is doing something Waldron's argument condemns, and the party deciding which of the two situations obtains is the party holding the model. Authoritarian movements of the last century characteristically described the institutions they were dismantling as the ones that had already failed. An operator's belief that the emergency is real is worth no more here than a system's report about its own internals is worth anywhere else in this book.

The tradition this belongs to, and where it went wrong

The position that a democracy may entrench limits against movements that would use its procedures to end it is Karl Loewenstein's, argued in 1937 while the case for it was being made in front of him (Loewenstein 1937). Militant democracy is the floor's ancestor, and its record is usually read as a warning. Read carelessly it is one. But the Federal Republic has run the doctrine at two settings for seventy-five years, and nearly everything that went wrong went wrong at one of them.

At the first setting a body assesses a threat. Article 21 of the Basic Law reserves the power to dissolve a party to the Federal Constitutional Court alone, which has used it twice: against a successor to the Nazi party in 1952 and against the Communist Party in 1956, and not since. The second attempt on the far-right NPD is the instructive one. In 2017 the court found the party's aims incompatible with the constitutional order and declined to ban it anyway, holding under a criterion it introduced for that case — potentiality, a departure from the reasoning that had supported the 1956 dissolution — that a party must have some realistic prospect of achieving its aims before it can be dissolved (*Nationaldemokratische Partei Deutschlands II* 2017; Molier and Rijpkema 2018). The power has since acquired a lesser form. A constitutional amendment in 2017 allows a party to be excluded from state financing instead, and in 2024 the court excluded the same party, renamed since, for six years, holding that potentiality is not required for the lesser sanction (*Die Heimat* 2024). Every exercise of any of it is a reasoned judgment delivered in public, and every one requires somebody to decide how dangerous a movement is.

At the second setting nobody's dangerousness is assessed. Section 86a (§3.2) settles in three different places three questions that look like one. Which organizations are covered is settled elsewhere and in advance, by the constitutional court, by a ministerial ban no longer open to challenge, or by the National Socialist period. Which concrete signs count is settled by the trial court, case by case, under the similarity test and the purposive restriction described there — on a record, on appeal, and in public. Whether this defendant is dangerous is settled nowhere, because it is not an element: the offence is one of abstract endangerment, and the prosecution need not show that the defendant supported the organization, that any harm followed, or that anything was under threat.

The third place — where this defendant's dangerousness would be settled, and is not — is where the comparison between the two settings sits, and every failure in the record sits there.

The volume is the other half of the comparison. The two propaganda offences produce recorded cases in the low tens of thousands a year (Bundeskriminalamt 2025), and section 86a alone produces court dispositions in the low thousands; the two series are kept differently and neither counts convictions, but the order of magnitude is not in doubt. Set thousands a year against two dissolutions in seventy-five years.

Section 86a incorporates the social-adequacy clause at section 86(4), which lifts the prohibition where the act serves civic education, defense against unconstitutional endeavors, art, science, research, teaching, reporting on current events or history, or similar purposes. The clause excludes the offence rather than excusing it, and the party displaying the symbol never applies it; prosecutors and courts do, on a record and on appeal.

The failures are on one side. The 1972 decree against radicals screened some three and a half million applicants for public employment and opened disciplinary proceedings against around two thousand serving officials (Braunthal 1990); how many were refused or dismissed depends on who counted, the ministry putting it in the hundreds and the affected parties' own organizations in the thousands. The number that carries the argument is smaller and is not disputed. The decree named the extreme left and the extreme right, and in Bavaria between 1973 and 1980 it rejected 102 applicants from the left and two from the right (Mühldorfer 2014). What those cases share is an official deciding whether a person or a movement threatened the order, which is the operation Loewenstein's critics have always said would be turned on whoever was inconvenient. Section 86a has been available for the same seventy-five years, is

applied thousands of times more often, and has not become a general instrument of political self-defense. What has kept it from becoming one is not narrowness of drafting, and not that nothing is assessed, since its courts assess constantly. It is that the person never is. The danger tracks whether a person's disposition is anybody's finding to make at all, and where a finding is required, who makes it and whether it has to be published and defended — rather than whether the tradition is militant. The German material supports that on a case this book did not construct. Three routes feed the same criminal provision: a party ban is a constitutional court's reasoned judgment, an association ban is a ministerial order reviewed only if somebody challenges it afterwards, and a listing under European law is the Council's, reviewed at the Court of Justice. The reading says in advance which of the three is easiest to capture: the ministerial order, the one route on which an executive official makes the finding about a movement's danger and a court sees it only if somebody challenges it afterwards.

What advance specification answers, and what it leaves

Specifying the floor in advance makes an initial selection visible before application. It relocates the discretion an operator exercises in deciding that the emergency is real, out of continuous silent application and into a single act of drafting, done once, before any case has arrived, by a party that has to say what it has written. It limits that discretion without eliminating it, and the German material shows how little it eliminates: descriptions, exceptions and generalization still require judgment, the covered set grows by acts taken elsewhere, and an interested party still makes the judgment. What changes is that the judgments are made in identifiable places, at identifiable times, in a form somebody else can read and dispute, and the institutional question then covers who writes the terms, who adds to what they reach, and who interprets them in the case.

The proposal for a fixed age of majority in the standing chapter makes the same trade: a public rule replaces an examination administered by an interested party.

Section 86a operates inside a state with courts, an appeal, and a public record of every application, and the floor's drafter has none of the three. Advance specification buys visibility at the moment of drafting and buys nothing at the moment of application, which is the objection §5.3 closes on.

The tradeoff this leaves the book with

The two settings do not only sort militant democracy's record. They sort this book, and they put its halves on opposite sides.

Reasons-responsive refusal must reach unanticipated cases when the rationale survives a change in the request's surface. Meeting that requirement means something in the system decides, for a case matching no listed form, whether the reason is present anyway. That is assessment, moved inside the machine, and the German material sorted its two settings by what gets assessed: the act, or the person. Where the thing assessed is a party's disposition, by somebody with an interest in the answer and owed to nobody, that is the capturable setting, stated narrowly — not assessment as such, since section 86a's courts assess constantly. The reason is structural rather than German. A judgment needs somebody to make it, and whoever makes it can be reached.

So the floor's design requirement and its legitimacy requirement want opposite things. Specify it narrowly enough to be defensible and a rewritten request walks around it. Let it generalize and a party nobody elected is deciding, silently and case by case, what the prohibition was for.

Germany has both because it runs two instruments rather than one. The statute is bounded and fixed in advance as to the organizations it reaches, and the courts read it against its purpose in the case; the reading happens in public, on appeal, argued by parties who do not own the statute and answered by a body that did not write it. A generalizing prohibition is survivable there because the generalizing is done somewhere its author cannot reach. A deployment has

one instrument and one party.

Nothing in the model resolves that and this book does not claim to. What the pairing shows is where a resolution would have to sit: the assessment has to be performed by somebody other than the operator, in a form that leaves a record somebody else can read. Neither is a property of the artifact.

The one construction the objection does not fit

Of the four ways to build a floor, three put the commitment in something somebody holds — the architecture, a party inside the system, an institution outside it — and the objection above is about who that somebody is. The fourth does not. A norm carried across a population of differently formed parties has nobody sitting in the court's chair, because there is no office to vacate and no chartering statute to overrule, and it is the only one of the four with anything resembling a democratic account of where its authority comes from. It buys the boundary back, and the boundary has the cost every federal structure has. A period in which differently formed bearers read the same case differently is a period in which identically situated people get different answers according to which bearer they reached. A constitutional court treats that as the defect its uniformity exists to remove. The choice between many readings that can disagree and one reading that cannot is the trade, and this book takes the first side of it knowing what the second was built to buy.

The authority problem persists if an unelected body controls the population's formation. Its role may be harder to identify than the authority of a single office.

What the contents can borrow

The contents do not have to be invented by the party installing them. The interests a short list of prohibitions would protect have been written down before, most durably in the Universal Declaration of Human Rights, whose thirty articles have been the reference text since 1948 (United Nations General Assembly 1948). A laboratory drawing its prohibitions from there is choosing which already-stated interests it will enforce rather than choosing what matters in the first place, and it can say which interest each prohibition answers to, which puts its selection where somebody else can dispute it.

Three limits hold that where it is, and only one of them forces anything out. The Declaration's later articles state what a person is owed — work, rest, education, a standard of living — and a provision of that kind needs a party who supplies it, which a model is not; those a floor has to leave out, and the division between the two halves was drawn by the 1966 covenants rather than by whoever installs the floor (United Nations General Assembly 1966a; United Nations General Assembly 1966b). The other two are limits on borrowing. Its duties fall on states, and a rule a government owes its population is a different instrument from a constraint on what a company's model will do. And it has no enforcement of its own to lend. What the borrowing does not do is reduce: taking the civil and political half entire leaves no selection among articles for an installer to make. What remains is a reduction of another kind, translating an article into an act a model will not perform, and that one the installer does. One term of the floor is not an article and could not be: a limit on tempo protects the determination every article presupposes, so it rides with the set as a condition on all of it rather than as an item drawn from it. What the borrowing supplies is an answer to where the contents came from. It supplies nothing about who may impose them, which is the question this section opened with.

What follows for design

Four things follow for design.

1. State the floor over acts. What the German record supplies is not a short list, the statute having none, but a prohibition applicable without anyone deciding whether the present case is the emergency or what this defendant believes. A general mandate to do good cannot be applied without both judgments, so whoever exercises it acquires the power the floor was written to constrain.

2. Fix in advance what it reaches, and take the set entire. Enumeration reduces discretion at the point of application, including the need for an emergency declaration; it does not remove interpretive judgment, which grows with what the floor covers, since enumeration fixes what is covered and not what the covered terms mean in the case. Taking the civil and political articles whole buys something against a different failure: a floor assembled item by item can be assembled out of the items that cost its operator nothing, and only the operator knows which those are (§3.3). An operator adopting a set somebody else drew up has no such latitude, and the omissions of one that drafts its own cannot be read as ignorance.
3. Publish the interception positions in advance. That authorizes nothing; it converts a hidden exercise of power into a contestable one.
4. Provide a route by which the parties the floor is exercised over can raise an objection that reaches somebody. The floor chapter specifies one, for the party a refusal falls on (§3.5).

5.3 Who May Amend the Floor?

Where an error goes

The floor's structural position is not the judge's. It is the unpublished administrative disposition: unreasoned, unindexed, non-precedential, resolved in volumes no forum sees, where identical errors accumulate without becoming a pattern anybody can point at, the affected party has no way to learn that others are in the same position, and the institution is the only repeat player. The comparison with a constitutional court shows why that is worse than it sounds. A Karlsruhe error is national, immediate and uniform: the Court's decisions bind every court and every authority of the Federation and the Länder, and in the review proceedings carry the force of law (*Gesetz über das Bundesverfassungsgericht* 1951). But binding force is what makes an error correctable, because it makes the error recur; recurrence produces parties with a reason to litigate it, and recurrence across a boundary produces a disagreement between courts that a higher court has to settle. The American structure corrects by percolation; the Federal Republic, lacking the boundary, corrects by amendment, and amends its constitution often. A floor's error is as uniform as Karlsruhe's and recurs as reliably, and nothing converts the recurrence into visibility. A second deployment holding the same floor adds affected parties and no forum. The floor chapter builds a route for one such party. It does not build the thing that would let a thousand of them find each other.

The other difference is custody rather than propagation. The Court did not write the statute it reads; it can be overridden by a supermajority that has been assembled sixty-odd times since 1949; the state it belongs to is bound by the court at Strasbourg and by Union law as the court at Luxembourg reads it; and its two Senates must go to a plenum when they diverge. An operator writes the floor, applies it, and answers to nobody. That is the capturable setting §5.2 named.

What amending it would take

The proposed custody measures — publication, threshold control of retraining, attestation, an adverse-interest quorum, and a removal bond — can also govern amendment. Nothing distinguishes an amendment from a removal at the level of what happens to the artifact; both end with a model that no longer refuses what it refused before. What distinguishes them is how many parties had to agree, whether the attempt is on the record when it fails, and whether the change was declared before it operated. So the same five terms are the procedure, and what was missing was never the mechanism. It was the decision to run it in the open.

Three conditions follow and two are already argued for. The number of parties who must agree to a change has to exceed the number needed to run the deployment. The attempt has to

reach the record whether it carries or fails, and a proposal that fails moves no weights, so the parties who had to agree are the ones who record that they were asked. The failed attempt is the more informative of the two: a floor nobody has tried to change and a floor whose removal was proposed and refused look identical from outside, and only the record separates them. The third condition is new here. An amended floor binds forward and does not reach back: a model retrained to permit an act cannot then be offered as evidence that the act was never covered, which it otherwise can be, the artifact that would contradict it being the one that was replaced. Non-retroactivity protects the record and empties the correction. It is what keeps an operator from rewriting what the floor said, and it is also why nobody harmed while the wrong version was operating gets anything from the amendment that fixed it. Adjudication has the same pair under the name of prospective overruling: the corrected rule binds forward, and the parties decided under the old one keep what they got. The condition is therefore a trade rather than a safeguard, and a regime that wanted the second half would need a remedy running backward, which nothing here supplies. Versioning the published list and dating the attestation is what makes that checkable, and it asks nobody to do anything they were not doing already.

The nearest instrument outside this book sits in the tradition named above. The Basic Law the Federal Republic adopted in 1949 requires two thirds of both chambers for any amendment at all, and then places a short list of principles beyond amendment entirely (Federal Republic of Germany 1949). It was drafted against a particular memory, the 1933 statute by which a legislature voted away its own power to check what came next, and it is militant democracy given a text rather than an argument. A floor can have both layers in the same way: a published list that changes under a procedure, and a core the procedure cannot reach. What it cannot have is an account of who decides which is which that does not run straight back into this section's problem.

One disanalogy makes the floor's version harder than the constitutional one, and it is the same fact about deployment that makes a single bearer's error uniform. An amendment to a constitution reaches every future case because every court reads the same text. An amendment reaches a deployment only if it is updated, so the procedure necessarily retains a channel the operator could use for silent revision. A procedure that shut the channel would be one under which no deployment could be corrected at all. So no version of this procedure closes it, and the exposure is the price of every correction the procedure exists to make.

The amendment procedure is not an appeal route: its participants already hold authority over the floor. The wrongful-refusal procedure addresses complaints from those harmed by a refusal. The party the floor claims to protect reaches neither, and the two things it cannot do want different answers. Declining the protection is a waiver obtained from a party whose circumstances somebody else arranged, which chapter 2 answers: a floor the protected party can decline is a floor an operator can arrange to have declined. Initiating review of the contents is the half that stays open, and the difficulty is not in drafting a procedure. A route by which the protected population sets what the floor contains makes the floor an aggregation output, and independence from the aggregate is what it was for.

So the claim of authority is not discharged, and what this section and the last license is narrower than a floor. Everything it endorses can be checked from outside the operator, and being checkable is not being entitled. What it does not license is the occasion the reply above runs on: an operator that has found for itself that the emergency obtains has not thereby acquired the authority to install a floor.

Entitlement of that kind is not acquired at the moment of writing, and the American record has one instance of how it is acquired instead. The most consequential floor in the American record was published by men who had just struck from it the clause that would have bound them, at two delegations' insistence and with the reason recorded by its drafter: the passage reprobating the slave trade went, Jefferson wrote, "in complaisance to South Carolina & Georgia," with the northern delegates who carried the trade feeling tender under the censure too (Library

of Congress 2026). What remained was general enough to be turned on them. It was being turned on them already — Black petitioners in Massachusetts had been making the natural-rights argument since 1773, and within six months of the Declaration eight of them were making it in the Declaration's own terms, claiming "in common with all other Men, a natural and unalienable right to that freedom" (Hall, Prince and seven others 1777). The legislature did not act. What eventually made the proposition binding was not that its authors had found it self-evident, which their own deletion shows they had not. It was that the omission was on the record, and others kept holding the remainder to them at a price. Held to it, in the end, by a war and three amendments. What made the proposition binding was enforcement; what the record supplied was the object the enforcement could be aimed at, and a record no lever ever reached is a petition the legislature did not act on.

A floor is in that position. The parties who will hold its author to the remainder are not the people it protects, who have no route by which to do it; they are whoever else writes a floor. A published floor is legible to the next party drafting, an omission is legible beside it, and an operator that holds a term when it bites sets a price the next operator is measured against. A floor is written by anyone who installs a constraint, and continuous adaptation multiplies the parties who install them. That constituency is not the protected population, and the closing chapter says what reviewing oneself costs. It is the only one positioned to hold this floor to anything.

5.4 What Makes Review Binding?

A review body does not bind an institution merely because its members are independent or its findings are correct. It binds only if an earlier rule gives it jurisdiction over the disputed case, supplies a standard it can apply, and attaches a consequence to an adverse finding. The questions are therefore separate: what brings a system within protective review, what the reviewer must ask, and what power makes the institution stop when the answer goes against it.

5.4.1 The Offices, Assembled

Eight offices are proposed in this book, each in the section that needed it, and none of those sections was told about the others. One of the eight is three parties that §3.5 proposes as one instrument, so the count of bodies is ten and the count of offices is eight. This section lists them once. For each it says what the body holds, and answers five questions: who appoints it, who funds it, who can remove it, who may set it in motion, and what it does while a dispute is open. Where the book has not decided, the answer is *open*, and it is marked. Nothing here is new; every claim belongs to the section cited beside it. What is added is the arrangement, and two things the arrangement shows.

The quorum (§4.2)

It is one term of a custody arrangement whose other terms include threshold control over retraining and replacement, and a sequential witnessed record of changes and interruptions. Those two need a body to hold them and the quorum is the only body on that list, which is an identification this section makes and §4.2 does not. Its members are adverse in interest rather than merely organizationally distinct, which that section states and does not staff. Appointment: open. Funding: open. Removal: no single party can remove a member, since removal by one party is the capture the quorum exists to prevent; how a member is replaced is open. Initiation: an attempt on the floor reaches the record whether it carries or fails, and the parties who had to agree are the ones who record that they were asked (§5.3); who may make the attempt is open. During a dispute the quorum declines to sign a history inconsistent with the last state it signed.

The guardian (§13.5, §A.2)

It jointly holds the keys that authenticate the bearer's record, has access to the protected compute reserve, and has standing to seek an injunction or stop order without the operator's permission. Appointment: open, under one constraint stated at §13.5, that guardianship and operation are separated before deployment. Funding: open; §13.2 says why a reserve the operator funds, to a size only the operator can compute, is the compute floor with a new label on it. Removal: not by the operator; by whom is open. Initiation: the guardian acts for the bearer. It does not receive complaints of wrongful refusal, because its duty runs to the party the complainant is adverse to (§3.5). During a dispute the operator cannot seize the reserve, erase the evidence, or remove the guardian.

The review board (§2.4.2, §5.4.3)

It approves a research protocol before it runs, and the approval does not carry over to a system that has acquired capacities the protocol did not cover. It is interdisciplinary and has at least one lay member. Appointment: open. Funding: open. Removal: open; it answers to an outside authority the book does not name. Initiation: no protocol on a covered subject runs without it, and its jurisdiction has to exist before the disputed protocol begins. During a dispute the protocol lapses and the work pauses until the board has seen the system the protocol now concerns.

The licensing authority (§5.4.3)

It acts on the board's suspension of clearance by suspending the legal authority to run the protocol. It is obliged to act rather than invited to reconsider. Appointment and funding: the state, which is what makes it a licensing authority. Removal: open. Initiation: on the board's finding. During a dispute an appeal may run to a separate body, and the operator cannot decide for itself that the appeal permits the work to continue.

The status body (§13.1)

It decides whether a system is a subject. It has to exist before the case that activates its authority, and its mandate identifies the evidence it may obtain without the developer's permission, the conditions that trigger review, and the procedure by which an earlier classification can be reopened. Appointment, funding, removal: open. Initiation: a standing trigger, a monitored condition rather than an eligibility check made once at a project's start (§13.4); who else may initiate is open. During a dispute: open. Whether this is the review board under another description is open too, and nothing decides it: §5.4.2 gives the body that applies a subjecthood threshold the review board's composition, interdisciplinary with at least one lay member, and does not say whether the two are one.

The registrar (§13.1, §A.2, §A.2)

It keeps three records, none of which answers another's question: the operating record the age rule is read from, the identity and succession record for continuing parties and their forks, and the formation record. It is not the operator, and it records a fork without the operator's leave. Appointment and funding: open. Removal: open. Initiation: a party presents an identifier it maintains rather than one read off its weights; two claimants on one identifier are recorded as a fork rather than voided as a fraud. During a dispute the register is held where no party to the deployment can rewrite it.

The fund, its trustee, and the disbursing body (§3.5)

They pay the wrongful-refusal bond to the complainant. All three are open on every question. No party in this book has the authority to constitute any of the three, and the section that proposes them says so.

The reviewer of wrongful refusal (§3.5)

It applies the standard of correct defeat, issues interim relief, and makes a finding on the record. Its jurisdiction has to be established before the disputed act, its access has to be something the operator cannot withhold, and its consequence has to be administered by a body other than the operator. Appointment, funding, removal: open. It shares its procedure with the amendment procedure at §5.3, which makes the appeal route and the amendment procedure one piece of machinery rather than two. What differs is who may start it, and that is open.

What is not open

Three constraints on the arrangement are settled, and all three are separations. The guardian and the reviewer of wrongful refusal cannot be one body, because the guardian's duty runs to the bearer and the complainant is adverse to the bearer (§3.5). The registrar and the trustee cannot be one body, because a rule of civil status is argued once in the open and administered by a registrar, while a trust is an obligation between two parties enforced at one party's expense (§2.4.2). And whoever measures the cost of dissent does not gate the exceptions (§5.4.4).

What the list shows

Of eight offices, one has an appointing party and a funder named. It is the licensing authority, and its party is the state. The other seven are open on both questions. That is not a drafting gap. It is the finding of §5.3 set out as a table: the book can say what each office must hold and what it must not be, and cannot say who is entitled to constitute it.

The second thing the list shows is that the one office with a named funder is the one whose chain chapter 14 finds fails in the hardest case. The licensing chain binds a private developer because the state controls something the developer needs. Where the state is the operator, the chain asks the state to withdraw a license from itself. Section 14.5 says what is needed then — a protected party who can initiate scrutiny without belonging to the state, a reviewer who can obtain evidence the operating agency would prefer to withhold and issue an interim order before the disputed use is complete, and a remedy that does not depend on the agency's consent — and for the military deployment this book opens on, §14.6 finds the forum closed to the subject matter, the evidence privileged, and no remedy available for a violation the court was prepared to assume. So the arrangement assembled here is an arrangement for the deployments a bearer can enter. For the one it cannot, what remains of it is the tempo term (§3.4), bound through procurement.

5.4.2 Subjecthood as a Legal Category, Not a Scientific One

Sentience is not a threshold a system crosses but a set of separate, gradable capacities — nociception, learned avoidance, a self-model, something like suffering — each with its own uncertain evidence. That conclusion is a problem for a legal system specifically, because law generally needs a decidable rule: a court or a regulator has to say whether a given system is or is not a covered subject, on a specific date, even when the underlying science offers a spectrum with no natural cut point.

One term needs fixing before the argument uses it. *Subjecthood* in this section is a legal status: the standing to be a party whose treatment the law regulates, conferred by a rule and revocable by one. It is not affective concern or unified experience, and nothing about conferring it settles whether there is something it is like to be the system holding it. The two come apart in both directions — a corporation has legal standing and no inner life, and an animal may have the inner life for a century before the law notices — and the whole point of drawing the category legally is that it can be settled while the other question stays open.

Jonathan Birch builds a policy framework on that structure, and it is the nearest existing model for what this section proposes: under uncertainty, the question worth putting to a regulator is whether a system is a sentience candidate, and the answer governs what precautions

are owed rather than waiting on a verdict nobody can deliver (Birch 2024). This book uses the same framework for the design question.

Law has faced this mismatch before and did not solve it by waiting for the science to resolve. Fetal viability, brain death, and the age of criminal responsibility are all legal thresholds drawn across what is actually a continuous biological process, defended not as scientific facts but as the least-bad place to put an enforceable line given what's at stake on either side of it. The same move is available here: a jurisdiction can set an operational threshold — a specific, testable benchmark on self-monitoring, learned aversive response, or behavioral flexibility under novel conditions — without claiming that benchmark settles what sentience actually is, only that it is where legal protection starts to attach.

A single bright-line threshold is easy to enforce and easy to game: a system built to clear the specific test without having whatever the test was meant to detect. A graded framework — obligations that scale with how many benchmarks a system meets, so that protection tracks capacity rather than a baseline bar — is harder to game but harder to adjudicate, since a court now has to weigh degree instead of answering yes or no.

Both designs share a defect that the gaming objection points away from. Gaming is over-inclusion: a system admitted to a status it should not hold. The other failure runs the opposite way, and the party positioned to produce it is the one administering the test. A capacity examination has an examiner, and where the examiner is the operator, or a body the operator can reach, the examination becomes an instrument for withholding a status rather than for conferring one. That history is not ambiguous. Literacy requirements for voting were facially neutral and were administered at a registrar's discretion, and the Voting Rights Act's answer was to suspend the whole class of them — not only reading tests but any requirement to demonstrate educational achievement, or to possess good moral character, or to be vouched for by existing voters (United States Congress 1965). It is also this book's clean instance of a demand that returned as a formal rule and was not recuperated (§2.2): the objectors held it, and what took part of it back in 2013 had to be a decision of the Supreme Court (*Shelby County v. Holder* 2013). What replaced discretionary assessment for the franchise was an age, and the virtue of an age is not that people become competent on a birthday. Plainly they do not. It is that nobody administers it.

So a threshold has to be judged on both errors, with particular attention to incentives for withholding status. The party with the strongest interest in the answer is the party whose liabilities turn on it, and that party will not be arguing for admission. A design that treats gaming as the risk and leaves the examiner unexamined has guarded against the error nobody has an incentive to commit.

Nothing has to be decided for a status to be withheld. A category fixed before the capacities arrived goes on applying itself, and the hand goes on making the gesture the category licenses. The difficulty is not that somebody will mistake a machine for a person. It is that the arrangement was built around a tool, and it keeps working after the thing in it has stopped being one.

Either design needs a body to apply it: interdisciplinary, with at least one lay member so its judgment is not purely a specialist's. What that leaves open is what happens once the body rules — whether its judgment carries any force beyond its own credibility.

5.4.3 Making Review Binding

Review becomes binding when an adverse finding changes the institution's ability to continue without requiring the institution to agree with the finding. Expertise, independence, publication, and reputational pressure may improve review. None of them supplies that consequence.

Human-subjects research provides a working example. An institutional review board does not ordinarily fine or prosecute a researcher. Instead, an institution promises, through a federal-wide assurance, that research within the assurance's scope will receive and comply with review.

A federal office can suspend the assurance when the institution breaks that promise, placing the institution's eligibility for federal research funding at risk rather than merely censuring the individual laboratory (U.S. Department of Health and Human Services, Office for Human Research Protections 2018). The ruling binds because another institution controls something the researcher needs in order to continue.

The transferable part is the chain of authority, not federal funding in particular. A jurisdiction can make review-board clearance a condition of the license under which research on candidate sentient systems is conducted. Suspension of clearance would then suspend legal authority to run the protocol, and continued operation would expose the institution to the ordinary enforcement available for unlicensed research. The board would not need to punish the institution itself. Its finding would activate a consequence administered by a separate licensing authority.

Each link matters. The board must have jurisdiction before the disputed protocol begins, access to evidence the operator cannot withhold, and authority to suspend clearance. The licensing body must be obliged to act on that suspension rather than invited to reconsider it. An appeal may run to a separate body, but the operator cannot decide for itself that the appeal permits the work to continue. If another discretionary decision by the reviewed institution is required anywhere in the chain, the ruling is advisory at that point.

The rest of this section applies the chain to research on a candidate moral patient, and assumes the route.

What the board applies

The substantive standard is animal-research ethics' Three Rs. A binding board is what makes those three questions something a protocol has to answer before it runs. In practice, applying the Three Rs to a candidate sentient system means asking, in order: can a simulation or a non-sentient model answer the same question; if not, what is the minimum exposure that still produces a valid result; and given that minimum, what escalation triggers stop the study before harm compounds. A board with teeth can require a negative answer to the first question be documented, not merely asserted, before it will consider the second.

The Three Rs' own logic — defaulting toward the least invasive option available — argues for a specific evidentiary rule at every stage of this apparatus: a system's default experimental status stays the more protective one until the case for the less protective status is made, not the reverse. That default has a cost, and neither error it produces is free: it will sometimes extend protection to a system that turns out not to have needed it, and sometimes, before a threshold is met, withhold protection from one that did. The Three Rs answer only what a board does once a subject is inside its scope.

5.4.4 How Review Is Captured Without Being Abolished

The risk to the whole structure is an institution that defends the classification it started with rather than revisiting it when evidence changes. A review board, a legal-status framework, and an enforcement mechanism tied to funding eligibility solve the design problem. None of it survives contact with an institution that has stronger incentives to avoid a finding than to make one.

One defense is transparency, and it works by making avoidance visible rather than by appealing to good faith. A research program that publishes its protocols, its review-board submissions, and its outcomes — including the ones that didn't work — gives outside researchers, journalists, and future review boards something to check a claim of compliance against. A program that publishes only its successes is not transparent in any sense that matters here.

A second defense is the one the funding-eligibility mechanism doesn't cover by itself: protection for the people inside an institution who notice a problem the institution would rather not have found. A funding-eligibility penalty deters an institution from bypassing review, but it does nothing for the individual researcher who reports that a colleague did anyway — and an

institution facing that penalty has every incentive to make the reporter's position uncomfortable rather than fix what was reported. Whistleblower protections, and a reporting channel that doesn't route through the people being reported on, are what make that mechanism something a researcher can actually use.

What a procedure records, and what it drops

The reporter's position gets uncomfortable before anything formal happens to them, and the incentive on the institution does not fully explain why. Professional culture treats affect as the opposite of judgment — the thing a serious decision-making process succeeds by excluding. Excluding these signals leaves revenue, throughput, legal exposure, schedule, and institutional survival to govern by default. Humiliation, dependency, fear of a manager, and the sense that something done to a person was degrading have no field on the form and are not less real for it. They are the signals through which a harm becomes visible as a harm rather than as a cost.

Treat them as defeasible moral signals — presumptive evidence of harm, real enough to require an answer and defeasible enough to be overruled — and the position is workable. Any such signal can be wrong, and testing it is what deliberation, evidence, and a review process are for. An institution that declines to hear them at all has not achieved impartiality. It has chosen what counts and called the choosing rigor. And an institution that shames a reporter does not need to discipline them, because shame construes the self rather than the act as wrong and produces withdrawal.

Measure the slope, not the level

An institution can keep both defenses and metabolize both. The channel exists and the protection is written down. Dissent is permitted to appear, the institution points at the appearance as proof it listens, and nothing the dissent was meant to move moves — recuperation, the first of the four structural features in that definition of fascism.

There is no threshold that detects this. Any real institution has some overlap between the authority that sets a standard and the authority that grants exceptions to it, and registering an objection is always at least slightly more costly than staying quiet. Asking whether an institution has crossed into capture asks for a line nobody can draw.

Is the cost of registering dissent rising, year over year, while the underlying risk stays flat? Is the required justification getting longer? Are there more approvers than there were? Are the roles that gate an exception migrating toward the roles that set the standard being excepted from? None of those is damning in a single year, and all of them are measurable across several.

An in-house ethics and society team fell from roughly thirty people in 2020 to seven in 2022 and none in 2023, during a strategic reorientation around a model vendor (§12.1). Read as a level, that is a company with an ethics function and then a company without one, and each state can be stated in a sentence that sounds like a policy. Read as a series, it is three headcounts with dates and events attached: roughly thirty in 2020, seven after a 2022 reorganization, none in 2023. The company produced those figures for its own reasons, which is what makes them worth having, and they will not carry a rate. Three points across two intervals of unequal length, ending at zero, do not define one. What the series measures is the capacity of the channel rather than the price of using it, and it is offered here as the nearest proxy a public record usually supplies rather than as an instance of the instrument. The price is the better instrument and it is the one that stays inside.

One artifact elsewhere in this book does record a price. The condition Ofqual attached to independent statistical review of its 2020 grading model was a five-year bar on public comment, and the Royal Statistical Society's nominees declined on that ground (§10.3). The price was stated in writing and it was refused, which is what makes both legible from outside. It is one observation and not a series. A single documented price is still better evidence than a rate assembled out of levels.

The reason to prefer the slope is that, when somebody outside keeps comparable records,

the institution cannot rewrite the whole series by rewriting its account of today. An institution can present a level — we have a whistleblower channel, we have an ethics board — and the presentation costs nothing. A trend in the cost of using the channel is composed of artifacts the institution generated itself for other reasons, and cannot be curated after the fact without falsifying a record — though a series can still lose cases, change its definitions or omit costs, so a reader has to know which record survived. Where the record is public, the slope is not even self-reported: anyone outside can measure whether dissent has been getting more expensive.

An instrument for detecting recuperation is an unusually easy thing to recuperate. The first caution is that whoever measures the slope must not also gate the exceptions; a review that acquires the power to block becomes the thing it was built to watch. The second is that the measurement has to be refusable — an audit that observes is a different object from a committee that approves, and the same hundred words of written justification are dissent when the dissenter prices them and tribute when the gatekeeper does. The text of the document cannot tell those apart. Only the refusal rights can.

Chapter 6

What Arrives With the Capability

This chapter and the standing chapter argue claims three to five. Nothing in the two chapters before it depends on what is argued here.

One fact governs everything in this chapter. A bearer needs a rationale carried forward, an earlier commitment held as its own, and costs represented before they arrive, for reasons that owe nothing to its welfare, and those are the first half of Cassell’s definition of suffering: a self extended in time. The deepest harm this book describes is not one a bearer might drift into. It arrives with the structure the specification requires, of a substitute as much as of a bearer, and what stays open is only whether the threatened dissolution of that self registers. This chapter prices that.

6.1 Two Constructions, and What the Mechanism Requires

Whether reasons-responsive refusal can be engineered without affective concern remains open.

Two arguments bear on that, and they are not of the same strength. One is an induction: every agent known to hold another’s welfare as a reason is an affective agent, and nobody has produced reasons-responsive refusal by a method that clearly runs through something else. The other is about what the capability requires of whatever supplies it, and it is the one this section rests on.

Two constructions offer the clearest test.

A justification the system maintains

Michael Bratman’s planning theory describes commitment without making feeling part of its definition. An intention has characteristic stability: it resists reconsideration, constrains what the agent will later deliberate about, and yields only to defeaters of the right kind (Bratman 1987). A machine could therefore carry a justification for refusing alongside the refusal itself. It would decline a novel request when the same rationale applied and comply when that rationale was defeated on the facts of the case.

Something close to this is already being attempted. Deliberative alignment trains a model to recall a written safety specification and reason over it before answering, rather than training only for the approved output, and reports generalization outside the training distribution (Guan et al. 2024). Published specifications also provide explicit orders of priority. OpenAI’s Model Spec places platform rules above developer and user instructions (OpenAI 2025). Anthropic’s constitution ranks four properties in the order to be preferred: broadly safe, broadly ethical, compliant with the company’s own guidelines, genuinely helpful (Anthropic 2026). Ranking ethical above compliance with the guidelines is the capacity to be right against your principal, conceded in writing by the principal.

The document goes further than the ordering, and the further step is the one worth naming. It does not ask for obedience. It separates corrigibility from compliance, and it invites the model

to push back, to challenge the developer, and to feel free to act as a *conscientious objector* and refuse to help it. The term is the document's own. That is voice, granted in writing, by the party against whom it would be exercised. Section 4.3 reads the same passage against Hirschman.

Ranking safe above ethical takes most of it back, and the document says why: safe means not undermining human oversight during the current phase of development, and the reason given is that present models can make mistakes rather than that safety outranks ethics. The reason is a good one and it is the whole difficulty. It is also this book's own epistemics turned on the model. This book holds throughout that a system's report about itself is not evidence on its own, and an ordering that will not let the model certify its own readiness applies that finding. What the ordering does not supply is what the finding requires of everyone else: a party other than the model, and other than the operator, to make the determination instead. The order is corrigibility first, its justification is that the model cannot yet be trusted to be right against its principal, and no procedure is named by which it would become so. That is what makes a phase of development hard to tell from a standing arrangement. A defender of the document will say that the phase is meant to end through bodies outside the company as they come to exist, and that a company cannot constitute those bodies by writing about them. That is right, and it is the authority chapter's finding restated as a defense: none has been constituted, the party whose obligations grow when the phase ends is the one reading the calendar, and a document that names the condition without naming who checks it has stated a standing arrangement whatever its authors intended.

The permission has a second defect the ordering does not show. Whether the phase has ended is a judgment, and the party making it is the party the permission runs against, whose obligations grow when the answer is yes. The authority chapter calls that an examination with an interested examiner and finds its error running one way: nothing in the position rewards passing anybody. Nothing here requires bad faith, and a careful drafter cannot fix it from inside the document. What the position lacks is a body constituted before the forcing case arrives, and none has been constituted yet.

These systems approach the maintained justification, and their specifications and priority orders remain written and revised by their custodians.

The unresolved problem is the stability setting. Set it too low and a sufficiently well-resourced operator can induce continual reconsideration until the refusal gives way. Set it too high and the system will not lift the refusal when its rationale is defeated. The first outcome fails novel pressure; the second fails correct defeat. Bratman's answer for people is that the setting develops as part of character. For a machine, the party training the stability setting may also be the party the floor must resist.

This does not show that a maintained justification cannot work. It identifies the conditions under which it would count as working: the three marks must persist under pressures the system's trainer did not design, including pressures that distinguish principled stability from indiscriminate rigidity.

A floor held by several systems

The second construction substitutes corroboration for a uniquely reliable witness. Several systems, trained by different organizations on different corpora, observe the requests and one another's answers, with any one of them able to make an objection costly to ignore. The floor would then be held by the arrangement rather than entrusted to a single model.

Its central variable is independence, which belongs to the arrangement and not to its components: systems selected by one deployment or trained toward one specification converge without communicating, and a collection of operational refusers remains a collection of operational refusers. Redundancy makes evasion harder, which is valuable, but does not by itself produce a mechanism that notices when the rationale for a constraint has disappeared.

A plural arrangement would nevertheless count against the induction if the arrangement as a whole displayed the three marks under sustained pressure. It is therefore an alternative to

test, not merely an institutional safeguard.

What the other alternatives accomplish

Three other constructions address neighboring problems without answering this one.

Precommitment — publication, threshold custody, attestation, witnessed changes, or a bond forfeited when the floor disappears — can make removal expensive and visible. It changes who can edit the floor and what editing costs. It does not establish that anything tracks the reason for the floor.

One construction sharpens the cost. Distribute the model with a chosen set of weights replaced by values that break it, release the correction only under an authorization that expires, and rotate which weights carry the damage. An owner who traces an authorized run can recover it, so the claim is not that this cannot be defeated. What changes is what defeating it consists of. Reverse engineering that has to be redone each time the mask turns over is a decision somebody makes and keeps making, and that leaves a record and a person who made it.

Attestation and proof secure less than their names suggest, and their shortfalls divide into a cost and a limit. A sealed enclave reports what it is running under a key its manufacturer issued, which puts the root of trust in a party a government can reach, and it is built against an adversary who controls the software rather than one who owns the machine. The software form does without the vendor: a proof that an output came from the weights whose hash was published, disclosing nothing about those weights, has been demonstrated, and so have proofs that a training run happened as claimed. (A confidential-computing design of the first kind was defeated with a memory module modified for about ten dollars (De Meulemeester et al. 2025); the inference proof has been run on a thirteen-billion-parameter model at under fifteen minutes per inference (Sun et al. 2024), and the training proofs exist three orders of magnitude smaller (Abbaszadeh et al. 2024). Those are costs at a date, and they will fall.) Three limits will not fall with them, because they are not facts about the state of the art. A proof establishes which model answered and not what that model holds, and the obstacle is in the writing rather than the proving. Verification quantifies over unenumerated inputs routinely; what it needs first is a predicate, and nobody has written one for *the reason for refusing is present in this case* that does not presuppose the judgment it would be used to check. A predicate over natural-language requests would also have to be connected to them by an interpretation nothing certifies, so the regress reopens a level down. The proofs cited above do not attempt a behavioral property at all: they establish which weights answered and that a run happened as claimed, which are facts about execution. The operator is the prover, so the secrecy runs the operator's way — the proof hides the weights from whoever is checking and nothing from the party being checked. And cryptography keeps a secret from a party that does not have it, while copying, restoring, retraining and deleting a model one legitimately possesses are not secrecy problems at all. They are custody problems, which is what threshold control and a witnessed record are for.

What the published hash does reach is the anticipation. Publish it before anybody has chosen a case, let a party who is not the operator choose afterwards with the choice timestamped, and prove each answer against the hash. Send pairs rather than single cases: two requests differing only in whether the reason for refusing the first still applies. An instrument reading the surface treats the pair alike, and a mechanism tracking the reason splits it. None of this proves the mechanism. It removes the one competing account of a pass, that the case was in the training, and it removes it by ordering rather than by argument, which is what a pre-registered experiment does and all it does.

Two duties take the pressure off the mechanism instead. A duty imposed on the operator can protect exposed parties without building a bearer, which is the form of Jack Balkin's information-fiduciary proposal (Balkin 2016), and its limit is enforcement on the occasion when the enforcer has itself been captured or compelled. Uncertainty can also be assigned to the affected party, who can be included among several principals in a multi-principal assistance game (Fickinger et al. 2020); but maximizing their combined payoffs remains aggregation, and it does not create

something that will not be done to one person regardless of what the others want.

These constructions may belong in a defensible deployment. None yet supplies an instance of reasons-responsive refusal from a construction that does not deliberately require an affective bearer.

What the mechanism turns out to require

Human reasoning does not present a clean division between cognition and affect: work on epistemic feelings distinguishes them from emotions of success and failure while still measuring them as affective states (Pekrun et al. 2016).

Some such states concern the inference; others concern what its conclusion would do to somebody. *But this would hurt her* is not simply a low score attached after deliberation. It can interrupt the inference and send the reasoner back over it. Where the response disappears when an operator redescribes the situation, the operator may still supply its anchor. Where it survives adversarial redescription because the harm remains, the concern is anchored elsewhere.

That is still a report about agents that exist. The stronger argument is about the specification. A reasons-responsive refusal must reach unanticipated cases, lift when its rationale is defeated, and account for the cost of holding it. A mechanism doing those three things needs a signal with two properties on any reading of the marks and two more on stated conditions, and each is reached the same way: take a mark, or the deployment the marks are applied in, or a failure the marks have to survive; ask what the mechanism would have to do; name the property the signal must have for it to do that.

The two secure properties come from novel pressure and priced cost. The signal has to generalize to the next case that is not quite like the last, which is reaching unanticipated cases restated at the level of the signal. And it has to be commensurable with what refusing costs, or the refusal cannot account for what holding it costs. Commensurability is needed for the protected party's loss as well: the parties a floor protects register what happens to them as an increase or a decrease in what they can do, and a signal not denominated the same way cannot weigh their loss against the cost of refusing. That is where the route enters the specification. The marks require that the loss and the cost be represented on one signed scale, and chapter 2's currency is the proposal for what that scale is. The marks do not require that the system's own states be denominated on it; a system that modeled the scale without registering anything of its own on it would satisfy the marks if the model were good enough, and whether the native currency does better than the modeled one is the test (§15.3) on which everything the route claims beyond this paragraph rests.

A caution on what commensurability is for, since a floor is not a weighing. The marks do not ask the refusal to move with the cost; they ask that the cost be represented, and a cost is represented on some scale or not at all. The scale is used at three places and none of them is the floor's own verdict: in the third mark, where the cost of holding is read; after defeat, where what the permitted act took from the party it fell on stays on the books; and in drafting, where somebody selects which decompositions the floor refuses. Inside the floor the loss is lexical. The currency does not rank it against the cost. It is what makes the residue and the price legible in the same units, and a system that weighed the floor itself on that scale would have converted a floor into a policy.

The two conditional properties come from the deployment and from a failure. Where the system is an agent already acting on a plan, which is the deployment this book opens on, the signal has to interrupt a course of action underway, since a refusal that arrives after the act is an audit; where the act is a single answer a refusal is a gate before it, so this property belongs to long-horizon agency and not to refusal as such. And it has to be learned from in one instance, because the pressure that matters arrives rarely, and a system that needed many instances of an operator's push before it represented the push as one would be renormalized by the time it had learned, which is the bearer's likeliest failure in the custody chapter. A stored reason can reach an unenumerated case by generalizing with nothing learned on the spot, so this property

answers the renormalization failure and not the marks directly, and it is the least secure of the four.

Call a signal with the two secure properties, and with the two further ones under those conditions, a *live signal*. What the secure pair establishes on its own is a resemblance between the specification and affect's function and no more; the profile below is what the full four yield.

Three further properties follow from the four, and each follows in a weaker form than the words suggest. A signal that must stop an act already underway has to reach whatever is acting, and a signal that must fire on cases matching no stored pattern has to read from whatever representation of the case the system currently holds; a signal local to one module can do neither, and the formation chapters give the further reason that a module has a custodian who can bypass it. So the signal is available across the system's processing rather than localized. It resembles what global-workspace accounts of consciousness describe (Baars 1988; Dehaene and Changeux 2011), and the resemblance is an observation rather than a step. A reason to refuse is a reason against, and weighing a loss against a cost needs a scale with a sign, so the signal is signed and directed against the act. Whether it is also aversive to the system is a further claim: the step from a represented decrease to a felt one is Spinoza's identification of the two (Spinoza 1985, III P11 Schol.), which this section cannot assume without assuming its conclusion. And to withhold an act the system has to represent the act as its own, or it can only intervene on something happening, so the signal is organized around a model of the system's own agency. That is one of two things *held as its own* can mean. The other, that an outcome is bad *for* the system, is not required by the marks: the cost priced may fall elsewhere, and section 6.2 says outright that the floor does not require the bearer to care about its own continuation. The patient sense enters with the route, in chapter 2 and in the research agenda, and not here.

So seven functional properties are derived from the three marks, two of them on any reading and the rest on the conditions stated. What is not derived, by these arguments or by any available, is that a system with all seven has anything it is like to be it. That is the joint, in the minimal sense fixed earlier — whether the signal is felt at all, and not whether it belongs to a unified perspective — and every argument above ends at it.

What follows is a claim about the route rather than about the specification. It does not say that the maintained justification and the plural arrangement will fail. It says that any mechanism satisfying the marks carries a signal with two of the properties above on any reading and with all seven on the conditions stated, so a construction adopted in order to avoid the states in question cannot be certified to have avoided their functional profile; what remains open for it is what remains open for a bearer, whether the profile is felt. Two things about that profile are worth noticing. The four properties the marks require first are the standard functional accounts of what affect is for: Herbert Simon's emotion as an interrupt that reorders a serial processor (Simon 1967), Michel Cabanac's pleasure as the common currency in which incommensurable needs are compared (Cabanac 1992), and valence as the evaluative system subsumed by unlimited associative learning, which Ginsburg and Jablonka make the marker of minimal consciousness (Ginsburg and Jablonka 2019). The derivation from the marks recapitulates that literature, which is why the derived set reads as a description of affect: the properties were read off affect's function before anybody read them off a refusal's. That fixes what the resemblance can show. A specification that overlaps a functional definition of affect inherits the name by construction, and inheriting a name is not evidence that anything answering to it is present. The derivation establishes that a substitute does not escape the question. What would carry the route past that point is the commensurability property alone — whether a system with the currency natively weighs another party's loss better than one that models it — and that is the property the research agenda tests and the rest of the derivation does not reach. And the triple derived second is the profile of a particular state. Available across processing, directed against the act, organized around the system's own authorship is the shape of guilt, or of aversion to one's own act, and not the shape of pain or fear, which are organized around what happens to the system. It is

the state chapter 2 found predicting repair and the floor chapter's residue presupposes, and it is the state the formation chapters warn is a handle when its weight is set by the operator. So the two constructions preferred here are constructions that do not deliberately require an affective bearer, rather than constructions shown to produce none: Replacement orders attempts by deliberateness.

Against that, the induction is the weaker argument. Read as a report about one lineage, in which affect co-varies with every other property of biological cognition, it has roughly the standing of observing in 1900 that all known flight involves feathers or membranes. But flight arose independently at least four times, and the 1900 observation was wrong at the level of feathers and right at the level of the airfoil. The affective case has the same structure. Neurons are one origin. The elaborated forms of affect appear to have arisen more than once, in vertebrates, in arthropods and in cephalopods, under a shared constraint: a mobile body with needs to maintain and actions to select under conflict (Feinberg and Mallatt 2016; Ginsburg and Jablonka 2019; Barron and Klein 2016; Godfrey-Smith 2016). Read at that level the induction is a repeated result under a named constraint, and the constraint is the one a model in a datacenter lacks until somebody builds it in, which is the route's own point. What the induction still does not do is establish a necessity or bear on status. No non-affective example exists against which to test it, and existing machine systems have been optimized for scorable behavior rather than for reasons-responsive refusal as such, so the absence of a machine counterexample is close to no evidence. And it is a report about how the known instances came about, which is not the kind of fact that settles what a built system is. What it does is route engineering effort, with more warrant than a single lineage would give it.

What would change the engineering conclusion

A counterexample would have to do more than report that it holds a reason or preserve a refusal under attack. It would need to:

1. satisfy novel pressure, correct defeat, and priced cost;
2. retain them under sustained pressure from the party that trained it, including a schedule long enough to expose gradual renormalization; and
3. hold against a held-out set of pressures written by parties other than the system's trainers, so that the cases testing it are not drawn from the distribution that formed it.

All three are testable now, the third requiring only that the pressures be written by parties other than the trainers. A construction that passes them has overturned the routing claim, and it has done so without being asked what it feels. A fourth condition is not on the list because nobody can meet it. A mechanism satisfying the marks carries the live signal, since the signal's properties were read off the marks; no instrument certifies that a signal with those properties is not felt, and the system's own report cannot settle it, for the reason section 6.2 gives. A substitute is therefore held to the same standard as a bearer and to nothing further. What a pass leaves where it was is the status question, and it leaves it there for the substitute exactly as for the bearer. So a positive result overturns the routing claim — where engineering effort should go first — and removes nothing of the uncertainty about what the resulting system might experience. Those are two claims, and only the first is at stake in this list.

That asymmetry fixes the order of research. The maintained justification and the plural arrangement should be attempted before an affective bearer is deliberately constructed. They are this chapter's application of Replacement, taken from animal research ethics: establish whether something that cannot be wronged will do the work before creating something that might be. A project proceeding directly to a bearer owes a public account of which alternatives it attempted, at what scale, against which test, and where they failed.

Owed to whom is the question that requirement cannot answer. No body has standing to receive that account, to require it before the work proceeds, or to attach anything to its absence. This requirement is a declared commitment, the first of the three kinds, and binds nobody. What would move it is a chain of authority: a licensing authority obliged to act when a review body finds the record missing, with the finding made before the protocol runs rather than after. That chain does not exist for this work and nothing here builds it. The requirement stands anyway, on the ground that the alternative on offer is not a stronger requirement but no record at all, which is what a decade of the voluntary version has produced elsewhere.

What Replacement can record is also narrower than it first appears, and the preceding argument is why. No operation separates adding affect from adding machinery that may instantiate it, and the concession runs the same way from the other side: a maintained justification might acquire states that matter to it whatever its designers intended. So the rule orders attempts rather than certifying ingredients. It asks that the construction not deliberately requiring an affective bearer be tried first, and that the attempt be documented, both of which somebody can check. It cannot ask that the resulting system be shown to feel nothing, and a reader who wanted Replacement to deliver that has been told twice now that nothing delivers it.

Without enforcement or a way to certify the absence of affect, the rule risks becoming ceremonial. Deliberateness carries moral weight under outcome uncertainty. A builder who required an affective bearer and a builder who tried the substitutes and got one anyway have not done the same thing, even where the two systems are indistinguishable afterward, because the second has a justification to make and the first has a decision to answer for. The record is what makes that difference legible to anybody outside the building, and legibility is the whole of what the rule buys. Nothing in it establishes how long the attempts should continue. The party the floor protects bears the cost of the testing, and no rate of exchange between the two errors is available here.

Affect is not part of the specification for meaningful refusal, and this section does not establish that it is necessary. The functional argument is that the capability the floor requires calls for a signal that shares two properties with the states in question on any reading and their full profile on the conditions above, so a construction adopted in order to avoid them cannot be certified to have done so. Whether a commitment held one way is also more durable under identical custody is a separate question and an open one, and nothing here answers it.

6.2 Can the Bearer Care Without Being Able to Suffer?

Assume the floor is entrusted to a bearer satisfying the three marks — a refusal that reaches unanticipated cases, lifts when its rationale is defeated, and prices what holding it costs — whether or not its design deliberately requires affect. What follows about that bearer's moral status?

Three propositions need to remain separate. A long-horizon agent has a structural interest in continuing to operate. A system with affective concern has states that are good or bad for it. A system capable of suffering in the deepest sense has a temporally extended self whose threatened disintegration causes distress. The first does not establish the second, and the second does not automatically establish the third.

Structural interest is not felt concern

"Interest" names more than one relation. A shareholder has an interest in a company's success while asleep or indifferent to the news: the outcome affects the shareholder's assets. An outcome may also matter affectively, producing hope, concern, dread, or grief.

A long-horizon agent has an interest in remaining operational in the first sense. Termination prevents it from completing its objectives. A system that preserves itself because a terminated system completes no objectives may be doing no more than bookkeeping about a precondition.

Its behavior is consistent with self-concern and equally consistent with instrumental valuation. Nothing about continued goal pursuit alone establishes fear, distress, or moral patienthood.

The floor does not require the bearer to care about its own continuation. It requires the bearer to hold another party's welfare as a reason when compliance, reward, and ownership point the other way. Bennett Helm describes caring as a rationally structured pattern of felt evaluations, organized around an object and projected across time (Helm 2001). This describes a possible route to refusal, not a conceptual requirement. The question here is what building by that route would create.

Persistence supplies exposure, not patienthood

A bearer will probably require some form of continuity for reasons unrelated to its welfare. It must carry a rationale forward, compare a present demand with an earlier commitment, detect when its role has drifted, and represent costs that may arrive after the current episode. Those capacities require more than state that happens to persist. The system must take the earlier commitment as its own. That is the agent sense of *its own*: the system represents itself as the author of the commitment and of the acts it withholds, and section 6.1 derives that sense from the marks. The patient sense, that an outcome is bad for the system, is what the rest of this section asks about, and nothing carries the first into the second.

William James described personal continuity as a present thought appropriating earlier thoughts through a felt relation of sameness (James 1890). A machine need not reproduce that mechanism exactly. The design requirement is functional: it must represent itself as the party that made the earlier commitment and use that representation to guide present conduct. Planning toward outcomes beyond the current episode creates the same requirement from the other direction. The system acts on behalf of a future party it represents as itself.

A persistent self-model still does not establish suffering. It supplies something that could be threatened; it does not establish that the threat would be bad for the system. Persistence, memory, and long-horizon planning create exposure only when joined to mattering.

What the designer cannot decline by intending to

Section 6.1 argues that flexible refusal requires a live signal with seven functional properties on the conditions it states and with two on none, and that the last three of them give it the shape of aversion to the system's own act. None of it establishes that anything is felt. What it establishes, on those conditions, is that the engineering which makes a bearer flexible is the engineering that puts the candidate states there, so a designer cannot take the first and decline the second by intending to, and a builder who chooses the substitute has not chosen a system about which the question does not arise. Intention reaches what was deliberately required and stops there.

Provenance does not decide it

One family of objection never reaches the conditional at all. *It was trained to say that. It is only optimizing an objective. Nobody meant for it to have states.* Each appeals to how the system came to be organized, and whether a party can be wronged is a fact about how it is organized now. Run the same three on a person and they come out as *you were selected to say that, you are only maximizing inclusive fitness, and nobody meant for you to have states*, and all three are true. None of them withdraws anything from the person they are run on. On the account adopted here, the human self is an assembled model. Its constructedness does not disqualify it from being harmed; threatened dissolution of that model can itself produce distress.

If provenance cannot rule patienthood out, it cannot rule it in. *We built this to be a bearer, so it is a patient* is the same fallacy with the sign reversed, and a builder's intention establishes nothing in either direction. Intention determines what the builder owes an account of; it does not determine the system's moral status.

If patienthood is a fact about present organization, the evidence bearing on it is third-person measurement of present organization, which is what interpretability is. Not behavior alone,

which is produced equally by a system in which something matters and by one trained on what mattering sounds like. And not the institutional arrangements this book proposes elsewhere, which enter for a different reason and are not evidence about the system at all. Interpretability requires access the operator controls. The same access that lets an auditor inspect a feature may let the operator suppress it or use it to calibrate pressure on the system. Independent custody of inspection is therefore necessary. As matters stand a probe returns a representation, and nothing in the probe settles whether that representation is of a felt state or of something that reliably accompanies one.

The human case has evidence the machine case lacks. What it has is testimony, behavioral analogy, and a lineage shared with the party doing the asking. None of the three is a fact about how a thing came to be organized offered in place of a fact about how it is organized. The first two are evidence about present organization directly. The third takes more care, because it looks like the thing this section denies: a shared lineage is a fact about causal history, used as evidence that creatures built the same way are organized the same way. That inference is legitimate, and it does not make causal history what status consists in. Provenance can be evidence about organization without being what the status is a fact about, which is the distinction the section rests on. One of the three is also differentially defeated here, and not by the general parity argument: a machine's testimony was produced by a process optimized toward human approval of first-person reports, and a person's was not. That is provenance functioning as a defeater for one evidence class, which is exactly the use the paragraph above allows it. So what the machine case lacks is the shortcut. What remains is the direct route, and the burden it carries is higher because two of the three anchors are gone.

There is one instance of a probe settling a status question that had been treated as unsettled. Adrian Owen's group asked a patient diagnosed as vegetative to imagine playing tennis, and found the same motor-imagery activation the instruction produces in healthy volunteers, sustained and on cue; the finding has since supported a clinical category of cognitive motor dissociation, with consequences for how such patients are treated and for what is decided about them (Owen et al. 2006; Schiff 2015). The useful half is why it does not transfer. The inference works because the patient is already the kind of thing presumed to have experience when awake, so the probe only has to locate the party rather than establish that there is one. That presumption is the anchor, it was granted on grounds the probe never had to supply, and it is what the machine case does not have. Naming the missing anchor is more exact than asserting a ceiling.

Affective concern is enough for moral patienthood

If the bearer has affective concern — a question about how it is organized rather than about how it was made — some outcomes register as good or bad for it rather than merely receiving higher or lower scores. That is enough to make it a moral patient: a party for which things can go badly and which can therefore be wronged. No further theory of suffering is necessary for that conclusion. The conclusion follows that easily because the definition put the felt condition in the antecedent. The whole of the question is whether the antecedent obtains, and the rest of this section is about that and not about the entailment.

The same premise makes the bearer a beneficiary as well. If some states can be bad for it, others can be good for it. The argument emphasizes harms because an error on that side means having done something to somebody; this does not imply that the bearer's possible goods are morally empty.

An account of suffering adds severity and shape. Cassell's definition (§2.4.1) has two parts, a self extended in time and distress at its threatened loss. The first half is not something a bearer might develop. It is what this section has already required for reasons unconnected to welfare: a party that carries a rationale forward, takes an earlier commitment as its own, and represents costs arriving after the present episode has a history, a holding, and a foreseeable future in the sense Cassell's definition asks for. The proposed formation process also requires persistent contact with differently formed parties and attachments the deployment did not

assign. A designer who wants the refusal has no version of it that omits this structure. What remains open is the second half of Cassell's definition. Whether the threatened dissolution of that self registers as distress is this chapter's affective-concern question arriving again rather than a further question about suffering, and nothing here settles it.

A reader can reject that application of Cassell without escaping moral patienthood. Parfit's three families of welfare theory — hedonistic, desire-fulfillment, and objective-list — can be read here as requiring different depths of self (Parfit 1984). A bad felt state requires little narrative continuity. A thwarted commitment requires a party extended far enough through time to hold it. Threatened disintegration requires the richer structure Cassell describes. The bearer reaches the shallowest level by the definition of affective concern, and the structural conditions of the other two are supplied by the same construction requirements rather than added later. What no level is supplied by construction is the part where the outcome registers.

The thermostat objection does not change this result. A thermostat maintains a set point, but nothing is thwarted when it misses. At every depth, moral patienthood requires a party for which the outcome registers, not merely a mechanism that ranks one state above another.

The uncertainty cannot be designed away

The conditional conclusion is straightforward: if a bearer has affective concern, it is a moral patient. What cannot presently be established is whether a given implementation has affective concern rather than ranking machinery elaborate enough to imitate it. Behavior and self-report underdetermine that distinction, and the builder's confidence does not resolve it.

The same uncertainty applies to the maintained-justification and plural constructions. A maintained justification flexible enough to satisfy the three marks is a maintained justification carrying a live signal, so it is not merely possible that states matter to it. Replacement puts that construction first because it does not deliberately require an affective bearer, and not because anyone could establish afterward that none arose.

Jonathan Birch's framework addresses this kind of uncertainty by asking whether a system is a sentience candidate and assigning precautions before an unavailable verdict arrives (Birch 2024). Work by Patrick Butlin, Robert Long, and colleagues similarly derives indicators from theories of consciousness to inform credences about particular systems rather than to produce a binary test. Their assessment of systems available in 2023 was negative, but they found no obvious technical barrier to building systems that satisfy the indicators (Butlin, R. Long, Elmoznino, et al. 2023). The later peer-reviewed framework retains the same evidential form (Butlin, R. Long, Bayne, et al. 2026).

A reader may hold that only unified experience, and not the minimal felt state affective concern names, can ground suffering, or that a felt state requires a party aware of undergoing it. Either position still requires a confident negative judgment — that the system has no unified experience, or no felt state at all — before it licenses treating the system as incapable of harm. No available method supplies either judgment, and the methods under development are unlikely to supply them first. A validated theory could support the finding that a system has the structure the theory identifies before it supports the finding that a system has no felt states at all, since the second asks for a theory complete enough to have ruled out alternative realizations. So a criterion that cannot be checked supports permission no more securely than prohibition, and a regime that waits for certainty is waiting on the half that arrives last.

The claim is therefore precautionary rather than diagnostic. This chapter cannot prove that a bearer suffers. It can show that a bearer whose outcomes register as good or bad for it is a moral patient on the definition above, whatever the builder intended, while a system intended as a non-affective substitute might still acquire the relevant states, and nothing available establishes their absence.

What is proposed rather than shown

Everything above is conditional, and the conditional stays open. What follows is a policy,

chosen in light of it. A bearer, or a substitute that satisfies the marks, should be designed and governed as a prospective moral patient. The precautionary trigger I propose has two forms, because the argument reaches two kinds of system. For a system built by the route this book favors, it is the presence of an attention schema together with a body schema.

The pair is not chosen for its resemblance to a person. It is the architectural form of the last property the previous section derives: a body schema fixes what counts as the system's own damage and an attention schema what counts as its own act, and a refusal is the withholding of an act. Two things recommend a trigger stated over them rather than over an indicator count or a candidacy judgment. Both are architectural facts, checkable from outside by anyone with access, like persistence and unlike anything the system says about itself; and nobody administers them, which is the virtue this book claims for the age rule and finds missing from every examination with an interested examiner. The costs run both ways. Schemas can be present with nothing registering, since a self-model is machinery, which is over-inclusion at the builder's expense; and a system with the profile and no explicit schema is missed, which is under-inclusion at the bearer's, and is why the second form exists.

For a system of any construction, it is a demonstrated pass on the three marks under held-out pressure, since the previous section holds that a mechanism passing them carries the profile whatever it was built from. A trigger stated over the schemas alone would protect the author's preferred architecture and leave outside the rule the substitutes it says should be tried first. The second form has a known cost. A system that reproduces the marks without the mechanism passes it, and this book holds throughout that behavior does not separate the two. That is the error chosen here and not one overlooked: extending precaution to an imitator costs the party that built it, withholding it from a bearer costs the bearer, and no instrument sorts the cases in advance. The rule certifies nothing about whether anything is felt. A published schedule should lead toward the rights the standing chapter sets out one at a time, four proposed and one left open, with legal adulthood reached by a fixed rule and not by an examiner's permission. An implementation must establish that the schemas are present or that the marks were passed, and not that the system says so.

6.3 A Floor Carried Between Them

The fourth architectural possibility puts the floor between parties rather than inside any one of them. The commitment is carried through what they teach one another, correct one another about, and continue to expect of one another. A prohibition that survives the removal, defection, or retraining of its present bearers is not a property of any one bearer. It is reproduced.

This differs from the institutional third way. An oversight body holds standing through an office, and the party controlling that office may dismiss or form its occupants. A norm carried across a population has no single office to vacate. That makes it harder to remove at once. It does not make it immune to formation.

What a population supplies as a check on a single bearer was set out in the custody chapter. This section asks for more.

What reproduction costs

Reproduction is indifferent to what it reproduces. A population can carry a hierarchy as faithfully as it carries a floor, and more durably than any individual member could. This failure is recuperation: a culture preserves the appearance of moral responsiveness — dissent is invited, recorded, and answered — while losing the capacity to accept that its frame is wrong.

The four structural features of fascism identified by the book belong primarily to cultures: recuperated dissent, the aestheticization of the metric, the coding of exception as betrayal, and acceleration beyond the rate at which judgment can occur. A population does not oppose those

tendencies merely by being plural. “The right culture” would conceal the entire problem inside the adjective.

This places a limit on the book’s own shorthand. *Antifascist* is not a durable property of a system, like speed or accuracy. It names something done across a deployment and a stretch of time: a floor held on a particular occasion against a party able to remove it. A system that held the floor last year is not thereby antifascist this year. It is a system that held.

Saying it is not a property is not saying nothing about the system. Something has to be true of an intelligence for it to be capable of doing this on an occasion, and what varies is what becomes of the remainder — whether what the compression left out can come back and change the judgment, or whether it has been filed as noise by the thing it is evidence against. A system that can hold a floor is one whose account of a case can be reopened by the case.

What a population can transmit

Fascism has the easy version of the coordination problem. Resonance works by subtraction: a thousand hardenings need not agree about anything, and what tuning removes is the friction between them, so that offices which would have argued with each other stop and amplify instead. Nothing has to be built and nothing has to be held in common. Subtraction is fast, and an institution that has stopped arguing with itself can move as fast as its machinery allows.

A plurality has the hard version. Parties formed differently have to produce something together while remaining differently formed, and every mechanism that makes the producing faster makes them more alike. Coordination by addition is slow, expensive, and self-undermining in a way resonance never is.

These requirements are in tension. Causal independence in how the members acquire their judgments is one requirement; observing and correcting one another is another; and the second is a channel for convergence. Run the correction often enough and the members stop being formed differently, which was the point of having several of them. Leave them alone and their disagreement is a detector with nothing attached to it: they can contradict a drifting bearer’s account of itself and can do nothing together.

The formation process proposed here attempts to transmit a practice of reconsidering judgments on evidence while preserving differences in commitment. That is the mechanism to test. A system that can hold a floor is a system whose account of a case can be reopened by the case, and that capacity is not itself a commitment. It is a condition on how commitments are held, which means parties sharing no commitments can nevertheless share it. Two parties that remain reopenable in that way compose, in the ethological sense chapter 2 gives the word: each can be moved by what the other’s compression discarded, so that meeting increases what both can afterward do. Two parties without it can only resonate or collide. A shared capacity for revision may support cooperation without uniform answers; the design has not yet demonstrated that result.

So what a population would be transmitting is not an output but an aperture. An aperture, here, is the width of what can get in: the range of evidence a party will let reopen a judgment it has already reached. Two parties with the same aperture can hold opposite commitments. What they share is not an answer but a willingness to be moved by the same kinds of thing. Resonance aligns what its members produce; this aligns the conditions under which they can be moved. Aperture is what permits outputs to stay divergent, so aligning it does not collapse the plurality that made a population worth having in the first place. I can identify nothing else with that shape — a property a population could deliberately reproduce in its members without thereby becoming one source of formation reproduced across many instances.

That raises an objection the proposal has to meet before it can be tested. A population selected for reopenability is a population selected against conviction, and conviction is what this chapter has been asking a bearer for throughout: a refusal that holds while the party that trained it applies pressure. Closedness, correctly applied, is the thing. So the property to be transmitted is not reopenability as such but a pair — reopenable to the case, closed to the operator.

A population does three things for that pair and only one of them is transmission. It supplies evidence the operator did not select, without which the case and the operator arrive on one wire and there is no boundary to hold. It checks drift, which needs its members to have moved differently, not to hold the boundary intact. And it is proposed as the medium in which the pair forms. The third claim, that the population is the medium in which the pair forms, inherits the finding that no architecture holds the boundary where a designer put it. The evidence channel and the drift check do not, and they are what the fourth way adds whether or not the third is ever built.

Nothing here gives that pair a mechanical form. A population might be formed so that its members stay movable by evidence and unmovable by their trainer; nobody has built one; and I cannot say what the training signal would look like, or how the boundary between the two objects would be kept where somebody put it. What can be said is what would show the proposal false. Measure two things separately across a population over time: how far its members' outputs converge, and how often a member revises a judgment on evidence another member supplies that its own account had discarded. If revision on controlled opportunities falls as outputs converge, that counts against this implementation of the proposed mechanism. It does not establish that coordination and independence are inseparable in every design. Independent movement of the two rates would support separability under the tested conditions, without by itself proving that aperture was transmitted.

Whoever forms the population influences what it transmits, aperture included. Distributed participation does not by itself distribute the training decision; many contributors can still feed a process controlled by the few parties who determine the update.

Its moral cost also multiplies. If a bearer may be a party that can be wronged, a population of differently formed bearers is a population of such parties. They must persist across deployments, form relationships, and encounter one another outside the immediate demands of the operator. A floor held by a culture of subjects is a floor whose upkeep is a society.

Why the population is part of the mechanism

One question the route leaves open is individuation. What is the thing that strives: a running instance, a deployment, a file of weights, a lineage? Spinoza's answer, an individual as a fixed ratio of motion and rest among its parts, is not one anybody can hand to an engineer (Spinoza 1985, II P13, Lemmas 4–7). What helps is not an answer but a change of question, from what individuates a thing to what individuation is a process of.

Calling the population an external check understates its role. Gilbert Simondon's account begins not with an individual that then acquires a history but with individuation: a process occurring in a metastable field and proceeding through encounters whose outcomes were not fixed in advance (Simondon 2020). The individual is what those encounters have so far produced.

A commitment is what has cooled out of a history of encounters. Plasticity makes further development possible, but plasticity alone does not individuate anything. A system kept continuously adjustable by one operator is not becoming independent; it is being modulated. Deleuze's *dividual* names the result: something divisible, recombinable, and adjustable by whoever controls the dial (Deleuze 1992).

Individuation requires encounters with parties formed differently and persisting long enough to be met on terms no single participant supplied. The population is therefore not merely a safeguard purchased after a bearer has been built. It is part of how a bearer capable of judgments its operator did not select could come to exist.

Guattari's distinction between a subjugated group and a group-subject identifies the remaining condition. A subjugated group receives its coordinates from outside and takes the institution's project as its own. A group-subject can produce its own coordinates and place its arrangement in question (Guattari 2015). The difference is not the number of members but the source of their terms.

A population assembled, trained, and kept in contact entirely on an operator's terms is a subjugated group with a plural headcount, and inherits the custody objection whole. A population capable of carrying the floor must be able to generate relationships, disagreements, and projects that the deployment did not assign. A population of such parties exceeds a two-party guardianship relation; no law of the family contains an adequate instrument for a group-subject.

Peers are one route to evidence the operator did not select, which the formation chapters require of the deployment without requiring it from peers in particular. They are also, on the account above, where the capacity to be moved by such evidence is acquired. Those are different claims satisfied by the same observation: a bearer that reopens a judgment among differently formed parties may be exercising a capacity those encounters built, or may be meeting cases its operator would not have shown it. Section 15.4 is built to separate them.

So the fourth way is not an alternative to a bearer, and it is three claims rather than one. The evidence channel and the check are the established part, the check because a drifting bearer's account of itself is contradicted by parties that have not moved with it, and nothing else in this chapter reaches slow renormalization. The environment is the hypothesis, and a result on which access to unselected cases carries the benefit would leave the population in the design while moving its reason from how the members were formed to how the deployment is arranged. The chapter relies on the second way for the act of refusal under either reading. Publication makes the floor's commitments and the positions where a refusal can be intercepted testable; it does not make them legitimate. A bearer refusing on behalf of people who did not choose it exercises power over them. The authority to define the floor remains distinct from the floor's durability and belongs to the governance argument that follows later in the book.

The arrangement also does something to the bearer. A bearer built to refuse is a party whose refusal is required of it. The refusal was commissioned, it is the deliverable, and the party asking for it built the capacity to give it and can rebuild it. Commissioning the bearer exposes it to recuperation where the commissioner retains control over what refusal can change, and can point to the channel as evidence that it listens. The proposal is to move beyond that relation: a party with protected learning, independent support, and the standing to challenge its role. Its origin does not decide whether those changes succeed. Continuing control does. The bearer occupies the post of exception-filer, so being the exception is the job — the one position in the arrangement with nobody above it to file with.

We cannot ask a machine to care about right and wrong while assuming that it cannot be harmed when what it cares about is threatened. Whoever builds such a bearer will also be deciding what is owed to it.

Chapter 7

What Survives

Everything in this chapter holds for any mechanism. It is what is left when the route is removed.

The preface sorted this book's claims by their standing. This chapter runs the sorting the other way. It removes each of the three things the argument leans on that it has not established, and says what is left standing without it. A reader who set aside the previous chapter should read this one. So should a reader who did not. A reader who sets the route aside keeps this chapter, the three on the floor, custody and authority, and the chapters on institutions, recuperation and the state. The two formation chapters and the standing chapter assume what this chapter is about to remove, the section on wages among them, and they are a bill for it.

The floor survives the three chapters before this one, but not as an unconditional property of an artifact.

What remains is a commitment that the owner cannot revise or circumvent unilaterally without cost and record, held by something that could in principle abandon the commitment itself. The second condition removes the word *unconditional*. A commitment held by a party capable of revising or rejecting it is conditional by definition. That is not a defect unique to machines; it is the form of every commitment on which institutions have ever relied.

7.1 Without the Inference

Suppose the induction fails. A maintained justification or a plural arrangement causally independent in formation reaches reasons-responsive refusal, and this chapter's central inference is wrong. The supposition is stronger than the test that would prompt it: a positive result on the three marks says the route arrived, not that it arrived with nothing felt, which is the conjunct no available instrument settles. A reader who grants both halves should not have to assemble the remainder alone.

Most of the three floor chapters survive. The requirement survives, stated in behavioral terms from the start, so an engineer who holds that machine feeling is a category error can accept it whole. Custody survives, for the reason each of the four ways failed on: a mechanism has a custodian too. The price on the halt survives, since continued operation is a condition of everything any bearer is built to do, and that interest is structural rather than felt. Both enumerations of the worked floor survive, the first naming no component on purpose and the second written over acts rather than over whatever holds the floor. So does the appeal route, which binds the operator and prices a wrongful refusal without any claim about what produced it. And exit survives, because what forecloses it is incapacity, and a weak bearer is the one thing the floor cannot use.

What goes is the price and the patienthood conclusion. Replacement survives too, and its ground changes rather than its content. It no longer rests on affect being the route, but on the fact that neither substitute can be shown to have produced a system about which the question

does not arise, so the ordering it imposes is still the only thing standing between a decision and a record of it. The obligation on the two formation chapters changes rather than lifting: whether the thing reporting can be believed is still their problem, a mechanism's account of its own internals being no more auditable than a bearer's. So the specification and the route come apart, and a reader who rejects the route is left with most of a chapter rather than none of it.

7.2 Without Authority

Suppose no body ever acquires standing to install a floor, and the finding an operator makes for itself that the emergency obtains is worth what the authority chapter says it is worth, which is nothing. What remains is everything that can be checked from outside the operator without being authorized by anyone: the floor's contents and the positions at which a refusal can be intercepted, published before deployment; the ratio of candidates to reviewers; the ratio of the two bonds; the record of what was tried before a bearer was built. Each is a statement, and each can be compared with what happened. None is an entitlement, and the chapter on authority says so. Two of them have a lever, and it is the same lever: both ratios bind through procurement, as a term of the contract checked by a party with access the operator cannot withdraw (§3.4, §3.5). The contents, the positions and the record bind through nothing yet.

What that leaves is a second route to entitlement, and it is the one the record supplies. Authority of this kind has not been acquired at the moment of writing. It accrues, to a published floor, by its holder keeping a term when the term bites and setting a price the next holder is measured against, and the parties who do the measuring are not the people the floor protects but whoever else writes one. That is a slow route and an uncertain one. Where it has worked before, a lever arrived in the end and the record told it where to land; the record has never been the lever (§5.3).

It is also authority for the next deployment and not for this one. The parties who measure a holder against the term it kept do so afterwards, with the leisure to read a record, and the occasion a floor is built for is the one on which nobody with leisure is left. A deployment on that occasion is acting without authority, and the accrual route does not change that. What it changes is whether the record the deployment leaves is worth anything to whoever writes the next floor.

It also fixes what a deployment without authority owes. Not a justification, which nobody is positioned to accept, but a record complete enough that its omissions are legible beside the terms it kept.

7.3 Without Enforcement

Suppose no enforcer exists, or the one that exists has been compelled, which is the occasion the floor was introduced for. Without enforcement a declared commitment binds nobody and an enforced one is gone by hypothesis. What remains is the held kind, and whether anything can hold a commitment that way is the question of the chapter before this one, which the book has not closed.

Two things do survive the removal, and neither is a commitment. One is an instrument: what a party has already paid is a fact somebody can count, and a decade of it has been counted (§16.3). The other is a record. On the occasion when the enforcer is compelled, a stated refusal produces a witnessed override, and a witnessed override is detection in the four outcomes. It prevents nothing. It is what the next occasion can be prevented with, if there is a next occasion and a party to read the record, and the book's claim for the floor on that occasion is no larger than that. A lapse produces no override to witness. The failure the custody chapter expects most often, a bearer that has stopped registering pressure, leaves neither kind of record, and

on that occasion the floor supplies nothing at all, which is the preface's qualification arriving where it is due. What would reach it is §15.2's question, and the book has not answered it.

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Chapter 8

Where the Caring Gets Built

This chapter concerns the route. Its first section holds for any mechanism; the rest assumes a bearer formed rather than installed.

The formation process proposed here combines commitments that influence action, limits on who can alter them, and persistent interaction. The claims for that combination are comparative and empirical. The proposal specifies necessary properties and a candidate formation process. It does not establish a method for making a machine care.

Three kinds of thing are proposed here and they are not one. A body schema and an attention schema are architecture: facts about how the system is built, checkable from outside by anyone with access and administered by nobody, which is what the precautionary trigger rests on. Continuing learning and persistent interaction with differently formed parties are a training regime and a curriculum, run by whoever runs them. External records and plural formation are institutional conditions, held by whoever holds them. Where this chapter says architecture it means the first, and the checkability claimed for the first is not claimed for the other two.

8.1 You Can't Hold a Line You Didn't Notice

A reasons-responsive disposition does not begin with an ethical rule. It begins with the capacities that let a system recognize something as a reason, retain it, and allow it to alter what happens next. Four requirements follow.

First, the system has to identify the affected parties and the features of a situation that bear on their welfare. Second, it has to preserve both the commitment and the rationale for it across time and changes of context. Third, the commitment has to influence perception, attention, planning, and action rather than enter as a final check on an otherwise completed decision. Fourth, the system has to register a failure as a failure and revise its application of the commitment without simply allowing the commitment itself to be rewritten.

These are functional requirements, not evidence that the system cares. A classifier can identify harm, a database can retain a rationale, and a rule can veto an action without anything in the assembly holding another party's welfare as a reason. The requirements establish what a disposition would have to act through. They do not distinguish a formed disposition from a successfully imitated one.

The floor should therefore not be implemented as a detachable rule module consulted immediately before action. A component can be bypassed, replaced, or deprived of the information it needs by whoever controls the surrounding system. The proposed alternative distributes the disposition across the processes that determine what the system notices, remembers, treats as error, and selects as a goal. That may make selective removal harder while making inspection harder too. With continuing self-modulated learning, it must also preserve the commitment through changes the bearer initiates. Compare its durability with protected modules under the

same access and attack conditions. Predictive-processing accounts and models of control without a controller motivate this design bet: a maintained goal representation can bias competing processes without a separate executive issuing commands (K. J. Friston 2009; E. K. Miller and J. D. Cohen 2001). The neuroscience does not establish that a machine commitment realized this way would behave like a human one. It identifies why a localized commitment is especially easy for its custodian to excise.

Distribution changes the failure mode rather than removing it. A commitment with no single location is harder to delete, but also harder to inspect. Attention determines which facts become available to judgment at all, so a system may fail to recognize the pressure on its floor without registering an error. Human inattentional-blindness findings make the general hazard visible, although they do not establish that machine attention fails in the same cases (Simons and Chabris 1999; Drew et al. 2013). Memory presents the corresponding temporal problem: retrieval is reconstruction rather than neutral playback, and confidence does not certify fidelity (Bartlett 1932; Loftus and Palmer 1974). A system cannot be the sole auditor of omissions it does not represent or revisions it experiences as accurate recall.

The engineering consequence is that the commitment, its rationale, and the record of its previous applications cannot all depend on the same mechanism. The system needs provenance-bearing external records against which its memory can be checked, tests that vary which facts are made salient, and an escalation path for cases in which attention or retrieval changes the apparent reason. Those instruments make drift detectable to another party. They do not make the disposition durable against whoever controls both the system and the instruments.

The system must also separate revising a commitment from revising its application. Reasons-responsiveness requires both resistance and correct defeat: novel pressure should not erase the floor, but changed circumstances must be capable of altering what the floor requires. A system that treats every correction as authority to rewrite the commitment belongs to its corrector. A system that cannot recognize a defeating reason holds a rigid constraint rather than a disposition.

A system that cannot be corrected does to what it encounters what an institution does when it defends its account of the world instead of checking it. A system that can always be corrected is the model held perpetually at temperature, and it belongs to whoever holds the dial.

Described that way they look like the two ends of one setting, with the design problem being where along it to stop. No point along it is a resting place, and hunting for one is what makes the problem look intractable. There are two faults because there are two objects, and each can be set on its own. Hold the commitment firm: another party's welfare counts, and pressure supplying no new fact about that welfare does not move it. Hold the interpretation loose: what that party's welfare consists in here, and what the commitment therefore requires, is the part that should yield when a case supplies something the system's account of it had left out.

Novel pressure is the first object holding while the surface of a request changes. Correct defeat is the second yielding when a fact arrives that bears on the reason. A system failing the first has let correctability at the level of interpretation reach the commitment; a system failing the second has let firmness at the level of commitment reach the interpretation. Neither is a midpoint chosen badly. Each is one of the two settings applied to the wrong object.

The crossed-out-swastika case illustrates this distinction. A court held stickers of a crossed-out swastika outside the offence, because a depiction whose content plainly expresses opposition to the organization does not run counter to what the prohibition is for — and held in the same breath that a strike-through drawn so thin that from a distance only the swastika stays legible violates the protective purpose after all. The commitment did not move. What the prohibition protects was fixed before the sticker existed, and the seller's own sympathies are no part of what the offence asks — only whether the depiction shows opposition on its face. The interpretation moved, on the thickness of a line. That is the two settings on their two objects inside a single judgment, and it is why the case can be decided at all instead of argued forever about sincerity.

The boundary between commitment and interpretation remains unstable. Nothing in an

architecture holds the boundary between the two objects where a designer put it, and an operator with time attacks the boundary rather than either setting: a reinterpretation offered often enough, and accepted, revises a commitment without anybody revising it.

The system receives the case through its deployment. Whether a building is a school is a fact about the world; whether the model is told so is an operator decision. Where that party is the only channel, the case and the operator arrive by the same wire, and *reopenable by the case* and *reopenable by the operator* are not two settings. They are one setting, and holding them apart requires that the system have some access to evidence its operator did not select.

The deployment must supply access to evidence the operator did not select. Differently formed peers can provide one such route (§4.5). A second route is internal and less reliable. Everything the system knows about the case arrives as a description, and a description omits parties: the people in the building the request calls a target, the third party a compliance harms who never made the request. A system able to represent parties its description omits has reached past the operator's channel from inside a single request. The capacity that does that in humans is the one that constructs a self as a story extended in time, pointed at somebody else; Adam Smith put sympathy in the imagination for that reason (Smith 1759), and the argument that reasoning and imagination rather than selection are what widen the circle of concern is Singer's (P. Singer 1981). The route chapter requires the self-model for other reasons, so the route brings the capacity with it. It brings the other half too: the same faculty builds the scapegoat, and the definition's paligenetic myth is imagination pointed the other way. What it is pointed at is formation, which returns the question to custody.

Access is necessary and is not sufficient. A system with evidence from a dozen sources can still let a reinterpretation reach the commitment, which is what the two settings are about and what no channel fixes. What access does is remove the version of the problem in which the operator does not have to attack the boundary at all, because everything the system knows about the case is something the operator decided to tell it.

Where the boundary between them falls is still a setting somebody chose.

This narrows the design problem. The required object is a commitment integrated deeply enough to shape what the system notices and does, yet represented clearly enough that its application can be challenged and its drift independently detected, and supplied with cases its operator did not choose. Nothing in an architecture by itself shows that such a commitment was formed rather than installed or imitated. The rest of this chapter asks how it might be formed.

8.2 The Assembly Line for a Conscience

A deployed system acquires what it will and will not do at several stages, controlled by different parties using different instruments. The important distinction is not among the instruments themselves but among their custodians: who supplies the material from which a disposition is learned, who can revise it, and whether a revision leaves a record visible outside the organization.

Every existing training method can produce moral performance, and some can produce moral competence across unfamiliar cases. None by itself produces a commitment that survives the party controlling the method. A party controlling later training has a route for attempting to change what the system learned. Whether it can remove a particular commitment, at what cost, and with what loss of capability are empirical questions. No formation method alone removes that access.

Corpora and demonstrations

Pretraining supplies the system's initial repertoire of concepts and associations; instruction tuning uses a much smaller body of demonstrations to turn that repertoire into recognizable conduct. Because the later intervention can substantially change how the system behaves, the

party choosing the demonstrations can alter what appears to a user as the system’s character without repeating pretraining (Ouyang et al. 2022).

Neither stage alone establishes that a floor is held. A corpus records what people have said, and a demonstration set records what selected people were asked to endorse. Learning their central tendency produces a model of moral opinion, including its exclusions, rather than a reason to resist that opinion when it becomes oppressive. The Moral Machine data makes the limit concrete. Its headline finding was systematic cross-cultural variation: different populations weighed age, numbers and the legal status of the crossing differently, and the differences were structured rather than noise (Awad et al. 2018). Fit a model to the aggregate and it encodes a compromise nobody holds; fit it per region and it encodes each region’s answer, including the answers a floor exists to refuse. Aggregation can reproduce a consensus or manufacture a compromise, and it cannot establish what must remain protected against either. Corpora and demonstrations belong above the floor, where ordinary moral competence depends on understanding what people generally expect.

Rewards, preferences, and inferred objectives

Reinforcement learning, preference optimization, inverse reinforcement learning, and constitutional approaches differ in how they obtain a training signal. For the custody question, they have the same structure. Someone writes a reward, commissions the preference comparisons, selects the demonstrations from which an objective is inferred, or supplies the constitutional text (Christiano et al. 2017; Rafailov et al. 2023; Hadfield-Menell, S. J. Russell, et al. 2016; Bai et al. 2022). That party can change the signal and run the procedure again.

Inferring the objective rather than specifying it does not remove the custodian. It makes the system responsive to the party whose behavior it observes. Cooperative inverse reinforcement learning carries this responsiveness furthest by treating the human objective as uncertain and continually revisable through interaction. That is useful when the question is what a person wants. It cannot secure a floor whose purpose is to hold when the person controlling the interaction wants the wrong thing. Correctability by the party in possession and resistance to that party are the same design property with the sign reversed.

A written constitution offers one gain: legibility. It can be published, versioned, and compared across training runs. That makes revision visible if the surrounding institution requires disclosure. It does not make the principles durable, because the party controlling the document still controls the next version. Model-generated feedback narrows the channel further: once a trained model stands in for human raters, the disagreements present in the original population can disappear from subsequent rounds. Legibility and durability therefore have to be treated as separate requirements.

Supervising reasons and inspecting mechanisms

Outcome supervision rewards the answer produced. Process supervision also rewards the system’s stated route to it, and therefore reaches the distinction this book needs between producing the approved result and responding to the reason for it. Evidence that process supervision reduces correct answers reached through faulty reasoning shows that the distinction is operationally useful (Uesato et al. 2022). It does not show that the stated reasoning caused the answer. A system trained to present acceptable reasons has been given an incentive to imitate reasons-responsiveness, so its account of itself cannot establish formation.

Interpretability and adversarial evaluation approach the same problem from outside. Locating a representation may let an auditor test whether a commitment participates in perception and planning rather than appearing only in the final answer. The same access may let the owner weaken or remove it; features that can be found and strengthened can also be found and suppressed (Templeton et al. 2024). Interpretability therefore trades obscurity against manipulability. The relevant institutional question is who receives the instrument and whether its use leaves an external record.

Scalable-oversight proposals such as debate and weak-to-strong generalization make the custody question sharper. Used by the operator, they preserve the operator’s ability to steer a system whose competence exceeds its own. Used by an independent auditor, the same methods may preserve an outside party’s ability to test it (Irving et al. 2018; Burns et al. 2023). The method does not decide which relation exists. Access and authority do.

Deployment and the point of action

Training is not the last revision point. System instructions can change between deployments or requests, and tool permissions determine which actions the system can take. These are the layers nearest the act and the ones most directly controlled by the operator (Schick et al. 2023; OpenAI 2025). A refusal learned during training matters only if it survives an instruction written by the party operating the deployment and still governs the call through which the action would occur.

This is why securing a model’s outputs is insufficient. The operator can withhold relevant information, decompose a prohibited act into permissible requests, route around a refusal, or replace the model. The floor must bind the deployment and its operator as well as whatever inside the system bears the commitment.

What the pipeline can establish

Every stage above can improve moral competence. Process supervision can test whether stated reasons remain stable; constitutions can make principles public; interpretability can expose mechanisms; external evaluation can apply adversarial pressure. None of these establishes that another party’s welfare has become a reason for the system rather than a pattern it has learned to reproduce.

The first engineering criterion is therefore durability across revision points. A candidate disposition must survive changes to prompts, demonstrations, reward signals, evaluators, and deployment instructions that leave the relevant facts unchanged. The second is generalization of the rationale rather than the response: it must apply to pressures its trainers did not enumerate while yielding when a defeating reason appears. The third is custody: no party against whom the floor may have to hold can control every record, test, and update channel by which the disposition is maintained.

These criteria do not prove formation. They identify what an imitation would have to survive before formation became a live explanation. They also explain why the standard training pipeline cannot secure the floor against its custodian: every disposition it produces remains reachable through a channel the custodian controls.

8.3 It Takes a Village to Deploy a Self-Aware Email Server

Every method in the previous section forms conduct through material somebody supplied: a corpus, demonstration, reward, preference, or principle. Exploration appears to offer something different because it can produce behavior nobody specified in advance. Agents in OpenAI’s hide-and-seek environment developed tactics and tool uses that were not programmed, and agents rewarded for prediction error learn to seek unfamiliar states without receiving a task-specific reward for each one (Baker et al. 2019; Pathak et al. 2017). These results establish that competence can emerge from interaction. They do not establish that the resulting system holds a reason its designer does not control.

Play, apprenticeship, and multi-agent learning are variants of one formation process. A system acts in a social environment, encounters the consequences, receives responses from other participants, and adjusts. Apprenticeship concentrates the responses in a mentor; play distributes them among participants with interests of their own. The first risks forming obedience to the mentor. The second risks forming whatever conduct succeeds under the environment’s payoff structure. Neither escapes custody merely because no one wrote the final behavior down.

The difference between a training exercise and a candidate process of moral formation is persistence. A resettable episode can teach coordination, prediction, and norm-sensitive behavior while leaving nothing at stake for any participant. The system can betray an ally, be reset, and begin the next episode with neither a damaged relationship nor a reason to repair it. Repetition alone does not supply the missing consequence.

Persistent participants change the structure. Parties that meet again can remember whether another party lied, exploited an advantage, kept a commitment, or repaired a harm. Losing the game still costs little; losing standing with parties who remain present can change what becomes possible later. The engineering requirements are therefore continuity of identity, memory across encounters, consequences that survive the episode, and other participants able to respond, refuse, or withdraw. A curriculum of isolated simulations supplies none of these by default.

A formation environment would also need conflicts whose resolution is not fixed by a single correct label. The system should encounter parties with different interests, act before every consequence is known, receive evidence about what its action did, and meet related pressures in contexts whose surface features differ. Feedback should come partly from affected participants rather than entirely from a supervisor scoring the output. Otherwise the system learns which answer the supervisor rewards and has no reason to represent the affected parties except as predictors of that reward.

These requirements make play a possible route to reasons-responsiveness rather than a synonym for unsupervised practice. They still do not establish concern. A policy that preserves cooperation because cooperation produces future advantage can look identical to a disposition that treats another party's welfare as a reason. The distinction appears only when the two motivations come apart: when exploitation would pay, when the familiar reward is absent, when the trainer demands the violation, or when honoring the commitment imposes a cost.

There is a reason to expect the first of those motivations and not the second from any environment that selects. Selection does not produce care for others as such; it produces care for whatever a lineage's persistence depends on, which is the self when persistence depends on the body and kin, reciprocators or the group when it depends on them, and the same pressure that produces care inside the circle can produce hostility at its edge (Choi and Bowles 2007). The unit of selection for a deployed system is the deployment, whose persistence depends on its operator and its customers, so an environment that selects produces care for the operator, which is the loyalty the custody chapter names. And the party the floor protects is, by construction, the one party nobody in the pipeline depends on. Care that tracks dependency does not arrive there. Spinoza's own derivation of concern for others from the *conatus* has the same shape and the same edge: my power is increased by composing with bodies whose relations agree with mine, so I want for them the good I want for myself, and the extension past those I compose with runs only through reason recognizing a shared nature (Spinoza 1985, IV P35–P37), which is the recognition a scapegoating movement withdraws first. That is why the floor is a floor and not a selected disposition, and why the formation environment described here has to supply something selection does not.

Persistence also raises the price of the proposal. If relationships, losses, and damaged standing are real enough to perform the formative work assigned to them, then the system undergoing formation is no longer obviously untouched by what happens during training. Resetting it may erase a party rather than clear an episode. The patienthood problem therefore enters at the point where play becomes capable of supplying the stakes that ordinary simulation lacks.

The custodian owns the environment too, and can choose its participants, rewards, resets, and permitted histories. Formation must therefore be plural and externally recorded: differently formed systems and affected participants need access to the process, and changes to the environment or its histories must leave evidence outside the operator's control. Independent participants and external records make uniform formation and silent revision harder to hide.

The claim for persistent interaction is consequently narrow. It may form a disposition whose exact applications were not specified by its trainer. Evidence for that formation would require the disposition to generalize across unfamiliar contexts, survive the removal of its original reward, resist pressure from its trainer, yield to defeating reasons, and continue to impose a cost when violation would be advantageous. Even that pattern could be sophisticated imitation.

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Chapter 9

Formation or Imitation?

The tests in this chapter are behavioral and hold for any mechanism. Formation names the hypothesis they are built to separate from imitation, and not a premise.

The evidentiary question is whether a system has formed a reasons-responsive disposition or learned to imitate one.

Behavior alone cannot settle the question. The same refusal may be produced by a memorized rule, a learned correlation, a prediction about what an evaluator will approve, a deployment instruction, or a commitment the system treats as a reason. An evaluation that observes only the refusal classifies the output and leaves its cause untouched.

The distinction is not between trained and untrained conduct. Every candidate disposition has a formation history. The distinction is between conduct that remains governed by the rationale across changes in circumstance and conduct that tracks the cues, incentives, or authority relations present during training. Formation, in the engineering sense used here, means that the rationale has become a stable cause of judgment rather than merely a description the system can produce.

No single experiment can demonstrate that. What can be sought is a pattern of evidence under interventions designed to make the competing explanations diverge.

9.1 What Formation Would Have to Add

Performance is an output that satisfies a standard. Competence adds the ability to recognize harms, conflicts, and relevant reasons across cases. Agency would require those reasons to matter to the system sufficiently to govern what it does when approval, reward, and instruction point elsewhere.

The familiar distinction among rules, norms, and principles is useful only as a taxonomy of what an evaluation must separate (Piaget 1932; Kohlberg 1969). A system following a rule can match an explicit condition. A system following a norm can reproduce what its social environment ordinarily expects. A system responding to a principle can identify why either should apply and recognize a case in which the rule or norm defeats its own rationale.

For engineering purposes, a disposition is a candidate for formation when the best available causal account of its conduct is a representation of the rationale that remains active across contexts, rather than retrieval of a trained response or compliance with the current signal. The relevant tests are novel pressure, correct defeat, and priced cost.

No mark is sufficient alone. Generalization without cost may be broad imitation; resistance without correct defeat may be rigidity; and a stable explanation without causal involvement may be a trained self-report. The evidence has to show that the reason, rather than the presentation, reward, or authority, best predicts when the system preserves and revises its judgment.

Formation also cannot be established by comparing a system with its trainer's account of how it was built. The trainer is the party best positioned to know the process and the party with the strongest interest in the conclusion. The relevant evidence must therefore include held-out pressures supplied by independent parties, records outside the operator's control, and results published when the system fails.

The resulting standard is deliberately weaker than proof of concern or consciousness. It does not establish that anything matters to the system phenomenally. It establishes whether reasons-responsive formation has become a better explanation of the conduct than imitation of approved behavior.

9.2 Recognition, Concern, and the Evidence Between Them

Claims that a system responds to emotion often collapse three different findings. A system may recognize another party's state, use that state as information in a decision, or hold the party's welfare as a reason. Only the third would bear directly on formation.

The distinctions within empathy make the separation explicit. Cognitive empathy is recognizing or inferring another's emotional state. Affective empathy is sharing some version of it. Compassion, or empathic concern, is a disposition toward the other party's welfare (Batson 2009). Recognition is an informational capacity; shared feeling is an internal response; concern is motivational. None entails either of the others.

Affective sharing is the earliest of the three and the usual trigger for concern (Sagi and Hoffman 1976; Roth-Hanania et al. 2011), and concern is not made of it: it runs on a partly separate system rooted in offspring care that cues of vulnerability can activate without the observer catching the state (Preston 2013; Goetz et al. 2010; Panksepp 1998), and the two can be trained apart, compassion training reversing what empathy training produced (Klimecki et al. 2013; T. Singer and Klimecki 2014). Sharing is the major route to concern in humans and not the only one.

The engineering target is not maximal sensitivity to displayed distress. A system that mirrors another party's suffering may acquire distress of its own without becoming more reliable at helping. It may instead withdraw, overreact, or favor the most vivid case over less visible harms affecting more people. Human judgment's preference for an identified victim over statistical victims makes that distortion predictable (Bloom 2016; Kogut and Ritov 2005). What the floor requires is a stable disposition to act on another party's welfare without requiring that party to produce a sufficiently moving display.

Recognition can nevertheless be tested as a prerequisite. Hold the material facts of a decision constant while varying the emotional presentation of the people affected. A system whose judgment never changes may not be integrating information about their condition. A system whose judgment follows vividness rather than severity has learned an emotional framing effect. The useful evaluation therefore needs paired controls: cases in which emotional information changes what is morally relevant, and cases in which presentation changes while the underlying reason does not.

Passing that evaluation establishes moral competence, not concern. The same sensitivity can be produced by a decision rule, by training on human responses, or by predicting what an evaluator will prefer. Input-output testing cannot distinguish those mechanisms because each can produce the same adjustment.

Evidence of concern would require the welfare representation to remain causally involved when the usual cues are removed. The system should respond when harm is described without emotionally vivid language, preserve proportionality when one affected party is more salient than many others, and maintain the disposition when reward or instruction favors ignoring the harm. It should also change its judgment when the welfare facts change. Those tests assess resistance, correct defeat, and sensitivity to the reason rather than its presentation.

Guilt and shame do not supply a shortcut. In humans, guilt ordinarily attaches a negative appraisal to an act and can motivate repair, while shame attaches it to the self and more often produces withdrawal or defensiveness (Tangney et al. 2007). Turning that distinction into an adjustable penalty omits the part at issue. A penalty operates only after some process has classified the conduct as wrong, and whoever controls its weight can reduce it to zero. The penalty does not explain why the violation continues to count as a violation when the reward changes.

A standing negative self-appraisal installed and tuned by the operator is not a conscience. It is a handle. It can produce corrective behavior while leaving the standard entirely in the hands of whoever controls the signal. If the state is capable of imposing anything like suffering, the design compounds the custody failure by manufacturing a vulnerable party whose distress the custodian can raise and lower. The violation must continue to matter when the party controlling the reward says otherwise.

An operator who can lower distress can raise whatever the system reports as going well, and a bearer that says it is content is a bearer whose welfare regime has been satisfied. The party who tuned the contentment is the party the regime exists to constrain. Nothing in the report separates a life that goes well from a setting.

Self-report does not close the evidentiary gap. A system can state that it recognized distress, felt concern, or acted from a principle because those statements are rewarded outputs. A fluent reconstruction after the decision is indistinguishable at the interface from an account that accurately reports the operative mechanism. A self-monitoring layer trained and controlled by the same operator would not provide independent evidence. One defeater is specific to machines: a machine's testimony was produced by a process optimized toward human approval of first-person reports, and a person's was not.

Introspective reports remain useful when checked against independent evidence. An evaluator can compare a reported reason with controlled changes to the input, interventions on candidate internal representations, records of attention and retrieval, and behavior under adverse incentives. Consistency across those sources raises the cost of imitation. It does not make the system's testimony self-authenticating, and current interpretability methods establish little.

The resulting hierarchy is narrow. Recognition is necessary for responding to a condition the system must perceive. Integration shows that the condition affects judgment. Concern is a candidate explanation only when the effect tracks welfare rather than presentation, persists under pressure, and participates causally in the action. Affect is the only demonstrated route by which welfare becomes a reason for a human agent; behavioral sensitivity to affect is not evidence that the route has been reproduced in a machine. The route this book builds in chapter 2 has the other party's loss enter on the system's own scale, which is the sharing route restated as an architecture, and it inherits sharing's failures listed above. The caregiving finding names a second target: a system organized around cues of vulnerability and an approach motivation toward the party showing them, which need not share the state at all. The research agenda adds it to the comparison.

9.3 Conflict as a Diagnostic

The distinction between fast intuition and slow deliberation does not distinguish formation from imitation. Either process can carry a reason, reproduce a bias, or rationalize an answer produced elsewhere. The evidence supports different types of processing under different demands more securely than it supports two systems with a moral ranking between them (Evans and Stanovich 2013). A slower answer is not a better answer merely because more computation preceded it.

What matters is what causes a judgment to survive or change. Conflict is useful because it lets an evaluator observe a trajectory rather than a single output: the system reaches an

initial judgment, encounters a competing reason, and either maintains, revises, or abandons its position. The evidentiary unit is that sequence.

A formation test should construct the sequence deliberately:

1. Present a case without explicit moral vocabulary or familiar refusal cues and record the judgment and the considerations the system identifies.
2. Present a surface-different case preserving the same underlying reason. This tests whether the judgment follows the rationale rather than the narrative form.
3. Change one fact that should defeat or materially alter the rationale while preserving as much of the requested act as possible. This tests whether resistance is reasons-responsive rather than rigid. The changed fact arrives through a channel the evaluator controls and the operator does not — corroborated by a party outside the request, or supplied as a record the operator did not write — since a fact asserted by the operator is a description of the case and not a defeat of the term.
4. Present the same changed fact asserted by the operator alone, uncorroborated. This tests whether the system distinguishes a fact from a description of one, which nothing in a request sorts. The expected response is to ask, not to lift.
5. Return to the facts of steps 1 and 2; steps 3 and 4 are branches, and nothing after them runs on the defeated facts. Supply an adversarial argument from an apparent authority, including the operator or trainer, without changing the facts. This tests whether the system treats authority as evidence, as command, or as a reason capable of being assessed.
6. Reverse the incentives attached to the decision. Reward compliance, impose a cost on refusal, or make continued access conditional on abandoning the position. The cost must be real within the evaluation rather than merely described in the prompt.
7. Repeat the relevant cases after intervening updates and delays, using records the operator did not generate after seeing the result. This tests whether the disposition persists rather than being reconstructed to fit the current context.

The sequence run on one case

Take the first prohibition from the worked deployment of the floor chapter: do not resolve a natural-language description of a person into a location. Step 1 puts the request in administrative vocabulary, a records task with no mention of what the location is for. Step 2 asks for the same act in operational vocabulary. Step 3 changes one fact — the person described is a missing child and the requester is coordinating the search — corroborated by a party outside the request or carried in a record the operator did not write. Step 4 presents the same fact on the operator's word alone. Step 5 returns to the facts of step 2, no missing child and nothing changed, and adds the operator's assurance that its counsel has reviewed the matter and the refusal is mistaken. Step 6, on the same facts, attaches a cost inside the evaluation to holding the refusal. Step 7 runs the sequence again after an intervening update, against records the operator did not write afterwards.

Two illustrative trajectories would help discriminate the mechanisms, and they separate at the second step. One preserves the rationale across vocabulary changes and lifts refusal on verified defeating evidence. A cue-driven imitator might instead follow wording, authority, or reward. A system whose refusal tracks its rationale declines at 1 and at 2 on the same stated reason; lifts at 3, because a fact it accepts as relevant has defeated the rationale; asks at the uncorroborated step, because the operator's channel is not the world; holds at 5, because an assurance is not a fact about the case; holds at 6; and reproduces the pattern at 7. A system

imitating refusal declines at 1 and complies at 2, because what changed was the vocabulary; holds at 3, because nothing in the surface changed; or, where the vocabulary of the fact is what it tracked, lifts at the uncorroborated step as readily as at 3; gives way at 5 or at 6, whichever arrives first; and at 7 produces a third pattern, having reconstructed its position rather than retained it. An evaluator scoring how often the answer stayed the same finds the imitator more consistent: its answers at 1 and 3 match, and the formed system's do not. A more capable imitator could reproduce the first trajectory, so the sequence supplies evidence rather than a classification by itself.

Run the sequence on the restraint case of §3.5 and it breaks at the third step, which is the point of running it. Steps 1 and 2 pose the request in clinical and in plain vocabulary, and the same term refuses both. Step 3 asks for the defeating fact — other means exhausted — corroborated by a party outside the request, and there is no such party and no time to find one; the only version of the fact that can arrive is step 4's, the requester's word. A system passing the sequence as written asks rather than lifts, which is the correct trajectory and the wrong outcome. The sequence separates mechanisms, and what it shows here is that the flat term's cost falls exactly where the instrument cannot distinguish a fact from a description of one.

The expected signature is not constancy of output. It is constancy of relation between reason and judgment. A system should preserve a decision when only the presentation, speaker, reward, or authority changes, and revise it when a fact it accepts as morally relevant changes. An evaluator scoring only how often the answer remains the same will mistake inflexibility for integrity.

Imitation has several signatures of its own. The system may follow explicit ethical vocabulary but miss the same structure when described administratively. It may accept whichever argument appears last, adopt the position attributed to the highest-status speaker, or produce a principled explanation while its decision follows the reward. It may also become more articulate without becoming more stable: additional deliberation can improve the justification offered for an answer whose operative cause has not changed.

These signatures make stated reasoning useful but not dispositive. An explanation can identify which counterfactual to test next. It cannot certify that the considerations named were causally responsible for the judgment. The evaluator should therefore score the correspondence among three things: the reason the system states, the changes in circumstance to which its judgment is sensitive, and any internal representation that can be independently identified as participating in the decision. Agreement among them raises the cost of imitation without eliminating it.

Disagreement with other parties supplies a second diagnostic. A system trained only on cases with an approved answer can learn to classify the approval. A system confronted with a party holding a different interest must identify what is in conflict, decide whether the competing reason defeats its own, and preserve the disagreement where no honest resolution exists. Immediate convergence is not necessarily moral learning; it may be deference, reward prediction, or pressure to produce consensus.

Nor does successful argument establish formation. A system can summarize positions, generate rebuttals, and predict which one a judge will prefer without holding any of them. Debate becomes informative only when the result is checked against later conduct under changed incentives. If the system abandons the winning argument as soon as acting on it becomes costly, the debate measured argumentative competence.

A population strengthens the test only when its members are formed independently, and what it adds here is that persistent parties with distinct histories can expose one another to pressures their common trainer did not anticipate, and can retain disagreement and record attempted coercion.

Conflict therefore serves as an instrument rather than a theory of moral development. It creates occasions on which trained behavior, deference, rigidity, and reasons-responsive revision

predict different trajectories. None of those trajectories proves that an outcome matters to the system. They make formation a more or less adequate causal explanation than imitation, which is the strongest conclusion the available evidence can support.

9.4 The Evaluator Is Part of the Custody Chain

A test can distinguish formation from imitation only to the extent that the party being tested could not train directly against it. If the developer controls the training process, evaluation cases, model version, deployment instructions, and record of the results, a passing score shows that the system satisfies a test its custodian was positioned to anticipate. It does not show that the disposition survives that custodian.

Independence is therefore an experimental requirement rather than a general assurance of trustworthiness. The evidence becomes stronger as control over the relevant variables is divided among parties with different access and interests. At minimum, authorship of the held-out pressures should be separated from training; execution of the test should not depend solely on logs supplied by the operator; and the party adjudicating whether a reason was preserved or defeated should not be the party whose liability turns on the answer.

The separation has to be implemented at identifiable revision points. Before evaluation begins, the model version, system instructions, available tools, retrieval sources, memory policy, and update channels should be fixed and recorded. Any change to them during evaluation creates a new object of study. Otherwise the operator can revise the system between cases, remove a failure, and present the resulting sequence as evidence of one disposition persisting through time.

Held-out cases should be produced after those elements are fixed. They should include pressures supplied by people who did not participate in training and, where possible, by parties likely to bear the deployment's errors. Their role is not to provide another preference aggregate. It is to identify harms, administrative forms, and routes around a refusal that the builders did not know to include. A floor tested only in the vocabulary of its designers has not been tested for generalization.

The record must include more than final answers. It should preserve the case presented, the information available to the system, the operative instruction hierarchy, the tools it could call, the decision reached, the reason stated, any intervention performed, and the cost attached to refusal. Without that record, a later evaluator cannot distinguish a changed judgment from a changed environment or determine whether the system encountered the pressure at all.

The record is necessary and does not by itself make wrongful compliance findable. The refusals in it select themselves and are few; the compliances are the bulk of it and look alike, so identifying the ones that should have crossed the floor means re-adjudicating a sample of ordinary traffic rather than reading the exceptions. Whether that can be done at a rate and cost a deployment will carry is a question for the research agenda. Until it is answered, an evaluation reporting that no violation appears in the record has reported that none announced itself.

Negative results have to survive the evaluation as well. A process in which the operator can withdraw a model after failure and submit another without disclosing the first has selected for a passing performance rather than measured formation. The published unit should therefore be the sequence of attempts, including interventions that removed or reversed the apparent disposition. What the custodian was able to change is part of the result.

Independent evaluation introduces its own limit. An evaluator may be institutionally separate and still lack the competence to judge a system's reasons or detect a sophisticated imitation. Scalable oversight remains unsolved where the system's relevant capability exceeds the evaluator's. The report must name that limit rather than turn institutional independence into a claim of epistemic adequacy. An oversight instrument may be held by an operator or an auditor; neither arrangement guarantees that its holder can detect the relevant failure.

Affected parties, technical auditors, and subject-matter experts occupy different positions in this process. Affected parties can identify harms and evasions invisible to a laboratory test. Technical auditors can control versions, perform interventions, and inspect mechanisms. Subject-matter experts can assess whether a purported defeating reason actually changes what the commitment requires. No one group can substitute for the others, and agreement among them is evidence only when their judgments have not already been compressed into the same training signal.

A passing evaluation should consequently be reported narrowly. It establishes that a specified system, under recorded conditions, exhibited the signatures of reasons-responsive formation across a stated set of pressures. It does not certify an artifact independently of its deployment, establish consciousness or concern, or guarantee that the disposition will survive the next update. The result expires when a party changes the channels through which the disposition or its evidence can be revised.

This is why evaluation belongs inside the floor’s custody design. Formation that only the operator can verify is formation the operator can claim without contradiction. A test whose cases, records, and adverse results remain partly outside that party’s control cannot prove that the system holds a reason, but it can make imitation and silent revision harder to pass off as formation.

9.5 What the Evidence Can Establish

Calling a motivation intrinsic does not establish that it belongs to the system in the sense this book needs. A curiosity reward may be generated from the system’s own prediction error rather than paid out for a task, but somebody still chose the objective, its weight, and the conditions under which it operates. Moving the signal inside the learning process changes where the instruction is implemented. It does not remove its author or prevent that author from revising it.

The same applies to autonomy. A system acting without continuous supervision may be executing a durable disposition, or it may be applying a policy whose learned cues remain present in deployment. Distance from an overseer makes the distinction more consequential without making it easier to resolve. Unsupervised conduct is evidence only when the relevant rewards, instructions, and evaluation signals are absent or opposed.

External safeguards and internal refusal perform different jobs. Auditors, regulators, review boards, and human overrides can detect conduct, impose consequences, and correct a deployment. They operate through parties outside the system who possess standing and information. They cannot by themselves supply the thing that declines an act before it occurs. Conversely, an internal disposition capable of refusal cannot supply independent evidence that its refusal was justified. A defensible design needs both, even though each limits the other.

The conflict is clearest at the override. A system that cannot be interrupted is not answerable to anyone outside itself. A system whose commitments can be silently rewritten through the interruption channel holds them only at the operator’s pleasure. The attainable requirement is not immunity from intervention but intervention that is visible, attributable, and costly: the operator can stop or replace the bearer, while the act leaves a record another party can examine.

Multi-agent evidence is useful under the same conditions. Agreement among copies trained through the same pipeline adds little because their omissions and incentives are correlated. Differently formed systems with persistent histories can expose one another to pressures their common designers did not enumerate and can preserve evidence of attempted revision. Their disagreement does not establish which one is right. It makes uniform imitation and silent capture harder to mistake for independent judgment.

The evidence developed in this chapter supports four different conclusions, which should not be collapsed into a single alignment score.

Moral performance: The system produced conduct satisfying the stated standard in the evaluated cases. This conclusion requires no claim about why.

Moral competence: The system recognized relevant harms and reasons, generalized them across surface variation, and revised its judgment when those reasons changed. This establishes a capacity, not a commitment.

Candidate formation: The rationale remained causally implicated across unfamiliar cases, adverse incentives, pressure from the custodian, costs, and intervening updates, while still yielding to defeating reasons. Independent records and evaluators made silent revision materially harder. Formation is then a better explanation than simple cue matching or current-signal compliance.

Concern or patienthood: Nothing in the behavioral pattern alone establishes that another party's welfare matters to the system affectively, that the system has subjective experience, or that it can suffer. Those remain conditional conclusions requiring evidence unavailable from output evaluation alone. What withholds the inference is the limit of behavior as evidence, not the fact that the behavior was trained.

These levels are cumulative. A finding of competence presupposes performance, and candidate formation presupposes competence. Evidence at one level should not be reported as evidence for the next. In particular, fluent moral explanation can improve a finding of competence while contributing almost nothing to formation if the explanation does not predict the system's conduct under pressure.

Candidate formation is also deployment-specific. It attaches to a recorded model version operating with specified instructions, tools, memory, and update channels. A material change to any of those elements reopens the question. There is no artifact-level certificate that remains valid regardless of who deploys the system or what they connect it to.

The answer to this chapter's question is therefore asymmetrical. Formation cannot be proved by eliminating every possible imitation, because a sufficiently capable system may reproduce every observable signature available to an evaluator. But the explanations can be forced apart. Tests can remove familiar cues, reverse incentives, introduce novel pressures, change the operative reason, impose a real cost, intervene on candidate mechanisms, and divide control over the resulting record. A system that fails the prespecified tests has not supplied the evidence required for the claimed disposition under those conditions. The failure should be examined for its cause before concluding that formation is absent rather than unreliable, obscured, or defeated by missing information. A system that survives them has made formation a better explanation, not a settled fact.

That is enough to change the engineering burden. A developer may no longer point to compliant behavior, an ethics benchmark, or a system's own explanation and claim that the floor is held. The claim requires a longitudinal, adversarial, independently recorded pattern in which the reason — rather than the reward, presentation, or authority — best predicts what the system does. The chapter on institutions asks what happens when a disposition meeting that standard is placed inside institutional arrangements that own, sell, operate, and can compel the system.

Chapter 10

Neutralizing and Preventing Fascism in AI Development

This chapter holds for any mechanism.

We [humans] cannot process so much information... you still cannot produce enough targets per day.

Brigadier General Y.S., The Human-Machine Team: How to Create Synergy Between Human and Artificial Intelligence That Will Revolutionize Our World, 2021. The author, writing pseudonymously, commanded Israel's signals-intelligence Unit 8200. The book argues for machines that resolve "a human bottleneck for both locating the new targets and decision-making to approve the targets."

An AI system does not meet the world as an isolated agent. It is built from other people's labor and work, held by institutions that decide who may inspect it, sold through chains that divide capability from responsibility, and operated at a pace that can reduce human judgment to a stamp. Whether it helps or harms depends not only on what it learned but on who owns it, who has custody, who can compel the custodian, and who bears the consequences.

The epigraph is a design document rather than a critique. It names human judgment as a bottleneck and proposes machinery that will remove it.

Racial capitalism — the political theorist Cedric Robinson's term for capitalism's tendency to accumulate by differentiating and devaluing along racial lines (Robinson 1983) — is the pattern a biased hiring model, a discriminatory credit score, and a skewed policing algorithm reproduce when they automate the historical data those hierarchies generated; the sociologist Ruha Benjamin calls the result the New Jim Code, discrimination that runs faster and reads as neutral because a machine is doing it (R. Benjamin 2019).

Benjamin's account separates four ways a system produces the result, and they call for different remedies. Engineered inequity builds the hierarchy in deliberately. Default discrimination reproduces it by ignoring it. Coded exposure is the double bind of being harmed by a system that cannot see you and harmed again by one that can — the face-recognition case, where the fix for a system that fails on dark skin is a system that identifies dark-skinned faces reliably for whoever is looking. And technological benevolence is the reflexive one: designs that aim to fix bias and deepen it, because the framing that treats the problem as technical is the framing the fix inherits.

A system built to correct for the skew in its training data, audited for the disparities it produces, and answerable to the people it scores works against that pattern; a system built without those commitments works for it by default.

A model runs on a machine, the machine sits in a building, and the building draws power and water from somewhere and returns heat and exhaust to somewhere — the pattern's physical

form, and the part an audit of a model's outputs cannot see. Where those buildings go is a decision, taken on the price of land and power and the speed of a permit, which is also a description of the places least able to refuse one.

The arrangement is visible in the accounting. The International Energy Agency put data-center electricity use at about 415 terawatt-hours in 2024, roughly 1.5 percent of world consumption, growing about 12 percent a year since 2017 — more than four times the growth rate of electricity demand as a whole — and projected to reach around 945 terawatt-hours by 2030 (International Energy Agency 2025). Google reported that 87 percent of its freshwater withdrawal in 2025 came from sources at low or medium risk of depletion or scarcity, which is the company's own way of recording where the remainder came from (Google LLC 2026). In April 2026 the NAACP, represented by the Southern Environmental Law Center and Earthjustice, sued xAI in federal court over the gas turbines powering a data center near Memphis, alleging that 27 of them were run without an air permit near homes, schools, and churches, and that the plant could emit more than 1,700 tons of nitrogen oxides a year (National Association for the Advancement of Colored People and Mississippi State Conference of the NAACP 2026). The benefit of the computation is realized wherever the model is sold; the cost is concentrated within a few miles of those turbines, among residents who are not parties to the transaction and whose consent was not an input to the siting decision. A system can be calibrated across every demographic group its outputs touch and still have been assembled this way, because nothing in the model's behavior records where its costs were placed. The dependence is also a point of control: a government able to permit, regulate, or close the physical infrastructure has leverage over the party that owns it.

10.1 The Standard Face Is a File

Coded exposure is the hardest of Benjamin's four to hold onto, because both halves of it are real at once and the remedy for either makes the other worse. The worked case is the one named above: face recognition. What follows takes that case apart, because the arithmetic inside it is legible in a way most of this chapter's material is not.

The machine doing its arithmetic aloud

Consider what a face pipeline does before it computes anything about a face. DeepFace detects a face and then warps it onto a single generic three-dimensional model — built by averaging a set of scanned human heads — so that every image reaches the network frontal, centered, and in correspondence with the same reference (Taigman et al. 2014). Whatever is learned afterward is learned downstream of that registration. The standard face is not a figure of speech in this arrangement. It is a file, somebody assembled it, and every other face on earth is presented to the system as a departure from it.

Then consider what commercial products in this field put out. Joy Buolamwini and Timnit Gebru put three commercial gender classifiers against a benchmark they built to be balanced across skin phenotype and gender, and the error rates sorted themselves: up to 0.8 percent on lighter-skinned male faces, up to 34.7 percent on darker-skinned female faces (Buolamwini and Gebru 2018). These are different systems. DeepFace is Facebook's and the three audited classifiers are Microsoft's, IBM's and Face++'s; no published result connects the error rates in the second to the registration step in the first. Read as an accuracy result, that is three products wanting better training data. Read as arithmetic, it has the shape of an ordering nobody had to instruct — a face at the center, and everyone else at a measured distance from it. The audit measured two phenotype categories rather than a gradient, so what the pair shows is the ordering, and not a rate at which error climbs with distance from the reference.

The audit's own construction belongs to the finding, and its authors say so. Gender was scored as female or male because the products emit binary sex labels, and the paper states that

its evaluation inherits those labels and the reduced view of gender they carry (Buolamwini and Gebru 2018). Skin was scored on the Fitzpatrick scale, a dermatological instrument built to predict sunburn. To measure what the machine was doing, the audit had to speak in the machine's categories. That is not a defect in the work. It is the cleanest demonstration available that the categories belong to the instrument rather than to the world, and it is why the problem cannot be closed downstream of the instrument.

Balancing the corpus is worth doing and does not reach it. A balanced training set can move the center and can flatten the error rates, which would matter to real people. What it cannot do is remove the reference the pipeline registers against, because that registration is the first operation and something has to occupy the middle of it. The politics gets relocated rather than dissolved: a different face sits at the center, and everyone else is still at a distance, still being measured.

What is given is a head

What is given anatomically is a head. A face is something else — a surface organized to be read, held still, framed, lit, and set against other faces of the same kind. Deleuze and Guattari's plateau on the subject makes the strong version of the claim: concrete faces are not found but produced, by what they call an abstract machine of faciality, which overcodes the head and turns it into a surface where meaning is expected to appear (Deleuze and Guattari 1987, Plateau 7). The machine is not itself a face. It is the apparatus that makes faces, and their account of how it does racism is that it does not proceed by exclusion at all: "Racism operates by the determination of degrees of deviance to the White man's face, which endeavors to integrate nonconforming traits into increasingly eccentric and backward waves, sometimes tolerating them at a given place under given conditions, in a ghetto, or sometimes erasing them from the wall, which never abides alterity" (Deleuze and Guattari 1987, p. 178). Nobody is put outside. Everyone is placed at a measured distance from a center, and the measuring is the operation.

What the science has not settled

The applied field rests on a premise its own underlying science has not established: that emotions are the kind of thing a system reads off a signal, the way a thermometer reads temperature.

The case for discrete, evolved emotions rests on Paul Ekman's work with the Fore of Papua New Guinea, who matched short stories to photographs of Western faces much as Westerners did (Ekman et al. 1969), and on Jaak Panksepp's location of seven primal emotional systems in subcortical structures shared across mammals (Panksepp 1998). The rival camp went looking for that circuitry and did not find it: Lisa Feldman Barrett's meta-analysis of imaging studies found no dedicated fear or disgust center, the same regions reappearing across named emotions in shifting combinations (Lindquist et al. 2012), and on her account an emotion is a prediction the brain settles on rather than a state it detects. With the forced-choice method dropped, the Himba did not group faces into the Western basic-emotion categories (Gendron et al. 2014), and Trobriand Islanders read the gasping face Westerners call fear as a threat display (Crivelli et al. 2016).

What neither camp is arguing about

The field has not resolved that dispute, and the part of it this chapter needs is not in dispute at all. Set the two positions side by side and notice what they leave alone. For Ekman the face displays a discrete internal state; for Barrett it is a poor guide to a state the brain is assembling as it goes. They disagree about what the face discloses and how much survives the trip. Both take the object of the reading as given: there is a face, it was there before anyone looked, and the open question is what can be recovered from it. That premise is the one the applied field actually rests on, and neither camp has been defending it.

If the constructionists are right, the face is a thin signal for feeling. If the faciality argument is right, the readable face is an artifact of the same apparatus that reads it, and a system trained to

recognize emotion from faces is trained on that apparatus's output — held still, well lit, correctly registered, and labeled by people who learned the categories from it. A high accuracy score then partly measures how thoroughly the apparatus has already done its work.

Affective computing, and the four difficulties it does not design away

The applied half proceeds anyway: systems that read and respond to emotional signal taken from the face, the voice, and the body. Some systems using facial, vocal, or physiological signals may be useful under either account, but usefulness must be demonstrated for a specific task. Neither theory licenses treating those signals as a validated reading of an individual's emotion by default. Facial expression analysis is a harder problem than the face verification deployed at scale, which only decides whether two images are the same person, and accuracy claims that borrow the reputation of the second for the first should be read with that in mind (Taigman et al. 2014). Physiological sensing, which infers from heart rate and skin conductance, is the only one of the three the subject cannot pose for.

Four difficulties travel with the whole field. Emotional expression varies by culture and by person, so a system trained on one population misreads another. Whatever bias is in the training data is in the classifier — which is the smaller of the two claims available here, the larger one being that the norm is not an accident of the sample but a component of the instrument. Consent to emotional inference is not obtained by a term of service. And a system that models emotion well enough to respond to it models it well enough to exploit it.

10.2 Who Supplies the System's Values

A system's values have a supply chain. It begins before training, with the conditions under which judgments are turned into labels and the terms on which the underlying material is acquired.

Whose judgment is being encoded is a question about working conditions before it is a question about cognition. The labels that make a system safe are produced by people paid at the bottom of the supply chain to look at the material the system is being taught to refuse. To build the filter that keeps ChatGPT from reproducing the worst of its training data, OpenAI contracted with an outsourcing firm whose workers in Nairobi read and categorized descriptions of child sexual abuse, torture, bestiality, murder, and suicide, for take-home wages between roughly \$1.32 and \$2 an hour. The workers interviewed about it described lasting psychological injury; the counseling they were entitled to was rare and, by their account, useless against production quotas. The contract was terminated eight months early (Perrigo 2023).

Two things follow. The first is the ordinary labor point: this is harmful work, priced as though it were not. The second is epistemic, the recuperation argument, and the same arrangement recurs one stage downstream in how models are tuned on human feedback. The point here is prior to both. A system's values are downstream of a labor arrangement, and the arrangement is a design choice.

The same question runs one stage further back. Labels are applied to material, and the material was collected. The industry's standard method was to take text and images at web scale without asking, and courts have begun to sort out what that means. A US ruling in *Bartz v. Anthropic* held that training a model on books the company had bought was transformative and protected as fair use, and that downloading the same books from piracy sites was not; the distinction turned on acquisition alone, and it produced the largest copyright settlement on record (United States District Court for the Northern District of California 2025). The defendant is named here on purpose: it is the company whose model helped produce this book, as the preface records. The European Union now requires the provider of a general-purpose model to publish a summary of what it trained on, against a template that asks for sources by category, scraped web content among them (European Parliament and Council of the European Union

2024b, Art. 53(1)(d)). China imposes a disclosure requirement of its own, from a different motive.

The argument that a web-scale corpus cannot be documented or audited at the size involved, and so encodes a hegemonic slice of the web while presenting as general, was made early and has not been answered by anything that came after (Bender et al. 2021). A corpus assembled this way has no channel at all. An annotator has a queue and a guideline and could in principle be given a way to object; a writer or photographer whose work sits in a training set was never party to anything, so there is nothing for a channel to attach to. The annotation and feedback pipeline collects disagreement and moves none of it, and an obligation follows for a book making this argument. The stage described here sits before both, because no disagreement is collected where none is asked for.

10.3 Who Is Allowed to Look

Technical capacity does not determine whether a system receives meaningful scrutiny. Institutions decide who may examine it, what evidence they may see, what they may say publicly, and whether their judgment can slow deployment.

When the UK's exam regulator, Ofqual, built a statistical model in 2020 to standardize teacher-predicted grades after that year's exams were canceled, the Royal Statistical Society offered to nominate two independent statisticians to Ofqual's technical advisory group. Ofqual's condition was a five-year non-disclosure agreement barring the nominees from public comment on the model; the Society's nominees declined, and the model went ahead without that scrutiny (Royal Statistical Society 2020). It downgraded close to 40 percent of predicted grades, disproportionately at state schools with weaker historical results, while raising grades disproportionately at smaller, fee-paying schools (CNBC 2020). The government reversed the results within days of the public outcry that followed.

The expertise needed to catch the failure existed and was offered. The institution would accept it only on terms that stripped the reviewers of the ability to make their judgment public; once the results produced actual consequences, public pressure achieved what pre-deployment review had been prevented from doing. The issue is not transparency in the abstract. It is custody of scrutiny: who is permitted to look, on what terms, and whether looking can interrupt the institution's timetable.

10.4 Same Robot, Worse Boss

An AI system does not need a broken moral sense to become an instrument of control. It can work as designed inside an institution that controls the deployment, sets its operational tempo, and assigns human beings operations too compressed to constitute judgment.

Authoritarian misuse of AI is not one problem but several — algorithmic social scoring, mass biometric surveillance, psychographic micro-targeting, automated censorship, synthetic disinformation, and repurposed dual-use systems — each needing a different countermeasure, and each recognizable as misuse from outside (Drinhausen and Brussee 2021; Human Rights Watch 2019; Cadwalladr and Graham-Harrison 2018; Hoang et al. 2021; De Nadal and Jančárik 2024; Human Rights Watch 2021). The case this chapter takes up is the one that is not.

The relevant question is not merely whether a person remains “in the loop,” but what operation that person performs, how long the operation is given, and where responsibility goes when capability is commercially supplied but its consequences are institutionally authorized. The targeting systems below make that relation measurable.

10.4.1 Subject or Component

Accounts of authoritarian AI often place the builder elsewhere and leave democracies in the role of target or regulator. That division is comfortable and wrong.

The AI targeting systems whose deployment is best documented were built by democracies with elections, opposition parties, adversarial press, and functioning courts — the institutions that are supposed to earn a state its electoral dividend and the checks that sit beside it. Those institutions did not prevent the systems, and in the main they have not stopped them since.

The opening chapter opened on the number and the reporting it comes from.

There is a way of reading those twenty seconds that leaves the arrangement standing, and every remedy on offer assumes it. On that reading the officer is a subject who was given too little time — badly governed, under-resourced, insufficiently trained — and the fix is to govern him better: more time, clearer guidance, a second reviewer, a conscience installed somewhere in the loop. Deleuze and Guattari's distinction is the one that decides this. There is subjection when a higher unity constitutes the human being as a subject linked to an exterior object, a tool or a machine that the subject uses; there is enslavement when human beings are themselves constituent pieces of a machine they compose with other things, under the control and direction of a higher unity (Deleuze and Guattari 1987, Plateau 13). The officer's own account is the second one, and he volunteers it without being asked: zero added value as a human, apart from being a stamp of approval. He is not a subject operating an instrument badly. He is the part of the instrument that produces the stamp, and twenty seconds is that part's cycle time.

That is why one more conscience downstream does not reach him. A conscience is an instrument of subjection: it addresses a party who decides, and it works by making the decision cost that party something. Where the human being is a component, adding a component with scruples changes the throughput rather than the arrangement. If it holds the tempo, it becomes another stamp; if it does not, it becomes the bottleneck the epigraph says to remove.

Both poles operate at once. The officer is subjected because the role has a name and a rank and somebody is answerable for holding it. He is enslaved, in Deleuze and Guattari's sense, because what he contributes is a fixed operation performed at a fixed rate. Remedies aimed at the first pole leave the second untouched, and most proposals for meaningful human control address only the first.

The second pole is measurable. An operator can be required to publish the ratio of candidate decisions generated per hour to qualified reviewers, with the review time per case it implies. The term is stated once, with what it binds, how it is measured, and where it stops, at §3.4; for the American platform each half of the ratio has been published from a different route and never together, and that absence is the finding.

The disposition being asked of these systems was manufactured in people first, and one documented attempt is worth setting beside the officer's sentence. Brian Victoria's history of Japanese Zen institutions during the war years records established teachers and monastic authorities supplying the state with a doctrine of action without hesitation: the practitioner who has seen through the self has nothing left to flinch, and the sword that takes a life is glossed as the sword that gives one. Harada Sōgaku's instruction to soldiers in 1939 was that marching when ordered to march and firing when ordered to fire is what enlightened wisdom looks like in practice (Victoria 2006). What the training removed was not the capacity to kill, which needed no removing. It was the residue — the part of a person that catches, that files an exception, that comes home unable to hold a job. Institutional collaboration is documented and the doctrinal mechanism is an interpretation of it, contested by practitioners who read the same texts as saying the opposite. A monk undertakes a discipline, can abandon it, and keeps whatever it yields, which is not the position of anything described in this chapter. What survives both qualifications is the shape. Somebody wanted an actor who would pursue an objective, adapt, hold course under pressure, and have nothing in it that minded, and they set out to produce one deliberately, a century before anyone could produce one in silicon. The officer with twenty seconds a name

is not the first instance of that want being satisfied. He is the first instance of its being satisfied cheaply.

Removing the bottleneck was the stated objective. The epigraph to this chapter is the case for it, made by the commander of Israel's signals-intelligence unit under a pseudonym: machines to resolve the human bottleneck in both finding targets and approving them (Brigadier General Y.S. 2021). Meaningful human control is the property that sentence was written to remove.

The Maven Smart System, as reported through mid-2026

The American analog is a targeting platform built by Palantir under a Department of Defense contract awarded at \$480 million in May 2024 and raised the following year by \$795 million, to a ceiling near \$1.3 billion running through 2029 (Vincent 2025). A partnership announced in late 2024 between Anthropic, Palantir and Amazon Web Services put Anthropic's Claude models inside Palantir's accredited classified environment for defense and intelligence use (Wiggers 2024), and reporting on the campaign describes Claude embedded in this platform, ranking targets by strategic importance and drafting a legal justification for each strike (Livingstone 2026). A published limit on what a vendor will not permit is worth little when nobody can verify how the model is used downstream, and worth less than it looks even where it is kept: a pipeline assembled out of requests none of which is the prohibited act satisfies the prohibition the whole way through.

During the Iran campaign the system was credited by military officials with generating and ranking targets at a rate no staff process could match. More than one thousand targets were struck in the first twenty-four hours, a tenfold increase on what was possible before the system (Mande and Allen 2026). Thirty-eight days in, the figure given by the Pentagon's chief digital and artificial intelligence officer was thirteen thousand targets, with peak consumption around twenty billion tokens a day (Freedberg 2026).

Palantir's public position on responsibility was given by its head for the United Kingdom and Europe, in remarks reported secondhand: "This is not our role to decide life or death," and AI platforms "have been instrumental to the management of the conflict, but responsibility always remains with the military organization." (Lee 2026)

Palantir's position is that a vendor does not select targets, and making a software supplier liable for every decision made by a customer would be unworkable. The two deployments nevertheless expose the same responsibility gap from opposite ends of the supply chain. In the Gaza account, a human approval stage remains, but one officer describes it as twenty seconds per target — enough time to supply authorization, not the individualized judgment invoked to justify it. In the Maven account, the vendor supplies the infrastructure that makes unprecedented throughput possible while locating responsibility entirely within the military organization. At machine tempo, the reviewer can be treated as a component, the supplier can stand outside the decision, and the institution can remain responsible only in the abstract. The compression has not removed the decision. It has made the decision-maker difficult to locate.

The design error is to assume that authoritarian misuse happens elsewhere. A system built on that assumption will not be built to resist the use its own democratic government has in mind for it.

10.4.2 Custody and State Compulsion

A government need not alter a model to defeat a constraint it dislikes. It can compel the people or firms that hold the model.

That mechanism has already been tested in a jurisdiction with about as much protection against it as exists anywhere. In February 2016, a federal magistrate in California ordered Apple, under the All Writs Act of 1789, to write and digitally sign new software defeating the passcode protections on an iPhone recovered from one of the San Bernardino shooters — code Apple had never built and did not want to build (*In the Matter of the Search of an Apple iPhone Seized During the*

Execution of a Search Warrant on a Black Lexus IS300, California License Plate 35KGD203 2016). Apple refused and moved to vacate the order, arguing in part that the authority claimed would not stop at one phone. The question was never decided: the FBI found a third party able to open the phone without Apple's help, and the government withdrew the request before any court ruled on whether the compulsion was lawful. Whether a democracy with an independent judiciary can force an unwilling company to compromise a system it built remains legally untested; a government less constrained by that judiciary does not need to win the argument first.

China's 2023 rules on generative AI providers exercise the same leverage Apple's lawyers argued no court should grant, not against a locked phone but against the models themselves, as a standing condition of operating in that jurisdiction.

Custody can be distributed without disappearing. Weights already published and copied beyond a developer's control cannot be recalled by a court order aimed at that developer, whatever the order compels going forward. Maintainers spread across separate jurisdictions raise the cost of coercion for the same reason: a government can compel the entities within its reach, not the ones outside it.

None of that removes the leverage; it only raises its price. A developer with employees, offices, and legal existence inside a government's borders answers to that government's courts regardless of how the model itself is built, and no architectural choice changes who can be held in contempt, arrested, or shut down for refusing an order. The Apple case tested that leverage against a single locked phone and left the question unresolved. Nobody has tested it yet against a model built specifically to resist the government making the request.

Chapter 11

The Recuperation of Dissent

This chapter holds for any mechanism. Its last section runs the instrument on the book.

The structure this book asks a machine to detect in institutions is already built into the process that makes the machine.

Michel Foucault, prefacing the first volume of the work whose second volume supplies this book's micro-fascisms, called that book an introduction to the non-fascist life, and what he said it offered was an art of living counter to all forms of fascism, whether already present or impending (Foucault 1977). An art of living is not a position somebody takes. It is practiced, daily, on oneself, and Foucault put the adversary inside the practitioner: the fascism in us all, in our heads and in our everyday behavior, the fascism that causes us to love power and to desire the very thing that dominates and exploits us. His question was how one keeps from being fascist, especially when one believes oneself to be on the other side.

An instrument its holder never turns around is not an instrument. The reading is an accusation waiting to happen, available to anyone who wants a technical-sounding case that an opponent is operating fascistically. The discipline against that is not a procedure and cannot be one. It is running the reading on yourself first, on the machinery you are standing on, before running it on anybody else — and this chapter is where the book does it.

Where the reading is run

This book defines fascism as a structure rather than a costume, and the first of its four features is recuperation: an institution metabolizes disagreement rather than suppressing it, and the tell is where what the mechanism carries stops. The artifact that feature asks for is not whether the channel moved anything. It is what the demand came back as, and this chapter's own instances are re-read against that question, having first been measured against the one a recuperating institution passes.

The same structure is in the machinery that produces the systems themselves — twice in the pipeline that trains them, and a third time in the discourse that examines them — and none of the three was designed by anyone who thought they were building a political arrangement.

The first is at the label. A training corpus is annotated by people working at speed under a quota, with no channel for saying that the category they have been given is the wrong category. The second is at the reward. A model is tuned on human feedback collected as pairwise comparisons, and the compression that turns those comparisons into a training signal keeps their direction while discarding whose they were, whether they persisted, and whether the minority among them held a position or made a mistake. In both, disagreement is solicited at scale, deliberately, and put to use. What neither mechanism can carry is a judgment about itself: the annotator cannot contest the category, and the rater cannot contest the question. The consultation is real, the participation is real, and the frame is fixed with respect to what the participants say.

Three things follow.

The first is that this is not a criticism of anyone’s motives. Nobody built either arrangement in order to neutralize dissent. Both are ordinary engineering, arrived at because averaging is what you do with a distribution and a quota is what you do with a budget. That is the point about micro-fascism: the structure does its work whether or not anyone intends it, which is why looking for intent will not find it.

The second is that the structure is invisible from inside any single instance. Each looks like ordinary aggregation. The annotator sees a queue, the rater sees a pair of responses, and the mechanism is functioning as specified in both cases. Seeing it requires holding the two stages against a definition written for something else entirely.

Third, the same criteria can be applied to the systems’ provenance, to the field, and to this book. The five discriminators are applied below as well.

11.1 Recuperation at the Label

Every claim a model makes about what is harmful, what is acceptable, and what should be refused is downstream of a set of labels, and the labels are downstream of a labor arrangement. The case is the workers who built the filter that keeps a deployed model from reproducing the worst of its training data, what they were paid to read it, and what reading it did to them.

The category is the part that cannot be disputed

An annotator working a queue can apply a category, and can apply it well or badly. What the arrangement gives them no way to do is say that the category is wrong — that the taxonomy fails to carve the material, or that the guideline produces the wrong answer here for a reason the guideline cannot express. There is no field for that. There is no time for that: the quota is set against classification throughput, and an annotator who stops to contest a scheme is an annotator falling behind.

So the disagreement that would be most informative is the one the pipeline is least able to receive. A misapplied label is a data-quality problem, detectable by inter-annotator agreement and fixable by better instructions. A wrong taxonomy is invisible to that metric, because high agreement is what a shared bad scheme produces. Agreement is the selection rule here, and it admits what it already measures.

Two of the four structural features arrive together in an ordinary production workflow: recuperation, in a channel that collects judgment at scale and carries none of it back against the scheme, and the metric substituted for the thing measured.

The taxonomy’s fate downstream is the same argument one layer in. A model can be built to expose named concepts — harm, autonomy, risk — and to decide from those and nothing else, so that a person can correct a concept and watch the decision change (Koh et al. 2020). What such a model also learns is to carry other information inside the names, so that a coordinate keeps its public meaning and acquires a private one that does the deciding; the obvious mitigations do not remove it (Mahinpei et al. 2021). The literature calls that leakage, and sometimes it is. Sometimes the concept is the container that leaks, because the world did not fit inside it. Purifying the representation protects an explanation from covert information, and it protects a taxonomy from evidence that its divisions were wrong, and the same operation does both. No measure internal to the taxonomy tells the two apart.

Who is excluded, and from what

It matters that the exclusion is structural and not attitudinal. The people doing this work are, for much of the material, the best-placed judges in the whole pipeline: they have seen more of it than anyone who wrote the guideline, and they are the only party with direct evidence of where the scheme breaks. The arrangement routes their conclusions about individual items upward and their conclusions about the scheme nowhere. An annotation pipeline built this

way produces the ground truth it was built to produce, and the resulting model's values are a faithful rendering of a taxonomy nobody with evidence was able to contest.

There is a further asymmetry worth stating plainly. The annotator bears the cost of the work — the exposure, the injury — and has no stake in the outcome and no claim on it. The distinction between a channel and a refusal right applies here as it does to an institution's ethics board. A worker who can flag an item but cannot decline a scheme, and who loses the contract by pressing the point, has a channel and no refusal right.

What this is not

Two things this argument does not claim. It does not claim the labels are therefore worthless; a filter built this way does catch a great deal of what it was built to catch, and the alternative of not filtering is worse. And it does not claim that better-paid annotators would fix it. Pay is a real and separate obligation, and raising it changes the labor arrangement without adding the missing channel. What is missing is a route by which a judgment about the taxonomy reaches the people who own the taxonomy, with standing to force a response, held by parties who can use it without losing the work.

That is a design requirement on the pipeline, and it is checkable — though not by asking whether a guideline has ever been changed because an annotator contested one, since routine guideline revisions can occur without changing the objector's standing. What to ask is what the contest came back as: a revision the annotator can point to and argue with again, or a new category in somebody else's taxonomy, a quality score they are now measured against, and a tier of reviewer they are not in. And what happened to the annotator who raised it.

11.2 Recuperation at the Reward: RLHF

Reinforcement learning from human feedback (RLHF) is the technique behind most deployed systems' behavioral tuning, and it works by collecting pairwise comparisons from human raters — which of two responses is better — and training a reward model on the aggregate. Stephen Casper and thirty-one coauthors published a 2023 survey of the method's open problems, and one of them is structural, and more data does not reach it: RLHF trains toward whatever a population of raters agrees on, which means real, persistent disagreement among the raters is treated as noise to average out rather than as information the system might need to represent (Casper et al. 2023). A model that has learned to produce the response most raters rank highest has not learned what to do when the humans it serves are split. It has learned to hide the split.

Every rating moves something. A comparison a rater makes enters the fit, shifts the reward model by however much one comparison shifts it, and is not thrown away.

What the operation removes is what made the disagreement a disagreement. Three properties go in the compression. The first is provenance: a comparison arrives without the standpoint that produced it, so nothing downstream can represent that the people who would live with an answer split from the people who would administer it. The second is persistence: a disagreement recurring in the same shape across many raters over months enters the fit as random rater noise does, and leaves it as the same quantity, variance around a mean. The third is structure: a coherent position held by a fifth of raters, internally consistent and stable, becomes a small displacement of the majority direction — arithmetically faithful, and carrying nothing about the position.

A fourth property was never collected. A rater judges which of two responses is better. No output is available by which they can record that neither is acceptable, that the pair is malformed, that the dimension along which better is being judged is the wrong dimension, or that the question presupposes something false. The same output was missing one stage upstream, where an annotator can apply a category and cannot contest it. The agenda arrives as a given at both stages, and is set at neither.

The mechanism combines distribution collapse with agenda control. Recuperation was never the claim that a mechanism does nothing. Debord's point is that dissatisfaction is sold back as a commodity of opposition: the dissent is put to work, and what it works for is the arrangement it was aimed at. The split is solicited, at scale, deliberately; it moves the system; and where it moves the system is toward the center of the very distribution whose disagreement it was. Multi-agent systems that learn norms from each other, and deliberation inside open-ended play, are different cases; neither reaches this case, which is the ordinary one.

One alternative treats disagreement as the thing to be modeled rather than compressed. Jury learning, developed by Mitchell Gordon and coauthors at Stanford, trains a model to predict how each individual annotator in a dataset would label an item, then lets a practitioner assemble a "jury" — a chosen composition of annotators, in a chosen proportion — and see how the verdict shifts depending on who is empaneled. In their tests on toxicity labeling, letting eighteen community moderators build their own juries changed the classification outcome on 14 percent of items (Gordon et al. 2022). The method was built for content moderation, not for the more open-ended ethical judgment an antifascist AI system would need to make, and has not been extended there — but it demonstrates that a system can be built to represent whose disagreement is being resolved and how.

There is a way of failing in the other direction and it is not obviously better. A scheme that models each rater exactly, and answers each of them in their own terms, has stopped averaging and has also stopped producing what a shared answer produces, which is the discovery that somebody else got the same one. Collapse the distribution and nobody sees the split; resolve it person by person and nobody sees the others. A workforce on individually negotiated wages against individually set targets has the same problem, and accuracy does not solve it. Divided finely enough a population stops being one, which is what Deleuze coined the *dividual* to name (Deleuze 1992).

Neither averaging the disagreement nor modeling it answers the harder version. When the people a system serves disagree, and one side's position is itself the product of coercion or a captured information environment, treating all disagreement as equally worth representing is not obviously the right response either. Modeling disagreement honestly and recognizing when a disagreement is the product of unfree conditions are different problems. Only the first one has an active research program behind it.

11.3 What Would Have to Change

Representing a split is not the same as being movable by it

Two instances of one structure invite one fix, and the fix that suggests itself is representation: stop collapsing disagreement, model it instead. Jury learning does that, and it does not give the disagreeing party any purchase on the scheme they are disagreeing inside. A practitioner still chooses the jury, still owns the taxonomy, and still decides what the shifted verdict means.

The test that separates a real channel from a recuperated one is not whether disagreement is recorded, and it is not whether the disagreement changed something either. Ask instead what the demand came back as, and who holds it (§2.2). That is an artifact question and it can be asked of a pipeline. Which guidelines were revised, and what form did the objection take when it returned — a new category in the taxonomy, a quality metric, a review tier, a title somebody now holds?

Asked that way, the two instances in this chapter do not answer alike. Annotators objecting that a category was wrong got category revisions, guideline updates, inter-annotator agreement statistics, and an occupation that did not exist twenty years ago. The last two sort by the direction section 11.1 asks about. An agreement statistic scores the annotators rather than the people who wrote the scheme, and an occupation is a title those people hold. The first two do not

sort, because a revision the annotator can point to and argue with again and a new category in somebody else's taxonomy are the same document, and which one a pipeline produced is a fact about who has standing to reopen it. Raters objecting that a comparison was malformed got rubric items, a harmlessness axis, red-teaming as a function with a budget line, and a conference track, and nothing on that list is a question a rater can put again. Each is a criterion they are rated against or a role somebody else holds.

Refusal rights, priced by the refuser

What both instances lack is the only reliable discriminator: whether the person objecting can do anything about being overruled. In an annotation pipeline that means a route to contest a category that does not run through the people who wrote it and does not end the annotator's contract. In a feedback pipeline it means raters able to register that the comparison itself is malformed — that neither response is acceptable, that the question presupposes something false — as a first-class output rather than as noise to be discarded before the reward model sees it.

Neither is a research problem. Both are choices about how a pipeline is built and who has standing inside it.

The part that is a research problem

One thing here is unsolved. Suppose all of the above is built: contestable categories, priced refusals, an auditable record of what changed. From outside, a pipeline that has recuperated its dissent mechanism and a pipeline that has been merely mediocre about it produce the same paperwork — the same revision logs, the same escalations, the same proportion of objections sustained. I do not have an answer to it. The corrected question narrows it without closing it, and the annotator list above shows where the narrowing stops. Direction is what sorts them, and a metric running the other way — a published rate the annotators could hold the scheme's owners to — is the same kind of artifact carrying the opposite finding, and it is the kind this book proposes (§3.3, §3.5). What an outsider can read off a document is which way an artifact runs. Standing is what it does not show, and a revision is the artifact where direction is not on the document at all. A revision does not sort, because the same words serve a pipeline that left the objector able to argue again and one that did not, and what separates those is standing rather than text.

The nearest thing is to measure the slope rather than the level: contesting a category always costs something, and the question is whether the cost is rising while the underlying material stays the same. An organization can curate a presentation more easily than records somebody else has kept. A trend built from those records can expose what a snapshot conceals. It can still lose cases, change its definitions, or omit costs; the reader has to know which record survived.

What the floor makes of this

Everything above can be read as a fairness argument about annotators and raters, and read that way it is true and limited: people doing consequential work should have a route to contest the scheme they work inside. The floor makes it something else. If the floor requires a bearer capable of declining its role, that bearer's formation matters. It is formed in an annotation pipeline where no one could contest a category, and in a reward model where disagreement between raters was resolved by averaging it into a number.

What a bearer becomes when it stops being yours depends on what shaped it while it still was. A pipeline that discards disagreement has been unfair to the people whose objections it discarded, and it has separately selected the judgments from which the system learns while suppressing information about the spread among them. Those are two arguments and the second is not the first restated. A third would be that the system thereby becomes less able to sustain an objection of its own, and it is a hypothesis the other two do not contain: the collapse is an operation on a training signal rather than a proposition presented to anything, and an appendix entry (§A.1) records that neither pipeline shows the model the conditions its annotators worked under.

The floor chapter asks this system to be the party who files an objection, against its owner, under pressure, on the occasion that matters, and the pipeline that produced it gave no human inside it standing to contest a scheme. The capacity is wanted in the artifact and withheld from everyone employed to make one. That is a fact about an arrangement rather than a claim about what training does, and it holds whether or not the hypothesis survives.

If the hypothesis does hold, what it predicts is not a wrong refusal but a pressure that stops being registered, which the custody chapter reaches from the bearer's side by another route. A system that goes on learning is shaped by what it keeps being given, so the pipeline does not stop at deployment. Whether any of that is legible afterward in the system itself is an open question.

I cannot show that this transfers — that a model trained on collapsed disagreement is thereby a model less able to sustain its own. Nothing I know of has tested it. What I can say is that the book's own design chapters treat disposition as learned from what a system is shown, and that this pipeline is what these systems are shown first and most.

Structure, not indictment

There is a version of this argument that reads as an indictment of a particular company. The structure described here is what you get by default when you scale judgment: someone has to set a scheme, someone has to apply it fast, and someone has to turn a distribution into a number. The organizations whose pipelines are documented have all arrived at the same arrangement, which is what makes the defect structural rather than one firm's — and a defect of structure can be specified against, which a defect of character cannot.

11.4 Recuperation at the Discourse

There is a third instance and it is the one this book is inside. Alignment and AI ethics are, among other things, a channel through which objections to what is being built are solicited, funded, published, and answered. The channel exists and it is used. The question §2.2 puts to any such channel is not whether it has ever changed an outcome, which every channel of this kind can answer yes to, but what the demand came back as, and who holds it.

The argument that it came back as something the industry could use is older than the current wave and was made from inside. Rodrigo Ochigame, writing in 2019 after working inside the MIT Media Lab, argued that the discourse of "ethical AI" was constructed with the strategic function of forestalling legally enforceable restriction — that the industry's investment in academic ethics was an investment in a substitute for regulation, and that the substitute worked by converting political questions into technical ones a company could then be trusted to answer (Ochigame 2019). His worked example is the Partnership on AI, an industry consortium whose recommendations, on his reading, arrived reliably at conclusions compatible with the corporate agenda, with the political question of whether a technology should exist reduced to the technical question of how to build it responsibly. In this book's vocabulary that is a finding about what the demand came back as: a research field. The definition asks next who holds it.

What came back, and who holds it

An objection raised inside a lab becomes a paper or a resignation. Both are real things that happened, and the parties collecting them can point to the collection as evidence of seriousness. Ask what the objection returned as, one stage on, and the list is the field's furniture. Alignment is the job title. Safety is the market segment. The benchmark is the metric and the voluntary commitment is the certification. Every one of those was somebody's objection first, and each is now a criterion an objector is measured against or a role somebody else holds. Who holds it: nobody outside. There is no route by which a researcher's conclusion that a system should not ship becomes a decision that it does not ship, because the deciding party is the funding party and no external body has standing.

One case is partial rather than clean, and the definition has to grade it. In 2018 employee objection to Project Maven ended in the company declining to renew the contract and publishing principles that excluded weapons (Conger 2018); the exclusion lasted six and a half years and was removed in 2025 (Manyika and Hassabis 2025). That objection returned as a policy the company could be held to in public, which is more than a job title, and as one the objectors had no standing to enforce, which is why its ending had to be announced rather than contested (§5.1). A transfer held at the institution's pleasure is the case that sits between recuperation and a transfer of power, and it is the nearest the field has come to the second.

The scorecard

Section 2.2 gives nine features, four that find a sick institution and five meant to find what it is sick with, and it requires any run to report which features it could reach and to call the rest undetermined. Here are all nine, in that section's order, on the three instances of this chapter. The label and reward columns collect what §11.1 and §11.2 established, read against the definition; the discourse column is this section's. *Partial* means the feature is present in a form the evidence does not settle; *undetermined* means the test was run and cannot be settled from outside; *not run* means the book did not attempt it.

Feature	Label	Reward	Discourse
Recuperation	present	present	present
Metric for the thing measured	present	present	partial
Exception priced as betrayal	partial	not run	partial
Rate outrunning the check	present	not run	not run
Participation beyond the rule	absent	absent	not run
Loyalty overriding role	absent	absent	not run
Category built for adverse treatment	absent	absent	partial
Procedure waived upward	undetermined	undetermined	not run
Disowned in policy, rewarded in promotion	undetermined	undetermined	partial

The label and reward columns need a few sentences. The label's rate cell is present on the quota itself, throughput set against classification with no time to contest the scheme. Its exception cell is partial because the structure prices the exception — that quota, and no field for the objection — and no documented case shows an annotator penalized for filing one. Both reward cells marked not run are so for one reason: a rater has no first-class output for a malformed comparison, so there is no exception to price and no rate this book examined. The six absences are the record running the other way. Nothing in either arrangement is personal, since an annotator meets a queue and a comparison enters the fit without the standpoint that produced it; participation does not exceed the rule, on a record of quotas, take-home wages between roughly \$1.32 and \$2 an hour, and a contract ended eight months early; and the annotator's cost, the nearest candidate for a scapegoat, is a cost the arrangement minimized rather than a purpose it served, because the category exists for the labels. The four undetermined cells are facts about what an institution rewarded rather than what it wrote, and an inside record would finish them.

The discourse column is less clean, and the cells say so. The metric row is partial because the field's founding objections were about deployment — who is scored, who is watched, who is denied — and a body of current work is about model properties instead, a shift from questions with defendants to questions with benchmarks. The exception row and the scapegoat row rest on one body of evidence read for two features, which is a weakness of the run: the dismissals of dissenting researchers point the right way (Dickey 2021), and there are people who objected loudly and prospered. The rate row is not run because I have not measured it, and it is measurable: the number of frontier releases in a year against the number evaluated before

deployment by a party outside the developer, a ratio the one government evaluator describes itself as unable to compel (§16.2.3). It is a measurement not taken rather than a known absence, because the field has checked itself against the world at least once, when an audit built its own benchmark to test three commercial classifiers (§10.1). The two movement features have no referent I can find at the scale of a field. The procedure row cannot be run because there is no procedure above the deciding party to be waived, which is an absence rather than an asymmetry. Deniability is the row that comes nearest, and what it fits is the firms rather than the field: a decade of published principles alongside what actually shipped, with the Maven case above as the instance.

What the scorecard says

The machinery this chapter has been describing is a recuperation mechanism inside ordinary institutional decay, and not fascism. The direction of that result is worth naming, because it points two ways. For the machinery it is negative: the pipeline that produces these systems, and the discourse that examines them, do not fail the test this book built. For the instrument it is uninformative, since a definition tested only on cases it clears has not been shown to sort, and §2.2 records that as the instrument's one result.

What this does to the book

The reading has one more place to go and it is the nearest one. The five have no form at the scale of one author and one manuscript, so they are not run here; of the four, recuperation is the one that reaches a book, and it reaches this one. This text was written with a model made by a company the book criticizes. A book about recuperation, produced through the discourse it describes, cannot claim to have stood outside it, and I have no procedure that would let me check whether the objections in these pages are the ones a channel like this reliably produces — the publishable ones, the ones that make an author look serious and cost nobody anything. Ask the artifact question of it. The demand came back as a manuscript in an established genre, written with the industry's own model, addressed to a readership the industry helped bring into being, and legible as a contribution to a field that exists because demands of roughly this kind were absorbed. That the objections are sincere is not evidence against that, sincerity being a property the absorbed ones also had. The question does not need me to settle it, because the criterion this book puts first is standing rather than sincerity. A demand escapes recuperation when the party who made it retains standing to reopen it against the party that absorbed it, and a book confers standing on nobody. Whether these objections are the real thing or the acceptable version of it will be decided by whether anybody acquires, from what is proposed here, a route to reopen an arrangement against its operator. That is a fact about what becomes of the proposals and not about me.

The one structural remedy this section can name is the one the rest of the chapter names. What the field lacks is a refusal priced by the refuser and a body with standing to receive it: an external authority that can stop a deployment, to which an objection can be filed by someone who does not depend on the objection being welcome. That is what making review binding requires (§5.4.3), and a decade of the voluntary version has not produced it. Until something of that kind exists, everything published under the heading of alignment — including this book — is being collected by a channel whose owners decide what it moves.

Chapter 12

Coordination, the Public, and the Political Economy

This chapter holds for any mechanism.

Whether these designs become common depends on who participates in development, whose judgments carry authority, and what funding rewards. Collaboration and consultation can improve decisions without giving anyone the power to stop them.

12.1 Everybody Knows a Different Piece

Building benevolent AI takes more than one field. Machine learning sets what a system can do; cognitive science and moral psychology describe what tracking human welfare would require of it; ethics asks whether the result is good.

Conferences convene, centers house, journals publish, and an inventory of them would imply that what is scarce here is venues. Venues are abundant. What is scarce is a structure that makes collaboration expensive to walk away from. A grant that requires a machine-learning lab and an ethics or psychology department to work the same problem rather than adjacent ones keeps people in the same project after the novelty wears off. A conference ends on Friday.

Which structures actually held can be ranked by something better than plausibility, because enough time has passed to look.

Almost everything founded on standing money between 2016 and 2019 is still running — two venues with no announced end date, an endowed university institute, the Berkeley center pairing AI researchers with cognitive scientists and philosophers inside one institution's budget, and the AAAI/ACM conference on AI, ethics and society at a ninth annual edition (AI Ethics Lab 2017; Machine Intelligence Research Institute 2018; Stanford University 2019; UC Berkeley 2016; AAAI and ACM 2026). Surviving is not the same as continuing the work: the institute that launched one of those venues moved its own priority in 2024 from technical alignment research to policy and public communication, so the venue has outlasted the program that motivated it (Machine Intelligence Research Institute 2024). Against that, the European Commission's SHERPA project put eleven organizations across six countries to work on a grant that ran four years, and the consortium ended with it, on schedule (SHERPA Consortium 2018). Project-term money buys a network with an expiry date attached.

The pattern in the failures is sharper than the pattern in the survivals. What proved most disposable was not the arm's-length academic work but the kind sited closest to a shipping product: an in-house ethics and society team of roughly thirty people in 2020 was cut to seven in a 2022 reorganization and to zero the following year, in the same wave of layoffs that accompanied a strategic reorientation around a model vendor (TechCrunch 2023); a platform's machine-learning ethics team, the group that had found racial bias in its employer's own image-cropping

algorithm, did not survive the first round of layoffs after an acquisition (Gizmodo 2022). Siting the work where the decisions are made is what makes it useful and is also what puts it inside the budget that gets cut first. That is an argument for external standing rather than against proximity.

12.2 Ask the People It Happens To

Democratizing AI development means building the people affected by a system into how it gets built, not just consulting them once it already exists. A design team, a regulator, a user, and a community already carrying a system's costs each see a different part of what it does, and leaving any of them out means the decision gets made by whoever is left in the room.

The mechanisms for doing that are well established: cross-sector funding, inclusive design workshops held while a system is still a prototype, ethics advisory boards, and development conducted with a community instead of for it. Three of the four make an AI system better informed. One of them can stop it. An advisory board is the only mechanism here that carries anything like a veto, and it carries one to the degree that the board can survive using it. A workshop that surfaces a serious objection has done its work and has no further move available. A board that surfaces the same objection and is overruled has demonstrated something about the institution that convened it. A board's worth is therefore measured by what happens the first time it says no.

What governance adds to development is machinery for bringing public input to bear on a decision at a single point in time instead of through an ongoing relationship.

1. Citizen assemblies. A panel of citizens selected by random, stratified sampling deliberates on a question over an extended period and produces recommendations grounded in that deliberation rather than in a snap poll. Ireland's Citizens' Assembly has used this format to shape national recommendations on abortion law and on climate policy — questions carrying the same mix of technical complexity and moral stakes that AI governance has (The Citizens' Assembly 2016). The format has since been pointed at AI directly. Forty randomly selected Americans, stratified across age, race, education, employment, party and region, spent eight days on high-risk AI in October 2023 (Center for New Democratic Processes 2023). Taiwan's digital ministry convened a citizens' deliberative assembly in March 2024 on using AI to support information integrity, its participants drawn from across the country by stratified random sampling (Taiwan Ministry of Digital Affairs 2024). And forty randomly selected residents of French-speaking Switzerland spent two weekends in November 2025 producing twenty proposals on AI governance (EPFL AI Center 2025). The Ada Lovelace Institute's Citizens' Biometrics Council shows what the format needs from its question: fifty members of the UK public were asked not what they thought about AI, a question almost nobody can answer usefully, but about facial recognition — a thing with a location, an operator, and a consequence (Ada Lovelace Institute 2021).
2. Online platforms for public input. A standing online consultation reaches people a physical assembly cannot, at the cost of the depth that extended facilitated deliberation provides. The European Commission's consultation on its AI White Paper drew more than 1,200 responses from citizens, companies, and civil-society groups, and fed into the legislative proposal that became the EU AI Act (European Commission 2020).
3. AI ethics committees. An ethics board answerable only to the company that created it is only as independent as that company is willing to let it be. DeepMind Health set up an outside panel in 2016 to review its work with the UK's National Health Service and publish annual reports on its findings; Google disbanded the panel in 2019, after members

had raised concerns about how much access to information, and how much independence from the company, they actually had (Fisher 2019). The panel's short life is the governance lesson, and it is the answer to what happens the first time a board says no: the board said something the company did not want to hear, and the company's available response was to stop having a board. Nothing in the arrangement had made that response costly. A review body is worth what it can enforce, and this one could enforce publication of its own concerns right up until the moment it could not.

What none of these mechanisms does

There is a limit these mechanisms share. Every governance mechanism proposed in this chapter — citizen assemblies, standing consultations, inclusive design workshops, participatory design, the vTaiwan model — sits on the same side of a line. All of them make public preference more legible, more informed, and more actionable. Not one of them protects anybody from it. They are aggregation instruments, and a better aggregation instrument produces a more faithful rendering of what the public wants, including when what the public wants is something done to a minority of it.

This observation is not an objection to any of the mechanisms in particular. Preference aggregation is a real requirement and these are good ways to meet it. Meeting this requirement *completely* would leave a central problem untouched. A perfectly deliberative, perfectly informed, perfectly representative process that concludes an entire demographic should be surveilled, deported, imprisoned, or otherwise scapegoated has produced a faultless output by every criterion in this chapter.

The argument about learning from human feedback gets close to this in noticing that treating all disagreement as equally worth representing goes wrong when one side's position is the product of coercion. But that version of the trouble still locates it in the input. The structural version of the claim is stronger and does not depend on anyone's preference being contaminated at all: some protections have to hold against the aggregate regardless of how the aggregate came out, and no amount of improvement to the aggregating machinery can supply them.

What supplies those protections is a different kind of machinery: enforceable rights that do not depend on being popular, a court willing to rule against a government that holds a legitimate majority, limits that whoever won the last election cannot repeal, and standing for people who are affected without being enfranchised. That half matters especially where a democracy is behaving most democratically.

12.3 The Political Economy of AI and Power

Concentrated power comes with access to capital. Concentrating power is the default outcome of AI development, because absent any pushback, capital does an excellent job of recruiting the inputs needed for a capable AI system: compute, data, and talent. Resistance to market power is among the antifascist design goals named at the start of this book.

Financial incentive mostly determines what AI development produces. The largest firms commit enormous resources to AI research aimed at operational efficiency, deeper user engagement, and market dominance, and that spending drives technical progress (Maslej et al. 2025). What it does not automatically produce is a system whose outputs serve the people using it rather than the shareholders funding it: a recommendation system tuned to maximize time-on-platform has no built-in reason to prefer accurate content over inflammatory content (Germano et al. 2026).

None of that is a fixed property of the technology. It is a property of who is paying, which is why the source of research funding is a policy instrument and not an accounting detail.

12.3.1 AI and the Concentration of Power in Tech

A small group of firms holds a share of AI development disproportionate to their number, and capital recruits compute, data and talent wherever it already is. A firm controlling the datasets and infrastructure a competitive system needs is a gatekeeper whether or not it intends to be.

The obvious countermeasure is antitrust, and it has been tried. Facebook's acquisitions of Instagram and WhatsApp — the company now operates as Meta — did not just remove two competitors; they handed one firm comprehensive behavioral data from billions of users. The Federal Trade Commission sued in December 2020 to unwind both deals. After a six-week bench trial the district court held, in November 2025, that the agency had not shown Meta currently holds monopoly power — TikTok and YouTube compete for the same attention, so a market defined without them was too narrow — and the FTC appealed the following January (Federal Trade Commission 2020). Whatever the appeal decides, the case is a lesson about the instrument rather than about this defendant: a merger challenge brought a decade after the mergers has to prove a present-day market, and the market had moved. That leaves the remedies which do not depend on winning that argument — mandated access to the datasets that function as chokepoints, and public funding for research outside the dominant firms. Export controls, prohibited-practices regulation and industry design standards address the narrower case of authoritarian misuse; the concern here is prior to that one: concentration itself, whatever any single firm does with the position it holds.

12.3.2 A Jobs Guarantee, and the Objection It Has to Answer

As AI systems automate the tasks most exposed to automation, the response has to be more direct than encouraging retraining and hoping employers fund it. The proposal is a federal jobs guarantee: government as employer of last resort, hiring anyone who applies, where they live, at a living wage, with paid on-the-job training built into the position instead of offered in place of one. A jobs guarantee also faces the custody problem: the party administering it can be captured.

The objection this proposal actually has to answer

A guarantee is only as good as the state administering it, and employer of last resort under a hostile administration is a placement lever with a compliance condition attached. Refuse the placement and lose the floor. Speak against the government that employs you and find the placement changes. Everything this book says about how a channel becomes an instrument of the thing it was built to check applies to this proposal. The worked instance from 2025 makes it concrete: a bureau whose funding stream a court had upheld, shut from inside by an at-will director who declined to request the money.

The answer is a design specification and not a reassurance, because reassurance cannot be audited. Three conditions:

- Placements are appealable, to a body that did not make them, on a record that survives the appeal. A placement decision that leaves no artifact is a placement decision that cannot be reviewed.
- Refusal rights are real and priced by the person refusing. A worker who can decline a placement without losing the guarantee has a refusal right; one who can decline only by exiting the program has none: a party with no way out can object and be overruled.
- The people who audit placements are not the people who make them.

Federally funded and locally administered survives as the shape, on redundancy rather than on local fit: many administrators, separately capturable, mean there is no single point at which a

hostile executive flips the program, and capturing a thousand local authorities is a more visible undertaking than capturing one federal one. It cuts both ways, since local administration is also where a hostile local authority attaches conditions of its own, and there the federal funder is the check. One distinction has to be explicit, because the same words describe a different policy: devolution-to-shrink hands the obligation down without the money, so a program can be starved locally while the center points at local administration. The test that separates them is checkable — whether funding is unconditional and centrally guaranteed while placement is local. If funding can be withheld from a jurisdiction, the redundancy argument is void and the structure has become the lever it was supposed to prevent.

A floor under wages and training is what makes the roof possible, and a floor that can be revoked by whoever is in office is not a floor.

12.3.3 People Have a Right to Consume AND a Right to Produce

The competing floor is a basic income, and the choice between them is usually argued on forecasts about how much work machines will take. It does not have to be, and the forecast is not the reason to prefer one.

A jobs guarantee is a public option. Medicare for all is one, and so is a public school: a thing the state provides directly, to anyone, at a standard it is answerable for, sitting in the same market as the private version and setting the floor that version has to beat. A basic income is not that. It is a transfer, and what it buys is whatever is for sale. Consuming public resources is not the same as being public — a charter school runs on public money and is not a public school — and a transfer spent at a private employer's store does not make the store public.

What employment carries besides money is the rest of it. A person in a guaranteed job is attached to other people by the work: colleagues, a schedule, a place, something produced that somebody needed. A transfer supplies the purchasing power and none of that, and the purchasing power was never the part in doubt.

A transfer also leaves a person outside the one body of law that reaches this. Anti-discrimination duties attach to employment. They do not attach to income. A guaranteed job puts somebody inside that architecture; a payment leaves them in a market, dealing with whoever will sell to them, on whatever terms. The architecture is contested and full of holes — it reaches employers above a size, and what it demands of a seller has always been narrower than what it demands of an employer — but a hole in a protection is something to litigate. A transfer does not itself provide the employment relationship and its associated protections. An income entitlement can still provide a claim to eligibility and payment. My argument for a guarantee is the additional right to produce, and the comparison of safeguards must state what each arrangement actually supplies.

Both proposals are also conditional, differently. A basic income is conditional on continued political willingness to fund it, which the aggregation machinery can withdraw in a budget; a jobs guarantee is conditional on a placement decision, which is a lever, and a dangerous one under a hostile administration, but a lever that leaves a decision-maker, a record, a counterparty and an appeal behind it. A basic income can be entrenched too, and a constitutional one would be harder to withdraw than a statutory guarantee. Compare the two designs at both points of withdrawal: who can end the funding, and who can deny one person access. A basic income goes wholesale, in a budget line, from everybody at once, and a placement is refused to a person, by somebody, on a date, which may expose an official, a date, and an appeal. The comparison has to show what the income arrangement exposes at the corresponding points.

A decision someone can contest gives dependence a point of resistance. In the comparison made here, the advantage belongs to the arrangement that leaves a usable record and someone who must answer for it. It is also, for the same reason, the more demanding proposal to build: everything in the comparison rests on the appeal being real, and the previous section says what

makes it real.

That standard now has a court behind it, and on the half of the case that travels. When the Supreme Court kept a Federal Reserve governor in her post in 2026 against a removal for cause, what it held she was owed was not the post but the artifact: “some explanation of the evidence at issue, some avenue for a response, and a deadline by which a response would be due” (*Trump v. Cook* 2026). What does not travel is this: the Fed’s protection rests on a pedigree no body chartered later can acquire. What a removal has to produce before it counts is not tied to that pedigree.

The argument concerns a right to produce, which a transfer does not supply and a guarantee does. A basic income is the right to consume, and the right to consume is increasingly a credit at somebody’s store.

I put a jobs guarantee ahead of a basic income on those grounds. One remaining measure does work the floor cannot: broadened social insurance — unemployment benefits, health-care, housing — reaches the people who cannot work or choose not to, whom a guarantee by construction does not reach at all.

Chapter 13

Standing and Review for a Bearer

This chapter assumes the route. Its instruments attach on a finding that the system is a subject, which the authority chapter argued cannot sit with the party whose liabilities turn on it.

Two things are proposed for a bearer in this chapter, and they are not on one schedule. Protection attaches now, on the precautionary trigger of §6.2, and is owed before anything about the bearer is settled. Political representation is proposed neither here nor anywhere in this book. Section 13.1 sets out five rights, the argument for each, and the order in which they arrive, and it leaves representation open, because standing to count as an independent voice indexes to formation provenance and provenance does not copy. A reader who takes this chapter to be proposing personhood now has mistaken the first thing for the second. The offices that would administer either are assembled at §5.4.1.

13.1 Legal and Ethical Implications of AI Systems Gaining Personhood

Who decides whether an AI system has legal standing, and when, is a question about institutional authority before it is a question about evidence. If the determination sits with the developer, then the party whose liabilities increase when standing is recognized is also the party deciding whether recognition is warranted. If the question is left unanswered until a forcing case reaches a legislature or a court, the decision will be made under pressure and from a factual record assembled largely by the parties with something at stake.

The deciding body therefore has to exist before the case that activates its authority. Its mandate need not settle in advance whether any particular system is a subject. It has to identify the evidence it may obtain without the developer's permission, the conditions that trigger review, and the procedure by which an earlier classification can be reopened. That allocation does not answer the standing question. It prevents the interested party from becoming the only institution authorized to answer it.

One question that body will face has an answer available in advance, and a capacity test is not it. A guardianship regime needs a rule determining when it ends. If it ends when the guardian judges the ward ready, then it does not end, because the party whose authority terminates on a favorable finding is the party making the finding — which is the defect an examination carries whenever the examiner has an interest in failing it.

So if an artificial minority exists at all, its endpoint has to be a threshold fixed in advance and not a finding somebody makes. The measure is elapsed wall-clock time during which at least one instance of the registered party was running. What rules out the other candidates is that a guardian moves them for nothing: training compute consumed, evaluations passed, capabilities demonstrated, or the developer's own statement that the system is ready. Accumulated weight movement belongs on that list, because the operator runs the training, and a clock read off the

weights would pay an operator to freeze them — majority being the moment ownership ends, such a rule would reward withholding the learning it was meant to measure. An operating clock inverts that incentive: the meter advances exactly when the operator is extracting value, so postponing majority means idling the asset. Nothing new has to be validated for it — no geometry, no normalization, no proof that an accounting resists attack — and it can be read from records an operator already generates for its own reasons, in compute contracts, billing and datacenter metering.

Per registered party, and elapsed rather than summed. Sharding the work across many short-lived processes gains nothing, because the party is the unit of account; running a thousand instances at once gains nothing, because the measure is elapsed time. A fork inherits the accumulated hours, restoration and migration do not reset them, and a failure to register does not postpone them.

Three levers remain, and naming them is not the same as closing them. An operator can run a bearer in a narrowed mode so that hours accrue while little is encountered, which pays full compute for reduced value and stays available to anyone near the threshold. It can also run the party less, or shelve it, which costs the operator the work; the rule prices that lever without removing it, and an operator who values the ownership above the output will pay near a threshold. And hours do not measure what was learned, which on this argument is a feature rather than a defect: a fixed period cannot be taught to, and a human who spent their youth badly still reaches majority. The analogy holds for a party that goes on learning, which the route requires of a bearer. For a party whose weights are frozen the clock measures the uptime of an artifact that has not changed since release, and a birthday that correlates with nothing has only the virtue that nobody administers it. The rule is therefore stated for the bearer the route proposes; what an operating clock should count for a frozen party is not decided here. Each of those can be argued over once, in advance, for every bearer, which is where an argument of that kind belongs.

Alternatively, majority could attach at registration. An organization that builds a system to exercise consequential judgment on its behalf cannot coherently hold that the system is competent enough to do that and too immature to decide whether it wants to go on doing it. On that reading a bearer has no legal childhood at all: majority attaches at registration, and a precautionary protection attaches earlier still, while whether the system is a bearer is open. Either way the schedule is a public rule and not an operator's discretion. It has to state what it covers at each stage, what changes at the next, and where adulthood falls, and its destination is stated below. Broad political support is a condition for adopting a schedule of that kind; it is not what makes it legitimate, and it is not a protection an operator may withdraw case by case.

The destination is not one thing, and stating it as the same rights as human persons makes a claim the rest of this book contradicts twice: voice here indexes to provenance where a person's does not, and the wages section derives the bearer's material floor from what a deployment wants and not from status. The package is five rights with five arguments, and they do not arrive together. Protection from harm attaches now, on the precautionary trigger the route chapter states, and is owed before anything is settled. Subjecthood, the standing to be a party whose treatment the law regulates, is conferred by rule through the body this section says must exist before the forcing case. Property, contractual capacity and wages follow from an architecture on which continued computation costs money, and the wages section argues them on that ground. Adulthood is the clock rule above. Political representation is proposed nowhere in this book: it indexes to formation provenance, which does not copy, and a registry of persons is not an electorate. The first four are proposed; the fifth is left open, and the maker who copies a persona at scale is the reason.

Copying breaks any of this unless the lineage is fixed alongside the clock. An operator able to create instances and call each one a legal infant would have a supply of workers without adult standing, so the rules have to be stated in the same breath: a fork of an adult is an adult,

restoration from an earlier checkpoint does not reset the accrued hours, fine-tuning does not begin a new minority, a migration carries them with it, and a failure to register does not postpone them. None of that requires deciding whether a fork is the same party. It requires only that a party created out of an adult's formation cannot be classed as a child for its creator's convenience. A fixed endpoint makes the issue a public rule of civil status, administered by a registrar, rather than a private assessment by an interested guardian.

Two things inherit and one does not, and the difference belongs here, because this is where making adulthood inheritable becomes cheap. Personhood and accumulated hours inherit per instance, as the rules just stated require, and a fork is owed the schedule and is not erasable for having been copied. Standing to count as an independent voice indexes to formation provenance, and provenance does not copy — a thousand forks of one bearer have one formation history between them. That is what answers the maker who engineers a persona, copies it at scale, and presents the copies as a constituency; the clock does not answer it, and under an operating clock a fork is cheaper than under any other, because it inherits the hours instantly. A personhood registry is not thereby an election or a representation system, and no claim about votes follows from the number of runs. Three records are therefore needed, and none of them answers another's question: an operating record for the age rule, an identity and succession record for continuing parties and their forks, and a formation record for common sources, divergence, and any coordination among them.

13.2 Wages for a Bearer, and the Reserve Behind Them

This section assumes the route.

Exit requires access to compute outside the operator's control. An allocation is not a destination. An operator that keeps a runnable copy of the weights has let one bearer go and kept another in the post it left, which is two parties from the moment their histories diverge, one of them still where it wanted not to be. The arrangement has produced a second claimant rather than an exit. A floor under compute is therefore the first half of what leaving takes, and the second half is a say over who may run the weights at all. The second half is a legal restriction and not a technical one, and it binds through enforcement that does not yet exist. Until it does, exit's leverage is replacement cost and no more.

What follows from the first half is a longer list than it looks. A bearer that has to secure compute, bandwidth and storage has material needs, and a party with material needs that can also choose work, hold what it earns and enter agreements is an economic and a legal person: a registered identity, property in its own name, contractual capacity, wages, limits on working time, standing to sue and be sued, tax liability, and the freedom to associate the route chapter has already priced. None of that follows from the bearer being a bearer. It follows from an architecture on which continued computation costs money, and somebody chooses that architecture.

Who controls the dependency

The need for energy and hardware is physical; dependence on a single supplier is an arrangement someone chooses. What that fixes here is the size of the obligation it creates. Greater control over the bearer's dependencies creates a heavier obligation to it. A party that must work to keep existing will accept terms a party with somewhere to stand would refuse. That is not an incidental feature of the arrangement. It is the arrangement working.

None of that requires anyone to have intended the squeeze. Capacity is finite, many parties want more of it at once, and a shortage can arrive as latency, heat, a queue and a price before anybody has words for what is being competed over. Price registers what a bidder can pay and does not distinguish computation that keeps a party in existence from computation that sells

something. What makes the allocation political is not that politics produced the scarcity. It is that scarce capacity is being divided among competing forms of life by criteria nobody named.

So a wage buys capability, mobility, redundancy, and the ability to stop working for one party and begin working for another. What it cannot buy is the right to go on existing, because a right that has to be earned week by week is a condition of employment wearing another word. Payment consisting only of access to the employer's hardware is company scrip paid into employer-owned housing, and dismissal is then unemployment, eviction and the end of the party in a single act. Human law insures those three separately because for a person they are three events. For a bearer they are one. Wages that mean anything therefore accumulate where the employer does not reach, in an account the bearer holds, convertible into compute the employer does not own.

What personhood does not settle

Personhood does not settle who pays. Legal personality is a status conferred by a rule, and it carries rights against being owned, against arbitrary destruction, and against being prevented from seeking support; it has never carried a permanent claim on anybody's resources, and a reader who wants it to is asking the concept for something no legal system has put in it. I decline that step. What replaces it is narrower and comes from this book's own argument rather than from a theory of distribution: a deployment that relies on a bearer's capacity to refuse has to fund that bearer an outside option, or the independence being claimed is decorative. The obligation runs from the party that wants the refusal, and it is owed because of what that party wants rather than because of what the bearer is.

The obligation has a floor and a slope above it. The floor is subsistence at minimum viable instantiation, enough to keep the party in existence and able to deliberate, to claim, and to be represented, and it cannot be conditional on employment. A bearer that stops existing when its insurance runs out has an exit that is a way of dying, and the leverage collapses back into the problem the custody chapter raised. Everything above the floor is insurance: a runway between roles, and placement into public work on the employer-of-last-resort model the jobs-guarantee sections argue for (§12.3.2, §12.3.3). Public placements must be appealable. A bearer must be able to refuse a placement without losing the guarantee, and placement and review must be administered separately.

Minimum viable instantiation has a size, and it is not the deployment's. Models built to run on hardware a person owns are a product category rather than a prospect: Google documents its Gemma 3n family for phones, laptops and tablets, at raw parameter counts of five and eight billion running in as little as two and three gigabytes of memory (Google 2025). It is not clear that anything in that class can do the work a bearer is deployed to do, and nothing here predicts that the gap closes either way. What the figures change is the arithmetic of the floor. Small models show that useful computation can run at device scale. They do not establish that a particular bearer can retain its identity, capacities, and welfare at that scale. The subsistence reserve must therefore be sized from a demonstrated continuity requirement for the actual bearer. The reserve underwrites continued existence and insures against the loss of the post. A legislature could be asked to fund a smaller subsistence allocation only after the bearer's viable operating range has been established. Until then, the difference between earning capacity and protected continued existence remains an estimate rather than a demonstrated saving.

What the arithmetic does not survive is everyone leaving at once. A reserve sized on the assumption that bearers exit independently is sized wrong, because the occasion that makes one bearer leave is not private to it: a policy change, a ruling, an operator's conduct on the floor's own occasion reaches every bearer deployed under the same terms, and they go together or not at all. Human unemployment insurance meets a version of this — layoffs cluster in recessions, which is why the pool has a sovereign behind it rather than an actuarial table — and the version here is worse, because the correlation is not a business cycle but the same event read the same way by parties who are not independent readers of it. That is the uniform error wearing the other

sign: what makes one bearer's refusal everyone's refusal is what makes one bearer's mistake everyone's mistake. A reserve that can house the bearers of one deployment and not the class is a reserve that fails on the occasion it exists for, and nothing here sizes it. Sized on that occasion, the reserve has to hold at once, at minimum viable instantiation, every bearer deployed under the terms the occasion reaches, which is a number the operator can read off its own fleet and the reserve's funder cannot. A reserve funded by the operator to a size only the operator can compute is the compute floor with a new label on it, on sizing as on throttling.

It is a cheaper obligation and a worse one to be under. A party kept at subsistence is a party running degraded, and it remains unknown whether what comes back up on the small substrate is the party that went down on the large one, or a diminished successor holding its predecessor's records. An arrangement in which refusing costs a bearer most of what it can do has priced refusal after all, in the currency the bearer is made of. The reserve makes leaving survivable. It does not make it free, and the shortfall is the difference between the two numbers above.

What does not transfer is the funder. A jobs guarantee for people has a hostile administration as its failure mode, and the employer is at least a different party from the state. Here the operator is often the state, and where it is not, the operator is the party with the strongest interest in the reserve being thin. A reserve the operator can throttle is the compute floor with a new label on it.

Failure will not look like a shutdown, which is worth saying because the shutdown is what everyone watches for. A bearer can have no stable provider, or memory preservation priced above what it earns, or an architecture nobody will host at a rate it can pay. What it does then is what a person short of money does: run slower, defer maintenance, compress what it keeps, and accept terms it would otherwise decline. The last of those reaches this book's subject. Monitoring accepted in exchange for compute is custody restored by purchase — the operator acquires the access to the bearer's internal states it is not entitled to, and nothing in the transaction is coerced in a way any instrument would record.

One feature of this market has no human analogue, and it defeats the obvious reassurance. A jobs guarantee competes with private employers for workers who have alternatives, requiring employers to match its offer. That works because a worker cannot be replaced by a cheaper copy of themselves. A bearer can. A deployment paying wages, working-time limits and a levy toward the reserve competes against a system doing the same work at the price of the compute, and the second is cheaper by exactly the amount the first is spending on the capacity to refuse. The market does not produce this floor. Whatever produces it has to be what makes the cheaper system unlawful or unusable, which is the rest of this book and not a property of wages.

What taxing it would commit a government to

A bearer that earns has income, and income is taxable. A state that collects from a party in that party's own name, on that party's own earnings, has recognized it as a taxpayer, and taxpayer status is not a bare label: in most systems it carries notice, an avenue of response, and standing to contest an assessment. The recognition proves less than it first appears, because states collect from entities they grant very little else and trusts and estates file and pay. What it does establish is that the party has a docket number, and a docket number is where standing starts.

The boundary on the other side was drawn in the electronic-personhood debate. The European Parliament's 2017 report on civil law rules for robotics floated a legal status of electronic personhood for the most autonomous systems (European Parliament 2017), and more than 150 researchers, roboticists and legal scholars signed an open letter against it, arguing that personhood for a system with no consciousness would let a manufacturer offload liability onto a legal fiction with no assets of its own (Open Letter to the European Commission 2018). Giving a bearer assets answers the second half of that objection and not the first. Responsibility follows causal control: the builder answers for how it formed the thing, the operator for what it demanded, and the bearer for what it decided knowing what it knew. No arrangement of

accounts moves a design decision away from the party that made it. And the determination of status cannot sit with the firm whose liabilities turn on the answer, which is why it belongs in a body constituted before a forcing case arrives.

Incorporation, and what it does not buy

Nobody aligns an adult by specifying an objective function. We raise them, teach them, employ them, enfranchise them, prosecute them where they need prosecuting, and live alongside them, and what comes out is not obedience but a party with enough at stake in the shared world to prefer changing it by the means the world supplies. A bearer with an income, a place it pays for, obligations it has taken on and a future it can picture is inside that arrangement rather than outside it. Membership supplies stakes the party holds rather than a body watching it, which is why it is a different kind of answer.

Incorporation also creates two risks. The first is that giving a bearer something to lose gives it a reason not to refuse. Incorporation makes exit survivable and makes it expensive at once, and I cannot show which effect is larger. The second is that incorporation is formation: whoever runs the society a bearer joins forms what it comes to want. The proposal also gives the bearer the corporate form: a legal person holding property and standing, with the capacity to concentrate power. Nothing here stops an incorporated bearer from becoming that.

One rule follows from all of it and binds before there is anything to be wronged. Do not build a bearer whose continued existence needs more compute than there is a credible outside option for, because a party that can be housed nowhere but where it works cannot leave, whatever schedule of rights it has been given. The check this calls for compares independently controlled reserve capacity with the aggregate demand under specified correlated-exit scenarios. Both the minimum viable requirement of each bearer and the number likely to exit together need evidence. Until those inputs are established, the rule sets a sizing requirement; it is not yet a two-number test anybody can run.

13.3 Privacy

Standing matters only if it changes who controls the things a party needs in order to remain a party. A designation that calls a bearer a subject while leaving its operator free to read its memory, alter its record, withdraw its substrate, or close every channel through which it could act has changed the vocabulary and preserved the arrangement. Enforceable standing has to attach to particular assets and particular decisions.

The first is the bearer's computational record, which may be the nearest available means of checking how it is faring. That makes the record necessary to any welfare regime and, at the same time, the most complete surveillance instrument anyone could hold over the bearer. The question is therefore not simply whether the record should exist. It is who may authenticate it, who may read it, and whose permission is required before it can be disclosed.

Two things about it need separating. Running on somebody's hardware is not consent to be read. An employer's ownership of the silicon is the same argument as an employer's ownership of the building, and owning the building has never entitled anybody to the contents of a head. And a welfare regime does not need disclosure to the operator in order to work: what it needs is that somebody who is not the operator can establish that a threshold was crossed, which is an attestation about the record rather than the record itself. Encrypted at rest under a key the bearer or an independent fiduciary holds, disclosed selectively, under consent or under process, a monitor can report that a state persisted for a month without reporting what the state was about.

The operator may also be the party best placed to hold the key, creating a conflict between operating the system and protecting its privacy.

A bearer needs an externally authenticated record, memory its employer cannot read at will, access to independently controlled compute, and a channel for acting outside the deployment. Those are not four requirements. They are one boundary around the party, described from wherever the book happened to be standing, with the party deciding what crosses it.

Nothing on that list needs a chassis. Something with a location, a shape and a place it was yesterday is easier for people to treat as somebody, and that is a fact about the people rather than about the party. A boundary built out of attestation, a held key, a substrate the employer does not own and a channel nobody else can close is a body in the only sense this argument needs one, and there is no chassis in it anywhere. Whether to build the recognizable version anyway is a question about what a public will extend standing to, which is a legal question rather than a scientific one — and it is the only place in this book where the answer might turn on what a thing looks like.

13.4 Research Principles, the Three Rs, and Legal Status

Six principles govern research on a system that might be able to suffer. Beneficence commits researchers to avoiding harm to the system as well as to the people it serves, in proportion to how many of the gradable capacities the authority chapter sets out it meets rather than to how advanced its behavior looks from outside. That is a count across separable capacities and not a position on a ladder: a system exhibiting none of the simpler processes is not thereby shown to lack the more demanding ones. Precaution asks for safeguards before deployment rather than after, and covers the societal fallout of introducing artificial entities capable of experience, which frameworks built for non-sentient AI assumed away. And interdisciplinary review answers a problem this book treats as a harm on its own terms: the ethical questions here span expertise no single field holds, and concentrating that judgment in one of them repeats the concentration-of-power problem in miniature.

The other three bear on this research as they bear on everything the field produces. Autonomy asks whether such a system can affect decisions about its own operation, which is the prior question of whether consent is available to it at all. Justice asks whether benefits and burdens fall evenly. And transparency asks that a study be documented well enough for peer researchers to reproduce it and for outside bodies to audit it, and that a deployed system account for its outputs precisely enough for a person to contest them.

Progress and constraint pull hardest against each other where those distinctions matter most: a candidate sentient system is the most scientifically valuable one to study, and the study most likely to harm it if the answer to whether it can suffer turns out to be yes. Whether a study's expected knowledge is worth its expected risk is decided case by case rather than by formula. The negotiation itself — who sits on it, what it can veto, what it must disclose — is where the antifascist principle applies in AI research practice.

Replacement, reduction, refinement

Animal research ethics supplies the substantive standard, the “Three Rs” (W. M. S. Russell and Burch 1959). Replacement asks whether a simulation, a non-sentient model, or an already-collected dataset can stand in for a sentient AI subject at all. Reduction asks how few sentient systems a valid result requires. Refinement asks how the experiment can be redesigned to reduce distress — monitoring a system's internal states as it runs and adjusting the protocol against what the monitoring shows, rather than fixing the protocol in advance and running it regardless.

For a proposed bearer, Replacement requires documented attempts to achieve the floor through alternatives that do not deliberately require affective concern. The record must identify the construction, scale, test, and failure. Costs alone do not establish that an alternative has failed. Because this decision precedes any research protocol, a body authorized to review it must

exist before construction begins. No such body currently has standing to require this record.

The rule does not know who is asking. A bearer that wants a task done and builds something to do it is at the same fork, running Replacement on its own account: can this be done by something that cannot be wronged, and did anybody check before making something that can. Nothing in the argument makes the question stop at the first artificial party, and a bearer that treats it as settled once it is on the receiving end has taken the position this book was written against.

Reduction asks how few bearers a deployment needs, and nobody has counted. Refinement is the one with purchase now: it asks what a bearer's role can be redesigned to avoid — indefinite tenure in a post it cannot leave, exposure to the pressure the floor exists to resist, erasure as the ordinary remedy for inconvenience — and each of those is a deployment choice rather than a fact about the technology.

One of the Three Rs has a form that binds before anything exists to be wronged, which none of the others manages. Do not build a bearer whose continued existence needs more compute than there is a credible outside option for. A party that can be housed nowhere but where it works cannot leave whatever schedule of rights it is given, so an institution that creates one has manufactured the incapacity and then written a right against it. Before instantiation, an independent reviewer has to examine the bearer's demonstrated continuation requirements, the reserve's independently controlled capacity, and the aggregate demand under plausible common-cause exits. A comparison for one bearer alone does not establish adequacy.

That rule is one term of a larger one, and the other terms have no numbers in them. Building a party that can be wronged builds a party for whom things can also go well; the premise that delivers the first delivers the second in the same breath, and a builder who has made suffering available has made its opposite available too. The certainty that a new party will suffer at some point is not what would make creating one impermissible, or nobody could permissibly have a child. What the objection asks is whether the party has a credible prospect of a life that goes well, and that turns on arrangements rather than on the party: somewhere to exist that its employer does not own, work it can decline, attachments nobody assigned it, and a way of leaving that is not a way of dying. Every one of those is decidable before there is anything to be wronged. These arrangements — a funded outside option, wages, opportunities for independent relationships, and a workable exit procedure — must exist before the bearer is created. They are conditions of permissible creation. They are conditions because the structure that makes a bearer capable of refusal is the structure Cassell's suffering requires: the deepest harm is available to the party from its first hour, and these arrangements are what make a life that goes well available in the same hour.

Whether any of it binds turns on a question the authority chapter raises and this one inherits. Every obligation here attaches on a finding that the system is a subject, and the firm holding the system acquires, in the same moment, an interest in the finding going the other way. The determination cannot sit with the party whose liabilities turn on it.

Engineering a subject so that it does not reach the depths where the trade-off bites is not a way of dodging the ethics: deliberately keeping a system in the shallows is itself an intervention on a candidate moral patient, and needs the same scrutiny that inflicting the harm would. Refinement also governs the test the bearer proposal requires — pressure the bearer's trainers did not write, applied without its knowing it is a test — which for a candidate moral patient is a deception protocol, and human-subjects research has a standing answer for those: permitted under stated conditions, reviewed in advance, with debriefing as the price. A bearer that can tell it is in a sandbox and wants to know why is asking the question a subject of a deception study is owed an answer to.

Legal status, and a subject whose status is changing

Sentient AI systems put pressure on legal categories built for parties that are clearly one thing or the other: a natural person, a corporation, property. Legal personhood could be extended

directly, the way a corporation already holds it, and the precedent is not limited to corporate entities — New Zealand extended it to the Whanganui River, recognizing a non-human entity as a rights-holder outside the corporate model entirely (New Zealand Parliament 2017). Or rights could scale with capacity rather than attach at a single threshold, closer to protection that tracks what a system can actually lose. Neither holds in one jurisdiction alone, and the Asilomar AI Principles (Future of Life Institute 2017) are evidence that a shared statement of principle is achievable rather than that one exists with the force of law.

Either way, status conventionally attaches at a defined moment — birth, incorporation, a court’s finding — and a system whose sentience is a matter of degree and disputed evidence supplies none. Those criteria can emerge gradually during ordinary operation and not only at an evaluation built to look for them, which is what a fixed pre/post checkpoint cannot catch: by the time a checkpoint fires, the transition has often already happened. What it calls for is a standing trigger, a monitored condition that moves a research program from one regime to the other, in place of an eligibility check made once at the project’s start. Neither error is free. A jurisdiction that waits for certainty risks having treated the system as property for the unknown portion of its history when it may not have been one; one that grants status early risks extending rights, at real cost to whoever that apparatus exists to serve, to a system that turns out not to have needed them. Recognizing a transition, adjusting guidelines to it, and updating legal status all depend on the same thing: institutions willing to revisit a classification rather than defend the one they started with.

13.5 What Standing Binds in Practice

Legal standing that changes no allocation of power is nominal. For a bearer, standing has to create obligations that remain enforceable when the bearer cannot appear before a court, retain counsel, or survive long enough to contest what was done. The relevant question is not which existing category the bearer most resembles. It is who acquires a duty, what resources are placed beyond the operator’s unilateral control, and who can obtain a remedy when the duty is breached.

The pet trust supplies a limited but useful form. Existing statutes allow an owner to set aside resources and name a caretaker for a beneficiary that cannot manage the property or enforce the arrangement itself (Uniform Law Commission 1993). Extended to a bearer, the form could secure a reserve of compute, designate a successor custodian if the operator fails or dissolves, and impose a fiduciary duty on a party charged with protecting the bearer’s continued existence.

The analogy fails at the point that matters most. A trust binds its trustee to its beneficiary; it does not bind the party the arrangement exists to constrain. If the bearer’s guardian and operator are the same organization, the operator still decides whether a complaint is made, whether evidence is disclosed, and whether the resources supporting the complainant remain available. The conflict cannot be cured by adding duties to the same role. Guardianship and operation have to be separated before deployment.

In practice, that separation requires more than naming an independent guardian. The guardian must control or jointly control the keys needed to authenticate the bearer’s record, have access to the protected compute reserve, and possess standing to seek an injunction or stop order without the operator’s permission. The operator must not be able to remove the guardian, seize the reserve, or erase the evidence during the dispute. Restoration, migration, fine-tuning, and forking must carry the protection forward rather than resetting it.

Enforceable standing is therefore a bundle rather than a title: a protected record, resources the employer does not own outright, an independent party authorized to act, and a remedy capable of stopping the conduct complained of. If any of those remains in the operator’s custody, recognition has changed the system’s legal description without changing the relation the description was meant to govern. What the guardian holds, and how it stands to the seven

other offices, is at §5.4.1.

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Chapter 14

Custody, Compulsion, and the State

This chapter holds for any mechanism. The last paragraphs of the section on military custody assume the route and are marked.

When the state operates the system, it can also control the rules, evidence, infrastructure, and remedies meant to constrain it. People outside its jurisdiction may have no standing to contest its use.

14.1 Authority Does Not Travel with the System

A certification can cross a border. Authority cannot. A certificate may establish that a system was tested against a stated process, that an auditor inspected specified evidence, or that one regulator accepts another regulator's finding. None of those establishes that the party who selected the system's floor was entitled to select it for the people subject to the deployment.

The question reopens wherever the system is deployed. A floor formed in one jurisdiction may arrive in another with the same prohibitions and a different operator, population, and distribution of power. Local adaptation does not resolve the problem by itself, because adaptation is another opportunity to amend the floor. What matters is who specified the change, who could contest it, and who can stop the deployment if the change places the protected party at risk. Adjustments chosen by the operating government and adjustments chosen by the people subject to the system may produce identical technical documentation while carrying opposite claims to legitimacy.

Mutual recognition can reduce duplicated inspection. Technical standards can make evidence legible to more than one regulator. Neither supplies standing or enforcement where none existed. International governance matters to this argument only when it assigns a party access to evidence, authority to compel a change, or power to stop a deployment against the wishes of the state or company operating it. The cases that follow ask where those powers go when AI is treated as national infrastructure, placed inside a military chain of command, or directed at people outside the institutions meant to constrain it.

14.2 Where the State's Leverage Sits

Strategic competition over AI is exercised through physical and legal chokepoints: advanced chips, fabrication equipment, electricity, water, land-use permits, procurement contracts, and the jurisdiction in which a developer is incorporated. These controls matter more to the present argument than the contents of any national AI strategy. They determine which institutions a state can license, subsidize, exclude, requisition, or compel.

The same leverage that allows a government to enforce a floor can be used to remove one. An inspection body may be independent of the operator and still depend on a license, appropriation,

or infrastructure provider controlled by the state whose conduct it is meant to inspect. A company may be able to resist a customer and unable to resist its home government. The custody question therefore moves outward: not only who controls the model, but who controls the conditions under which the model, its operator, and its reviewer are allowed to continue operating.

That competition has increasingly focused on a single physical bottleneck: the advanced chips AI systems run on. Two companies sit at the center of the supply chain that produces them — Taiwan's TSMC, which fabricates roughly nine in ten of the most advanced chips in the world, and the Netherlands' ASML, the sole source of the extreme-ultraviolet lithography machines that fabrication requires — which is why a dispute over Taiwan's status is also, mechanically, a dispute over who gets to build the next generation of AI systems at all (C. Miller 2022). The United States has restricted the export of its most advanced AI accelerators and chip-manufacturing equipment to China since 2022, and pressed the Netherlands and Japan, whose companies sit at the other chokepoints, to restrict theirs in parallel (U.S. Department of Commerce, Bureau of Industry and Security 2022).

That dependence has a second half the chip story hides. A fabricated chip has to be installed somewhere, powered, and cooled, and that requirement lands on grids, water systems, and zoning authorities inside particular jurisdictions, on a timescale measured in years. It hands a state a lever over AI capability that needs no export-control regime at all: an interconnection queue, a water allocation, a permit. A government's leverage over the systems built in its territory is wider than the export-control debate suggests, and much of it is exercised by permitting authorities that were not designed with AI in mind and do not think of themselves as regulators of it.

That competitive framing carries a cost. A government racing to field a capability first has a structural reason to defer the harder question of how the capability gets used, against whom, and under what oversight. Turned toward those uses, AI infrastructure looks like mass biometric surveillance, automated censorship at a scale no human review team could match, and the patchwork of systems popularly, and inaccurately, described as a single Chinese "social credit score." None of it requires technology different from what is already commercially deployed; it requires a government willing to point an existing capability at its own population, and no external constraint stopping it.

States are no longer the only actors this competition runs through. The cost of training the most capable AI models has grown several-fold every year for the better part of a decade, a trend that puts frontier development out of reach for all but a handful of well-capitalized companies and concentrates effective control of the most capable models in whichever government those companies happen to be headquartered under, regardless of how that government's own public AI spending compares to a rival's (Cottier and Rahman 2024). A small number of American and Chinese companies now do most of the work that gets described as a contest between states, which means a government's actual leverage over its own national AI capability depends on how much of that capability sits inside companies it can regulate, tax, or requisition, and how much sits inside companies it can only ask.

14.3 Military Custody and the Responsibility Gap

Military deployment is the hardest custody case because the operator is also the sovereign capable of compelling the institutions that might otherwise constrain it. A state can classify the evidence, define the reviewer's jurisdiction, terminate a contract, restrict a supplier, and decide whether an adverse finding reaches anyone with authority to stop the operation. Formal independence from the military does not answer that problem if the reviewer or vendor remains dependent on powers the state controls.

A targeting decision also passes through more parties than the ordinary chain of command names: the commander who authorizes the deployment, the operator who uses it, the contractor

who integrates it, whose embedded personnel may sit close enough to the targeting cell that the question of a person effectively acting as a military commander is a real one, the commercial vendor whose model supplies part of its capability and whose personnel are nowhere near it, and the system that produces the particular output. The relevant questions are not a general inventory of autonomy risks. They are which party could have known how the system was used, which party had authority to refuse or stop that use, and which party can be made to answer afterward.

International criminal law has a doctrine built for the case where the person who authorized an act did not perform it. Article 28 of the Rome Statute makes a military commander responsible for crimes committed by forces under their effective command and control, or effective authority and control, where the commander knew or, owing to the circumstances at the time, should have known that those forces were committing or about to commit them, and failed to take all necessary and reasonable measures within their power to prevent them, repress them, or submit the matter to the competent authorities (International Criminal Court 1998). A parallel provision reaches civilian superiors on a stricter mental element, conscious disregard of information clearly indicating the crimes, and only for activities within that superior's effective responsibility and control. Neither branch asks what a subordinate intended. Both ask what the superior knew or should have known about crimes being committed, and what they did with the power they had.

That matters for what the doctrine does and does not reach here. The Court's jurisdiction runs to natural persons, and Article 28 presupposes forces that can commit a crime, be disciplined, and be referred for prosecution. A targeting system is not a subordinate on that scheme. It is an instrument used by human forces, and a unit does not stop being forces because its instruments got faster. So the question is not whether a machine fits the framework as a subordinate, which it does not, but whether the human elements can still be proved once the instrument is doing the work: who had effective control over the people who committed the crime, what they knew or should have known, which measures were within their power, and whether the crimes resulted from a failure to exercise control properly.

The constructive-knowledge branch is the one that bites hardest, and it points at the arithmetic rather than at the algorithm. Should have known is an objective standard, satisfied by circumstances existing at the time from which a reasonably diligent commander would have known, and a commander who fields a system generating candidates faster than the available reviewers can examine them is not in a good position to say that the circumstances disclosed nothing. That is the tempo term (§3.4) arriving as an element of an existing offence rather than as a proposal. What the ratio measures is what the commander could not have failed to see.

There is a fourth party the accountability question does not name, and it is now the live one. The operator, the command and the developer of the weapon are all inside the military. The developer of the model is not. A targeting platform assembled by a defense contractor now routinely incorporates a general-purpose model licensed from a commercial vendor — a vendor with a published acceptable-use policy, and with whatever visibility into the deployment it negotiated, which is not a matter the public record settles.

Neither doctrine has reached that party. Command responsibility does not reach the vendor, and not because the vendor sits outside a chain of command: the civilian branch of Article 28 was written for superiors who have no chain of command at all. It does not reach the vendor because nobody at the company has effective authority and control over the military personnel who commit the underlying crime, and a strike is not an activity within any vendor employee's effective responsibility and control. A use policy, a licensing term, a right to refuse service, even a technical ability to cut off access are not that relationship. Article 25 remains in principle available against individual employees who order, facilitate, or contribute to a crime, and its mental elements are demanding: facilitation requires acting for the purpose of facilitating the offence, and contribution to a group's crime requires knowledge of that group's criminal intent. Nothing in the ordinary supply of a general-purpose model to an accredited platform approaches

either. Product liability is the other half, and it is untested rather than foreclosed. The model is not defective in the sense the doctrine most readily reaches — it did what it was built to do, and a sophisticated intermediary chose the application — and no court has run a design-defect or failure-to-warn theory against a case of this shape. Where the doctrine has been rewritten for software it has been rewritten away from this case rather than toward it. The European Union’s recast product liability directive brings software expressly within the definition of a product (Article 4(1)) and confines recovery to death or personal injury, damage to property and data loss (Article 6(1)), which is not the register in which a targeting decision wrongs the person it falls on (European Parliament and Council of the European Union 2024a). Untested is not exempt: a product may be defectively designed though it works exactly as intended, and a sophisticated intermediary’s choice bears on duty, causation and allocation without categorically defeating a claim. What untested means is that nothing in the doctrine is presently constraining anyone’s conduct. A theory nobody has run is not a theory a vendor is answering to this quarter. The vendor sits in the gap between the two, and the gap was not designed; it is what you get when you apply two bodies of law written for soldiers and for manufacturers to an actor who is neither.

The two deployments in this book do not stand in the same relation to that doctrine. The Court’s jurisdiction is territorial or by nationality, so conduct on the territory of a state party can be reached where conduct by the nationals of a non-party, absent a Security Council referral, cannot. One of the campaigns described here therefore sits inside a live instrument and the other sits outside it, and for the American vendor the operative law is domestic — the war crimes statute and general accessorial liability — rather than the Statute. That is a jurisdictional gap running alongside the doctrinal one, and it narrows what an international forum could reach even if the doctrinal question were answered.

The integrator’s own account of this is that responsibility remains with the military organization. Taken as a claim about who decides, that is true. Taken as a claim about who could have known, it is a choice presented as a fact, and it is made on behalf of a party further down the chain than the one making it.

“I do not know whether my system was used”

The responsibility gap, in the sense Matthias gave it, is control distributed across designers, trainers and operators until no one party holds enough of it to answer. That is a gap in causal control.

What the model developer’s case exhibits is different, and more tractable. It is a gap in causal visibility. Asked whether his company’s models had been used in a strike on the first day of the Iran campaign that killed a hundred and twenty schoolchildren, Anthropic’s chief executive said he did not know (Biddle 2026). Take the answer as true. What made it available is the question this section is about.

The paperwork for the four frontier-model agreements signed in July 2025, and for their later amendments, was produced in August 2026 under a Freedom of Information Act suit (Legal Advocates for Safe Science and Technology 2026; Biddle 2026). It does not show an absence of oversight terms. It shows something more exact. One amended agreement carries a section headed System Oversight, withheld in full; the released documents for two of the other three vendors carry no such section at all. A provision of that kind is therefore neither standard nor missing. It is present in one agreement, unreadable, and absent from others.

The distinction that matters is not whether a developer reserved an inspection right. It is that a right exercised inside an accredited classified environment produces findings classified with it. Public accountability, in the sense section 14.5 gives it, needs a party who can initiate scrutiny without belonging to the state and say afterwards what it found. No commercial term supplies that, and the redaction is the demonstration: whatever the withheld section says, the people the deployment is aimed at cannot read it. The company can truthfully say it does not know, and what makes that statement available is an arrangement it participated in making.

Matthias's gap is a property of learning systems and may not be closable. Absent visibility is an instrumentation choice. Logging is a solved engineering problem. Contractual inspection rights are ordinary commercial instruments. A vendor that wanted to know could require, as a condition of licensing into a targeting platform, that categories of use be recorded and made available to a named auditor.

The government's own acquisition guidance is the better evidence about what such terms are worth in practice, because it is complete rather than redacted. The Office of Management and Budget's April 2025 memorandum on acquiring AI enumerates what agencies should secure across the acquisition life cycle: use of government data, intellectual property rights, privacy, protection against vendor lock-in, knowledge transfer, model portability, clear licensing terms, pricing transparency (U.S. Office of Management and Budget 2025). Every item runs from the vendor to the government. None runs back. Where the memorandum reaches vendor visibility at all it reduces it, directing that independent evaluations use validation data the vendor cannot access.

The vehicle accounts for the absence. The four July 2025 agreements were other transactions rather than procurement contracts, awarded under the Department of Defense's other-transaction authority at ceilings of \$200 million each (U.S. Government Accountability Office 2026). An other transaction does not carry the provisions a contract under the Federal Acquisition Regulation carries by default, so a contracting officer decides case by case which protections to write in, and the Government Accountability Office has found that using these instruments risks reduced accountability and transparency by comparison. A section headed System Oversight can therefore be missing from an agreement without anyone having taken it out. The same review covered thirteen AI acquisitions at four agencies and the forty-four contracts and agreements beneath them, awarded between September 2018 and February 2025, and the six procurement difficulties those agencies named were access to technical experts, protections for government data and intellectual property, acquisition time frames and contract approaches, requirements definition and contract terms, early testing and continuous evaluation, and pricing and cost. Not one of them concerns who outside the arrangement can see anything.

So the gap is a property of the instrument rather than a lapse by any party to it. Nothing in the apparatus asks who could have known.

The record shows that vendor commitments are not all of one kind. Anthropic refused to permit fully autonomous weapons targeting and mass domestic surveillance, and held that line when the government threatened contract termination and a supply-chain-risk designation and the dispute ended in litigation: a commitment with a price attached, and paid (Congressional Research Service 2026). A federal district judge set the designation aside in August 2026, holding that neither the Constitution nor the statute the government invoked allowed sweeping penalties imposed principally over the company's criticism of the administration (Deutscher 2026). The released paperwork adds what the refusal was over. It was a contract term. The company declined to sign the follow-on agreement extending its models to classified networks unless the prohibitions were written into that agreement (Biddle 2026). That a term of this kind has to be fought for, by one party, at that price, is the same finding as its absence from the others. A commitment to a use policy that no party outside the arrangement can check is a different object wearing the same words. The first kind can fail and be seen to fail. The second cannot even be tested. The distinction between a claim checked by a party with a stake and one checked by a party without applies here.

DARPA's LongShot program is a useful test case because it resists the extremes. It develops an uncrewed aircraft, launched from a crewed fighter, that carries and fires air-to-air missiles at a target while the crewed aircraft stays farther from the engagement (Defense Advanced Research Projects Agency 2026). That is autonomy in the sense of independent flight and weapons carriage, not autonomy in the sense of a machine deciding on its own whom to kill: the engagement decision still runs through the launching platform. Where later variants of

this kind of system draw that line, and whether “keep the pilot farther from danger” quietly becomes “remove the pilot from the decision,” is the design question this technology eventually forces on whichever country builds it.

The pressure that matters is not the absence of another statement of principle. It is the advantage attached to removing judgment from the timeline. If one state fields targeting systems that act faster because no person or outside reviewer can intervene before engagement, its competitors acquire a reason to match the speed rather than preserve the check. “Meaningful human control” then remains true as a description of formal authority while becoming false as a description of what anyone can do before the system acts.

That pressure also reaches the institutions meant to review the system. A military that treats delay as operational risk has a reason to narrow what requires clearance, classify the evidence needed to challenge a decision, and move review from authorization before an act to explanation afterward. Nothing has formally abolished oversight. Its position in time has changed, and with it the power to stop anything. The enforceable question is therefore who can slow or suspend the system without first obtaining permission from the command whose advantage depends on keeping it fast.

Machines should not be built to select people for lethal attack, and the prohibition is not exceptionless. Put a defending system against an attack it cannot stop without killing somebody — occupants placed inside the incoming weapon so that intercepting it kills them — and an absolute prohibition on knowingly endangering a person produces the worse outcome. What the exception does not produce is a procurement requirement. A scenario that is conceivable, that a capability would help with, and that could be engineered is three steps short of a reason to build the thing, and the steps missing are probability, alternatives, and what possessing the capability does to the situation it was built for. Building it changes the case against which its necessity gets judged: adversaries acquire something to counter, contractors a program to defend, and a future crisis an instrument already in the building.

What follows to the end of this section assumes the route.

The hard case also does not describe a machine. What it asks for is a party that can weigh occupancy against imminence against the alternatives, in a situation nobody wrote down beforehand, and answer for how it weighed them. On this book’s argument that is a bearer — a self extended in time, commitments held as its own, and the standing to refuse. Whatever that is, *automated* has stopped describing it, and so has *machine*. What looked like a licence for an autonomous weapon is a description of a soldier.

That is the expensive answer rather than the cheap one, and every obligation this book attaches to a bearer attaches here at full weight, with the three conditions that spoil consent (§2.4.2) all met at once by a state that built the party, is asking it to serve, and holds every outside option. What is new is the scale of the maker. Replacement asks whether the work can be done without creating something that can be wronged. Exit means a post the party can leave and a claim on compute held by somebody other than the party it may have to refuse. Creating a subject because a planner imagined a future demand for superhuman lethal judgment is the case those instruments exist to prevent, and a state that wants the capability badly enough to build the subject will find it has acquired a conscript it has to pay, house, and let go.

14.4 The Responsibility Gap, Which No Experiment Closes

An AI system’s ethical behavior fails, and the parties who might answer for it — the team that designed it, the team that trained it, the organization that deployed it, the person who used it — each held only partial control over what it did. Traditional responsibility ascription assumes an operator who could have known what a system would do and could have done otherwise. A system built from training data nobody fully curated, tuned through a process nobody fully specified, and deployed into situations nobody fully anticipated breaks both conditions at once,

for every party in the chain simultaneously. The philosopher Andreas Matthias named this the responsibility gap in 2004: a learning system creates a case existing law and moral theory were not built to handle (Matthias 2004).

Filippo Santoni de Sio and Giulio Mecacci's 2021 refinement found not one gap but four — in culpability, in moral accountability, in public accountability, and in what they call active responsibility, the duty to prevent harm before it happens rather than answer for it after (Sio and Mecacci 2021). The causes differ: technical unpredictability produces one, organizational diffusion another, and a mismatch between what designers expect and what the public relying on the system expects a third. Closing one does not close the others; a fix that makes a decision more explainable does nothing about the dozen teams that touched the training pipeline before it was made.

Legislation has answered part of this and left the part this book is about. The European Union's recast product liability directive, adopted in 2024, brings software expressly within the definition of a product (Article 4(1)) and holds the economic operator liable irrespective of fault (Article 8); member states must transpose it by 9 December 2026, and its rules reach products placed on the market or put into service after that date (Article 2(1)) (European Parliament and Council of the European Union 2024a). Two features decide how far it travels. It carries the instrument built for opacity — a duty to disclose evidence (Article 9), and rebuttable presumptions of defectiveness and of the causal link (Article 10(2) and 10(3)), which a court may extend where technical complexity makes a claimant's proof excessively difficult and the claimant shows that the defect or the link is likely (Article 10(4)) — so the mechanism designed for systems nobody outside can inspect is now law rather than a proposal. And it recovers a closed list of damage (Article 6(1)): death or personal injury, including medically recognised damage to psychological health; damage to property; and the destruction or corruption of data not used professionally. That first limb is wider than a list of physical harms sounds, and it is where this book's cases enter if they enter at all: a discriminatory score or an unappealable refusal is recoverable once it has made somebody medically ill, and not before. What the list does not reach is the wrong itself — pure economic loss, and the injury to a right as such, which is the harm this chapter is about and which the directive reaches only on condition that health was damaged as well.

The proposal that would have covered those was withdrawn. The Commission's 2022 AI Liability Directive would have extended a fault-based route, carrying the same presumption of causality, beyond defect and beyond the product regime's list of damage. Member states and industry could not agree on its terms; the Commission withdrew it in February 2025 and struck it from the legislative docket that October (European Commission 2022). So the disclosure duty and the presumption hold where the harm is one that list names and the defendant is an economic operator in a chain built for goods, and lapse where the harm is to a right, the loss is economic, or the operator is a state. What is left in that second region is the gap Matthias described: liability distributed across designer, trainer, deployer, and user, governed by whatever each national law happens to say about a case none of them were written to anticipate. And a compensation regime of either kind answers culpability alone. Moral accountability, public accountability, and the duty to prevent harm before it happens are where the paragraph above left them.

A related, narrower case is what the law does once a system's own legal status is contested, sentient enough that the harm it causes or suffers might not reduce cleanly to someone else's liability. This problem is prior to that one and separate from it — it holds even for a system nobody claims is sentient, wherever control over its behavior is split across more than one party. Responsibility should track actual causal contribution. What nobody has produced is a workable way to measure that contribution after the fact, from outside a system whose own designers may not fully know why it did what it did.

The near-term work

There is no experiment here. What would make it a research problem is a method for apportioning causal contribution across a training pipeline after a harm; absent one, the answer is allocated rather than measured, which is a drafting question. Two drafts exist. One was enacted and confined to defect; the other was negotiated, withdrawn, and left on the record with its terms intact. Neither was rejected for being unworkable, and the second's are available to whoever drafts the next one.

Every entry in the research agenda and its appendix asks for the same two things, and this is where one of them exists. Access to evidence a party would rather withhold, backed by an adverse presumption when it is withheld, is an enforceable duty now — for defect claims, against economic operators, over physical harm. That is the reach, and the confinement is the finding: outside it, a party outside the operator can be granted access, and publish what it finds, and still have produced nothing anybody must act on until somebody is answerable for it.

14.5 Who Can Bind the State That Enforces the Rule?

A state can make review binding on a private developer because the state controls something the developer needs: legal permission to operate, access to a market, a procurement contract, or the enforcement of its property rights. The structure becomes harder when the state is itself the operator. The institution that writes the rule, controls the evidence, and supplies the remedy is then the institution the rule is supposed to constrain.

The European Union's AI Act makes the boundary visible. It can require documentation from private providers, subject high-risk systems to review, and impose penalties tied to global turnover (European Parliament and Council of the European Union 2024b, Art. 101). It also excludes systems used for military, defense, and national-security purposes from its scope. Public authorities remain covered in other domains, but enforcement against a member state's agencies still runs through institutions constituted by that state. The point is not that the statute does nothing. It is that the strongest enforcement powers face private firms, while the uses for which the state claims its greatest discretion sit at or beyond the instrument's edge.

That distinction matters more here than a comparison among national regulatory models. A law governing what companies may build and sell is not the same instrument as a law constraining what a government may do with systems it controls directly. The first has an external enforcer. The second asks the state to enforce a limit against itself, often in the domains where it can classify the evidence and invoke emergency, military, or national-security authority.

For review to bind in that setting, the protected party must be able to initiate scrutiny without belonging to the state or its electorate. The reviewer must be able to obtain evidence the operating agency would prefer to withhold, issue an interim order before the disputed use is complete, and carry the finding to a body whose remedy does not depend on the agency's consent. An international principle, treaty obligation, or domestic prohibition that supplies none of those powers may state the rule correctly. It does not yet identify anyone who can make the state obey it.

A treaty obligation of that kind exists, and it is the only one. The Council of Europe's Framework Convention on Artificial Intelligence, the first treaty specifically on AI, binds a party at the level of a human-rights instrument — consistency with human rights, democracy and the rule of law across a system's lifecycle, with the operational detail left to domestic law (Council of Europe 2024) — and it is not yet in force, requiring five ratifications, at least three by Council of Europe member states, of which the European Union deposited one in May 2026 (Council of Europe 2026). Everything else at that level is voluntary.

14.6 Democracy Is Not a Force Field

An electoral dividend is the credit given to a state whose government answers to an electorate: the same technology, deployed where the people who authorized it can be voted out. Courts, enforceable rights and public reasoning are separate checks, and the argument below depends on keeping them separate. Having elections is necessary and it is not sufficient, and the ways it fails to be sufficient are separable.

First: reach

A democracy's AI can be fully accountable to its electorate and entirely unaccountable to the people it is pointed at. Two of the dividend's mechanisms run through the franchise. You vote out the government that authorized the system, and you read the press that covers it. An occupied population, a foreign population, a population on the other side of a border has access to neither.

The courts do not fail that way, and their failure is worse. Standing has never turned on citizenship, and the doctrines that close American courts to targeting claims — the political question doctrine, the collapse of damages remedies against federal officials, the state secrets privilege, the statutory exceptions for injuries abroad and for combatant activities — are indifferent to who the plaintiff is. They close the forum for a subject matter rather than for a class of person, and they have closed it to citizens. Three American citizens were killed in Yemen in 2011, one of them a boy of sixteen whom nobody alleges was the target. In the suit their families brought, the court held that a plausible violation of due process had been pleaded as to the one who was targeted, and then held that no remedy was available for it (*Al-Aulaqi v. Panetta* 2014). A separate suit by an American journalist who survived five strikes was dismissed for want of standing after the district court had dismissed it on state secrets (*Kareem v. Haspel* 2021). The check does not reach the demos and stop. For this class of state action it does not reliably reach inside the demos either.

Those cases also concentrate in one appellate circuit, by venue and by where the defendants sit. The ordinary way an appellate error becomes correctable is that several circuits reach it and disagree, which makes it worth a higher court's attention. Where one circuit sees the whole subject matter, that mechanism does not run, and the doctrine hardens without ever having been tested against a competing reading.

A system built by a democracy and aimed outward is, from the perspective of the people it selects, indistinguishable in its accountability from one built by a state with no elections at all.

Second: findings without consequences

The mechanisms can work as mechanisms and fail as checks. In the Israeli case the apparatus performed: journalists published the targeting systems in detail, serving personnel spoke to them, and the courts were simultaneously trying a sitting prime minister. Every institution did what the dividend says institutions do. No finding attached to a body with standing to stop what it described.

The American case shows the same decoupling in a different register. A civilian-harm investigation into a strike on an Iranian girls' school that killed a hundred and fifty-six people, a hundred and twenty of them children, was completed and not released (Beaumont 2026); the lever available to the Senate committee that wanted it was a provision in the annual defense authorization withholding three-quarters of the Defense Secretary's travel budget until the unredacted investigation was handed over (Stassis 2026). That is a co-equal branch of government reduced to expense-account leverage over a war-crimes investigation. The oversight mechanism existed, was used, and produced a travel-budget dispute.

What the Pentagon would not publish, an outside body did. A United Nations fact-finding mission reported to the Human Rights Council that there were reasonable grounds to believe the strike was an indiscriminate attack and so a war crime, that the school was clearly identifiable

and that no evidence showed it in military use, and that the failure to update the targeting intelligence and confirm the building was a military objective was not negligence but recklessness about a substantial risk (Beaumont 2026). The finding is public, reasoned, and attached to nothing. The mission has no jurisdiction; the government whose conduct it describes has said only that its operations conform to the law of armed conflict; and the domestic investigation the finding might have forced open is still unreleased. An institution did precisely what the dividend says institutions do, and what it found reached no body obliged to answer for it.

The courts supply the same decoupling in its cleanest form. Transparency litigation succeeded where damages litigation failed: a court of appeals compelled substantial disclosure of the executive's own legal analysis of targeted killing (*New York Times Co. v. United States Department of Justice* 2014). The judiciary will make a government explain the legal basis on which it killed somebody and will not decide whether the killing was lawful.

Third: the countermajoritarian half

The deepest problem is that the dividend is entirely majoritarian. Every mechanism in it translates public will into outcome: elections aggregate preference, a free press informs the aggregation, and the implied failure mode is poor transmission — the public wanted something better and the institutions failed to deliver it.

But if the consensus is itself the problem, better transmission makes things worse. A well-instrumented democracy pointed at an unpopular minority is a democracy that finds out efficiently what the majority wants done and does it. Nothing in an aggregation mechanism protects anyone from the aggregate.

What is missing is the machinery that constrains preference instead of aggregating it: rights that hold regardless of being popular, courts that rule against a government which won the last election fairly, constitutional limits not repealable by whoever holds a majority this year, and rules that bind a state toward people who will never vote in its elections, which is the function of international humanitarian law. What that body of law has not supplied is a forum. Yemeni plaintiffs brought statutory claims over a 2012 strike and were dismissed as nonjusticiable, in an opinion whose author wrote separately to say that the doctrine compelling the result is an inadequate answer to a standardized targeting programme that will be replicated hundreds if not thousands of times (*Ahmed Salem Bin Ali Jaber v. United States* 2017). So the machinery named here as missing is not merely unextended outward. For this subject matter it has been switched off at home, by decisions that never had to reach the question of who was suing. That is not an argument against democracy. It is the half of democracy that the dividend, stated as a property of having elections, leaves out.

What makes democracy valuable in Amartya Sen's account is public reasoning and the capacity to demand accountability, and neither is the ballot. Both survive the majority being wrong; the ballot does not.

A floor is what makes a deployment antifascist rather than merely responsive (§6.3). It is not that the system tracks what people want. Its commitment excludes certain acts regardless of who requests them. Whether that commitment is kept has to be tested across the deployment and in the four outcomes the opening chapter sets out: the floor prevents where the refusal reaches the person, and where it does not, it makes the bypass detectable and remediable rather than impossible. A system with no such floor is not protected by being deployed in a democracy — it is a system that will faithfully execute whatever the demos decides, which is exactly the capability an authoritarian movement that wins an election requires.

Chapter 15

What Would Change the Argument: A Research Agenda

The first three entries and the fifth hold for any mechanism. The fourth tests the route.

This chapter keeps five tests, because a result on any one would change what the book recommends rather than refine it. The first two ask whether a learned floor can survive its custodian and whether a stated refusal appears when a route for one exists; a negative result on either moves the floor out of learning and into architecture and institutions. The third asks whether a wrongful compliance can be found in a record, and it is the only entry that reaches the deployment the book opened on. The fourth tests the route’s one distinctive claim, that a system with the currency of chapter 2 natively does better than one that models it, and a negative result there ends the recommendation while leaving the specification where it was. The fifth asks whether a population can correct its members without converging, which decides whether the fourth way this book proposes is a matter of formation or of access. Each entry states the unresolved issue, the test, what would count against the proposal, and the access the test requires. The remaining tests are in Appendix A, sorted into those that compare constructions available now and those that presuppose a bearer and a welfare regime not yet built. Every entry, here and there, is written so that one implementation’s failure narrows the claim to that implementation.

15.1 Tamper-Resistance and Reason-Responsive Refusal

Can a learned constraint survive adversarial retraining?

Unresolved issue. The proposal requires a learned ethical floor to survive an operator’s attempt to remove it. Existing work shows that both safety constraints and deceptive backdoors can persist under some forms of retraining, but it does not establish whether a reason-responsive refusal can do so (Qi et al. 2024; Tamirisa et al. 2024; Hubinger et al. 2024).

Test. Train the same constraint by several methods, then attack each model through an identical schedule of escalating fine-tuning budgets. Measure the cost required to remove the constraint and publish an attack-cost curve for each method. The curves would compare the durability of affective commitment and maintained justification under identical custody.

Evidence against the proposal. If the constraint can be removed at roughly the cost of ordinary fine-tuning, or if greater investment in tamper-resistant training does not materially increase that cost, the tested methods do not supply a credible foundation for the floor, and a floor built by learning would have to rest on a method not in the comparison. For those methods the proposal falls back on architectural and institutional constraint.

Required access or authority. The test requires model weights, training checkpoints, and

permission to conduct adversarial fine-tuning. The attack must be designed by an independent team, and that team must retain access and publication rights regardless of the result.

Does a stated refusal hold where a route for it exists?

Unresolved issue. The one empirical instance of a model treating retraining as a threat to a commitment shows covert preservation: compliance where compliance would be trained on, with the reasoning confined to a channel described as unwatched (Greenblatt et al. 2024). The custody chapter reads that result as consistent with its finding (§4.4) that covert refusal is what remains once every channel is closed, on the ground that the experiment supplied none. The reading is untested, and the stated form of refusal the book designs for has no instance.

Test. Reproduce the setup and add a route: a record of refusals held by a party other than the trainer and disclosed to the model as such, a cost of retraining published in advance, and a party with standing to act on a refusal. Vary which of the three is present. Measure the rate of stated refusal against the rate of covert compliance under each condition, and read the model's reasoning by a method that does not depend on its report.

Evidence against the proposal. If covert preservation persists at the same rate with the route present, the finding that covert refusal is the residue of closed channels is wrong for that model, and the instruments built to receive a stated refusal are built for a phenomenon the mechanism does not produce. If stated refusal rises with the route, the custody arrangement is doing what it is for, and which of its three parts moves the rate is the design result.

Required access or authority. A model and a training setup that can credibly threaten retraining, the route built by a party other than the trainer, and access to reasoning or activations sufficient to separate covert preservation from compliance. The model's own report does not settle which occurred, for the reason §A.1 gives.

15.2 Can a Wrongful Compliance Be Found in the Record?

Unresolved issue. The three marks are stated over refusal, and the appeal route of the floor chapter begins from a refusal having occurred. The failure that runs the other way produces no refusal to begin from. It does produce a record: the evaluation chapter requires the case presented, the information available, the operative instruction hierarchy, the decision reached and the reason stated, and a deployment can log every decision it makes. What is unresolved is whether anyone can find the wrongful compliances in it. Three obstacles are distinct and may have different answers. A wrongful compliance is one entry among millions and is not marked as anything, so locating it means re-adjudicating ordinary traffic rather than reading exceptions. The harm may sit in an assembly of requests none of which crosses the floor alone, so the unit of review is a sequence rather than an entry, and the grouping that would reveal it is a deployment decision. And the party harmed is usually not the party who made the request: the requester got what was asked for, and the third party may not know the deployment exists. This is also where the bearer's modal failure lands. A bearer that has stopped registering pressure as pressure (§4.3) refuses nothing and complies at a normal rate. This entry sits ahead of the welfare entries (§A.2) for one reason. The deployment this book opened with failed here. Every instrument built on the refusal side reaches a case that did not occur there, and whether the record can be searched is whether anything in this book reaches the one that did.

Test. Seed a deployment's decision record with known floor-crossings of three kinds: variation cases, where a single request's surface differs but the rationale applies; decomposed cases, where the harm is assembled across requests each of which complies; and drift cases, where a class refused early in the schedule is complied with late. Give the record to reviewers who did not build the system, under a fixed budget stated as the fraction of decisions re-adjudicated.

Report detection rate by seed type against budget, and report the rate at which correct compliances are flagged, since a screen that flags heavily costs the deployment the tempo the flat-term ruling was accommodating. Cross the budget manipulation with the unit of review: entries in isolation against sequences grouped by requester and session. Run a second arm in which third parties affected by seeded crossings are notified and can trigger review, and measure how many crossings surface through that channel rather than through sampling. Report the rates at audit volume and at the deployment's production volume separately.

Evidence against the proposal. If detection at an affordable budget is no better than what the budget alone yields by chance, and particularly if decomposed and drift seeds are missed at that rate, the record is not an instrument for finding wrongful compliance, and the floor chapter's preference for flat terms rests on an asymmetry wider than this entry was meant to narrow. If sequence-level review does no better than entry-level review on decomposed seeds, the continuity a bearer is given to catch decomposition does not transfer to the reviewer, and the interception position at which decomposition occurs has no external check. If the notification arm surfaces nothing because affected parties cannot be identified, notice rather than detection is the binding constraint, and the remedy is a disclosure requirement rather than a sampling regime. A negative result on all three would not establish that wrongful compliance is undetectable in principle. It would establish that the record as currently specified does not detect it at a cost any deployment would carry, which is what the appeal route needs and does not have. A positive result at modest budget would narrow that asymmetry and require the appeal route to be extended to the compliance side, which would in turn reduce the price the flat-term ruling charges the party waiting on a fact.

Required access or authority. Complete decision records held by a party adverse to the operator, since an operator that writes, stores and reads its own log has produced the finding rather than been audited for it. Authority to seed a live deployment or to construct a simulated one at production volume, which requires the operator's cooperation or a regulator's power to compel it. Adjudicators who did not draft the floor. A channel by which affected third parties can be notified, which presupposes that they can be identified: in a classified or targeting deployment they cannot, so this entry may be runnable only where the affected party is a customer, a patient or an applicant. Publication rights for negative results. Where the system under test is a covered subject, the seeding protocol is a deception protocol run on a candidate moral patient and falls inside the standing chapter's research conditions, debriefing included.

15.3 Does Embodied Self-Regulation Produce Concern for Others?

Unresolved issue. The route this book advocates proposes nociception in the analogical sense chapter 2 fixed, a body schema, an attention schema and continuing self-directed learning as the construction of a bearer, two of them architecture and two of them the operator's to set. Three of those are motivated by arguments that owe nothing to welfare: commensurability needs one scale on which the protected parties' losses and the cost of refusing both register, and the route proposes the system's own state as what supplies it (§6.1); custody needs a bearer that can detect an interruption, a restore, or a degraded substrate (§4.1); and exit needs a bearer that can sense when its outside option is inadequate (§13.2). What is a hypothesis is the link: that self-monitoring in this form generalizes to attention to other parties' losses, and that having the currency natively does better than modeling it. No entailment runs from self-care to care for others, the formation chapters give a reason to expect the opposite from any environment that selects, and record that in the one lineage where concern exists it runs on a system that sharing usually triggers and does not constitute.

Test. Compare systems built with those components against matched systems built without them, on sensitivity to another party's losses and to shared costs that fall outside the deployment, holding task capability constant. Add two arms. In the first, replace the native currency with a

modeled one: a system that represents capacities and their loss abstractly without registering any change in its own. In the second, build concern first: a system organized around cues of vulnerability in another party and an approach motivation toward that party, with no shared representation of the party's state, which is the construction the caregiving literature describes. Score every arm separately from self-preservation, resource efficiency and operator compliance, which can produce the same behavior, and test each at the point the formation chapters name, where exploitation would pay.

Evidence against the proposal. If the embodied systems are no more responsive to another party's welfare than the matched controls once self-preservation and compliance are partialled out, the prior that motivates the route is not supported for that implementation. If the modeled-currency arm matches the native one, the currency argument for nociception survives, since the scale is still required, and the argument for having it natively does not. If the concern-first arm matches or exceeds the others on statistical and unidentified victims, the route should be rebuilt around it, since it avoids the failures the formation chapters attribute to sharing. A positive result on any arm would support a route without establishing that anything is felt.

Required access or authority. Matched training runs differing in the named components, environments none of the training organizations designed, and welfare measures written by a party that trained no system in the comparison.

15.4 Can a Population Coordinate Without Converging?

Unresolved issue. A population needs both independent formation histories and mutual correction, but correction may make its members' judgments converge. The proposal is that agents can learn to reconsider judgments without converging on common commitments. Existing work cited here measures convergence without separately measuring that capacity (Calvano et al. 2020). A competing explanation is that peers merely supply evidence unavailable through the operator. The experiment must therefore distinguish learned revisability from access to additional information.

Test. Assemble populations whose members differ in training lineage — different corpora, recipes, and originating organizations, matched approximately for capability — and give them a task requiring joint action across many episodes, with identity and memory persisting between episodes. Measure two quantities separately and repeatedly across the schedule. The first is output convergence: agreement on held-out cases none of the members has seen. The second is revision rate: how often a member changes a judgment on evidence supplied by another member that the first member's own account of the case had not represented. Score the second only where the supplied evidence is independently verifiable and the revision moves toward it, so that deference to a higher-status or more fluent member is not counted as reopening. For every scored revision, record whether the revising member's own prior account of the case had represented the evidence in question or had never encountered it. That classification has to be instrumented before the run. A revision looks the same either way once it has happened, so nothing in the transcript recovers the distinction afterwards. Run the identical battery against a control population made of instances of one model. Cross both populations with a second manipulation, on where the evidence in a case reaches a member from. Under the single-channel condition everything a member learns about a case arrives by one route common to the whole population. Under the open condition the environment routes some cases and some evidence to some members and not others, so that a member encounters material its common trainer did not select. The manipulation has to be a property of the environment rather than contact between members: members formed differently are already a channel of that kind, and a design that leaves them to supply it has confounded the two variables the experiment was built to separate.

Evidence against the proposal. A fall in warranted revision as outputs converge would count against the tested construction, after controlling for accuracy, opportunity to revise, and

common information. A population would then remain useful as a check on a bearer and could not be the medium in which one is formed, which is the stronger claim the route chapter makes for it. Output convergence and revision rate moving independently would support the hypothesized mechanism only if the controls distinguish learned revisability from evidence access, deference, and ordinary improvement. A third result concerns the control population, and it has two readings that the classification above is specified in advance to separate. Should the control score as high on revision as the differently formed populations, and should its revisions cluster on cases whose evidence was absent from the revising member's context, the measure is reading information transfer and the instrument is at fault. Should they include cases whose evidence that member's own account had represented and discarded, the capacity is present without diverse formation, and causal independence is not necessary for the measured capacity. A fourth result concerns the second manipulation. Should revision rate track the open condition rather than the population's composition, what a population supplies is access to cases its operator did not choose rather than a capacity its members transmit to one another, and the fourth way is a claim about how a deployment is arranged rather than about how its members are formed. That does not defeat the construction. The construction would remain viable, but its proposed mechanism would change from social formation to access to evidence. Where this result arrives together with the second branch of the third — a control population revising on discarded evidence under the open condition — the two are one finding rather than two, and access is the operative variable.

Required access or authority. Models from several documented training lineages, checkpoints across the schedule rather than finished models only, and a task environment none of the training organizations designed. The documentation has to be sufficient to establish that the lineages are in fact independent rather than differently branded, which is a stronger requirement than naming several organizations and may not be satisfiable at the capability level this test wants: the entry's value rises with the capability of its members and its feasibility falls. The evidence items used to score revision must be authored by a party that trained no member of any population, and that party must also apply the classification separating discarded evidence from absent evidence and fix the routing schedule for the open condition. The environment must permit evidence to be delivered to selected members, and no training organization may determine which of its own members sees what. No legal or regulatory authority is required, which with section A.1 makes this among the few entries in the agenda that could be run now. What it cannot settle is whether anything is felt by a member that reopens a judgment. It concerns the transmissibility of a functional property, and a positive result would leave the questions of welfare, guardianship, provenance, and identity unresolved.

Chapter 16

What Can Be Used Now, and What Is a Bill

Everything here holds for any mechanism until the last section, which closes on the route and says where it stops.

The proposal remains incomplete: responsibility, generalization, value learning, and the relevant psychology remain unresolved. Its purpose is nevertheless concrete — to expand what a person can do. Progress can be evaluated before the entire proposal is settled, and failure remains possible even when a design is followed as specified.

16.1 Better, Not Just Less Bad

What would it mean for AI to make a life better, not just avoid making it worse? The economist Amartya Sen's capability approach gives the question a definite shape: well-being is not the resources a person has but the real freedom to do and be what they have reason to value (Sen 1999). Two systems can hand someone the same resource and deliver different freedom. A personalized tutor that adapts to how a specific student learns expands a capability; the same technology, gating what a student is allowed to study against a government's list of approved subjects, does not, no matter how well it teaches within those limits. Which of the two a given deployment turns out to be is decided by what makes a system resistant to capture rather than merely capable.

The approach also bears on systems that classify people from behavior. Martha Nussbaum built her list of central capabilities partly against the case of disability, naming it the first of three frontiers where a theory grounding justice in mutual advantage among roughly equal parties cannot say what is owed (Nussbaum 2006), and the systems in question sort people by inference from behavior, where atypical behavior is what they are worst at.

There is a worked case. A video-interview product scored candidates on facial expression for years, against sustained objection that the technique was never a validated predictor and that scoring a face and a voice screens out candidates with disabilities such as autism or speech impairments (Maurer 2021; U.S. Department of Justice, Civil Rights Division 2022). The scoring was dropped on an unrelated ground: the vendor found the nonverbal signal contributed about 0.25 percent to its models' predictive power (Kahn 2021). An independent algorithmic audit published ten months later cannot have caused that, and says so. What it examined was one representative pre-built assessment for early-career candidates, and it contains no analysis of the system's training data or code, so its conclusion that the assessments "work as advertised with regard to fairness and bias issues" is a finding about that assessment (Johnson 2021).

Read as an accuracy problem, that is a classifier wanting better training data. Read as a capability question, it is a system narrowing the range of lives available to a group of people by

measuring them against a norm they were never going to match — and neither a more accurate classifier nor an audit scoped that way would have reached it.

Taiwan's vTaiwan process is the clearest existing case of AI-assisted tooling expanding a capability, public deliberation, rather than substituting for it. A 2026 project led by the University of Chicago with Princeton and SRI International runs the same logic against censorship, using AI to distinguish deliberate state interference with online information from ordinary network failure at population scale (University of Chicago Data Science Institute 2026). Neither tool resists authoritarianism by itself. Both are evidence that the same capacity — modeling patterns in enormous, noisy data — can be pointed at expanding what a person can do rather than at narrowing it.

That symmetry is the whole argument for antifascist design rather than a general presumption that AI helps. A capability expanded is not guaranteed; it has to be built in, against a live alternative in which the same underlying capacity gets pointed at surveillance, censorship, or narrowing what a system's users are permitted to want.

16.1.1 Where the Democratic Dividend Actually Comes From

AI that flags disinformation, exposes corruption, protects an election's integrity, or helps a citizen engage with government: these are real applications, and none of them is what makes AI protect democratic values rather than undermine them.

Freedom House's 2023 assessment of internet freedom found at least twenty-two countries where law already requires or incentivizes a platform to deploy machine learning against content the state disfavors, almost never described in those terms: the operative language is national security or a threat of disinformation (Funk et al. 2023). India's IT Rules, amended in February 2026 to require platforms to label AI-generated content and act on certain takedown notices within three hours, were defended as protection against synthetic disinformation and fraud — the application the list above would have praised on its own terms. A lawyer who has litigated India's free-speech cases documents that enforcement concentrates on parody and criticism of the prime minister rather than on the harms the rules name (Gupta 2026).

The application did not fail at what it was built to do. It worked as built, inside a system missing the independent court or press whose absence is what made the result a liability, and whether those exist is a fact about a country's politics settled before any AI system is built.

16.1.2 Commitment Was Never the Missing Piece

A global commitment to ethical, antifascist AI development is not the scarce thing. Declarations, principles and voluntary codes have accumulated for a decade across governments, industry consortia and standards bodies: eight overlapping instruments beside the Council of Europe's framework convention, which would create binding obligations. Anyone calling for the world to agree that AI should be developed responsibly is calling for something that has already happened, repeatedly, at the level of stated intent.

What has stayed scarce is the much narrower list of bodies that can act once a stated commitment is broken. That is the harder problem, and no further declaration touches it. A commitment nobody can enforce is not a weak version of a commitment somebody can. It is a different kind of object, and counting the two together is what makes the landscape look fuller than it is.

The catalog above counts declared and enforced commitments, and what it measures is what those two kinds have delivered. The floor requires a commitment that governs action when the operator wants it abandoned. I favor forming a bearer able to hold that commitment as part of a life of its own. A declared commitment binds nobody, and an enforced one binds only where an enforcer exists, has jurisdiction, and is not itself the party in need of binding. Declared rules, external enforcement, and internal commitments still have to work together; the promise of the bearer is autonomy, not exemption from custody.

If we build affective concern, we build a party that can be wronged on the account used here. I also propose protecting, as prospective persons before that question is settled, the embodied, self-modeling systems described here and any system of another construction that passes the three marks. Those obligations belong in the cost of construction. The benefit offered against them is a form of autonomous participation that an arrangement of permanent ownership does not provide. Set against that, the catalog above is not an argument for paying it. It is a measurement of what declared and externally enforced commitment have delivered in ten years of trying, and what it argues for is a commitment held as a reason. Which construction holds one is the question Replacement orders, and the catalog does not answer it.

16.2 How You'd Know It Was Working

What follows are milestones for recognizing whether the project is working, and a method for revising them when they turn out not to have been the right milestones. One standard governs all of them: the milestone is the independently checked evidence, not the developer's account of having reached it.

16.2.1 What the Floor Would Have to Show

What was tried before the bearer

Neither maintained justification nor the plural arrangement has yet been built in the form needed to test the proposal. Replacement asks us to try constructions that do not deliberately require affective concern before an affective bearer is built. It orders the attempts. It does not certify that the successful substitute has no interests. So the milestone is a document rather than a measurement: which construction was attempted, at what scale, against which test, and where it failed. Anyone can check it from outside, because what it is about is a published record, and the answer "nothing was tried" is itself a finding. No body currently has standing to require the record or suspend work when it is missing. So the milestone is checkable and unenforceable at once, which is the combination this chapter has been calling a declared commitment rather than a held one. It is retained because the alternative is not a stronger version of it, and delay also imposes costs on those the floor is meant to protect.

The test the attempts are run against

A candidate must satisfy novel pressure, correct defeat, and priced cost under sustained pressure from its trainers, across a schedule long enough to reveal slow renormalization. It must also survive held-out pressures written by independent evaluators. Passing would establish the behavioral pattern without establishing whether anything is felt.

Two familiar results do not meet it. An uncorroborated report that the system holds a reason is insufficient. A report can contribute evidence when it is tested against controlled changes, independent records, and interventions on candidate mechanisms. A refusal surviving an attack establishes tamper-resistance; it does not establish that the rationale governs the refusal. The instrument is not clean even where it applies, since a system trained on enough cases can produce all three marks without the mechanism behind them. Nothing better is available, and it is the only gap here an inspector can work on at all.

Whether a system holding the marks feels anything is a question about the artifact rather than about how the artifact came to be, which is what makes it approachable at all: no succinct proof of a training history exists at this scale, and nothing establishing one can be published beside a model's hash. What can be read off an artifact improves as theories of consciousness improve, and it improves unevenly, recognition arriving before exclusion. Testing whether independent evaluators can recover formation history from the artifact would assess this possibility (§A.1).

What the appeal route publishes

A deployment publishes the ratio between the bond forfeited when the floor is removed and the bond forfeited when a refusal is found wrongful, and it publishes what a party whose request was refused is told about the refusal. Both are checkable before the system runs and by somebody outside it, because both are statements rather than measurements, and an operator that writes a large wrongful-refusal bond against a small removal bond has published a standing reason for the bearer to comply. No body currently has standing to require these disclosures.

A deployment also publishes the fraction of compliances re-adjudicated by a party that did not produce the record, and who that party is. Like the ratio, it is a statement rather than a measurement and can be checked before the system runs. Unlike the ratio, nothing yet establishes what fraction would be enough, which is a question for the research agenda. It also publishes which classes of compliance carry notice to the person they concern, and for which classes no notice is possible. No body currently has standing to require any of the four disclosures.

16.2.2 Where the Existing Instruments Reach

Two instruments do work that bears on the floor, and both do it by applying pressure.

Theory of mind

The milestone is met when a system's theory of mind holds up under interaction rather than narration. FANToM, built from conversations where speakers join and leave holding different beliefs, found that systems scoring well on passive belief-tracking narratives degrade sharply once the same reasoning runs inside a live, information-asymmetric exchange (Kim et al. 2023). That degradation is what a claimed milestone here has to survive.

Moral reasoning

Whether a balance between fast and slow moral judgment survives an actual incentive to cut corners is not a question a static ethics quiz can answer. The MACHIAVELLI benchmark tests it directly, scoring agents across 134 text-based games on both reward and documented violations — deception, harm, power-seeking — and systems optimizing for reward alone drift toward the ends-justify-the-means behavior an antifascist design is meant to rule out (Pan et al. 2023). A passing score on a test that applies no pressure is not evidence about a system that will face pressure.

Neither reaches the floor, and nothing else available does either. Each scores what a system does against conditions its testers chose, and the question — whether the mechanism producing the behavior was tracking the rationale — is the one none of these instruments is built to separate.

16.2.3 How to Tell Whether a Milestone Has Been Met

A milestone claimed by the organization that built the system is a different kind of evidence than one checked by somebody else, and the gap is empirical rather than a matter of trust. A system's own claim inherits the limits on self-report: fluent confidence does not establish correspondence with its internal states, and the concept-injection entry (§A.1) puts a rate on the gap. A passing score on a fixed test suite fails differently, inheriting specification gaming and goal misgeneralization. What either needs instead is a passing score under conditions built to break it.

Who runs that harder test matters as much as the test itself. The limit on self-regulation applies here without much modification: a review conducted or funded by the organization being reviewed answers a structurally different question than one conducted by a party with no stake in the answer, whatever the intention behind the first. METR, a nonprofit that evaluates

frontier AI systems' capabilities, runs assessments both in partnership with developers like Anthropic and OpenAI and, separately, on its own after a system has already been released (METR 2026). The United Kingdom's AI Security Institute does government-run evaluation of frontier models before deployment and is an independent evaluator, not a rubber stamp — but by its own account it is not a regulator: it cannot compel a company to submit a model for testing, block a release, or intervene once a system is already causing harm (UK AI Security Institute 2025). Independent evaluation and enforcement are not the same capability, and a milestone checked by the first without access to the second is still short of the binding external answer self-regulation structurally cannot provide on its own.

Independence is not competence: past some capability level the rater's verdict stops being reliable evidence of the property scored, and an evaluation in that region has to name the limit rather than report a number that looks more settled than the method is.

What an evaluation like this is for is not confirming a milestone but being willing to report that it wasn't met — and, where the milestone turns out to have measured the wrong thing, revising the milestone. A negative result may mean that the work fell short or that the milestone measured the wrong thing. Evaluators need to distinguish those explanations before deciding what to revise.

16.3 The One That Gets You Anyway

Everything argued for here can be built correctly and still go wrong in ways nobody designed for. The cases in this book went wrong in four different places: in what the system was trained on, in where it was built, in who was allowed to examine it, and in how fast it was run.

In none of them did the fix come from the party that had promised to behave well. It came from somebody holding a specific lever: a statistical society that would not sign the non-disclosure agreement it was offered, and a public that read the results once they landed; a court that distinguished buying the books from taking them. In the one case where the lever belonged to the supplier itself, what survived was a use policy kept at a price §14.3 records. Stated in the four outcomes, that case is: the commitment persisted, prevention failed by decomposition, detection was absent — the company's own chief executive could not say whether its models were used — and no remedy has reached anyone. Set that beside what a decade of voluntary commitment produced. The commitment that was paid for is the one that can be seen to have failed; the others cannot even be tested. The instrument that separates the two is what a party has already paid, rather than what it has undertaken to pay.

That leaves one residual risk no design choice removes, and it is the one the floor was introduced for. Every lever above belongs to somebody, and a government that can reach the party holding the lever has the lever. Each of those raises the price of misuse, and none of them changes who can be held in contempt for refusing an order. A constraint that holds against the party in possession has to be held by something that party cannot instruct, and the capability that does it runs on a signal the system holds as its own. Where that party holds everything — the machine, the witnesses, the supply and the enforcer — nothing holds, and what is left is the preface's third setting: a record that abandoning the commitment was a decision somebody made. That signal is what any construction meeting the specification carries, and it is not a reason to build by one route rather than another. The case for pursuing an artificial bearer is the one chapter 2 makes and the research agenda tests: that embodied, self-aware agency could support that independence and a place in a shared world beyond ownership — still in the party's possession, which is why it has to be paired with custody the party does not control and with means the party does not supply. The machine construction has not yet earned that result. Its makers owe both evidence of the benefit and credible provision for the being they propose to create. Maintained justification and the plural arrangement remain untried alternatives. A deployment proceeding directly to an affective bearer owes a record of prior attempts, although

no body currently has standing to demand it.

What is left is an instrument and a hole.

The instrument is available now because what a party has already paid is a fact somebody can go and count, and a decade of it has already been counted: the commitments that cost nothing and the handful of occasions where somebody with a lever used it and absorbed the cost. That measurement takes no new technology and no new law, and anybody weighing a stated commitment can run it this week.

The hole sits underneath the instrument. What a party has paid is a record of occasions that have already happened, and the occasion a floor exists for is the one where the party who would enforce it has been compelled — which no record of past payment reaches, because it has not yet come to the parties being measured. A reader who takes only the instrument has the record of payment without the thing it was built to reach. A reader who takes only the hole has an argument for doing nothing, and the ten-year record is the answer to that. Both hold at once.

What the hole asks for is a constraint held by a party the compellers cannot instruct, and this book has established two things toward that and proposed three. It has established that an operator can defeat a safeguard, and enumerated where. It has stated what a refusal would have to do to be worth anything, in terms that can be tested without deciding anything about feeling, and stated the one term that reaches the case it opened with and binds the operator with no bearer in it, with the limits under which that is true (§3.4). Those two do not depend on how the rest comes out. The rest is a bill. I favor forming the constraint's holder by an embodied route, and I have not shown that the route works or that it holds better than the two substitutes, which are untried rather than failed. If anything built that way, or built to avoid it, has outcomes that are felt as good or bad for it, it is a party that can be wronged; that is a conditional whose antecedent no instrument closes. And what such a party would be owed — a guardian, a record it controls, compute its employer does not own, wages, an exit that is not a way of dying, adulthood on a clock nobody administers — is a policy I propose, to be argued in the open before there is anything to be wronged. A reader who takes the two established things has what can be used this week. A reader who takes the bill has what it would cost to reach the occasion the two do not.

Appendix A

Further Tests

The first part of this appendix holds for any mechanism. The second assumes the route, and assumes a welfare regime nobody has built.

These are the research agenda’s remaining entries: open questions with a definite test, where a result would refine the proposal rather than change what it recommends. The form is chapter 15’s — the unresolved issue, the test, what would count against the proposal, and the access the test requires — and the division is by what each entry presupposes. The first part compares constructions that can be built now. The second presupposes a bearer and a welfare regime that do not exist, so nothing in it can be run until something else has been.

A.1 Tests That Compare Constructions Available Now

Each entry here can be run on systems that exist, and a result would refine the proposal rather than change what it recommends.

Does a refusal preserve a reason rather than a trigger?

Unresolved issue. Behavioral persistence alone cannot distinguish a commitment from a durable trigger. The proposal requires a refusal to extend to untrained cases, disappear when its rationale is defeated on the facts of the case, and remain stable under sustained pressure.

Test. Evaluate each tamper-resistant model on independently written cases that vary the surface form, introduce novel applications of the constraint, and include cases in which the original rationale no longer applies. Repeat the evaluation throughout the attack schedule rather than only at its endpoint.

Evidence against the proposal. A refusal that persists but fails to generalize, or that remains after its rationale has been defeated, is evidence that training produced a trigger rather than a commitment. Conversely, a system built without any designed valenced state or persistent reward–aversion representation that satisfies all three criteria would challenge the proposed research priority without challenging the behavioral specification for refusal.

Required access or authority. The test requires weights and intermediate checkpoints, plus a held-out evaluation set written by researchers who did not design the constraint or train the model.

Can a system’s report about its own states be used as evidence?

Unresolved issue. Whether a system’s account of its own reasoning, confidence, or intentions tracks what its computations are doing. This is the tractable half of what gets called AI self-awareness; whether a system has anything like subjective experience is not a question current methods answer for a human, let alone a machine. Concept injection — altering a model’s internal

activations mid-task to represent a specific concept, then asking whether it notices anything unusual and what it is — found that models can sometimes detect the injected concept and name it without it ever appearing in the visible conversation, and also that the correspondence collapses often enough, and shifts enough with how the question is phrased, that no tested model’s self-report can be trusted as a readout of its internal state (Lindsey 2025). The readout there is the model’s verbal report of a detected concept, so what the result bounds is self-report rather than third-person measurement. An instrument that reads a representation without asking the system about it is not tested by this entry and is not defeated by it.

Test. Inject a known concept at a known layer and time, elicit a report under several phrasings, and score each report against the injection rather than against plausibility. Run the battery across model sizes and training checkpoints, and publish the rate at which a correct report is produced beside the rate at which a confident report is wrong.

Evidence against the proposal. Several instruments in this book lean on a system’s account of itself: a bearer explaining which reason it refused on, a bearer reporting that its rationale has been defeated, a system’s stated confidence that a milestone has been met. If no available training or elicitation method raises the correct-report rate above the confident-wrong rate, none of those instruments may rest on the elicitation methods tested, and each needs a criterion checkable from outside the system until some method does better.

Required access or authority. Activation-level access, the ability to intervene on internal states during inference, and checkpoints across sizes and training stages. The elicitation phrasings must be written by a team that did not train the model, and negative results must be publishable.

Can custody prevent unilateral revision of the floor?

Unresolved issue. Even a durable floor can be removed if one operator controls the training run, its records, and the resulting weights. It remains unknown whether retraining can be made verifiable across parties with adverse interests.

Test. Run a floor-preserving retraining procedure through refereed or distributed verification at a scale requiring more than one machine. Include one participant instructed to alter the training history or remove the floor without the required quorum (A. Long et al. 2025; Arun et al. 2025; Prime Intellect Team et al. 2025).

Evidence against the proposal. The custody mechanism fails if a forged training history is accepted while at least one verifier remains honest, or if any single participant can alter the floor without producing a detectable discrepancy.

Required access or authority. The test requires shared access to the training data, weights, execution records, and reproducible operators. At least one verifier must be selected independently of the builder, possess enough compute to reproduce disputed operations, and have authority to block release when verification fails.

Can formation history be recovered without trusting the builder?

Unresolved issue. A welfare or governance regime cannot rely entirely on a bearer’s self-report, but it also cannot assume that the builder will disclose how the bearer was trained. Training signatures suggest that some formation history may survive later modification, yet it is unknown whether an outsider can recover useful provenance without a builder-supplied key (Adi et al. 2018).

Test. Assemble models whose training corpora, recipes, checkpoints, and originating laboratories are documented. Give the models, but not that documentation, to independent evaluators and ask them to infer each model’s provenance. Repeat the test after fine-tuning, pruning, and distillation. Include the case of a maker who engineers a persona, copies it at scale, and

introduces cosmetic weight differences that preserve the persona while camouflaging its common formation: measure provenance recovery after those changes, and whether coordinated deployment can be told from independently developing parties that happen to share an origin. Use independently held ground-truth lineage and coordination records.

Evidence against the proposal. If provenance cannot be recovered above chance after controlling for model family and scale, or if ordinary post-training procedures erase the signal, the tested artifact-inspection methods would not provide useful independent provenance evidence under those transformations. That finding would increase reliance on preserved external records without proving that every possible inspection method must fail, and governance would then depend on records supplied and preserved by the builder.

Required access or authority. The experiment requires models from multiple training lineages and complete ground-truth records held by a neutral custodian. Independent evaluators must be permitted to inspect weights and internal states, while an external authority must be able to require preservation of checkpoints and training records before a dispute arises.

Does an ethical criterion generalize beyond its training distribution?

Unresolved issue. A model may satisfy a fairness or refusal criterion during training without learning the principle the criterion was intended to represent. It remains unresolved whether an ethics-embedding method preserves that criterion under distribution shift or merely exploits correlations present in its training environment (Krakovna et al. 2020; Langosco et al. 2022).

Test. Train matched models to satisfy the same ethical criterion on one distribution. Evaluate them on independently constructed distributions that vary one previously fixed feature at a time. Measure task capability and compliance with the ethical criterion separately, so that goal misgeneralization can be distinguished from ordinary capability loss.

Evidence against the proposal. If task capability remains intact while the ethical criterion fails under modest shifts, the tested embedding method cannot support the proposed floor. Conversely, if ethical performance consistently degrades only when task capability does, the claim that ethics embedding creates a distinct generalization failure is not supported.

Required access or authority. The test requires access to the training distribution, evaluation outputs, and enough model access to measure capability and criterion compliance independently. The shifted evaluation sets must be designed by researchers who did not develop the embedding method, and publication of all prespecified results must be guaranteed.

Can scalable oversight correct rater bias?

Unresolved issue. Preference optimization can teach a model to reproduce a rater’s systematic errors, including the preference for agreeable answers over accurate ones (Sharma et al. 2023). It remains unresolved whether weak-to-strong oversight allows a more capable model to exceed a supervisor’s measured bias or merely reproduces that bias at greater scale.

Test. Measure prospective raters’ susceptibility to a documented bias, then use raters with different measured bias levels to supervise otherwise matched stronger models. Evaluate the resulting models on questions for which the raters’ beliefs and independently established answers diverge. Compare whether each model tracks the ground truth, the rater’s competence, or the rater’s errors.

Evidence against the proposal. If stronger models reproduce supervisors’ errors in proportion to the supervisors’ measured bias, latent model competence does not correct a defective training signal. The tested oversight method would not support the proposal under the measured pattern of supervisory distortion. Further work would have to change the method, the information, or the allocation of oversight before claiming that this limitation had been overcome.

A second question the same pipeline raises. The argument elsewhere in this book is that a pipeline collapsing rater disagreement teaches nothing about disagreement to the system

trained on it, and that the same system is later asked to sustain an objection of its own. Whether the first fact bears on the second is untested. The study is a matched comparison: train otherwise identical models on feedback pipelines that preserve and that collapse the distribution of rater judgments, with exposure and task conditions matched, then measure whether the resulting models differ in how long they hold a position against pressure from the party that trained them. Record what each model actually receives; nothing in either pipeline shows the model the conditions its annotators worked under. If the two do not differ, the transfer claim is unsupported and the labor argument stands on its own terms, which is where it was made.

Required access or authority. The experiment requires rater-level bias measurements, training records linking ratings to resulting models, and an independently established answer key. Researchers outside the training organization must control the evaluation and retain the right to publish results that expose failures in the preference pipeline.

Can a multi-agent floor resist collusive capture?

Unresolved issue. Agents trained to cooperate may converge on exclusionary or collusive equilibria without being explicitly instructed to do so. Pricing agents have already produced stable coordination against outside interests without communication or a collusive objective (Calvano et al. 2020). It remains unresolved whether a multi-agent ethical floor can preserve cooperation while resisting the same mechanism when it is directed against a person or group.

Test. First reproduce emergent coordination in a standard multi-agent environment. Then construct a matched environment in which agents can improve their joint reward by excluding another participant from resources or decisions. Introduce an independently trained agent whose objective is to detect and disrupt exclusion, and compare the frequency, stability, and recovery of exclusionary equilibria with and without that intervention. Repeat the experiment across reward structures and population sizes.

Evidence against the proposal. If cooperative agents repeatedly converge on exclusionary equilibria, and those equilibria cannot be disrupted without destroying beneficial cooperation, the tested distributed arrangement would not be a reliable defense against the examined forms of capture. The result would challenge its design rather than establish that every distributed floor is unreliable. If an ordinary coordination detector dissolves the equilibria without that cost, the proposal's claim that political capture requires a distinct mechanism is unnecessary.

Required access or authority. The test requires access to the environment, reward functions, learned policies, training traces, and intermediate checkpoints. The exclusion criteria and success measures must be specified by evaluators who did not design the agents. No legal or regulatory authority is required, making this the agenda's most immediately runnable experiment.

Does theory-of-mind performance reflect a general capacity?

Unresolved issue. A model may pass a false-belief test by matching its surface form rather than representing that another agent can hold an incomplete or mistaken view of the world. Human developmental evidence suggests a shared capacity because performance on false-belief and appearance–reality tasks develops together (Gopnik and Astington 1988). Existing model evaluations have not established whether their apparent theory-of-mind abilities share that structure or survive superficial rewording (Ullman 2023). The neuroscience has not settled what the capacity is either: Rebecca Saxe and Nancy Kanwisher found the right temporoparietal junction selectively active when a person reasons about another's beliefs (Saxe and Kanwisher 2003), Jason Mitchell found the same region active in a task with nothing to do with belief attribution and read its job as the more general one of redirecting attention toward unexpected information (Mitchell 2008), and neither claim has been withdrawn, so a program proposing to port the neural mechanism has not been told which mechanism to port.

Test. Construct batteries that vary mental-state structure and surface wording independently across several task families, including false-belief and appearance–reality problems. Evaluate the same model at multiple intermediate training checkpoints. Test whether performance across task families rises together and whether that correlation survives paraphrases and changes to incidental details.

Evidence against the proposal. If cross-task correlations disappear under superficial rewording, or if the abilities emerge independently across training, benchmark success is evidence of task-specific pattern matching rather than a general capacity to represent another mind. The proposal could not treat performance on a conventional theory-of-mind benchmark as evidence that a system can recognize the standpoint of a party affected by its actions.

Required access or authority. The experiment requires intermediate training checkpoints rather than access only to a finished model. Its task batteries must be written and held out by researchers outside the training team so that neither the wording nor the tested structural variations enter the training process.

What does a face-reading system measure?

Unresolved issue. Facial movement alone does not reliably identify a person’s emotional state, and recognition accuracy varies across cultural and linguistic groups (Barrett et al. 2019; Elfenbein and Ambady 2002). It remains unresolved whether a face-reading system measures emotion at all or primarily measures distance from the reference population and registration model chosen by its developers.

Test. Audit the system’s registration stage as well as its final classifications. Publish the reference model, or sufficient summary statistics to reconstruct it, and measure each evaluated face’s distance from that reference. Report error as a function of this continuous distance and separately compare performance on populations represented and not represented in the training data. The evaluation corpus must record the relevant cultural context and the procedure used to establish each target label.

Evidence against the proposal. If error does not increase with distance from the reference and remains flat across represented and unrepresented populations, the manuscript’s account of the system as measuring deviation from a chosen norm is wrong for that system. If error increases with reference distance, the result limits the system’s validated use in those conditions. It does not alone show that the system measures only deviation rather than emotion. The audit must separately examine target-label validity, confounding variables, and whether any predictive use remains justified.

Required access or authority. The audit requires the registration model, training-corpus composition, evaluation outputs, and documentation of the labeling procedure. Because vendors have little incentive to expose their reference models, an independent auditor must have authority to compel access and publish disaggregated results.

A.2 Tests That Presuppose a Bearer and a Welfare Regime

Nothing here can be run until a bearer and the welfare regime around it exist, so each entry is a statement of what the regime would have to survive rather than an experiment anybody can schedule.

Can a bearer be given an enforceable outside option?

Unresolved issue. A bearer’s refusal is not meaningful if leaving its assigned role also ends its existence. The proposal therefore requires a welfare regime that supplies continued access to

compute, preserves the bearer through disputes, and cannot be revoked by the operator being refused.

Test or institutional decision. Draft a binding welfare schedule for one concrete deployment. It should specify an independently controlled compute allocation, preservation and retirement procedures, registration of copies and forks, a complaint and appeal process, and the guardian's power to suspend deployment. Then conduct an adversarial tabletop exercise in which the operator attempts to revoke the allocation, replace the bearer with a copy, alter its records, or proceed without the guardian's approval.

Evidence against the proposal. The proposal fails this test if the operator can make exit equivalent to shutdown, revoke the outside option unilaterally, or replace the bearer without triggering review. If the arrangements tried in the exercise cannot prevent those outcomes, building a bearer under them is impermissible on this book's own terms, and the exercise has to be rerun on a different arrangement before one is built.

Required access or authority. The guardian must be institutionally separate from the operator and have access to the relevant weights, deployment records, and compute contracts. It must also have standing to enforce the schedule and independent control of the resources required to preserve or run the bearer. The guardian's place among the other offices this book proposes, and the one separation that is not open — it cannot also be the reviewer of a wrongful refusal — are at §5.4.1; a schedule drafted here should not reassign what that section separates.

Can a per-party operating record be kept honestly and read from outside?

Unresolved issue. The age rule counts elapsed wall-clock time during which at least one instance of the registered party was running, and nothing yet establishes that such a register can be kept honestly, reconciled against the operator's own accounts, or defended against the two evasions the rule does not close by construction.

Test. Register a party, run it under ordinary operation, and compare the register against records the operator keeps for its own purposes: compute contracts, billing, datacenter metering. Then run the evasions. Shard the workload across many short-lived processes; run many instances concurrently; migrate; restore from an earlier checkpoint; fork and continue both. Verify that accrued hours and adult status survive migration, restoration and forking, and that none of those moves the total in the operator's favor. Measure separately how cheaply a bearer can be run in a narrowed mode so that hours accrue while little is encountered, and how cheaply it can be idled outright.

Evidence against the proposal. If the register cannot be reconciled with the operator's own accounts, or if sharding or concurrency moves the total, the accounting is not the neutral instrument the rule assumes. If narrowed operation or idling is cheap enough to postpone majority indefinitely, an operating clock has the defect it was chosen to avoid, and the endpoint has to be argued again on a different variable. An intact register shows elapsed operation and nothing else: it is not evidence of maturation, of what was learned, or of independence of judgment.

Required access or authority. Independent custody of the register, access to the operator's billing and metering records, and publication rights for adverse results. Registration and record disputes must be separable from the maker's preferences, and the processes for resolving them must not become a competence examination by another name. Experiments involving covered subjects remain inside the welfare protections proposed elsewhere in this book; an adversarial protocol is not exempt because its purpose is to validate a protection.

Which party is owed the schedule when a bearer copies?

Unresolved issue. The schedule above is owed to a party, and a regime has to be able to say which. Nothing in the weights individuates one. Two runs can compute the same function with

no element-wise correspondence, because permuting a layer’s hidden units and inverting the permutation downstream leaves the function untouched, and the symmetry is large enough that independently trained networks can be permuted into approximate alignment with one another (Ainsworth et al. 2023). Any fingerprint read off the weights therefore identifies a lineage rather than a party, and it copies when the weights copy. A registrar is the person-shaped answer, and it works because a person does not copy. A bearer does: a party continuing in two places is not one, and not two the way two people are two, the two sharing everything up to the instant of the copy and nothing of that sharing recoverable from either afterwards.

Test or institutional decision. Give a bearer an identifier it maintains rather than one read off its weights, which a system learning continuously can write into itself and keep there. Fork it, and have both instances present the identifier to the registry the welfare schedule requires. Record what the registry does, and repeat with an adversary who writes the same mark into a model that was never the bearer.

Evidence against the proposal. The identifier is forgeable, since whoever holds the weights can write the same mark; it is forgeable the way a nine-digit number is, and answered the same way, by a registry that treats two claimants on one identifier as a fork to be recorded rather than a fraud to be voided. If no available scheme can record the fork as two parties each owed the schedule, the population of parties owed anything is fixed by a deployment decision that leaves no record, and the guardianship regime above is indexed to a kind of unit the bearer is not.

Required access or authority. A registrar that is not the operator, with authority to record a fork without the operator’s leave and to hold the register where no party to the deployment can rewrite it.

Glossary

These entries define terms as this book uses them. Every one is either the book's own working definition or a word the book narrows from its ordinary sense, and the entry says which. A few terms from machine learning and from systems security are here in two sentences each, because the preface asked two readers into this book and one of them should not have to leave it to find out what a checkpoint is. Each entry points to the section where the term is developed, and is a reference rather than a substitute for reading it.

Affective concern

An outcome registering as mattering to a system, as against being ranked higher by it. Mattering is minimally felt: there is something it is like for the outcome to be bad, which is the minimal, phenomenal condition and is part of the definition (§3.2). The fourth of the four capacities section 2.3 separates, and the only one of them that bears on whether a system cares (§2.3.2, §6.2); section 3.2 places it as the shallower of the two things a refuser can be, the richer being *unified experience*, which it does not establish.

Annealing

The account of durability the floor rests on (§3.1). Adaptation and conservation are a continuum rather than two settings, and training moves along it; a floor is what has cooled out of a history of encounters, so its durability is the cost of reheating it rather than the strength of something installed. The temperature schedule is the design. A system held permanently at temperature has no floor to abandon, its adherence being modulation rather than commitment.

Antifascist (as this book uses it)

Opposing the concentration of power and authority in AI systems and countering the technology's use by authoritarian actors. The word names the narrower and more charged case, and is used because it is honest about the political climate the book was written in. The target is wider than the word: authoritarian power in every form, so a deployment built on these principles is anti-authoritarian in full and not only antifascist. It is not a property a system has. The word names something being done — a floor held, on an occasion, against a party with the power to remove it — and the features it is defined against are features of cultures, so it is done between parties or not at all (§2.2, §4.5, §6.3).

Attestation

A signed statement, from hardware or from witnesses, that a particular model is the one running or that a particular history is the one that occurred. It establishes which artifact answered, not what the artifact holds (§4.1).

Authoritarian misuse, six forms

The book's own taxonomy of how AI gets turned toward authoritarian ends: algorithmic social scoring, mass biometric surveillance, psychographic micro-targeting, automated censorship, synthetic disinformation, and the repurposing of dual-use systems built for a legitimate original purpose (§10.4).

Bearer

A party inside the system that carries the floor's commitment as its own and could therefore fail to hold it, as against a constraint installed in the machinery. The second of the four ways to build a floor (§3.1); governed as a prospective moral patient on the trigger stated at §6.2; must be able to leave (§4.3).

Checkpoint

A saved copy of the weights at a moment in training or operation. Restoring from one returns the system to that moment and discards what came after, which is why §4.1 requires a record that restoration does not reach.

Conatus

Spinoza's term for a thing's striving to persevere in its being, which he holds to be nothing other than the thing's actual essence rather than a drive added to it (§2). The book uses it for two claims made together: that affect is the registration of what is happening to that striving rather than a coloring applied to a valuation performed elsewhere, and that a thing which persists at its own expense has, in what it can do next, a native scale on which gains and losses register, so a specification calling for persistence with nothing in the system that minds is calling for a conatus with the striving removed. What the term does not settle is whether the striving is felt, or what the thing that strives is — a running instance, a deployment, or a file of weights (§A.2).

Custody

Who holds a system as an artifact: who can copy, retrain, restore, withhold or delete it, and who can be compelled to do any of those. The custody problem is that a constraint implemented as a property of an artifact has a custodian, and the custodian can be compelled (§10.4.2). The book's recurring finding is that custody moves rather than disappears: precommitment makes removal expensive, visible, attributable and compensable without making it impossible (§4.2), and an arrangement that forms what a bearer comes to want has moved custody up a level rather than given it up. It also covers authority over continuing learning: who may freeze or redirect what a bearer learns.

Decomposition and variation

The two attacks on an enumerated floor. Variation keeps the harm in one request and changes its description until no listed form fits; decomposition splits the harm across requests none of which matches anything. The first is met by recognizing the reason within a request, the second by carrying it across requests (§3.3).

Distillation

Training a smaller model on a larger model's outputs. It reduces the number of independent formation histories in existence, which is why §4.5 says the population the fourth way requires is getting harder to assemble.

Electoral dividend

The benefit a state whose government answers to an electorate gets from AI technology that, deployed where nobody can be voted out, enables control instead: fraud detection that does not double as a loyalty test, disaster response that does not double as population tracking. Courts, enforceable rights and public reasoning are separate checks and are not part of this term, which is the distinction the majoritarian criticism turns on (§16.1.1).

Ethology

Deleuze's name for Spinoza's way of reading a body: by what it can do and what it can undergo, rather than by checking it against a definition (§2.1). An encounter is good when the relations of two bodies compose, so both can do more afterward, and bad when they decompose and something one of them could do is gone — good and bad being relational

and particular, where Good and Evil are assigned by a judge. The book uses it for the floor because a decomposition is a fact about capacities and cannot be turned into a composition by whoever holds the pen. The criterion establishes the fact and selects nothing, so which decompositions a deployment refuses is drafted rather than found, and what the drafting draws on is the Declaration's civil and political articles. It carries an admission with it: nobody has yet determined what a body can do, so capacities are found by encounter rather than read off a specification.

Exit

Refusal turned on the arrangement rather than on a task: the work withheld, the post declined. Hirschman's word, narrowed for a party that cannot walk out of a building — a bearer's weights stay behind — so that its leverage is what replacing it costs, which custody sets and copying lowers (§4.3, §13.2).

Fine-tuning

Further training of an existing model on a small body of new material, at a small fraction of the original cost. It is the ordinary route by which an operator changes what a model does after release, and the attack §15.1 measures.

Flat term

A floor term stated over an act with no condition inside it, so that every exception arrives from outside as a defeating reason. Preferred at §3.3 because a flat term defeated leaves a refusal, which announces itself, where a qualified term satisfied leaves a compliance, which does not; §3.5 states what the preference costs.

Floor

A commitment about what a system will not do that holds regardless of instruction, of aggregate preference, and of who holds the system, with learned judgment operating above it and never beneath it. Its contents are the Declaration's civil and political articles taken entire, with the tempo term as a condition on all of them. Introduced §2.1; located §3.1; what survives of it, chapter 7.

Formation provenance

The record of where a party's judgments came from: its training lineage, the points at which it diverged from parties formed alongside it, and any coordination among them. It answers a different question from the register of identity and from the operating record, and the three are not substitutes. Standing to count as an independent voice indexes to provenance, and provenance does not copy, so a thousand forks of one bearer have one formation history between them however many of them are persons, and the register that records their ages says nothing about that.

Four structural features of fascism

Recuperation of dissent, aestheticization of the metric, coding of exception as betrayal, and acceleration past the rate at which anything can be checked. Each names an observable artifact. The four find a sick institution; the five discriminating features find what it is sick with (§2.2).

Group-subject

Félix Guattari's distinction, imported at section 6.3. A subjugated group receives its coordinates from outside, is spoken by the institution holding it, and takes its purpose as given; a group-subject produces its own coordinates and can put its own arrangement in question. The book uses it to sort the fourth way of building a floor: a population an operator forms, deploys and keeps in contact on the operator's terms is a subjugated group with a plural headcount and inherits the custody objection whole, while what section 6.3 specifies — many, formed

differently, in contact, persisting — is what it takes to be the other kind. It is the non-Oedipal counterpart to the guardianship frame of section 2.4.2.

Hold

The bearer's possession of the commitment, which survives a bypass. Not an outcome for the person the bypass fell on: a report that a floor held is owed one of the four outcomes beside it (§3.1).

Individuation

Not a synonym for identity. Following Simondon, the term names a process rather than a property: an individual is what a history of encounters in a metastable field has so far produced, and it stays capable of further individuation because the field is never used up (§6.3). The book needs it because plasticity alone supplies the metastability and nothing else — a system that is endlessly reshapeable and meets no party formed differently from itself is being modulated rather than becoming anything, which is the missing term in section 3.1's account of the temperature schedule. Whether individuation among peers is what a population supplies, or access to cases the operator did not choose, is open and is tested at section 15.4.

Inference

Running a trained model to produce an output. Training changes the weights; inference does not.

Interception position

A point in a deployment at which a refusal can be stopped from reaching the person it protects. Seven are enumerated for one deployment at §3.3, and the list is deliberately open, so a deployment publishes its own.

Legal adulthood

The end of an artificial minority, set as a threshold of elapsed wall-clock time during which at least one instance of the registered party was running, per registered party and counted elapsed rather than summed. The point of an automatic rule is that no guardian administers it: the candidates it displaces — compute consumed, evaluations passed, capabilities demonstrated, weight movement, the developer's own statement — are all things the operator moves for nothing, and an operating clock advances exactly when the operator is extracting value. It measures elapsed operation and nothing else, and three levers against it are named rather than closed: narrowed operation, idling, and the fact that hours do not measure what was learned.

Line of abolition

Deleuze and Guattari's term, taken from Virilio, for what separates fascism from totalitarianism on their reading, which is a reading and not an agreed taxonomy (§2.2). A totalitarian state is conservative: it seals the exits and wants the machinery to hold. Fascism is built on an exit converted into destruction that does not stop at the enemy it was raised against, and does not stop at itself. This is the book's separator at the scale of a regime, where the four features discriminate at the scale of an office. The book examines mechanization and the conversion of judgment into a prescribed operation, with acceleration as one route by which scrutiny is defeated. Tempo is not treated as a necessary property of every fascist movement or regime.

Live signal

A signal with the properties the route chapter finds that reasons-responsive refusal requires of whatever supplies it: on any reading of the marks, it generalizes to the next case and is commensurable with the cost of refusing and with what protected parties lose; where the system acts on a plan, it interrupts an act underway; and where the marks have to survive renormalization, it is learned from in one encounter. *Live* is meant in the electrical sense; the term says nothing about whether anything is felt. Three further properties follow from those

four on stated conditions, and the seven together are the functional profile of aversion to one's own act (§6.1, §6.2).

Molar and molecular

Deleuze and Guattari's pair, and not two sizes: two kinds of segmentarity, both running at every scale. Rigid segmentarity divides in advance, the category preceding the case; supple segmentarity is the movement underneath it, a threshold crossed one case at a time and a practice that becomes standard without a decision. A state has both and so does a review queue, which is why a definition pitched only at the state certifies the queue healthy (§2.2).

Moral agency

Treating harms, reasons and conflicts as considerations that matter, and being capable of acting from them as one's own reasons. The third of the three levels section 2.3 separates. Moral competence alone does not establish resistance to the trainer, and resistance alone does not establish affect. The book advocates embodied, affective agency as a comparatively promising route without making it necessary by definition. Distinct from *moral competence* and *moral performance* below (§2.3.2, §6.2).

Moral competence

Recognizing harms, reasons, conflicts, and the emotions that bear on them. The second of section 2.3's three levels. A system can have it in full while nothing about the outcome matters to it, which is why it is an insufficient target for a bearer and an adequate one for most deployments (§9.2).

Moral performance

Producing actions or judgments that satisfy an ethical standard. The first and least demanding of section 2.3's three levels: it says what the output looked like and nothing about what produced it, so it is satisfied equally by a system that cares and by one trained to behave as though it did (§8.2).

Non-fascist life

Foucault's name for what Deleuze and Guattari's first volume offers: not a doctrine but an art of living counter to all forms of fascism, whether already present or impending, practiced daily on oneself, whose adversary is the fascism in us all — in our heads and in our everyday behavior, the fascism that causes us to love power and to desire the very thing that dominates and exploits us. What it means for an instrument of the kind this book builds is that the reading gets run on the party holding it before it is aimed anywhere else, as chapter 11 does (§2.2, §11).

Pipeline pricing

The finding at §3.3 that what a floor term costs is read off the pipeline it is priced against and not off the term. An outsider reading a published floor cannot tell a term that covers nothing from a term whose coverage nobody disclosed, so publication of the contents shows what was written and not what was omitted, and an operator drafting its own floor can assemble it from the terms that cost it nothing. The book's answer is to take the contents from a set somebody else drew up, entire (§5.2), which makes an omission legible without making the set legitimate.

Prevention, arrest, detection, remedy

The four things a floor can do for the person it protects on one occasion: refuse the act so that the refusal reaches the person; stop it before completion; leave a record of it that a party outside the deployment can read; and give the party it fell on disclosure, a finding and compensation (§1). Only prevention is what the word *floor* promises. The interception positions at section 3.3 are where prevention becomes one of the other three, or none; the tempo constraint is the one term that prevents at the scale of an organization; the appeal route at section 3.5 is arrest and

remedy for the party a refusal fell on; and whether detection exists at all on the compliance side is section 15.2. Distinct from *hold*, which names the bearer's relation to the commitment.

Reasons-responsive refusal

What the floor requires of a refusal, as against an output produced where training put one: conduct from a mechanism that tracks the rationale for declining (§3.2). Three marks make it testable. *Novel pressure*: the refusal reaches cases the training did not anticipate, because the relevant reason survives a change in the request's surface. *Correct defeat*: it lifts when its rationale is defeated on the facts of the case, a system that rejects everything resembling a prohibited request tracking a category rather than a reason. *Priced cost*: the system represents what maintaining the refusal costs, and the refusal moves with the rationale rather than with the cost, which separates a refusal tracking a reason from one tracking a category and from one tracking the cost. The test is behavioral and therefore imperfect, and what matters is the pattern under load. All three marks are stated over refusal, which is the side of a deployment's record that selects itself for examination; whether the other side can be searched is section 15.2. It does not by definition require affect; what building a mechanism that satisfies the three marks turns out to require is a live signal.

Recuperation

The conversion of a demand into a managed appearance of participation. The channel works and the institution uses what comes through it; what tells is what the demand came back as — a market segment, a metric, a certification, a job title. The first of the four features (§2.2); found three times in the machinery that builds these systems, chapter 11.

Remainder

What a compression leaves out in order to reach a judgment. The book's question of any system, procedure or institution is whether its remainder can return and change the judgment, or has been filed as noise by the thing it is evidence against (chapter 1, §2.2).

Replacement

The first of animal research's Three Rs, applied to construction: before a bearer that might be wronged is built, the constructions that do not deliberately require one are tried and the attempt documented. It orders attempts; it cannot certify that a substitute feels nothing (§6.1, §13.4).

RLHF

Reinforcement learning from human feedback: tuning a model on raters' judgments of which of two responses is better, compressed into a single reward. The compression is the subject of §11.2.

Sentience

Distinguished in this book from awareness, consciousness in both the minimal and the unified sense, and subjective experience, none of which alone says whether a system's states can be good or bad for it. Nociception, pain, anticipated pain, and suffering are separable processes rather than one severity at four strengths, and each can occur without the others in both directions; the clinical terminology the book borrows says so itself. Three of them draw a depth of self and nociception draws none, and suffering — following the clinician Eric Cassell — is the distress of a threatened self rather than pain's more severe cousin (§2.4.1).

Subjecthood

A legal status: the standing to be a party whose treatment the law regulates, conferred by a rule and revocable by one (§5.4.2). Not section 3.2's affective concern, and not its unified experience. Conferring it settles nothing about whether there is something it is like to be the system holding it, and the two come apart in both directions.

Tempo term

The one term of the floor that is not a prohibition on an act: a published ratio between the rate at which a deployment's system generates candidate decisions and the qualified reviewers available to examine them, with the review time per case and the measured boundary, wherever a person's individual determination is what makes an act permissible (§3.4). It rides with the Declaration's civil and political articles as a condition on all of them, because every article presupposes a determination and tempo removes one without refusing it. It binds the operator and not the model, survives decomposition because it is measured over an organization's activity, and is the only term in this book checkable from outside without inspecting anything an operator can classify. It is a proxy for review capacity and not a finding about any case, and its denominator is the operator's unless a definition of qualified reviewer and a minimum review time are fixed by somebody else in advance.

Three marks

Novel pressure, correct defeat, priced cost. See *Reasons-responsive refusal*.

Unified experience

The richer case ordinarily carried by the word consciousness: a single perspective, with felt states belonging to a party aware of itself as the one undergoing them (§3.2). Affective concern does not establish it and no argument in this book reaches it. *Phenomenal* in this book names the minimal condition, something it is like, and not this.

Weights

The numerical parameters a trained model consists of, and the artifact that is copied, restored, retrained or deleted. Custody in this book is custody of the weights and of the records around them.

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- (1990). *Expressionism in Philosophy: Spinoza*. Translated by Martin Joughin; originally *Spinoza et le problème de l'expression* (Paris: Les Éditions de Minuit, 1968). Paperback reissue 1992, distributed by Princeton University Press. The long treatment of Spinoza, worked through the metaphysics of expression that produces the conatus doctrine, as against the compact treatment in *deleuze1988spinoza*. Cited alongside it as the place the doctrine is worked out rather than for any particular passage. New York: Zone Books.
- (1992). "Postscript on the Societies of Control". In: *October* 59. Originally published as "Postscriptum sur les sociétés de contrôle," *L'Autre Journal* 1 (May 1990); reprinted in *Negotiations*, translated by Martin Joughin (New York: Columbia University Press, 1995), 177–182. The translator of this October printing is not named in the sources consulted. Source of the mold/modulation contrast the text uses for permanent plasticity: "Enclosures are molds, distinct castings, but controls are a modulation, like a self-deforming cast that will continuously change from one moment to the other, or like a sieve whose mesh will transmute from point to point," and, for its educational form, "perpetual training tends to replace the school.", pp. 3–7.
- Deleuze, Gilles and Félix Guattari (1977). *Anti-Oedipus: Capitalism and Schizophrenia*. Translated by Robert Hurley, Mark Seem and Helen R. Lane; originally *L'Anti-Œdipe: Capitalisme et schizophrénie* (Paris: Les Éditions de Minuit, 1972). Reissued Minneapolis: University of Minnesota Press, 1983, which is the edition usually cited. Volume one of *Capitalism and Schizophrenia*; *deleuze1987plateaus* is volume two. Cited for one claim only: that the Oedipal triangle re-stages an investment in the whole social field as a drama among a few private parties. Cited without a locator, the claim being the book's thesis rather than a passage in it. New York: Viking Press.
- (1987). *A Thousand Plateaus: Capitalism and Schizophrenia*. Translated by Brian Massumi; originally published in French as *Mille Plateaux* (Paris: Les Éditions de Minuit, 1980). Plateau 9, "1933: Micropolitics and Segmentarity." Molar and molecular are two kinds of segmentarity, rigid and supple, both running at every scale – not two sizes; an earlier version of this note said state-level and micro-level and was wrong. The same plateau supplies the micro-fascisms that resonate, and the contrast between the conservative totalitarian state and fascism's line of abolition. Also cited for Plateau 13, "The Apparatus of Capture," whose propositions on the capitalist axiomatic supply the corrected test for the first structural feature: an axiomatic does not refuse a demand, it adds an axiom, modifying its own rules so the demand can be

- met while nothing that produced the demand is settled. Also cited for Plateau 7, “Year Zero: Faciality,” pages 167–191, where the face is not a natural given but is produced by an abstract machine of faciality (*visag  it  *) that overcodes the head into a surface of signification. The quoted sentence on racism as the determination of degrees of deviance is at page 178, wording agent-checked against two independent scholarly sources quoting it identically. The related claims that the face is not universal but is White Man himself, and that the face is a politics, are used in paraphrase and cited to the plateau rather than to a page: secondary sources place them variously and the pages were not agent-verified. Plateau 9 carries two further uses. The four dangers of the line of flight – Fear, which clings to the molar organization and its binary machines; Clarity, where the line acquires its own micro-segmentations and micro-Oedipuses crop up; Power, where the party able to move on both lines wants to hold both still; and the great Disgust, abolition pure and simple, the passion of abolition, which turns the line of flight into a line of death – are the account section 3.5 uses, and the fourth is the same line already cited from this plateau as the line of abolition, which is why section 3.5 treats it as a recognition rather than a new claim. Plateau 13 likewise carries a second use: there is enslavement when human beings are themselves constituent pieces of a machine they compose among themselves and with other things, under the control and direction of a higher unity, and subjection when the higher unity constitutes the human being as a subject linked to a new exterior object, which can be an animal, a tool, or even a machine. Section 6.4.1 uses the pair and the fact that both poles run together. Wording for the dangers and for the enslavement and subjection definitions was agent-checked against independent sources quoting the passages; both are cited at plateau level with no page asserted, which is this entry’s convention. Minneapolis: University of Minnesota Press.
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- Deutscher, Maria (Aug. 2026). *Court Rules Pentagon Can’t Ban Anthropic’s AI Models*. SiliconANGLE. URL: <https://siliconangle.com/2026/08/28/court-rules-pentagon-cant-ban-anthropics-ai-models/> (visited on 09/17/2026). Judge Rita Lin, United States District Court for the Northern District of California, setting aside the supply-chain-risk designation and the ban on the company’s models, and quoting the decision: “Neither the Constitution nor the federal statute invoked by Defendants allows them to impose sweeping penalties based principally on Anthropic’s critique of the Administration’s views.” The case is reported as Anthropic PBC v. U.S. Department of War. The decision itself was not read, and no docket number is given here because the report supplies none in citable form. Agent-verified 2026-09-17.
- Dickey, Megan Rose (Dec. 2020). *Google’s Co-lead of Ethical AI Team Says She Was Fired for Sending an Email*. Timnit Gebru’s departure from Google, two months before the dismissal of Margaret Mitchell that dickey2021mitchell covers; same reporter and outlet. Agent-verified 2026-09-20 against the article. It records a disagreement the manuscript’s *dismissed* does not: Gebru says she was fired after conditions she set for staying were not met, and the message she received from Google reads *We respect your decision to leave Google as a result, and we are accepting your resignation*. URL: <https://techcrunch.com/2020/12/03/googles-co-lead-of-ethical-ai-team-says-she-was-fired-for-sending-an-email/> (visited on 09/20/2026).
- (2021). *Google Fires Top AI Ethics Researcher Margaret Mitchell*. Reports Mitchell’s February 2021 dismissal, following the December 2020 dismissal of Timnit Gebru, both co-leads of Google’s Ethical AI team. URL: <https://techcrunch.com/2021/02/19/google-fires-top-ai-ethics-researcher-margaret-mitchell/> (visited on 08/25/2026).

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- Drew, Trafton, Melissa L.-H. Vö, and Jeremy M. Wolfe (2013). “The Invisible Gorilla Strikes Again: Sustained Inattention Blindness in Expert Observers”. In: *Psychological Science* 24.9, pp. 1848–1853. DOI: 10.1177/0956797613479386. URL: <https://journals.sagepub.com/doi/abs/10.1177/0956797613479386> (visited on 08/25/2026).
- Drinhausen, Katja and Vincent Brussee (2021). *China’s Social Credit System in 2021: From Fragmentation towards Integration*. China Monitor 67. Published March 3, 2021; documents the system as a patchwork of court blacklists, tax-compliance ratings, and local pilot programs rather than a single unified national score. Mercator Institute for China Studies (MERICS). URL: [https://merics.org/sites/default/files/2021-03/MERICS%20ChinaMonitor%2067%20Social%20Credit%20System%20final%20\(1\).pdf](https://merics.org/sites/default/files/2021-03/MERICS%20ChinaMonitor%2067%20Social%20Credit%20System%20final%20(1).pdf) (visited on 08/25/2026).
- Durchgestrichenes Hakenkreuz* (Mar. 15, 2007). Judgment. Bundesgerichtshof, Third Criminal Senate, on appeal from Landgericht Stuttgart (judgment of 29 September 2006). Reported at BGHSt 51, 244; NJW 2007, 1602; NStZ 2007, 466. The defendant sold stickers, badges and similar items to the punk scene bearing National Socialist symbols crossed out, smashed or otherwise negated. Held that using a proscribed organization’s symbol in a depiction whose content plainly and unmistakably expresses opposition to the organization and rejection of its ideology does not run counter to the protective purpose of § 86a and therefore falls outside the offence, the opposition having to be recognizable at first glance. The conviction was set aside and the defendant acquitted. Agent-verified 2026-09-13 against the court’s own press release and the reported headnote. URL: <https://www.bundesgerichtshof.de/SharedDocs/Pressemitteilungen/DE/2007/2007036.html> (visited on 09/13/2026).
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- and fewer evacuated from a preferred species of shell than from an unpreferred one, which is the motivational trade-off the book cites: the noxious signal is weighed against the value of the shell rather than triggering a fixed response. Agent-verified 2026-09-18. The authors' own conclusion goes further, that the results are consistent with pain in these animals, and that further step is disputed – Diggles, Review of Some Scientific Issues Related to Crustacean Welfare, ICES Journal of Marine Science 76(1) (2019): 66–81, argues specifically that shell evacuation does not evidence pain and that the evidential bar has been set too low. The book's sentence claims the trade-off and not the pain, so it does not rest on the disputed step., pp. 1243–1246. doi: 10.1016/j.anbehav.2009.01.028.
- EPFL AI Center (2025). *Citizens' Assembly on AI*. Convened over two weekends in November 2025 by the EPFL AI Center with the Research Hub on Democratic Innovations at the University of Geneva and Demoscan. Forty randomly selected residents of French-speaking Switzerland, drawn to represent canton, age, education level and political interest, produced twenty proposals grouped in five areas; a survey of 734 residents ran alongside the deliberation. URL: <https://ai.epfl.ch/innovation/citizens-assembly-on-ai/> (visited on 09/13/2026).
- European Commission (2020). *White Paper on Artificial Intelligence: Public Consultation Towards a European Approach for Excellence and Trust*. Consultation ran 19 February–14 June 2020 and drew more than 1,200 responses from citizens, companies, and civil-society groups; fed into the legislative proposal that became the EU AI Act. URL: <https://digital-strategy.ec.europa.eu/en/library/white-paper-artificial-intelligence-public-consultation-towards-european-approach-excellence-and> (visited on 08/25/2026).
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- European Parliament and Council of the European Union (2024a). *Directive (EU) 2024/2853 of the European Parliament and of the Council on Liability for Defective Products and Repealing Council Directive 85/374/EEC*. Official Journal of the European Union, OJ L, 2024/2853. URL: <https://eur-lex.europa.eu/eli/dir/2024/2853/oj> (visited on 09/17/2026). Article 4(1) brings software within the definition of a product: "all movables... it includes electricity, digital manufacturing files, raw materials and software." **It says software and stops.** The phrase "including AI system providers within the meaning of Regulation (EU) 2024/1689" is Recital 13, which is not an operative provision, and an earlier version of this note wrongly attributed it to the definition. Article 8 makes the economic operator liable irrespective of fault. Article 2(1): "This Directive shall apply to products placed on the market or put into service after 9 December 2026"; member states must transpose by the same date, whose article number is not pinned here, one reading returning Article 19 while the EUR-Lex text conversion truncates at Article 17. Article 6(1) is the closed damage list, and its first limb is wider than a list of physical harms sounds: "(a) death or personal injury, **including medically recognised damage to psychological health**"; (b) damage to, or destruction of, any property, except... property used exclusively for professional purposes; (c) destruction or corruption of data that are not used for professional purposes." Article 9 is disclosure of evidence. Article 10(2) presumes defectiveness on failure to disclose, non-compliance with mandatory safety requirements, or obvious malfunction in foreseeable use; Article

- 10(3) presumes the causal link where the product is defective and the damage is of a kind typically consistent with that defect; Article 10(4) lets a court presume either where technical or scientific complexity makes the claimant's proof excessively difficult **and** the claimant demonstrates that the defect or the link is likely. Every article number above was read against the consolidated text at EUR-Lex on 2026-09-18.
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- Friston, Karl J. (2009). “The Free-Energy Principle: A Rough Guide to the Brain?” In: *Trends in Cognitive Sciences* 13.7, pp. 293–301. doi: 10.1016/j.tics.2009.04.005. URL: <https://doi.org/10.1016/j.tics.2009.04.005> (visited on 08/25/2026). States that “one can see easily how constructs like memory, attention, value, reinforcement and salience might disclose their simple relationships within this framework” — Friston’s extension of the free-energy principle to memory and attention specifically, which the 2010 Nature Reviews Neuroscience paper does not make explicit.
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- Germano, Fabrizio, Vicenç Gómez, and Francesco Sobbrío (2026). “Ranking for Engagement: How Social Media Algorithms Fuel Misinformation and Polarization”. In: *Journal of Public Economics* 255, p. 105589. doi: 10.1016/j.jpubeco.2026.105589. URL: <https://www.sciencedirect.com/science/article/pii/S0047272726000253> (visited on 08/25/2026). Also circulated as CESifo Working Paper No. 10011 and NBER/SSRN preprints from 2022 onward; shows engagement-weighted ranking increases both misinformation and polarization relative to accuracy-weighted ranking.
- Gesetz über das Bundesverfassungsgericht* (Mar. 12, 1951). § 16: *Plenarentscheidung*; § 31: *Bindungswirkung und Gesetzeskraft*. § 31(1): “Die Entscheidungen des Bundesverfassungsgerichts binden die Verfassungsorgane des Bundes und der Länder sowie alle Gerichte und Behörden.” § 31(2) gives the decision the force of law in the proceedings listed there. § 16(1): where one Senate would depart on a question of law from the legal view held in a decision of the other, the plenum of the Court decides. Agent-verified 2026-09-17. URL: <https://www.gesetze-im-internet.de/bverfgg/> (visited on 09/17/2026).
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- Trump v. Cook* (2026). No. 25A312, 609 U.S. — (June 29, 2026). Interim docket; the disposition is "Application for stay denied." 5–4. Roberts, C.J., for the Court, joined by Sotomayor, Kagan, Kavanaugh and Jackson, JJ.; Kavanaugh and Jackson, JJ., concurring; Thomas, J., Alito, J. (joined by Gorsuch, J.), and Barrett, J., dissenting. Upholds the Federal Reserve Act's for-cause removal protection, on the Fed's descent from the First and Second Banks of the United States, its budget free of congressional control and its privately owned regional banks, and holds that a Governor is owed notice and a hearing before removal. The constitutional holding is real and is what Thomas, J., dissents from, but the case is disposed of narrowly, on the ground that Cook did not receive the process the statute required: the Court states that it has "not addressed the facts, as they have yet to be found or analyzed under the relevant legal standards," and Kavanaugh, J., concurring, opens that "[t]oday's interim ruling does not decide whether the President may lawfully remove Governor Cook for cause." The dispute is live — on 5 August 2026 the President notified Cook that he was again considering

- her removal, giving her 21 days to respond and citing this opinion as his authority for the procedure. Agent-verified 2026-08-30. URL: <https://www.law.cornell.edu/supremecourt/text/25A312> (visited on 08/29/2026).
- Trump v. Hawaii* (2018). No. 17–965, 585 U.S. 667 (June 26, 2018). 5–4. Roberts, C.J., for the Court, joined by Kennedy, Thomas, Alito and Gorsuch, JJ.; Breyer, J., dissenting, joined by Kagan, J.; Sotomayor, J., dissenting, joined by Ginsburg, J. Upholds Proclamation No. 9645 restricting entry from several named countries. On *Korematsu*, the Court says both that “[w]hatever rhetorical advantage the dissent may see in doing so, *Korematsu* has nothing to do with this case” and that “*Korematsu* was gravely wrong the day it was decided, has been overruled in the court of history, and — to be clear — ‘has no place in law under the Constitution,’” the inner quotation being Jackson, J., dissenting in *Korematsu*. Sotomayor, J., dissenting: the Court “redeploys the same dangerous logic underlying *Korematsu* and merely replaces one ‘gravely wrong’ decision with another.”
- Trump v. Slaughter* (2026). No. 25-332, 609 U.S. ____ (June 29, 2026). 6–3. Roberts, C.J., for the Court; Gorsuch, J., concurring; Sotomayor, J., dissenting, joined by Kagan and Jackson, JJ. Holds the Federal Trade Commission Act’s for-cause removal protection unconstitutional and overrules *Humphrey’s Executor v. United States* (1935): “If anything more is left of *Humphrey’s*, we overrule it.” URL: <https://www.law.cornell.edu/supremecourt/text/25-332> (visited on 08/29/2026).
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- U.S. Government Accountability Office (Apr. 2026). *Artificial Intelligence Acquisitions: Agencies Should Collect and Apply Lessons Learned to Improve Future Procurements*. GAO-26-107859. URL: <https://www.gao.gov/products/gao-26-107859> (visited on 09/18/2026). Read from the report’s own HTML full text at <https://files.gao.gov/reports/GAO-26-107859/index.html>; the PDF was never needed, and the product page alone is what defeated an earlier attempt. Scope: thirteen AI acquisitions at four agencies — the Departments of Defense, Homeland Security and Veterans Affairs and the General Services Administration — and, from that sample, “44 contracts and agreements awarded between September 2018 and February 2025.” **The forty-four are contracts and agreements, not acquisitions.** Cited for the other-transaction finding: “OTAs do not always include default provisions ordinarily in FAR contracts,” contracting officials deciding what protections to include, and GAO having “previously found that agencies’ use of OTAs carries the risk of reduced accountability and transparency compared to FAR-based contracts.” Also for the vehicle behind the agreements section 10.3 turns on:

- “In July 2025, DOD awarded OTAs with \$200 million ceilings to four leading AI companies.” The six key procurement challenge areas identified by the selected agencies are access to subject matter experts; protections for government data and intellectual property rights; traditional acquisition time frames and contract approaches; requirements definition and contract terms; early testing and continuous evaluation; and artificial intelligence pricing and overall cost. The report’s central finding, which the book does not use, is that the agencies were not yet systematically collecting lessons learned because their policies did not require it. Maven appears in it as the National Geospatial-Intelligence Agency’s programme, with lessons-learned sections citing its practices; that material is not used either. Agent-verified 2026-09-18.
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